



BASE24 Remote BankingTM

Standard Interface Support Manual

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Preface

The ***BASE24 Remote Banking Standard Interface Support Manual*** describes the standard message format used by BASE24 Remote Banking for communicating with RBSI-compliant devices. The manual also describes the transactions supported for the Remote Banking Standard Interface, provides transaction flow examples, and documents conditions that a vendor needs to know in order to implement an RBSI-compliant device software application successfully with BASE24 Remote Banking. BASE24 Remote Banking includes the BASE24-telebanking, Enhanced Telebanking Application, and BASE24-billpay products.

The message specifications in this manual are based on the International Organization for Standardization (ISO) 8583-1993 standard.

Audience

This manual is intended for BASE24 Remote Banking customers and software application vendors who are responsible for implementing a device that functions with BASE24 Remote Banking.

Prerequisites

The structure and content of the BASE24-telebanking Standard External Message is patterned after the standard external message developed by ISO, which is described in the ISO 8583-1993 standard, ***Financial Transaction Card Originated Messages—Interchange Message Specification***. The BASE24-telebanking Standard External Message carries both BASE24-telebanking and BASE24-billpay information.

Because of the similarity between the BASE24 external message and the ISO standard, familiarity with the above document, commonly referred to as “ISO 8583,” is recommended.

Additional Documentation

The BASE24 documentation set is arranged so that each BASE24 manual presents a topic or group of related topics in detail. When one BASE24 manual presents a topic that has already been covered in detail in another BASE24 manual, the topic is summarized and the reader is directed to the other manual for additional information. Information has been arranged in this manner to be more efficient for readers who do not need the additional detail and at the same time provide the source for readers who require the additional information. This manual contains references to the following BASE24 publications:

- The ***BASE24 Base Files Maintenance Manual*** describes the screens and fields used to maintain the External Message File (EMF) and the Bank Table (BANK).
- The ***BASE24 External Message Manual*** provides default External Message File (EMF) settings for the Integrated Authorization Server, ISO Host Interface, and From Host Maintenance processes.
- The ***BASE24 Logical Network Configuration File Manual*** provides descriptions of assigns and params used by BASE24-telebanking and BASE24-billpay.
- The ***BASE24 Core Files and Tables Maintenance Manual*** describes the screens and fields used to configure the Remote Banking Standard Interface. It also provides detailed descriptions for customer, account, and personal data returned from BASE24 in several transaction responses.
- The ***BASE24-billpay Tables Maintenance Manual*** describes the screens and fields used to configure BASE24-billpay.
- The ***BASE24 Remote Banking Customer Service Support Manual*** describes the fields used to maintain the Loan Table (LOAN) and Miscellaneous Request Table (MISC) as well as columns in the Future Table (FUTR) and Recurring Table (RCUR).
- The ***BASE24 Remote Banking Standard Interface Message Specifications*** contains message layouts for each transaction supported by the Remote Banking Standard Interface.
- The ***BASE24 Remote Banking Transaction Processing Manual*** describes the controls for transaction processing and transaction flows for transactions received from an RBSI-compliant device.
- The ***BASE24 Integrated Server Transaction Security Manual*** provides detailed information on configuring PIN verification and dynamic key management for RBSI-compliant devices.

- The ***BASE24 ITS Kernel Configuration Manual*** contains information on using large messages.

This manual also references the ***NET24-XPNET Network Control Command Reference Manual*** for detailed information on the protocols supported by the XPNET system.

In addition to the BASE24 and NET24 manuals, this manual also contains references to the following ISO and American National Standards Institute (ANSI) publications:

- The ISO 8583-1993 standard, ***Financial Transaction Card Originated Messages - Interchange Message Specification***, documents a common interface for financial transaction messages between acquirers and issuers.
- The ISO 4217 standard, ***Codes for the Representation of Currencies and Funds***, describes currency codes.
- The ISO 3166 standard, ***Codes for the Representation of Names of Countries***, describes country codes.
- The ANSI X9.17-1985 standard, ***Financial Institution Key Management (Wholesale)*** and the ISO 8732-1988 standard, ***Banking-Key Management (Wholesale)***, describe dynamic key management techniques.
- The 1988 ANSI X3.38 standard, ***Identification of the States, the District of Columbia, and the Outlying and Associated Areas of the United States for Information Interchange***, describes state codes.
- The 1988 ANSI X3.31 standard, ***Structure for the Identification of the Counties and County Equivalents of the United States and its Outlying and Associated Areas for Information Interchange***, describes U.S. county and county equivalent codes.

Software

This manual documents standard processing for the following BASE24 Remote Banking product releases:

- BASE24-billpay release 1.3
- BASE24-telebanking release 1.3

Customer-specific code modifications (that is, CSMs) may result in processing that differs from the material presented in this manual. The customer is responsible for identifying and noting these changes.

Manual Summary

The following is a summary of the contents of this manual.

“Conventions Used in this Manual” follows this preface and describes notation and documentation conventions necessary to understand the information in the manual.

Section 1, “Introduction,” provides an overview of the BASE24 Remote Banking products, supported transactions, and business uses for the Remote Banking Standard Interface. It also discusses the ISO message standard, standard message variations, the use of large messages, supported protocols, and responsibilities for implementing the Remote Banking Standard Interface.

Section 2, “External Message Types,” describes each of the message types supported by the Remote Banking Standard Interface.

Section 3, “External Message Defaults,” documents the BASE24 Remote Banking Standard Interface message defaults.

Section 4, “External Message Data Elements,” explains the structures used in describing the BASE24 Remote Banking external message data elements. It also provides tables listing all of the data elements used for each message type.

Section 5, “Data Elements 1 Through 64,” contains descriptions for data elements 1 through 64 of the BASE24-telebanking Standard External Message.

Section 6, “Data Elements 65 Through 128,” contains descriptions for data elements 65 through 128 of the BASE24-telebanking Standard External Message.

Appendix A, “ISO Conversion Tables,” shows how processing codes are converted to and from ISO standard codes by the Integrated Authorization Server process.

Appendix B, “Transaction Flows,” presents a series of diagrams illustrating the transaction flow between an RBSI-compliant device and BASE24 Remote Banking for various type of transactions and processing conditions.

Appendix C, “Transaction Capabilities and Restrictions,” details the various capabilities, requirements, and restrictions of the Remote Banking Standard Interface.

Appendix D, “Recommended Transaction Sequences,” provides recommended transaction sequencing to perform various remote banking functions.

A glossary that contains definitions of terms used in this manual follows the last appendix.

Publication Identification

Two entries appearing at the bottom of each page uniquely identify this BASE24 publication. The publication number (for example, TB-DT100-01 for the ***BASE24 Remote Banking Standard Interface Support Manual***) appears on every page to assist readers in applying updates to the appropriate manual. The publication date (for example, 09/2000 for September, 2000) identifies the last time material on a particular page has been revised. This date can be used to verify that the reader has the most current information and is particularly helpful when dealing with HELP24 to answer operating questions.

The print record, located on the page immediately following the title page, is also helpful when verifying that a manual contains the most current information. An entry is added to the print record each time a *product* publication is modified in any way. Each print record entry includes the date of the change, a unique version number, and the type of change (new, update, reissue, or preliminary). Version 1 is assigned to a *new* manual. Later versions can be updates or reissues, depending on the changes being made. An *update* typically involves changes on a small number of pages throughout the manual. A *reissue* occurs when all pages of the manual are replaced because changes affect a large number of pages. A manual is identified as *preliminary* when it is released to customers on a limited basis, but it has not yet reached final form.

ACI Worldwide Inc.

Conventions Used in this Manual

This section explains how data elements are documented in this manual.

Documentation Template

The data elements used in the standard external message exchanged between BASE24 Remote Banking and an RBSI-compliant device or client are described in detail in sections 5 and 6. A standard format has been used for describing these data elements. The standard format is described below.

Format: States the attributes for the data element. The values used to represent the attributes are based on the ISO 8583-1993 standards and are shown below:

A = Alphabetic characters
N = Numeric characters
P = Pad characters, space
S = Special characters
AN = Alphabetic and numeric characters
AS = Alphabetic and special characters
NP = Numeric and space (pad) characters
NS = Numeric and special characters
ANP = Alphabetic, numeric, and space (pad) characters
ANS = Alphabetic, numeric, and special characters

For fixed-length fields, the above characters are followed by the number of characters in the field (for example, N 10 indicates that the field is a fixed-length, 10-position, numeric field).

For variable-length fields, the above characters are followed by two, three, or four dots and the maximum number of characters that can be carried in the field. The number of dots indicates the number of positions in the length field prior to the actual data. For example, A ..21 indicates that the field is a variable-length, alphabetic field, that can be from zero to 21 characters in length. The length field is two positions.

Note: All fixed length numeric data elements are assumed to be right justified with leading zeroes. All other fixed length data elements are left justified with trailing blanks.

An X indicates a prefix of “ ” or “+” for positive amounts, or “-” for negative amounts, and is always associated with a numeric amount data element. For example, X + N12 in the Additional Amounts (P-54) data element indicates a prefix of “ ”, “+” or “-”, plus 12 digits.

Date and time formats are shown using the following values:

YY or YYYY	= Year
MM	= Month
DD	= Day
hh	= Hour
mm	= Minute
ss	= Second
hh	= Hundredths of a second
mmmmmm	= Microseconds (millionths of a second)

Used by: States the BASE24 Remote Banking products that use the data element.

The above fields are followed by a description of how the data element is used. General information concerning the data element, which is the same for all BASE24 Remote Banking products, is presented first. Information that varies by BASE24 Remote Banking product is presented, by product, following the general information.

Section 1

Introduction

This section provides an introduction to the BASE24 Remote Banking Standard Interface.

The BASE24 Remote Banking Standard Interface provides a common interface for BASE24-telebanking, Enhanced Telebanking Application, and BASE24-billpay transactions initiated at different types of devices. This common interface allows customers to select the functionality that they require for their particular business application. BASE24-billpay and Enhanced Telebanking Application are optional add-on packages to BASE24-telebanking. The Enhanced Telebanking Application provides a subset of BASE24-billpay functionality. If a BASE24 customer has not licensed BASE24-billpay or the Enhanced Telebanking Application, the billpay or enhanced telebanking functionality described in this document is not applicable for that BASE24 customer. Customers who have licensed the Enhanced Telebanking Application should review all BASE24-billpay functionality described in this manual.

Implementing this functionality requires the cooperation of three separate parties—ACI, the device vendor, and the device owner. Each of these parties has certain responsibilities when implementing the BASE24 Remote Banking Standard Interface, but the device owner is primarily responsible for selecting the options that are applicable to the owner's business environment. The owner then works with the device vendor to develop the requirements and scripting for the device.

The BASE24 Remote Banking Standard Interface is based upon the ISO 8583-1993 standard, *Financial Transaction Card Originated Messages—Interchange Message Specification*. It is recommended that users of this manual obtain this ISO standard for reference.

Remote Banking Products

The BASE24 Remote Banking product family currently includes two products—BASE24-telebanking and BASE24-billpay. The BASE24-telebanking product provides online banking services (e.g., balance inquiries, check clearance inquiries, statement downloads and statement closing downloads, check stop payments, and immediate transfers between two accounts) to customers using remote devices (e.g., a touch-tone telephone or personal computer) or calling a customer service representative (CSR) or branch employee. The BASE24-billpay product provides automated payment and scheduled transfer services (e.g., immediate, future, and recurring payments, future and recurring transfers, and associated inquiries) to customers using remote devices or calling a CSR or branch employee. The BASE24-billpay product provides files that can be used as input to programs that generate checks and ACH output for payments; however, the BASE24-billpay product does not currently include these programs. The BASE24-billpay product also provides facilities for signing up new customers and accessing customer information from a CSR- or branch employee-operated device.

Optional Enhanced Telebanking product modules provide financial institutions and service providers the ability to support all BASE24-telebanking transactions as well as the subset of banking transactions that are provided with the BASE24-billpay product. These product modules consist of a subset of the BASE24-billpay software.

Enhanced Telebanking provides for the scheduling and real-time history of immediate, one-time future transfers and recurring transfers. The product modules monitor future and recurring transfers, and on the scheduled transaction date, automatically initiate the transfer transaction. When Enhanced Telebanking initiates a transfer, BASE24 routes the transaction to the appropriate authorizing entity for authorization. Various reports are available directly related to these transfer transactions.

Enhanced Telebanking also supports customer and account activation and maintenance transactions, financial institution and financial institution branch list inquiry transactions, loan application transactions, and miscellaneous request transactions.

The BASE24-telebanking software provides two functions used to support the services provided by both the BASE24-telebanking and BASE24-billpay products—customer verification and funds authorization. Customer verification involves identifying the customer and verifying his or her authenticity. Funds authorization involves checking the customer's account to ensure that sufficient funds are available to perform a transfer or payment.

Because the BASE24-billpay product relies on the BASE24-telebanking software to support its services, BASE24-billpay cannot be installed without BASE24-telebanking. The BASE24-telebanking product can be installed without the BASE24-billpay product; however, scheduled transfer capabilities and customer signup functions are available only if both products are installed.

This manual documents transaction processing for the BASE24-telebanking, Enhanced Telebanking, and BASE24-billpay products. Throughout this manual, references are made to BASE24, BASE24 Remote Banking, BASE24-telebanking, Enhanced Telebanking, and BASE24-billpay, as follows:

- BASE24 includes BASE24-telebanking, Enhanced Telebanking, and BASE24-billpay, as well as other BASE24 products.
- BASE24 Remote Banking includes BASE24-telebanking, Enhanced Telebanking, and BASE24-billpay.
- BASE24-telebanking indicates the BASE24-telebanking product with or without the Enhanced Telebanking or BASE24-billpay product. For example, the BASE24-telebanking Integrated Authorization process withdraws funds from a customer's account for a BASE24-telebanking transaction, Enhanced Telebanking, or BASE24-billpay transaction.
- Enhanced Telebanking indicates the subset of BASE24-billpay transactions described above.
- BASE24-billpay indicates the BASE24-billpay product only.

Business Uses

Because of the flexibility provided by the BASE24 Remote Banking Standard Interface transaction set, and the extensive set of transactions available, financial institutions or service providers can use BASE24 Remote Banking transactions in a variety of ways. The same interface can be used for both client applications and customer service representative or branch employee applications. By developing different client desktop applications that use the same interface, a financial institution or service provider can minimize the number of interfaces to support and maintain. In fact, some ACI vendor partners use the BASE24 Remote Banking Standard Interface for both their client and call center or branch banking applications.

Caution: Due to the comprehensive nature of the Remote Banking Standard Interface transaction set, thoughtful consideration should be given as to which transactions are implemented for each application. For example, some transactions, such as the customer add transaction, may be appropriate for use by customer service representatives or branch employees only.

Client Applications

The BASE24 Remote Banking Standard Interface has been expanded in this release to support all the functionality that existed in prior releases of the BASE24-telebanking and BASE24-billpay Customer Service interfaces, as well as support for some new search and update capabilities. By implementing this release, BASE24 Remote Banking customers can provide the following functionality from any remote client (e.g., web, personal computer, interactive voice response system, screen phone, kiosk, or personal digital assistant):

- The ability to add, update, activate, and deactivate customer ID relationships and customer accounts
- The ability to add, update, activate, and deactivate customer vendors, including the ability to list available vendors
- The ability to inquire on the balance of multiple accounts in one transaction
- The ability to stop payment on a check or range of checks
- The ability to download statement and statement closing information for an account
- Increased search capabilities for inquiry transactions to improve the response time and information provided

- The ability to update additional information on a scheduled payment or transfer such as account numbers
- The ability to submit a loan application or miscellaneous message to a financial institution
- The ability to inquire into a list of the available financial institutions and branches in the system

Call Center or Branch Banking Applications

BASE24 Remote Banking transactions can be integrated directly into existing customer service call center or branch banking systems. By building a Remote Banking Standard Interface directly from the call center or branch banking application, a customer service representative or branch employee can perform any remote banking transaction or inquire into payment or account history without using a Tandem Pathway or 3270 terminal. By implementing this release, BASE24 Remote Banking customers can also provide the following enhanced functionality that was not available in prior releases of the BASE24-telebanking and BASE24-billpay Customer Service interface:

- The ability to activate, deactivate, and delete customer accounts
- The ability to activate customer vendors
- The ability to inquire on customer vendor masks
- The ability to inquire on the balance of multiple accounts in one transaction
- The ability to stop payment on a check or range of checks
- The ability to download statement and statement closing information for an account
- Increased search capabilities for inquiry transactions to improve the response time and information provided
- The ability to update additional information on a scheduled payment or transfer such as account numbers

Remote Banking Transactions

The BASE24 Remote Banking products support a complete set of transactions, identifying each of these transactions by a six-character processing code. The first two characters of the processing code contain the transaction code, which indicates the type of transaction to be performed (e.g., 30 for an available funds inquiry transaction). The next two characters of the processing code indicate the type of the first account on which the transaction is to be performed (e.g., 10 for a savings account), while the last two characters indicate the second account on which the transaction is to be performed (e.g., 20 for a checking account). If an account type is not applicable for a transaction, 00 is used for the account type in the processing code. For more information on processing codes, refer to the description of the Processing Code (P-3) data element in section 5.

All transactions are considered to be standard or nonstandard. Standard transactions are defined by a default set of processing codes. These codes are contained in the Processing Code Definition File (PCDF) provided with the standard BASE24 Remote Banking products. All other transactions are considered to be nonstandard and use a value of 00 for each of the account types. Refer to the ***BASE24 Remote Banking Transaction Processing Manual*** for a complete list of default processing codes. Refer to the ***BASE24 Core Files and Tables Maintenance Manual*** for more information on the PCDF.

Nonstandard Transactions

Nonstandard transactions permit the customer or ACI to define transactions in addition to those defined in the set of default processing codes. Nonstandard transactions are categorized based on who is responsible for them, as follows:

- User-defined transactions can be defined by the customer. These transactions have a transaction code of Ux, where x represents a value of 0 through 9 or A through Z.
- Custom development transactions can be defined as part of customer-specific development work (i.e., RPQs) done by ACI personnel. These transactions can have a transaction code of Px, Qx, or Rx, where x represents a value of 0 through 9 or A through Z.
- Distributor transactions can be defined by the ACI regional marketing groups. These transactions can have a transaction code of Xx, Yx, or Zx, where x represents a value of 0 through 9 or A through Z.

- ACI-use-only transactions are reserved for use with products other than BASE24 Remote Banking. These transactions can have a transaction code of Wx, where *x* represents a value of 0 through 9 or A through Z.

Standard Transactions

This subsection describes each of the standard transactions that can be processed by the Remote Banking Standard Interface, including their capabilities and restrictions. Each transaction is identified by its descriptive name and its transaction code in parentheses. Each description also indicates whether the transaction is available when BASE24-telebanking, BASE24-billpay, and the Enhanced Telebanking Application are installed. Transaction descriptions are provided in alphabetical order by descriptive name, following the table summary below.

Detailed information on transaction capabilities and restrictions are provided in appendix C of this manual. Recommended sequencing for transactions is provided in appendix D.

The following table summarizes which transactions are available for each of the BASE24 Remote Banking Standard Interface product options. A check mark in columns three through five indicates that the transaction is available for that product option. Financial transactions (i.e., transactions that actually debit or credit one or more account balances) are denoted with an asterisk (*) in the Transaction Description column. Transfer and payment transactions are not considered to be financial transactions until accounts are actually debited or credited. Therefore, the transactions used to schedule transfers and payments are not considered to be financial transactions.

Note: Internal transactions are initiated from within the BASE24 Remote Banking system. No external message formats are required for these transactions.

Tran Code	Transaction Description	TB	Enhanced TB	BP
30	Available funds inquiry	✓		
3A	Check clearance inquiry	✓		
3B	Last count of debit/credit transactions inquiry	✓		

Tran Code	Transaction Description	TB	Enhanced TB	BP
3C	Last count of account by source transactions inquiry	✓		
3D	Last count of check transactions inquiry	✓		
3E	Last count of debit transactions inquiry	✓		
3F	Last count of credit transactions inquiry	✓		
3G	Last count of account transfer transactions inquiry	✓		
3H	Customer vendor list inquiry			✓
3J	Scheduled payments list inquiry			✓
3K	Scheduled transfers list inquiry		✓	✓
3L	History inquiry – payments and transfers		✓	✓
3M	Scheduled transactions list inquiry		✓	✓
3N	Customer account list inquiry	✓		
3P	Multiple account balance inquiry	✓		
3Q	Statement closing download	✓		
3R	Statement download	✓		
40	Transfer – immediate*	✓		
4A	Transfer – future (internal transaction)*		✓	✓
4B	Transfer – recurring (internal transaction)*		✓	✓
50	Payment – immediate (internal transaction)*			✓
5A	Payment – future (internal transaction)*			✓
5B	Payment – recurring (internal transaction)*			✓
90	PIN change	✓		
91	PIN verify	✓		

Tran Code	Transaction Description	TB	Enhanced TB	BP
9A	Schedule payment – immediate			✓
9B	Schedule payment – future			✓
9C	Schedule payment – recurring			✓
9D	Schedule transfer – future		✓	✓
9E	Schedule transfer – recurring		✓	✓
9F	Scheduled payment delete			✓
9G	Scheduled transfer delete		✓	✓
9H	Scheduled payment update			✓
9J	Scheduled transfer update		✓	✓
9K	Customer add		✓	✓
9L	Customer inquiry		✓	✓
9M	Customer update		✓	✓
9N	Customer vendor add			✓
9P	Customer vendor deactivate			✓
9Q	Customer vendor update			✓
9R	Master vendor list inquiry			✓
9S	Master vendor add (internal transaction)			✓
9T	Customer vendor request mailing			✓
9V	Loan application		✓	✓
9W	Miscellaneous request		✓	✓
9X	History inquiry – all transactions		✓	✓
A0	Check stop payment add	✓		

Tran Code	Transaction Description	TB	Enhanced TB	BP
B0	Customer account list inquiry (internal transaction)		✓	✓
B1	Bank list inquiry		✓	✓
B2	Bank branch list inquiry		✓	✓
B3	Cancel immediate payment (internal transaction) Note: BASE24-billpay processes this transaction as a reversal.			✓
B4	Miscellaneous requests inquiry		✓	✓
B5	Customer account add		✓	✓
B6	Customer account activate		✓	✓
B7	Customer account deactivate		✓	✓
B8	Customer account delete		✓	✓
B9	Customer account update		✓	✓
BB	Customer activate		✓	✓
BC	Customer deactivate		✓	✓
BD	Customer vendor activate			✓
BE	Customer vendor mask verify inquiry			✓
BF	Customer verify	✓		
BG	Customer select	✓		

Available Funds Inquiry (30)

The available funds inquiry transaction provides balance and account information on a single account or on two accounts simultaneously. This transaction is also referred to as a balance inquiry transaction. The number and type of balances returned for this transaction depends on the number and type of accounts on which the balances are requested.

The possible balances that can be returned are as follows:

Type of Balance	Description
Ledger 1	Displays the current ledger balance for account 1.
Available 1	Displays the current available funds for account 1.
Amount Owed	Displays the current amount owed for a loan account.
Amount Due	Displays the current amount due for a loan account.
Available Credit	Displays the current amount of credit available for a credit or line of credit account.
Ledger 2	Displays the ledger balance for account 2.
Available 2	Displays the current available funds for account 2.

Note: Some or all of these balances may not be displayed if they are not supported by the customer's financial institution.

The available funds inquiry transaction is a BASE24-telebanking transaction.

Bank List Inquiry (B1)

The bank list inquiry transaction returns a list of financial institutions configured in the Bank Table (BANK). Many of the BASE24 Remote Banking transactions require a valid financial institution identifier (FIID) to be provided with the transaction. This inquiry provides the customer with the ability to view the list of

FIIDs to choose from when signing up for remote banking service. It also allows some client server applications (e.g., web and interactive voice response system servers) to update a local database that can be used to provide valid FIIDs before sending transactions to BASE24. Refer to section 6 for a list of all the information returned in the transaction response for each financial institution.

Note: Some financial institutions may have multiple FIIDs due to acquisitions or mergers or because of different product offerings (e.g., insured versus uninsured accounts). BASE24 does not support transfers between accounts belonging to different FIIDs.

Because the customer does not have to be a customer of the financial institution, neither the customer ID or a customer account number are required for this transaction. If a customer ID is not provided with the transaction, BASE24-billpay uses the default customer ID from the Bank Table (BANK). This is the customer ID logged for the transaction.

If an FIID is not provided with the transaction, the default acquirer FIID specified in the BP-ACQ-FIID param in the LCONF is used. For more information on this param, refer to the ***BASE24 Logical Network Configuration File Manual***.

Information for up to 19 financial institutions can be returned in each response message if large messages are used. Otherwise, information for only up to four financial institutions are returned in each response message.

The bank list inquiry transaction is a BASE24-billpay transaction that is also provided with the Enhanced Telebanking Application.

Bank Branch List Inquiry (B2)

The bank branch list inquiry transaction returns a list of financial institution branches configured in the Bank Branch Table (BKBR). This inquiry provides the customer with the ability to view the list of valid bank branch numbers to choose from for a particular FIID when signing up for remote banking service. It also allows some client server applications (e.g., Web and interactive voice response system servers) to update a local database that can be used to provide valid bank branch numbers before sending transactions to BASE24. Refer to section 6 for a list of all the information returned in the transaction response for each financial institution branch.

Because the customer does not have to be a customer of the financial institution, neither the customer ID or a customer account number are required for this transaction. If a customer ID is not provided with the transaction, BASE24-billpay uses the default customer ID from the Bank Table (BANK). This is the customer ID logged for the transaction.

Information for up to 16 bank branches can be returned in each response message if large messages are used. Otherwise, information for only up to four bank branches are returned in each response message.

The bank branch list inquiry transaction is a BASE24-billpay transaction that is also provided with the Enhanced Telebanking Application.

Check Clearance Inquiry (3A)

The check clearance inquiry transaction allows a customer to determine whether a specific check for the customer has cleared. Customers must provide the number of the check and can optionally provide the amount of the check.

If the check has cleared, check details are returned in an approved response. If the check has not cleared, check details are not returned in an approved response. Refer to “Transaction History Response Data” in section 6 for a list of all the information returned in the transaction response.

The check clearance inquiry transaction is a BASE24-telebanking transaction.

Check Stop Payment Add (A0)

The check stop payment add transaction allows a customer to stop payment on a specific check or a range of checks. Customers must provide the number of a specific check or the high and low numbers for a range of checks as well as a date on which the stop payment request expires. Customers can optionally provide the amount and date of the check or checks to be stopped, the date and time the stop request was added, and a brief description of the stop payment request. If the date and time for the stop request is not provided, it defaults to the current date and time. Only one stop payment request can be made in each request message.

This transaction must always be routed to a host for authorization. If the host is unavailable, the BASE24 system denies the transaction.

The check stop payment add transaction is a BASE24-telebanking transaction.

Customer Activate (BB)

The customer activate transaction changes a customer's verify status from initial to verified, sets the begin date for the customer ID to the current calendar date and the end date to the value defined in the default customer data for the customer's financial institution, and sets the timestamp for the customer's verify status to the current calendar date. The begin date specifies the first date a customer ID can be used in processing, and the end date specifies the last date a customer ID can be used in processing. A customer must also have a verify status of verified for most transactions to be performed. Only one customer can be activated in each customer activate transaction. Refer to "Customer Response Data" in section 6 for a list of all the information returned in the transaction response.

If the customer already has a verify status of verified when this transaction is received, BASE24 Remote Banking checks a reactivation flag in the Bank Table (BANK) and a reactivation flag provided in the transaction request (see "Customer Activate Data" in section 5) to determine whether the activate request is allowed by the RBSI client. Both flags must allow reactivation for the activate request to be allowed. If so, the begin date, end date, and customer verify status timestamp are updated. If not, the transaction is denied. For more information on customer reactivation, refer to appendix C and the *BASE24 Base Files Maintenance Manual*.

The customer activate transaction is a BASE24-billpay transaction that is also provided with the Enhanced Telebanking Application.

Customer Add (9K)

The customer add transaction adds a customer to the database. Customer data is added to the Customer Table (CSTT), Personal Information Table (PIT), and Customer/Personal ID Relation Table (CPIT). Only one customer can be added in each customer add transaction. Refer to section 6 for a list of the information required in a customer add request and returned in a customer add response.

If a bank branch is not provided with this transaction, it defaults to spaces.

The customer add transaction is a BASE24-billpay transaction that is also provided with the Enhanced Telebanking Application.

Customer Deactivate (BC)

The customer deactivate transaction sets the end date for the customer ID to the current calendar date minus one, thereby effectively removing the customer from service. The end date specifies the last date a customer ID can be used in processing. After a customer has been deactivated, the Customer Activate transaction can be used to reactivate the customer, if customer reactivation is allowed by the customer's financial institution. Only one customer can be deactivated in each customer deactivate transaction. Refer to "Customer Response Data" in section 6 for a list of all the information returned in a customer deactivate response.

The customer deactivate transaction is a BASE24-billpay transaction that is also provided with the Enhanced Telebanking Application.

Customer Inquiry (9L)

The customer inquiry transaction retrieves customer ID and personal information about a customer from the BASE24 database. There are two ways that this transaction can be used, depending on whether a customer ID and personal ID are provided in the request. An FIID must always be provided with this transaction.

- If a customer ID and personal ID are provided, BASE24-billpay retrieves a row from the Customer Table (CSTT) using the customer ID and a row from the Personal Information Table (PIT) using the personal ID. The personal ID is optional. If the personal ID is not provided in the request, the value of the personal ID is assumed by BASE24 to be the same as the customer ID.
- If a customer ID and a personal ID are not provided, BASE24-billpay retrieves the default CSTT and PIT rows associated with the FIID. This allows the device to retrieve default customer information when signing up new customers.

Only one customer can be inquired upon in each customer inquiry transaction. Refer to "Customer Response Data" in section 6 for a list of all the information returned in a customer inquiry response. The customer inquiry transaction is a BASE24-billpay transaction that is also provided with the Enhanced Telebanking Application.

Customer Update (9M)

The customer update transaction updates customer ID and personal information that resides on the BASE24 database. The personal ID is optional. If the personal ID is not provided in the request, the value of the personal ID is assumed by BASE24 to be the same as the customer ID. The information that can be updated includes the following:

- Family name (last name)
- Given name (first name)
- Middle initial
- Title
- Street or mailing address
- City
- State
- Country
- Postal code
- Home phone
- Work phone
- Other phone 1 and description
- Other phone 2 and description
- Birth date
- Language
- Alternate contact
- Mother's maiden name
- Government ID
- Customer information (user-defined, 3 fields)
- Customer type
- Customer profile
- Branch
- Default account number and type
- Service fee account number and type

Only one customer can be updated in each customer update transaction. Refer to section 6 for a list of the information required in a customer update request and returned in a customer update response. The customer update transaction is a BASE24-billpay transaction that is also provided with the Enhanced Telebanking Application.

Customer Account Activate (B6)

The customer account activate transaction allows the customer to activate a customer account after it has been configured for a customer ID. Once activated, the account can be used in transaction processing. Only one customer account can be activated in each transaction and the customer ID must have a verify status of verified before the transaction is allowed.

This transaction updates the begin date for the account to the current date and sets the end date for the account to the value defined in the default Customer Account Relation Table (CACT) row for the customer's financial institution. Refer to "Customer Account Response Data" in section 6 for a list of all the information returned in a customer account activate response.

The customer account activate transaction is a BASE24-billpay transaction that is also provided with the Enhanced Telebanking Application.

Customer Account Add (B5)

The customer account add transaction adds a new customer account to the BASE24 Remote Banking database. Only one customer account can be added in each customer account add transaction. Refer to section 6 for a list of the information required in a customer account add request and returned in a customer account add response.

The customer account add transaction is a BASE24-billpay transaction that is also provided with the Enhanced Telebanking Application.

Customer Account Deactivate (B7)

The customer account deactivate transaction allows a customer to deactivate an account that is associated with a customer ID. Once the account is deactivated it cannot be used in transaction processing. Only one customer account can be

deactivated in each transaction. Refer to “Customer Account Response Data” in section 6 for a list of all the information returned in a customer account deactivate response.

This transaction updates the end date for the account to the current calendar date minus one day.

Note: BASE24 Remote Banking does not check the Future Table (FUTR) to determine whether there are any outstanding transactions that use the customer account before the account is deactivated. Therefore, it may be beneficial to perform a scheduled transactions list inquiry transaction to determine whether there are any outstanding transactions prior to using this transaction.

The customer account deactivate transaction is a BASE24-billpay transaction that is also provided with the Enhanced Telebanking Application.

Customer Account Delete (B8)

The customer account delete transaction allows the customer to delete a customer account that is associated with a customer ID from the database. Once the account is deleted, it can no longer be used in transaction processing. Only one customer account can be deleted in each transaction and the transaction is allowed only if the customer has an inactive verify status. Refer to “Customer Account Response Data” in section 6 for a list of all the information returned in a customer account delete response.

The customer account delete transaction is a BASE24-billpay transaction that is also provided with the Enhanced Telebanking Application.

Customer Account List Inquiry (3N)

The customer account list inquiry transaction provides a list of all the active customer accounts configured to the system (i.e., either the host or BASE24) for a given customer ID. Refer to “Account List Response Data” in section 6 for a list of all the information returned in a customer account list inquiry response.

Note: It is not possible to have two inquiries where one is routed to a host and one is routed to BASE24 for the same financial institution.

Up to 15 accounts can be returned in each response message for both large and small messages.

The customer account list inquiry transaction is a BASE24-telebanking transaction.

Customer Account Update (B9)

The customer account update transaction updates an existing customer account at any time, as long as the customer verify status is verified. Only one customer account can be updated in each customer account update transaction. The information that can be updated includes the following:

- Account qualifier
- Debit allowed flag
- Credit allowed flag
- Inquiry allowed flag
- Account description

Refer to section 6 for a list of the information required in a customer account update request and returned in a customer account update response.

The customer account update transaction is a BASE24-billpay transaction that is also provided with the Enhanced Telebanking Application.

Customer Select (BG)

The customer select transaction allows customers to select a particular person or customer ID in the following cases:

- When more than one person is configured for the customer ID provided in a customer verification request (e.g., when a customer ID is assigned to a business and several different employees are allowed to use the ID).
- When the full or partial customer name is provided in the customer verification request. This transaction is always returned when an exact match is found for a customer name entered in a customer verify transaction or when one or more names similar in spelling are found for the name entered in a customer verify transaction. For the later case to occur, the BASE24 system must use a name algorithm (e.g., Soundex). For more information on name algorithms, refer to the PIT-NAM-ALGO param description in the ***BASE24 Logical Network Configuration File Manual***.

Information for up to six customers can be returned in each customer selection response. Multiple customer selection request and response messages may be required before the correct customer is selected. Refer to section 6 for a list of all the information returned in a customer select request and required in a customer select response.

The transaction is not routed or logged, and BASE24 does not check the Institution Definition File (IDF) for customer service PIN processing information, even if the transaction source in the request is set to IB (inbound). No PIN processing occurs for this transaction.

The customer select transaction is a BASE24-telebanking transaction.

Customer Vendor Activate (BD)

The customer vendor activate transaction allows the customer to activate a customer vendor after it has been configured for a customer ID. Once activated, the customer vendor can now be used in transaction processing. Only one customer vendor relationship can be activated in each transaction. Refer to “Customer Vendor Response Data” in section 6 for a list of all the information returned in a customer vendor activate response.

This transaction can be performed if the customer vendor has a verify status of either verified or initial. If the verify status is initial, BASE24 changes the status to verified and updates the verify status timestamp to the current date and time. BASE24 sets the begin date for the customer vendor to the current calendar date and sets the end date to the maximum date allowed.

The customer vendor activate transaction is provided with BASE24-billpay only.

Customer Vendor Add (9N)

The customer vendor add transaction adds a customer vendor to a customer’s vendor list at any time, regardless of the customer’s verify status. Only one customer vendor can be added in each transaction. Refer to section 6 for a list of the information required in a customer vendor add request and returned in a customer vendor add response.

The customer vendor add transaction is provided with BASE24-billpay only.

Customer Vendor Deactivate (9P)

The customer vendor deactivate transaction sets the end date for a customer vendor to the current calendar date minus one, thereby effectively preventing the customer from sending payments to the vendor. The end date specifies the last date a customer vendor can be used in processing. Only one customer vendor can be deactivated in each transaction. The customer's verify status is not checked for this transaction. Refer to "Customer Vendor Response Data" in section 6 for a list of all the information returned in a customer vendor deactivate response

The customer vendor deactivate transaction is provided with BASE24-billpay only.

Customer Vendor List Inquiry (3H)

The customer vendor list inquiry transaction provides a list of all customer vendors associated with a specified customer ID.

Up to four customer vendors can be returned in each response message if large messages are used. Otherwise, only one customer vendor can be returned in each response message. Refer to section 6 for a list of all the information returned in a customer vendor list inquiry response.

The customer vendor list inquiry transaction is provided with BASE24-billpay only.

Customer Vendor Mask Verify Inquiry (BE)

The customer vendor mask verify inquiry transaction provides a list of all the vendor masks associated with a specific vendor and vendor branch. This transaction will help customers and customer service representatives to enter the customer account with the vendor correctly, thereby reducing the time required to add a customer vendor if account masks are used. Each vendor can have multiple masks. A customer ID and vendor must be provided with this transaction. The customer account with vendor can optionally be provided with the request. If the account with vendor is provided, BASE24 includes an indicator for each mask returned that indicates whether the account number passed or failed the mask edit check.

Up to eight vendor masks can be returned in each response message if large messages are used. Otherwise, only two vendor masks can be returned in each response message. Refer to section 6 for a list of all the information returned in a customer vendor mask verify inquiry response.

The customer vendor mask verify list inquiry transaction is provided with BASE24-billpay only.

Customer Vendor Request Mailing (9T)

The customer vendor request mailing transaction provides a means to generate a hard copy listing (Customer Vendor List Report) of a customer's vendor list that can be mailed to them.

The Customer Vendor List Report is described in the ***BASE24 Remote Banking End-of-Period Processing and Reporting Manual***.

The customer vendor request mailing transaction is provided with BASE24-billpay only.

Customer Vendor Update (9Q)

The customer vendor update transaction serves two purposes—updating customer vendor information or updating vendor information. Only one customer vendor relationship can be updated in each transaction.

This transaction allows you to update the following customer vendor information for all customer vendors, regardless of the customer's verify status:

- Account with vendor
- Budget category
- Customer vendor short name
- User information 1
- User information 2

This transaction also allows you to update the actual vendor fields (columns) in the Vendor Table (VNDR) and Vendor Branch Table (VNDB). The following fields in these tables can be updated by any customer for most vendors. The only exceptions are public vendors with a vendor verify status of verified. Customers

can not update any of these fields because these vendors are used by many customers. Therefore, the vendor and vendor branch information for these vendors is in effect *owned* by the system owner, not an individual customer.

- Vendor name
- Vendor branch name
- Address fields 1 and 2
- City
- State
- Postal code
- Phone
- Account with vendor
- Customer vendor short name
- Budget category
- Customer vendor user information (2 fields)
- Vendor branch information (2 fields)
- Vendor branch remittance information (2 fields)
- Vendor information (2 fields)
- Vendor remittance information (2 fields)

Refer to section 6 for a list of the information required in a customer vendor update request and returned in a customer vendor update response.

The customer vendor update transaction is provided with BASE24-billpay only.

Customer Verify (BF)

The customer verify transaction provides a customer, customer service representative, or branch employee the ability to review customer specific information to confirm if the customer selected is the correct customer. A customer ID, customer name, or customer account number is required in a customer verification request. If the customer cannot be determined from the information provided in the request because there is more than one customer who fits the criteria provided, or if a customer name is provided, a customer select transaction may be required to select the customer.

Refer to section 6 for a list of the information required in a customer verify request and returned in a customer verify response.

The transaction is not routed or logged, and BASE24 does not check the Institution Definition File (IDF) for customer service PIN processing information, even if the transaction source in the request is set to IB (inbound). No PIN processing occurs for this transaction.

The customer verify transaction is a BASE24-telebanking transaction.

History Inquiry – All Transactions (9X)

The history inquiry – all transactions lists all types of transactions (financial, nonfinancial, or maintenance) logged in the History Table (HIST) for a customer ID. The customer, customer service representative, or branch employee can specify the customer ID, and whether only approved, denied, or both types of transactions are returned, either by reference number or by business date. Transactions are returned in descending order only.

- If the Accumulated Maximum field in the request is set to a nonzero value, BASE24 will return only that number of transactions. For example, if the request indicates that only the last six items can be returned, BASE24 will return no more than six HIST rows, even if more HIST rows exist for the customer ID. This maximum applies to the total number of items that can be returned, rather than the number of items that can be returned in each response message for the transaction. Multiple response messages may be required to complete the transaction.
- If the Accumulated Maximum field in the request is set to zero, BASE24 can return all rows that are associated with the customer ID that meet the search criteria.
- Transaction responses are returned in descending order only if the MAX HISTORY RECORDS field for the customer in the Customer Table is set to a nonzero value. If this field is set to zero for the customer, then transactions can be returned in either ascending or descending order.

Information for up to seven transactions can be returned in each response message if large messages are used. Otherwise, only one transaction can be returned in each response message. Refer to “History Inquiry Response Data” in section 6 for a list of the information returned in a history inquiry – all transactions response.

Note: Since transfer – immediate (40) transactions are not logged to the HIST, they are not included in the response to this transaction.

The history inquiry – all transactions is a BASE24-billpay transaction that is also provided with the Enhanced Telebanking Application.

History Inquiry – Payments and Transfers (3L)

The history inquiry – payments and transfers transaction allows customers to inquire on all financial transactions in the History Table (HIST), both approved and denied. Customers can select only approved transactions, only denied transactions, or both, by either customer reference number or business date. The default selection is by business date and time for all transactions in descending order.

- If the Accumulated Maximum field in the request is set to a nonzero value, BASE24 will only return that number of transactions. For example, if the request indicates that only the last six items can be returned, BASE24 will return no more than six HIST rows, even if more HIST rows exist for the customer ID. This maximum applies to the total number of items that can be returned, rather than the number of items that can be returned in each response message for the transaction. Multiple response messages may be required to complete the transaction.
- If the Accumulated Maximum field in the request is set to zero, BASE24 can return all rows that are associated with the customer ID that meet the search criteria.
- Transaction responses are returned in descending order only if the MAX HISTORY RECORDS field for the customer in the Customer Table is set to a nonzero value. If this field is set to zero for the customer, then transactions can be returned in either ascending or descending order.

Up to seven payment and transfer items can be returned in each response message if large messages are used. Otherwise, only one payment or transfer item can be returned in each response message. Refer to “History Inquiry Response Data” in section 6 for a list of the information returned in a history inquiry – payments and transfers response.

Note: Since transfer – immediate (40) transactions are not logged to the HIST, they are not included in the response to this transaction.

The history inquiry – payments and transfers transaction is a BASE24-billpay transaction that is also provided with the Enhanced Telebanking Application.

Last Count of Account by Source Transactions Inquiry (3C)

The last count of account by source transactions inquiry transaction returns a list of transactions performed for a particular account at a particular source (i.e., an ATM device, remote banking device, POS device, or teller device) or any device. The term *count* represents the maximum number of transactions that can be returned in each response message. The transactions are returned in reverse chronological order.

Up to 15 transactions can be returned in each response message for both large and small messages. Refer to “Transaction History Response Data” in section 6 for a list of the information returned in a last count of account by source transactions inquiry response.

The last count of account by source transactions inquiry transaction is a BASE24-telebanking transaction.

Last Count of Account Transfer Transactions Inquiry (3G)

The last count of account transfer transactions inquiry transaction returns a list of all transfers to or from a specified customer account. The term *count* represents the maximum number of transactions that can be returned in each response message. The transactions are returned in reverse chronological order. This transaction displays all transfers recorded in the Transaction History File (THF). For more information on the THF, refer to the ***BASE24 Remote Banking Transaction Processing Manual***.

Up to 15 transactions can be returned in each response message for both large and small messages. Refer to “Transaction History Response Data” in section 6 for a list of the information returned in a last count of account transfer transactions inquiry response.

The last count of account transfer transactions inquiry transaction is a BASE24-telebanking transaction.

Last Count of Check Transactions Inquiry (3D)

The last count of check transactions inquiry transaction returns a list of all check transactions for a specified customer account. The term *count* represents the maximum number of transactions that can be returned in each response message. The transactions are returned in reverse chronological order. Customers can optionally specify the amount of a check to further refine the search.

Up to 15 transactions can be returned in each response message for both large and small messages. Refer to “Transaction History Response Data” in section 6 for a list of the information returned in a last count of check transactions inquiry response.

The last count of check transactions inquiry transaction is a BASE24-telebanking transaction.

Last Count of Credit Transactions Inquiry (3F)

The last count of credit transactions inquiry transaction returns a list of all transactions that credited a specific customer account. The term *count* represents the maximum number of transactions that can be returned in each response message. The transactions are returned in reverse chronological order. Refer to the ***BASE24 Remote Banking Transaction Processing Manual*** for a list of all standard transactions that can credit an account.

Up to 15 transactions can be returned in each response message for both large and small messages.

The last count of credit transactions inquiry transaction is a BASE24-telebanking transaction.

Last Count of Debit Transactions Inquiry (3E)

The last count of debit transactions inquiry transaction returns a list of all transactions that debited a specified customer account. The term *count* represents the maximum number of transactions that can be returned in each response message. The transactions are returned in reverse chronological order. Refer to the ***BASE24 Remote Banking Transaction Processing Manual*** for a list of all standard transactions that can debit an account.

Up to 15 transactions can be returned in each response message for both large and small messages.

The last count of debit transactions inquiry transaction is a BASE24-telebanking transaction.

Last Count of Debit/Credit Transactions Inquiry (3B)

The last count of debit/credit transactions inquiry returns a list of transactions that debited or credited a specific customer account. The term *count* represents the maximum number of transactions that can be returned in each response message. The transactions are returned in reverse chronological order. Customers can optionally specify a range of dates for which transactions are to be reported. Refer to the ***BASE24 Remote Banking Transaction Processing Manual*** for a list of all standard transactions that can credit or debit an account.

Up to 15 transactions can be returned in each response message for both large and small messages.

The last count of debit/credit transactions inquiry transaction is a BASE24-telebanking transaction.

Loan Application (9V)

The loan application transaction allows a customer to submit a loan application. The customer does not have to be an existing remote banking customer (i.e., have access to a customer ID). When BASE24 receives this transaction, the data is written to the Loan Table (LOAN), which can then be extracted for processing by another system. Refer to “Loan Application Request Data” in section 6 for a list of the information required in a loan application request.

A valid financial institution identifier (FIID) is required for this transaction. ACI suggests that a bank list inquiry transaction be performed prior to this transaction to obtain the FIID. Because the individual requesting the loan does not have to be an existing remote banking customer of the financial institution, neither a customer ID or a customer account number are required for this transaction. If a customer ID is not provided with the transaction, BASE24-billpay uses the default customer ID from the Bank Table (BANK). This is the customer ID logged for the transaction.

The loan application transaction is a BASE24-billpay transaction that is also provided with the Enhanced Telebanking Application.

Master Vendor List Inquiry (9R)

The master vendor list inquiry is a transaction that is used by both customers and customer service representatives to determine if a vendor is already included in the BASE24-billpay database. A customer ID is required for this transaction.

The transaction returns most of the information that resides in a Vendor Table (VNDR) and associated Vendor Branch Table (VNDB) row. The transaction allows users to find potential vendors that already reside on the database when setting up their initial customer vendors. This should result in fewer duplicate vendors being added to the database.

Vendors are searched for by vendor name in ascending or descending order. The default search direction is ascending. In addition, searches can be filtered to further refine the search. Refer to “Master Vendor List Inquiry Data” in section 5 for a list of all the information required in a master vendor list inquiry request.

Refer to “Master Vendor List Response Data” in section 6 for a list of the information returned in a master vendor list inquiry response.

Up to five vendors can be returned in each response message if large messages are used. Otherwise, only one vendor is returned.

The master vendor list inquiry transaction is provided with BASE24-billpay only.

Miscellaneous Request (9W)

The miscellaneous request transaction allows a customer, customer service representative, or branch employee to send a message to a financial institution. The transaction is logged to the database for retrieval at a later time.

Refer to “Miscellaneous Request Data” and “Miscellaneous Response Data” in section 6 for a list of the information required in a miscellaneous request and returned in a miscellaneous request response, respectively.

The miscellaneous request transaction is a BASE24-billpay transaction that is also provided with the Enhanced Telebanking Application.

Miscellaneous Requests Inquiry (B4)

The miscellaneous requests inquiry transaction allows a customer, customer service representative, or branch employee to review any outstanding miscellaneous requests for a given customer. Requests can be searched for in ascending or descending order by transaction source date. The default search direction is descending. Refer to “Miscellaneous Requests Inquiry Data” in section 5 for a list of the information required in a miscellaneous requests inquiry request.

Information for up to five miscellaneous requests can be returned in each response message if large messages are used. Otherwise, only one miscellaneous request can be returned in each response message. Refer to “Miscellaneous Requests Inquiry Response Data” in section 6 for a list of the information returned in a miscellaneous requests inquiry response.

The miscellaneous requests inquiry transaction is a BASE24-billpay transaction that is also provided with the Enhanced Telebanking Application.

Multiple Account Balance Inquiry (3P)

The multiple account balance inquiry transaction allows a customer to obtain the current balances of all configured accounts for a customer ID. The account number must be provided in the request for each account on which the customer wants to obtain a balance. Therefore, ACI suggests that a customer account list inquiry transaction be performed prior to this transaction to obtain the customer account numbers.

Up to 13 account numbers can be specified in each request message and up to 13 balances can be returned in each response message for both large and small messages. There is no limit to the total number of account balances that can be displayed in a multiple account balance inquiry transaction. The customer or CSR can continue to request to display account balances until there are no more applicable rows in the CACT.

Refer to “Multiple Account Information Request Data” and “Multiple Account Balance Response Data” in section 6 for a list of the information required in a multiple account balance inquiry request and returned in a multiple account balance inquiry response, respectively.

Note: It is not possible to have two inquiries where one is routed to a host and one is routed to BASE24 for the same financial institution.

The multiple account balance inquiry transaction is a BASE24-telebanking transaction.

PIN Change (90)

The PIN change transaction allows a customer to change the existing personal identification number (PIN) or select the initial PIN for a customer. The PIN CHANGE ALLOWED field in the IDF indicates whether customer service representatives or branch employees are allowed to perform this transaction on behalf of a customer.

The PIN change transaction is a BASE24-telebanking transaction.

PIN Verify (91)

The PIN verify transaction verifies that the personal identification number (PIN) entered by the customer is correct. The PIN REQUIRED field in the IDF dictates whether customer service representatives or branch employees are required to verify a customer's PIN.

The PIN verify transaction is a BASE24-telebanking transaction.

Schedule Payment – Future (9B)

The schedule payment – future transaction schedules the one-time payment of funds from a customer account to a customer vendor (also known as a payee or biller) on a specified date in the future. Only one future payment can be scheduled in each transaction.

The schedule payment – future transaction is provided with BASE24-billpay only.

Schedule Payment – Immediate (9A)

The schedule payment – immediate transaction schedules the immediate payment of funds from a customer account to a customer vendor (also known as the payee or biller). Only one payment can be scheduled in each transaction.

The schedule payment – immediate transaction is provided with BASE24-billpay only.

Schedule Payment – Recurring (9C)

The schedule payment – recurring transaction schedules the payment of funds from a customer account to a customer vendor (also known as a payee or biller) on a recurring basis (e.g., every two weeks, monthly, or quarterly) for a specified period of time (e.g., 12 months). Only one recurring payment can be scheduled in each transaction.

The schedule payment – recurring transaction is provided with BASE24-billpay only.

Scheduled Payment Delete (9F)

The scheduled payment delete transaction allows the customer to delete a previously scheduled payment. Only one scheduled payment can be deleted in each transaction.

The scheduled payment delete transaction is provided with BASE24-billpay only.

Scheduled Payment Update (9H)

The scheduled payment update transaction allows a customer to update a scheduled (future or recurring) payment. This transaction can also be used to skip the next payment. Only one scheduled payment can be updated in each transaction.

The following data fields can be updated in this transaction:

- Amount
- Next payment date
- Total number of recurring payments
- Recurring period (i.e., weekly, every two weeks, twice a month, monthly, every four weeks, every two months, quarterly, semi-annually, and annually)
- Vendor specific data
- From account number
- From account type

The scheduled payment update transaction is provided with BASE24-billpay only.

Scheduled Payments List Inquiry (3J)

The scheduled payments list inquiry transaction provides a list of all scheduled payments for a specified customer ID or application account.

Up to nine payments can be returned in each response message if large messages are used. Otherwise, only up to two payments can be returned in each response message.

The scheduled payments list inquiry transaction is provided with BASE24-billpay only.

Schedule Transfer – Future (9D)

The schedule transfer – future transaction schedules the movement of funds from one account to another account to be performed on a specified date in the future. The account debited by the transfer is the *from* account, while the account credited by the transfer is the *to* account. This type of transfer is defined as an internal transfer. Note that external transfers (i.e., transfers between two accounts owned by different financial institutions) can be performed as BASE24-billpay payments, where the *to* account is defined as the account with vendor and the financial institution owning the *to* account is defined as the vendor. Only one future transfer can be scheduled in each transaction.

The schedule transfer – future transaction is a BASE24-billpay transaction that is also provided with the Enhanced Telebanking Application.

Schedule Transfer – Recurring (9E)

The schedule transfer – recurring transaction schedules the movement of funds from one account to another account to be performed on a recurring basis (e.g., every two weeks, monthly or quarterly) for a specified period of time (e.g., 12 months). The account debited by the transfer is the *from* account, while the account credited by the transfer is the *to* account. This type of transfer is defined as an internal transfer. Note that external transfers (i.e., transfers between two accounts owned by different financial institutions) can be performed as BASE24-billpay payments, where the *to* account is defined as the account with vendor and the financial institution owning the *to* account is defined as the vendor. Only one recurring transfer can be scheduled in each transaction.

The schedule transfer – recurring transaction is a BASE24-billpay transaction that is also provided with the Enhanced Telebanking Application.

Scheduled Transfer Delete (9G)

The scheduled transfer delete transaction allows a customer to delete a previously scheduled transfer. Only one scheduled transfer can be deleted in each transaction.

The scheduled transfer delete transaction a BASE24-billpay transaction that is also provided with the Enhanced Telebanking Application.

Scheduled Transfer Update (9J)

The scheduled transfer update transaction allows a customer to update a scheduled (future or recurring) transfer. This transaction can also be used to skip the next transfer. Only one scheduled transfer can be updated in each transaction.

The following data fields can be updated in this transaction:

- Amount
- Next transfer date
- Total number of recurring transfers
- Recurring period (i.e., weekly, every two weeks, monthly, quarterly, semi-annually, and annually)
- From account number
- From account type
- To account number
- To account type

The scheduled transfer update transaction is a BASE24-billpay transaction that is also provided with the Enhanced Telebanking Application.

Scheduled Transfers List Inquiry (3K)

The scheduled transfers list inquiry transaction provides a list of all scheduled transfers for a specified customer ID.

Up to 10 transfers can be returned in each response message if large messages are used. Otherwise, only up to two transfers can be returned in each response message.

The scheduled transfers list inquiry transaction is a BASE24-billpay transaction that is also provided with the Enhanced Telebanking Application.

Scheduled Transactions List Inquiry (3M)

The scheduled transactions list inquiry transaction provides a list of all transfer and payment transactions for a specified customer ID that are scheduled to be performed in the future, including both one-time and recurring transactions.

Up to 8 transfers and payments can be returned in each response message if large messages are used. Otherwise, only up to two transfers and payments can be returned in each response message.

The scheduled transactions list inquiry transaction is a BASE24-billpay transaction that is also provided with the Enhanced Telebanking Application.

Statement Download (3R)

The statement download transaction allows customers to receive statement information on an application account that typically appears as part of a paper statement. This information includes the opening and closing dates and balances for a statement period, as well as a detailed breakdown of all transactions that were posted for the account during the specified statement period. Using the statement download transaction, customers can receive transactions and balances on a daily basis if they wish. Because this information is only maintained on a host, this transaction must always be routed to a host.

Customers must provide a valid account number or qualifier, and can optionally provide a start date or end date for the statement period. If the start date is omitted, the period starts with the earliest transaction available for the account. If the end date is omitted, the period ends with the most recent transaction available.

Refer to “Statement Download Response Data” in section 6 for a list of the information returned in a statement closing download response.

Up to three transaction items can be returned in each response message for both large and small messages.

The statement download transaction is a BASE24-telebanking transaction.

Statement Closing Download (3Q)

The statement closing download transaction allows customers to receive statement closing information on an application account that typically appears as part of a paper statement.

For non-credit accounts, this information includes the opening and closing dates and balances for a statement period, as well as a detailed breakdown of debits, credits, fees, and interest for the account.

For credit accounts, this information includes the opening and closing dates and balances for a statement period, the next payment due date, the minimum payment due, finance charges, debit adjustments, credit limit, and a detailed breakdown of all debits, credits, and fees for the account.

Because this information is only maintained on a host, this transaction must always be routed to a host.

Customers must provide a valid account number or qualifier, and can optionally provide a start date or end date for the statement period. If the start date is omitted, the period starts with the earliest transaction available for the account. If the end date is omitted, the period ends with the most recent transaction available.

Refer to “Statement Closing Download Response Data” in section 6 for a list of the information returned in a statement closing download response.

Up to 16 transaction items can be returned in each response message for both large and small messages.

The credit card statement closing download transaction is a BASE24-telebanking transaction.

Transfer – Immediate (40)

The transfer – immediate transaction allows a customer to immediately move funds from one customer account to another customer account owned by the same financial institution. The account debited by the transfer is the *from* account, while the account credited by the transfer is the *to* account. This type of transfer is defined as an internal transfer. Note that external transfers (i.e., transfers between two accounts owned by different financial institutions) can be performed as

BASE24-billpay payments, where the *to* account is defined as the account with vendor and the financial institution owning the *to* account is defined as the vendor. Only one transfer can be made in each transaction.

The transfer – immediate transaction is a BASE24-telebanking transaction.

Transaction Prerequisites

Some BASE24-billpay or Enhanced Telebanking payment and transfer transactions may require an inquiry to be initiated before the transaction can be performed. The inquiry response contains data that may not be known but must be included in the payment or transfer transaction request. The following table lists the BASE24-billpay transactions that require a preceding inquiry.

Note: Third-party billpay service providers may have other transaction prerequisite requirements.

BASE24-billpay Transaction	Prerequisite Inquiry Transaction
Schedule payment – immediate [*]	Customer vendor list inquiry
Schedule payment – future [*]	
Schedule payment – recurring [*]	
Scheduled payment delete [†]	Scheduled payments list inquiry
Scheduled payment update [†]	
Scheduled transfer delete [†]	Scheduled transfers list inquiry
Scheduled transfer update [†]	

^{*} The customer vendor list inquiry is a prerequisite transaction only if a customer does not know the vendor information that is required for a schedule payment transaction.

[†] The scheduled payments list inquiry and scheduled transfers list inquiry are prerequisite transactions only if a customer does not know the customer reference number that is required for scheduled payment delete, scheduled payment update, scheduled transfer delete, and scheduled transfer update transactions.

Although the BASE24 Remote Banking Standard Interface does not enforce prerequisite transactions for any transaction, certain transaction sequencing is recommended for some remote banking functions. These recommended transaction sequences are provided in appendix D.

Transaction Processing

BASE24-telebanking and BASE24-billpay are transaction-based products. A transaction request is generated at the device based upon the information provided by a customer, customer service representative, or branch employee interacting with the device. This request is sent to BASE24 Remote Banking and processed while the device waits for the response to be sent back. When BASE24 receives the transaction, the customer is verified and the transaction is routed to the appropriate authorizing entity based upon the processing code. Depending on the transaction type, the transaction can be authorized on BASE24, on a host, or BASE24 can stand in if the host is unavailable.

Note: Some Remote Banking transactions cannot be sent to a host for authorization because they require large messages, and large messages are not supported by the ISO Host Interface process. Refer to the ***BASE24 ISO Host Interface Manual*** for a list of the Remote Banking transactions that can be sent to a host.

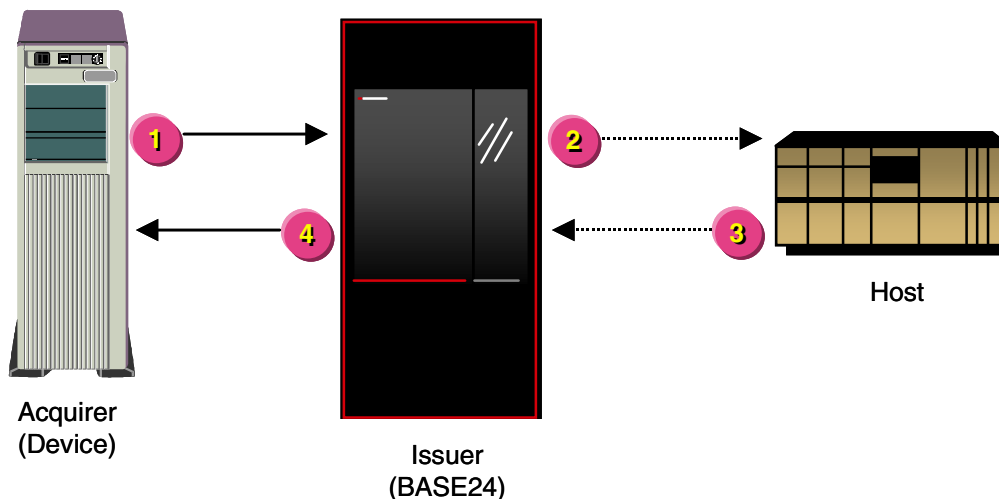
After the transaction is authorized, the response is sent to the device. If the transaction does not complete successfully, a reversal may be generated. Reversals are always generated for financial transactions (i.e., transfers or payments) and the following nonfinancial transactions if they do not complete successfully:

- PIN change
- Schedule payment (immediate, future, recurring)
- Schedule transfer (future or recurring)
- Scheduled payment delete
- Scheduled transfer delete
- Scheduled payment update
- Scheduled transfer update
- Customer activate
- Customer add
- Customer deactivate
- Customer update
- Customer account activate
- Customer account add
- Customer account deactivate

- Customer account delete
- Customer account update
- Customer vendor activate
- Customer vendor add
- Customer vendor deactivate
- Customer vendor update
- Miscellaneous request
- Loan application
- Check stop payment add

Reversals for all financial transactions are logged on BASE24 and are sent to the host. Nonfinancial inquiry transactions that do not successfully complete, and are subsequently reversed, are logged and sent to the host as determined by configuration parameters in the BASE24 database. You can configure some nonfinancial transactions to be discarded and not logged or sent to the host. For more information on configuring whether certain nonfinancial transaction reversals are discarded, refer to the ***BASE24 Remote Banking Transaction Processing Manual***.

The following diagram illustrates the general transaction processing flow. Steps 2 and 3 are optional for certain nonfinancial transaction reversals.



Customer and Account Identification

A transaction request must contain sufficient information to identify the customer and the type of transaction request. A personal identification number (PIN) is always required in a request originating at a remote banking device, but if the device is a *smart* device, the PIN can be verified locally at the device.

Note: Once the PIN is successfully verified for the first request in a multiple request session for a particular customer, the device can indicate that the PIN was verified locally in subsequent requests to avoid re-verification of the PIN on each request. See the “Call Processing” discussion below for details.

In addition to the PIN and a processing code, which identifies the type of transaction requested, a valid combination of customer ID, account number, account type and account qualifier is required. An account qualifier, also known as a short name, is used to identify an account number and account type. Account qualifiers are used to make entering data easier for the customer by allowing a short name to be entered instead of a complete account number and account type. The following combinations are valid:

- Customer ID/account number/account type
- Customer ID/account number
- Customer ID/account type (results in multiple account selection)
- Account number/account type
- Account number
- Customer ID/account qualifier
- Customer ID (if transaction is not associated with an account)

Although a customer ID is not required in the request, a customer ID must exist in the BASE24 database for each customer. If a customer ID is not entered, the account number must be unique within the BASE24 logical network and the account number is used to look up the customer ID. If account qualifiers are used instead of account numbers, a customer ID is required in the request. Note that sometimes the customer ID is referred to as the primary account number (PAN).

Call Processing

A customer can initiate several transactions in one call. Each transaction is sent as a separate request to BASE24. The customer may enter his or her PIN only once but the device must send the PIN in each request to BASE24 or indicate in the Point of Service Data Code (P-22) data element that the PIN has already been verified locally.

If BASE24 performs the authorization, the PIN is verified and the transaction is appropriately authorized based upon information contained in the BASE24 database. Advices for approved or denied transactions can optionally be forwarded from BASE24 to a host (see the “Transaction Processing” discussion provided earlier in this section).

Multiple Account Selection

If a transaction is initiated without an account and the authorization method requires an account, multiple account selection processing is optionally performed. Up to 15 accounts can be returned in each response message. If the customer has more accounts than can be returned in a single message to the device, a code is returned in the message indicating that more accounts exist. The customer can then request that additional accounts be returned. After the customer has selected an account or accounts, the transaction is authorized as in normal transaction processing.

If the transaction is a two-sided transaction (e.g., a transfer) and accounts must be selected for both sides, the *from* accounts, or account 1 accounts, are returned first to the device until either a customer has selected an account or all *from* accounts have been returned. At this time, the *to* accounts, or account 2 accounts, are presented to the customer. It is possible for both *from* and *to* accounts to be present in a single message to the device. However, *to* accounts can be present in the message only if there are no more *from* accounts to be presented to the customer.

Transaction Inquiry Continuation

BASE24 allows a customer to continue to request inquiry information after the response for the initial request is returned for the following inquiry transactions:

- BASE24-telebanking *last count of* transactions, customer account list inquiry, multiple account balance inquiry, statement download, and statement closing download transactions

- BASE24-billpay customer vendor list inquiry, scheduled payments list inquiry, scheduled transfers list inquiry, history inquiry – payments and transfers, scheduled transactions list inquiry, master vendor list inquiry, history inquiry – all transactions, bank list inquiry, bank branch list inquiry, miscellaneous requests inquiry, and customer vendor mask verify list inquiry transactions.

This allows customers to continue to retrieve additional information for the specified inquiry until a maximum limit is reached or there are no more matching records or rows in the database. A value containing information on how to reposition for more transaction information is returned to the device in each inquiry response message. If the customer requests additional information, this value is sent back unaltered to BASE24 in subsequent requests.

Security Overview

BASE24 Remote Banking performs the following security processing to ensure the security of messages received from an RBSI-compliant device:

- PIN encryption ensures that PIN information is maintained in a confidential, secure manner. PIN decryption and translation are used when an encrypted PIN needs to be verified.
- PIN verification ensures that the rightful customer is performing the transaction.
- Dynamic key management provides a means for automating the distribution of keys between BASE24 and the device.

PIN Encryption, Decryption, and Translation

PIN encryption, decryption, and translation functions are handled by the BASE24 processes that send transactions to and receive transactions from processing entities outside the system, such as an RBSI-compliant device. Encryption refers to the process of rendering PINs unintelligible for transmission and decryption refers to subsequently returning PINs to their original form. Translation occurs in situations involving more than one data format or key (e.g., an RBSI-compliant device that acquires a transaction uses one key and the external processor that authorizes the transaction uses another key). In this case, the data must be translated as necessary so that both entities can accurately read the data.

When a transaction is received from outside the BASE24 system, such as from an RBSI-compliant device, the PIN may have to be encrypted, decrypted, or translated from one encrypted form to another by BASE24.

Within BASE24 Remote Banking, PINs travel in the clear or are encrypted under a key either in software or using a hardware security module. When a hardware security module is used by BASE24 for PIN encryption, PINs are never in the clear within BASE24. When a hardware security module is not used for PIN encryption, BASE24 decrypts PINs upon entry to the system, and the PINs either remain in the clear or are re-encrypted in software until they are verified or sent to an external system.

For detailed information on PIN encryption, decryption, translation and the hardware security module models supported for BASE24 Remote Banking, refer to the ***BASE24 Integrated Server Transaction Security Manual***.

PIN Verification

PIN verification is the authentication of a customer (i.e., the user of a customer ID) by a transaction authorizer. BASE24 Remote Banking can verify PINs or PINs can be verified locally by the RBSI-compliant device before the transaction is forwarded to BASE24 Remote Banking.

PIN verification confirms that the PIN entered with a transaction is the actual PIN of the user involved in the transaction. If PINs are verified by BASE24, they can be verified in software or by a hardware security module. When verifying PINs in software, all associated PIN data required for verification can be in the clear or can be encrypted in software. When verifying PINs using a hardware security module, the PIN is carried in the system encrypted under a key known only to the hardware security module, and possibly the device that may need to know the clear value to encrypt the PIN.

A PIN is always required in a request originating from an RBSI-compliant device, unless the PIN is verified locally or the request is part of a multiple transaction session for a customer and the PIN was verified with the first request. See the “Call Processing” discussion provided earlier in this section for details.

BASE24 Remote Banking supports the following PIN verification methods:

- IBM 3624-format
- Diebold
- Identikey
- VISA PIN Verification Value (PVV)

For detailed information on PIN verification, PIN verification methods, and the hardware security module models supported for BASE24 Remote Banking, refer to the *BASE24 Integrated Server Transaction Security Manual*.

Dynamic Key Management

BASE24 Remote Banking supports dynamic key management as the means for automating the distribution of keys between BASE24 and the RBSI-compliant device. BASE24 Remote Banking always acts as the master key processor, and the device always acts as the slave key processor. The master key processor controls all key generation and the slave key processor has no responsibility for key generation. BASE24 Remote Banking can automatically change keys based on a number of factors. Thresholds can be established based on time, number of transactions, number of errors, number of consecutive errors, or any combination

of these factors. Operator commands are also available for changing keys as needed. Because the RBSI-compliant device always acts as the slave key processor, the device **cannot** request new keys.

The same cryptographic techniques used to protect PINs transmitted using a computer network are used to protect the integrity and secrecy of keys transmitted using the network. Encrypting keys before transmitting them protects their secrecy by preventing the unintentional or unauthorized disclosure of the keys.

Cryptographic Service Messages (CSMs) are used for moving key management data between BASE24 and the device when the keys are distributed. BASE24 Remote Banking sends new keys to the device in responses to authorization requests (financial or nonfinancial) from the device that contain PINs. Dynamic key management is supported for PIN encryption keys only; message authentication and data encryption keys are not supported.

Dynamic key management is based on the ANSI X9.17-1985 standard, *Financial Institution Key Management (Wholesale)*, and the ISO 8732-1988 standard, *Banking-Key Management (Wholesale)*. Readers of this manual should refer to these standards for further information.

For detailed information on dynamic key management, refer to the *BASE24 Integrated Server Transaction Security Manual*.

Threshold Configuration

New keys are generated and distributed when one of the following thresholds are met:

- A specified time duration elapses.
- A specified number of transactions are processed.
- A specified number of key synchronization errors occurs.
- A specified number of PIN translation errors, including key synchronization errors, occurs.
- A specified number of consecutive PIN translation errors occurs.

These thresholds are configured in the BASE24 database and are used in any combination. For example, if you want a new key to be generated every four hours, only the time duration parameter is set. If you want a new key to be generated when 200 transactions are processed, only the number of transactions parameter is set. If both parameters are set, a new key is generated when either

threshold is surpassed, whichever occurs first. Only transactions containing PINs count towards these thresholds; transactions without a PIN do not cause counters associated with the limits to be increased.

New keys can also be generated and distributed by an operator using a network control facility.

A key verification value (VVK) is used to verify the keys have been correctly exchanged.

Counters

Hexadecimal counters in the ANSI X9.17-1985 standard are used as a synchronization tool. In the BASE24 Remote Banking implementation of dynamic key management, however, these counters are purely informational and are not used for synchronization purposes.

The sender and receiver in a point-to-point environment maintain counters of which key is currently in use. The counter is used as a sequential identifier that is increased by one for each key exchange. If the sender's and receiver's counters do not match, the receiver should accept or adopt the sender's counter value. See the ANSI X9.17-1985 standard for more information on counters.

Supported Message Types

BASE24 Remote Banking supports the following types of messages between BASE24 and the RBSI-compliant device.

Requests

Requests originate at the RBSI-compliant device and can be financial requests or authorization requests. A financial request permits the application of the approved transaction amount to the customer's account for billing or posting. An authorization request allows approval or guarantee of funds by the issuer to the acquirer, but is not intended to permit the application of the approved transaction amount to the customer's account.

An example of a financial request is a transfer transaction. An example of an authorization request is an available funds inquiry or any of the transaction inquiries.

Responses

Responses can originate from BASE24 Remote Banking or from a host. BASE24 Remote Banking can be configured to stand in for the host if the host is unavailable.

Reversals

Reversals can originate from BASE24 Remote Banking or from the device. A reversal nullifies the effects of a previous financial or authorization transaction. The logging and sending of nonfinancial inquiry and PIN verify reversal transactions from BASE24 Remote Banking to a host can be configured in the BASE24 database. Reversals of all financial transactions and some nonfinancial transactions (e.g., PIN change, schedule payment, etc.) are always logged and sent to the host.

Administrative Notifications

Administrative notifications are used to notify either the issuer or the acquirer that an external ISO message cannot be read. In this case, the message is returned to the sender as a rejected message.

BASE24 Remote Banking can reject an external message sent by the device. Likewise, the device can reject an external message sent by BASE24 Remote Banking. Messages can be rejected for a variety of reasons, including the following: an invalid message type, an unsupported message type, an invalid value in a data element, and missing required data elements.

Rejected External Message Sent by Device

If BASE24 Remote Banking receives an external message from the device that it cannot process or reformat for internal use, it formats and sends an administrative notification message to the device. The original external message is returned, unmodified, to the device in the rejected message.

If the external message BASE24 Remote Banking rejects is a network management message containing Cryptographic Service Message (CSM) data, BASE24 Remote Banking continues to regard the key most recently accepted by the device as its current key.

Rejected External Message Sent by BASE24

If the device is unable to read a network management request response sent by BASE24 Remote Banking, the device can send an administrative notification back to BASE24 to reject the message.

The device can also send an administrative notification to reject an authorization or financial request response or a reversal advice response sent by BASE24 Remote Banking. If the rejected response to the device contained a Cryptographic Service Message (CSM), BASE24 Remote Banking sends a new key to the device in the next applicable response upon receipt of the administrative notification message from the device.

Network Management Messages

Network management messages (NMMs) are used to control system security and are initiated by the device. The device sends a network management request in response to an authorization or financial request response from BASE24 Remote Banking containing a new encryption key. The NMM request indicates whether the device has accepted or rejected the new key.

When BASE24 Remote Banking sends key data in an authorization or financial request response, it overwrites the oldest key with the new one. If the device then sends an NMM request indicating acceptance of the new key, BASE24 Remote Banking marks this latest key as the current key and the previous current key as the old key. If the NMM request indicates rejection of the new key, BASE24 Remote Banking continues to regard this latest key as the old key. The last key accepted by the device remains the current key. Regardless of whether the key was accepted or rejected, BASE24 responds to the device with an NMM request response.

The ISO Standard

The message formats used by the Remote Banking Standard Interface are based on the ISO 8583-1993 standard, ***Financial Transaction Card Originated Messages—Interchange Message Specification***, developed by the International Organization for Standardization (ISO). This specification defines an international standard for the exchange of electronic messages relating to financial transactions. This standard uses a concept called a *bit map*. In a bit-mapped message, each data element is assigned a position indicator in a control field, or bit map. The presence of a data element in a message is indicated by a one in the assigned position; the absence of a data element is indicated by a zero in the assigned position. The primary bit map is always present in a message and represents the most frequently used data elements. An optional secondary bit map represents infrequently used data elements. Each bit map is comprised of 64 bits.

The ISO 8583-1993 standard supports the use of both fixed-length and variable-length data elements. Fixed-length data elements are always of a set length. The length of variable-length data elements can vary according to the type of data carried in the element. Variable-length data elements contain a field length indicator that indicates the actual length of the data element (in bytes).

The ISO 8583-1993 standard also supports the concept of private data elements, which can be defined in any manner by an organization implementing the standard.

A message type identifier at the beginning of each message identifies the message version, message class, message function, and transaction originator.

For more information on the ISO 8583-1993 standard message format and the above concepts, refer to section 4.

ISO Standard Variations

The Remote Banking Standard Interface adheres to the ISO 8583-1993 standard whenever possible, but also varies from the standard in several significant ways.

First and foremost, the Remote Banking Standard Interface does not support binary data in the bit maps or in any data elements. All data in messages exchanged between BASE24 and an RBSI-compliant device must be in ASCII format. BASE24 substitutes alphanumeric characters for any message components defined as binary in the ISO standard. For example, BASE24 uses hexadecimal

characters to represent binary data in the bit maps, with each character representing four bits. Thus, each of the 64-bit maps are carried as 16 hexadecimal characters.

An example of a data element defined as binary in the ISO standard is the Personal Identification Number (PIN) Data (P-52) element. In the ISO standard, this data element is defined as 8 bytes of binary data. For the Remote Banking Standard Interface this data element is defined as 16 hexadecimal characters.

Another way in which the Remote Banking Standard Interface message formats differ from the ISO standard is the way in which the formats of some data elements are defined, as shown in the following table. The first column of the table identifies the data element, the second column of the table defines the format of the element in the ISO standard, and the third column of the table defines the format of the element for the Remote Banking Standard Interface.

Data Element	ISO Format	RBSI Format
Primary Account Number (P-2)	1–19 numeric	1–19 alphanumeric
Processing Code (P-3)	6 numeric	6 alphanumeric
Primary Account Number, Extended (P-34)	1–28 numeric	1–28 alphanumeric
Personal Identification Number (PIN) Data (P-52)	8 binary	16 alphanumeric
Key Management Data (S-96)	1–999 binary	1–999 alphanumeric and special
Account Identification 1 (S-102)	1–28 alphanumeric and special	1–19 numeric
Account Identification 2 (S-103)	1–28 alphanumeric and special	1–19 numeric

A third way in which the Remote Banking Standard Interface message formats differ from the ISO standard is the use of an external data block for certain transactions. A external data block is required for some BASE24-billpay or Enhanced Telebanking transactions because the transaction data will not fit within the defined field size of a data element. The Remote Banking Standard Interface uses an external data block when necessary for the Transaction Data, Private (S-125) data element. A flag in data element 125 indicates whether an external data block is present in the message. If the external data block is present in a

message, it immediately follows the data for the last data element in the message. The length of the data block is provided in a four-position length indicator field, immediately preceding the external data block.

Ordering the ISO 8583-1993 Standard

Within the United States, you can order a copy of the ISO 8583-1993 standard, ***Financial Transaction Card Originated Messages—Interchange Message Specification***, from the American National Standards Institute (ANSI). ANSI is the primary source and official sales agent for ISO standards in the United States.

The current address, telephone number, facsimile number, and Internet location for ANSI are as follows:

ANSI
11 West 42nd Street
New York, NY 10036
Telephone: (212) 642-4900, Facsimile: (212) 398-0023
Internet: www.ansi.org

Outside the United States, you can order a copy of the ISO 8583-1993 standard directly from the International Organization for Standardization at the following address or Internet location, or from the official sales agent for ISO standards in your country of business.

International Organization for Standardization (ISO)
1, rue de Varembe
Case postale 56
CH-1211 Genève 20
Switzerland
Internet: www.iso.ch

Large Message Support

The Remote Banking Standard Interface can utilize both small (4K) or large message (up to 10K) buffers. Some BASE24-billpay transactions always require large messages, while others can use small or large messages. For the latter group of messages, large message buffers allow BASE24 Remote Banking to send and receive more data in each message exchanged with an RBSI-compliant device than small message buffers, thereby improving message efficiency and response time.

Note: The external data block is used whether or not large message buffers are used in your BASE24 Remote Banking system. The external data block can carry more data when large messages are used.

The following BASE24-billpay transactions always require large messages.

Transaction Code	Transaction Message Description
9K*	Customer add
9L*	Customer inquiry
9M*	Customer update
9N	Customer vendor add
9P	Customer vendor deactivate
9Q	Customer vendor update
9T	Customer vendor request mailing
9V*	Loan application
9W*	Miscellaneous request
B1*	Bank list inquiry
B2*	Bank branch list inquiry
B4*	Miscellaneous requests inquiry
B5*	Customer account add
B6*	Customer account activate
B7*	Customer account deactivate

Transaction Code	Transaction Message Description
B8*	Customer account delete
B9*	Customer account update
BB*	Customer activate
BC*	Customer deactivate
BD	Customer vendor activate

The following BASE24-billpay transactions can utilize small or large messages:

Transaction Code	Transaction Message Description
3H	Customer vendor list inquiry
3J	Scheduled payments list inquiry
3K*	Scheduled transfers list inquiry
3L*	History inquiry—payments and transfers
3M*	Scheduled transactions list inquiry
9R	Master vendor list inquiry
B1*	Bank list inquiry
B2*	Bank branch list inquiry
B4*	Miscellaneous requests inquiry
BE	Customer vendor mask verify inquiry

* These transactions are included with the Enhanced Telebanking product modules.

A large-memory model toggle set when the Application Network Services object is compiled in integrated server processes, determines whether small messages or large messages are used in your BASE24 system. If the toggle is set, large messages are used. If the toggle is not set, small messages are used. For more information on this toggle and its impact on message size, refer to the ***BASE24 ITS Kernel Configuration Manual***.

Note: The above option is available for BASE24-telebanking software only. BASE24-billpay uses the same size of messages in responses that it receives in requests from BASE24-telebanking.

VRU Configuration Data (VCD) screen 18 contains several fields that dictate the maximum number of transaction items that can be returned in each response message from BASE24 Remote Banking. The range of valid values for these fields varies, depending on whether small or large messages are used. In general, more transaction items can be accommodated in each response message when large messages are used.

XPNET release 2.1 or greater supports the extended features of the Tandem D-series operating system that enable a process to use a large-memory model as well as a small-memory model. Integrated server processes running under control of the XPNET process can use either memory model. Using the small-memory model, objects can allocate space in the upper 64K bytes (32K words) of the data stack and use message buffers of 4096 bytes. Using the large-memory model, objects can allocate space within a flat segment. Flat segments allow memory space to be larger than 65,536 bytes, use message buffers of 32,000 bytes, and require a D30 or later operating system running on a Tandem NonStop Series/ RISC machine.

Supported Protocols

The Remote Banking Standard Interface can be configured to use any communications protocol supported by the XPNET system. Commonly used protocols include the following:

- SNA—Secondary LU Type 2
- TCP/IP
- X.25

The protocol selected by the customer depends on the line communications environment established between BASE24 Remote Banking and the RBSI-compliant device.

Authorized ACI personnel assist customers at installation or when a system is configured in determining the type of protocol best suited for the customer's particular requirements. For detailed information on protocols supported by the XPNET system and their usage, refer to the *NET24-XPNET Network Control Command Reference Manual*.

Implementation Responsibilities

Using the Remote Banking Standard Interface requires the cooperation of three parties: ACI, the customer, and the third-party software application vendor for an RBSI-compliant device. These responsibilities are outlined below.

Responsible Party			Task
Customer	Vendor	ACI	
✓			License the BASE24 Remote Banking Standard Interface from ACI.
	✓	✓	Sign a nondisclosure agreement with the software application vendor. ACI sends the vendor a nondisclosure agreement. The vendor must sign the nondisclosure agreement and return it to ACI.
✓	✓	✓	Receive the <i>BASE24 Remote Banking Standard Interface Support Manual</i>, <i>BASE24 Remote Banking Standard Interface Message Specifications</i>, and additional documentation from ACI.* The <i>BASE24 Remote Banking Standard Interface Support Manual</i>, <i>BASE24 Remote Banking Standard Interface Message Specifications</i>, and additional documentation are sent directly to the application vendor from ACI after ACI receives a signed nondisclosure agreement. The customer receives a separate copy of the <i>BASE24 Remote Banking Standard Interface Support Manual</i>, <i>BASE24 Remote Banking Standard Interface Message Specifications</i>, and additional documentation for their own use.
✓	✓		Review functionality of the Remote Banking Standard Interface. The customer and vendor must read the manual and understand the options available.

Responsible Party			Task
Customer	Vendor	ACI	
✓	✓		Develop a specification. The customer outlines the features that will be supported by the customer's RBSI-compliant devices, as follows: • Include transaction set supported. • Include message formats for all supported requests and responses. • Include PIN encryption and verification requirements. • Develop operator procedures for FIID database synchronization, if required.
✓		✓	Determine protocol requirements. The customer must decide on the type of line to be used in communication with the RBSI-compliant device and whether the line will be leased, dial, or network.
✓		✓	Review the specification with ACI. ACI reviews the specification developed by the customer to verify that the configuration decisions are complete and that the functionality desired is feasible.
✓	✓	✓	Develop implementation plan for the RBSI-compliant device, as follows: • Include vendor programming. • Include vendor test plan with ACI. • Include vendor delivery to customer.
	✓	✓	Complete vendor testing. The vendor and ACI test the RBSI-compliant device and the Remote Banking Standard Interface before delivery to the customer.

Responsible Party			Task
Customer	Vendor	ACI	
✓	✓	✓	Complete customer testing. The customer tests the RBSI-compliant device and the Remote Banking Standard Interface, along with the vendor and ACI.
✓			Review the RBSI-compliant device and software. The customer must accept the final product.
✓			Review procedures and make appropriate changes. The customer reviews the product and its procedures and makes any identified changes.

- * Vendors should also review the additional documentation listed in the preface of this manual to gain a full understanding of Remote Banking Standard Interface functionality and requirements.

Section 2

External Message Types

This section describes the various external message types supported between BASE24 Remote Banking and an RBSI-compliant device.

Supported Message Types

Message type codes are used to identify the general function of messages, and one type code is required in each message.

BASE24 Remote Banking supports the message types shown below for incoming and outgoing messages. The message types are divided according to ISO standard message classes.

Message Class: Authorization

The Remote Banking Standard Interface supports the following authorization message types. These message types are used for nonfinancial transactions.

Type	Description
1100	Authorization Request
1101	Authorization Request Repeat
1110	Authorization Request Response

Message Class: Financial Transaction

The Remote Banking Standard Interface supports the following financial authorization message types. These message types are used for financial transactions.

Type	Description
1200	Financial Transaction Request
1201	Financial Transaction Request Repeat
1210	Financial Transaction Request Response

Message Class: Reversal

The Remote Banking Standard Interface supports the following reversal message types for all financial transactions and certain authorization (nonfinancial) transactions:

Type	Description
1420	Acquirer Reversal Advice
1421	Acquirer Reversal Advice Repeat
1430	Reversal Advice Response

Message Class: Administrative

The Remote Banking Standard Interface supports the administrative message type of 1644 (Administrative Notification) for messages rejected by the RBSI-compliant device or BASE24 Remote Banking.

Message Class: Network Management

The Remote Banking Standard Interface supports the following network management message types. These message types are used for dynamic key management.

Type	Description
1804	Network Management Request
1805	Network Management Repeat
1814	Network Management Response

Interactive and Noninteractive Messages

The message descriptions that follow in this section identify the appropriate category for the message. There are two general categories into which messages fall: interactive and noninteractive. Interactive messages are those message types transmitted and responded to while a transaction is taking place. Noninteractive messages are those message types transmitted after a transaction has taken place and where there is no urgency implied for the response.

Issuers and Acquirers

The message descriptions that follow in this section identify the appropriate routing for the message. Message routing is documented in terms of issuer and acquirer.

An issuer is the party in a message exchange representing the transaction authorizer (i.e., who is, or is acting on behalf of, the institution that issued the customer ID or account).

An acquirer is the party in a message exchange representing the customer ID or account acceptor (i.e., who originally initiated the transaction).

With the Remote Banking Standard Interface, the RBSI-compliant device is always the acquirer and BASE24 Remote Banking is always the issuer. BASE24 Remote Banking can also act as an acquirer if a transaction is routed to a host for authorization; however, the Remote Banking Standard Interface is not involved in communicating with a host. The ISO Host Interface process is responsible for communicating with a host and uses message types based on the ISO 8583-1993 standard, *Financial Transaction Card Originated Messages—Interchange Message Specification*.

Authorization Messages

This subsection describes the authorization messages supported by the Remote Banking Standard Interface.

1100 Authorization Request

Category: Interactive

Routing: From acquirer to issuer

An Authorization Request (0100) message requests approval authorization for a nonfinancial transaction. It allows approval or guarantee of funds by the issuer to the acquirer, but it is not intended to permit the application of the approved transaction amount to the customer's account for billing or posting.

An Authorization Request Response (1110) message is expected in return for the 1100 message, either approving or denying the request.

1101 Authorization Request Repeat

Category: Noninteractive

Routing: From acquirer to issuer

An Authorization Request Repeat (1101) message is identical to an Authorization Request (1100) message, except that it denotes to the receiver that it is a possible duplicate message. An 1101 message is used when a response is expected to an 1100 message but is not received within a specified time frame.

An Authorization Request Response (1110) message is expected in return for the 1101 message, either approving or denying the request.

1110 Authorization Request Response

Category: Interactive

Routing: From issuer to acquirer

An Authorization Request Response (1110) message is returned in response to an Authorization Request (1100) or Authorization Request Repeat (1101) message to approve or deny the request.

Financial Transaction Messages

This subsection describes the financial transaction messages supported by the Remote Banking Standard Interface.

1200 Financial Transaction Request

Category: Interactive

Routing: From acquirer to issuer

A Financial Transaction Request (1200) message requests approval for a transaction that, if approved, can be immediately applied to the account of the customer for billing or posting.

A Financial Transaction Request Response (1210) message is expected in return for the 1200 message, either approving or denying the request.

1201 Financial Transaction Request Repeat

Category: Noninteractive

Routing: From acquirer to issuer

A Financial Transaction Request Repeat (1201) message is identical to a Financial Transaction Request (1200) message, except that it denotes to the receiver that it is a possible duplicate message. A 1201 message is used when a response is expected to a 1200 message but is not received within a specified time frame.

A Financial Transaction Request Response (1210) message is expected in return for the 1201 message, either approving or denying the request.

1210 Financial Transaction Request Response

Category: Interactive

Routing: From issuer to acquirer

The Financial Transaction Request Response (1210) message is returned in response to a Financial Transaction Request (1200) or Financial Transaction Request Repeat (1201) message to approve or deny the request.

Reversal Messages

This subsection describes the reversal messages supported by the Remote Banking Standard Interface.

1420 Acquirer Reversal Advice

Category: Noninteractive

Routing: From acquirer to issuer

An Acquirer Reversal Advice (1420) message reverses an earlier transaction or authorization. The RBSI-compliant device sends 1420 messages when an Authorization Request (1100) message or Financial Transaction Request (1200) message was approved by BASE24 Remote Banking, but the transaction did not complete as approved.

A Reversal Advice Response (1430) message is expected in return for the 1420 message to acknowledge the reversal.

1421 Acquirer Reversal Advice Repeat

Category: Noninteractive

Routing: From acquirer to issuer

An Acquirer Reversal Advice Repeat (1421) message is identical to an Acquirer Reversal Advice (1420) message, except that it denotes to the receiver that it is a possible duplicate message. A 1421 message is used when a response is expected to a 1420 message but is not received within a specified time frame.

A Reversal Advice Response (1430) message is expected in return for the 1421 message to acknowledge the reversal.

1430 Reversal Advice Response

Category: Noninteractive

Routing: From issuer to acquirer

A Reversal Advice Response (1430) message acknowledges an Acquirer Reversal Advice (1420) message or an Acquirer Reversal Advice Repeat (1421) message.

Administrative Messages

This subsection describes the administrative messages supported by the Remote Banking Standard Interface.

1644 Administrative Notification

Category: Noninteractive

Routing: From acquirer to issuer or from issuer to acquirer

An Administrative Notification (1644) message is used to notify either the issuer or the acquirer that an external ISO message could not be properly read. In this case, the message is returned to the sender as a rejected message.

BASE24 Remote Banking can reject a message received from an RBSI-compliant device if it cannot process or reformat the message for internal use. The original external message is returned unmodified to the device in the administrative notification.

The RBSI-compliant device can reject a message received from BASE24 Remote Banking if it cannot read the message. If the message contains a Cryptographic Service Message (CSM), BASE24 Remote Banking resends the key change indicator and the CSM to the device in the next applicable response upon receipt of the message from the device.

Network Management Messages

This subsection describes the network management messages supported by the Remote Banking Standard Interface.

1804 Network Management Request

Category: Interactive

Routing: From acquirer to issuer

A Network Management Request (1804) message is used to send dynamic key management messages.

This message is sent from the RBSI-compliant device to BASE24 Remote Banking to indicate whether the new inbound PIN key received in a new key message was determined to be valid. If the new key was determined to be valid, this message contains a Response Service Message in the request message. If the new key was determined to be invalid, this message contains an Error Service Message in the request message.

A Network Management Request Response (1814) message is expected in return for the 1804 message to acknowledge receipt of the request.

1805 Network Management Request Repeat

Category: Noninteractive

Routing: From acquirer to issuer

A Network Management Request Repeat (1805) message is identical to a Network Management Request (1804) message, except that it denotes to the receiver that it is a possible duplicate message. An 1805 message is used when a response was expected to a 1804 message but was not received within a specified time frame.

A Network Management Request Response (1814) message is expected in return for the 1805 message to acknowledge receipt of the request.

1814 Network Management Request Response

Category: Interactive

Routing: From issuer to acquirer

A Network Management Request Response (1814) message is returned in response to a Network Management Request (1804) or Network Management Request Repeat (1805) message.

The NMM ENABLED field on VRU Configuration Data (VCD) screen 1 must be set to a value of Y for the device interface instance to enable BASE24 Remote Banking to send this message.

Section 3

External Message Defaults

The data elements included in each incoming and outgoing external message exchanged between BASE24 and an RBSI-compliant device can be configured individually for each external message type. This configuration is accomplished using the External Message File (EMF), which specifies the combinations of data elements to be included in the external messages for each interface instance.

Data elements included in messages can be configured uniquely, based on the interface instance handling the message, the message type, and whether the message is incoming or outgoing. This gives interface instances the flexibility to modify the external message according to the data elements they need and those they do not.

The EMF is used to build the external information in the Interface extended memory table. The Voice Response Unit Interface object in the Integrated Authorization Server process uses this external message information. This object also has internal default settings that it can use when an EMF record has not been defined or the EMF is unavailable. Default settings have been established that specify the data elements contained in a message. An EMF record is not needed if these default settings are appropriate for external message processing.

EMF data element default settings vary according to interface type, interface instance (configured as the process name in the EMF), BASE24 product, message type, and message direction. Default settings for the Voice Response Unit Interface object in the Integrated Authorization Server process are documented in this section. Default settings for the ISO Host Interface and From Host Maintenance processes are documented in the ***BASE24 External Message Manual***.

Keep in mind that these message defaults do not imply fixed-length messages, only fixed groupings of data elements in each message. The data elements themselves can be fixed- or variable-length.

This section documents the Remote Banking Standard Interface external message defaults.

BASE24 External Message Codes

BASE24 products use the following codes to denote whether a data element should be present in its external message. These codes appear on EMF screen 1 and are used throughout this section.

- C = Conditional. The element is mandatory under certain conditions.
- M = Mandatory. The element is required in the message.
- ␣ = Not used (␣ represents a blank character). The element is not included in the message.

Mandatory

On incoming messages, a mandatory data element must always be present. If a mandatory element is not present in a message from an RBSI-compliant device, the message is rejected and returned to the device.

On outgoing messages, a mandatory data element is always present. If the Integrated Authorization Server process does not have the appropriate information for the element, the process fills the field with zeros or blanks or sets the length indicator to zero.

Conditional

Inclusion of the element is determined message-by-message, based on conditions cited in sections 5 and 6. An RBSI-compliant device must be prepared to send and receive conditional elements.

On incoming messages, a conditional data element must be present if the Integrated Authorization Server process requires the conditional element for processing. If it is not present in a message from an RBSI-compliant device, the transaction will be denied when the Integrated Authorization Server process determines it needs the information for processing. The data element descriptions indicate when a conditional element is required by the Integrated Authorization Server process. A conditional element is mapped to the internal message if the element is included in the incoming message, there is an equivalent internal message field, and the element does not contain zeros or spaces.

On outgoing messages, a conditional element is included in the message only if the element contains data and the data is valid. If the element contains blanks, it is not sent.

Not Used

Under certain conditions, a blank can be changed to an M or a C if the RBSI-compliant device needs to receive the element. However, the ability of the Integrated Authorization Server process to carry the value internally may dictate whether an element can be included in the message. Refer to the individual element descriptions in sections 5 and 6 for information on how the element value is handled internally by the Integrated Authorization Server process.

Message Direction

The IN-OUT-IND field in the EMF record indicates whether the record values apply to incoming messages (I), outgoing messages (O), or both incoming and outgoing messages (B). The incoming or outgoing direction is taken from the point of view of BASE24 Remote Banking.

Some messages have different defaults established for the incoming and outgoing versions of the messages while others use the same defaults for both. Users can, however, create separate incoming and outgoing records for any message type desired.

The message defaults described in this section are the same for both incoming and outgoing messages, unless a message direction is noted.

Administrative and Network Management Message Defaults

This subsection identifies the elements that are included or expected by default in administrative or network management messages. These messages are used for dynamic key management and administrative notification of a rejected message.

Data Element Defaults

The following table summarizes the BASE24 external message element defaults established for network management external messages.

Data Element		1644	1804/5	1814
P-1	Secondary Bit Map	M	C	C
P-11	System Trace Audit Number		M	M
P-12	Local Transaction Date/Time		M	M
P-24	Function Code	M	M	M
P-39	Action Code			M
P-56	Original Data Elements		M	
P-59	Transport Data		C	C
S-72	Data Record	M		
S-96	Key Management Data		C	C

MAC Defaults

BASE24-telebanking does not currently support message authentication between BASE24 Remote Banking and an RBSI-compliant device.

BASE24 Remote Banking Message Defaults

This section identifies the elements that are included or expected by default in BASE24 Remote Banking messages.

Data Element Defaults

The following table summarizes the BASE24 external message element defaults established for BASE24-telebanking external messages.

Data Element		Message Class					
		Nonfinancial Transaction		Financial Transaction		Reversal	
		1100/1	1110	1200/1	1210	1420/1	1430
P-1	Secondary Bit Map	C	C	C	C	C	C
P-2	Primary Account Number (PAN)	C	C	C	C	C	C
P-3	Processing Code	M	M	M	M	M	M
P-4	Transaction Amount	C	C	M	M	C	C
P-11	System Trace Audit Number	M	M	M	M	M	M
P-12	Local Transaction Date/Time	M	M	M	M	M	M
P-17	Capture Date		C		C		
P-19	Acquirer Country Code	C	C	C	C	C	C
P-22	Point of Service Data Code	C		C			
P-24	Function Code	M	C	M	C	M	
P-25	Message Reason Code	C		C		M	
P-30	Original Amounts				C		
P-32	Acquiring Institution ID Code	C	C	C	C	C	C
P-34	Primary Account Number, Extended	C	C	C	C	C	C
P-38	Approval Code		C		C	C	C

Data Element		Message Class					
		Nonfinancial Transaction		Financial Transaction		Reversal	
		1100/1	1110	1200/1	1210	1420/1	1430
P-39	Action Code		M		M		M
P-44	Additional Response Data		C			C	
P-48	Additional Data, Private	C	C	C	C	C	C
P-49	Transaction Currency Code	C	C	M	M	C	C
P-52	Personal Identification Number (PIN)	C		C			
P-54	Additional Amounts		C		C		
P-56	Original Data Elements					M	
P-59	Transport Data	C	C	C	C	C	C
P-62	Recurring Transaction Data	C	C			C	
P-63	Source Code	C	C	C	C	C	C
S-73	Action Date	C	C			C	
S-95	Card Issuer Reference Data		C			C	
S-96	Key Management Data		C		C		
S-98	Payee	C	C			C	C
S-102	Account Identification 1	C	C	C	C	C	C
S-103	Account Identification 2	C	C	C	C	C	C
S-104	Transaction Description		C			C	
S-123	Account Qualifiers	C	C	C	C	C	C
S-124	Message Format Type ID	C	C	C	C	C	C
S-125	Transaction Data, Private	C	C				
S-126	Multiple Account Select Data	C	C	C	C		

MAC Defaults

BASE24-telebanking does not currently support message authentication between BASE24 Remote Banking and an RBSI-compliant device.

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Section 4

External Message Data Elements

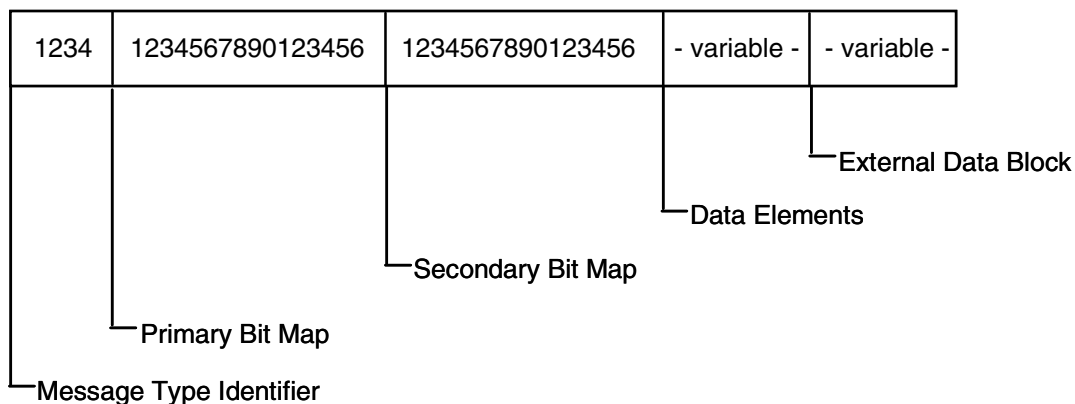
This section describes the external message components and structure, explains how bit maps are used by BASE24, and explains the fixed-length and variable-length structures used in describing the BASE24 Remote Banking external message data elements.

The ISO standard includes a number of private use data elements that can be defined by any organization that uses the standard. This section describes how one such data element has been uniquely defined for BASE24 Remote Banking.

This section also provides a series of tables that specify which data elements are mandatory or conditional for each supported transaction type.

External Message Components and Structure

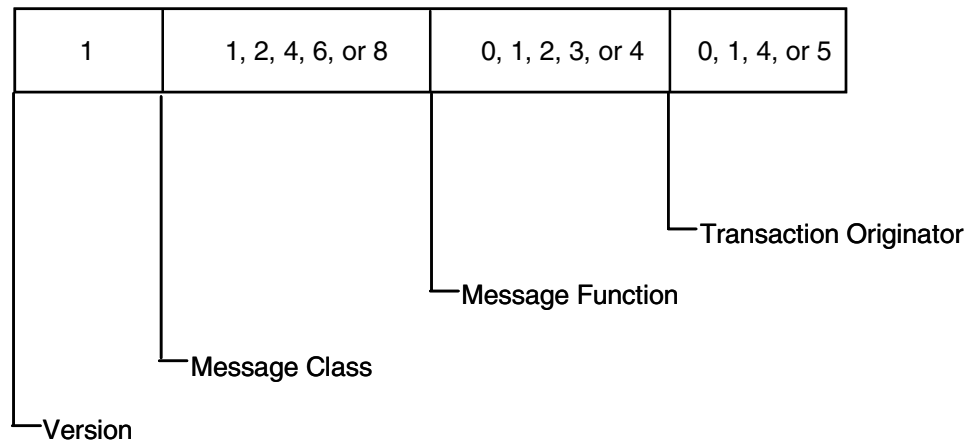
Each message supported by the BASE24 Remote Banking Standard Interface follows the ISO 8583-1993 standard and is constructed in the following sequence: message type identifier, a primary bit map, an optional secondary bit map, and data elements in the order they are represented in the bit maps, and an optional external data block. Note that the external message supported by the BASE24 Remote Banking Standard Interface contains ASCII data only and fixed-length messages are not supported. The figure below shows the order of the components in the message.



The BASE24 Remote Banking Standard Interface message format differs from the ISO 8583-1993 standard format in one respect. For certain transaction requests and responses, an external data block is appended at the end of the data elements. Refer to the definition for data element 125 for a list of requests and responses that use the external data block.

Message Type Identifier

The first message component is the message type identifier (MTI). The MTI is a four-digit field that identifies each message version number, message class, message function, and transaction originator. Every message begins with a message type identifier. The following figure shows the components of the message type identifier.



Version Number

The first position of the message type identifier structure is the version number. The valid value for BASE24 Remote Banking is always 1, which designates the version as ISO 8583-1993.

Message Class

The second position of the message type identifier structure is the message class. Valid values for BASE24 Remote Banking are as follows:

- 1 = Authorization
- 2 = Financial
- 4 = Reversal
- 6 = Administrative
- 8 = Network management

Message Function

The third position of the message type identifier structure is the message function. Valid values for BASE24 Remote Banking are as follows:

- 0 = Request
- 1 = Request response
- 2 = Advice
- 3 = Advice response
- 4 = Notification

Transaction Originator

The fourth position of the message type identifier structure is the transaction originator. Valid values for BASE24 Remote Banking are as follows:

- 0 = Acquirer
- 1 = Acquirer repeat
- 4 = Other
- 5 = Other repeat

Message Types

The following are the external ISO message type identifiers, comprised of the version number, message class, message function, and transaction originator, that are supported between an RBSI-compliant device and BASE24 Remote Banking:

1100/1101—Authorization Request/Repeat Request
1110—Authorization Request Response

1200/1201—Financial Request/Repeat Request
1210—Financial Request Response

1420/1421—Reversal Advice/Repeat Reversal Advice
1430—Reversal Advice Response

1644—Administrative Notification (Message Rejects)

1804/1805—Network Management (NMM) Request/Repeat Request
1814—Network Management (NMM) Request Response

Transactions Supported by Message Type

The following BASE24-telebanking transactions are supported for Authorization Requests/Repeat Requests:

- Available funds inquiry
- Check clearance inquiry
- Check stop payment add
- Last count of debit/credit transactions inquiry
- Last count of account by source transactions inquiry
- Last count of check transactions inquiry
- Last count of debit transactions inquiry
- Last count of credit transactions inquiry
- Last count of account transfer transactions inquiry
- Customer account list inquiry
- Multiple account balance inquiry
- Customer verify
- Customer select
- PIN change
- PIN verify
- Statement download
- Statement closing download

The following BASE24-billpay transactions are supported for Authorization Requests/Repeat Requests:

- Customer vendor list inquiry
- Schedule payment – immediate
- Schedule payment – future
- Schedule payment – recurring
- Schedule transfer – future
- Schedule transfer – recurring
- Scheduled payments list inquiry
- Scheduled transfers list inquiry
- Scheduled transactions list inquiry
- Scheduled payment delete
- Scheduled transfer delete
- Scheduled payment update
- Scheduled transfer update
- History inquiry – payments and transfers
- History inquiry – all transactions
- Customer activate
- Customer add
- Customer deactivate

Customer inquiry
Customer update
Customer account activate
Customer account add
Customer account deactivate
Customer account delete
Customer account update
Customer vendor request mailing
Customer vendor activate
Customer vendor add
Customer vendor deactivate
Customer vendor update
Customer vendor mask verify list inquiry
Bank list inquiry
Bank branch list inquiry
Loan application
Master vendor list inquiry
Miscellaneous request
Miscellaneous requests inquiry

The BASE24-telebanking transfer – immediate transaction is supported for Financial Requests/Repeat Requests.

Primary Bit Map

The second message component is the primary bit map, which is required for every message and contains the most frequently used data elements. It consists of a bit map of 64 bits that signify the presence (denoted by 1) or absence (denoted by 0) in the message of the data element associated with that bit. The 64 bits are converted to and from 16 bytes of display data using hexadecimal equivalents. To convert 64 bits to 16 bytes, the 64 bits are first divided into 16 groups of four. Then, each group of four bits is assigned a hexadecimal equivalent according to the table shown below. It is the hexadecimal equivalents that are carried in the bit map.

Conversion Table			
Bit Value	Hex Value	Bit Value	Hex Value
0000	0	1000	8
0001	1	1001	9
0010	2	1010	A

Conversion Table			
Bit Value	Hex Value	Bit Value	Hex Value
0011	3	1011	B
0100	4	1100	C
0101	5	1101	D
0110	6	1110	E
0111	7	1111	F

Secondary Bit Map

The third message component is the secondary bit map, which contains the less frequently used data elements. It consists of a bit map of 64 bits that signify the presence (denoted by 1) or absence (denoted by 0) in the message of the data element associated with that bit. The secondary bit map (bits 65–128) is optional based upon the message. If the secondary bit map is present, it is signified by a value of 1 in data element 01 of the primary bit map. Like the primary bit map, the secondary bit map is also represented as 16 bytes of display data using hexadecimal equivalents. Refer to the description above for the primary bit map for a description of this representation.

Data Elements

The fourth message component consists of the data elements. Each data element is associated with a particular bit in the primary bit map or secondary bit map.

External Data Block

After the data elements it is possible to have an external data block. The external data block contains a four-byte length field at the beginning and a block of data for the defined length following. The external data block can carry large blocks of data. The intent of the external data block is to carry large message data, however, it is not restricted to carrying just large message data. Data element 125 has a field defined to indicate whether the external data block exists for a message. This indicator exists for a set of defined transactions. See data element 125 for additional information.

Bit Map Parsing Example

The following is an example of the message type identifier and bit maps carried in the external message for an available funds inquiry transaction request message. These components always precede the actual data element data in an ISO message. This example assumes that PINs are not verified locally, the RBSI-compliant device uses transport data, and the transaction is entered by a customer service representative on behalf of a customer.

```
110060300501000010220000000000400000013JOHNQCUSTOMER301000  
12345698103123595961050061510C10811123456789010123456789ABCDEF  
14TRANSPORT DATA02IB191234567890123456789
```

The first four digits of the message (1100) identify the message as an authorization request. The next 16 characters (6030058100001022) are hexadecimal characters that signify the presence or absence of data elements in the primary bit map. The following 16 characters (0000000004000000) are hexadecimal characters that signify the presence or absence of data elements in the secondary bit map.

Each hexadecimal character carries the values for four bits, with each bit representing the presence or absence of a data element.

The first hexadecimal character in the primary bit map carries the values for data elements 1 through 4, the second character carries the values for data elements 5 through 8, and so on up to the 16th character, which carries the values for data elements 61 through 64.

The first hexadecimal character in the secondary bit map carries the values for data elements 65 through 68, the second character carries the values for data elements 69 through 72, and so on up to the 16th character, which carries the values for data elements 125 through 128.

In the above message example, the first character in the primary bit map is 6. Since 6 has a bit value of 0110, and is the first character in the bit map, it indicates that data elements 2 and 3 are present in the message and that data elements 1 and 4 are absent from the message. The next character in the primary bit map is 0. Since 0 has a bit value of 0000, and is the second character in the bit map, it indicates that data elements 5 through 8 are absent from the message.

The following table shows how the entire primary and secondary bit maps are parsed for this example.

Data Elements	Bit Map Value	Hex Bit Values	Data Elements Present
1–4	6	0110	2, 3
5–8	0	0000	
9–12	3	0011	11, 12
13–16	0	0000	
17–20	0	0000	
21–24	5	0101	22, 24
25–28	0	0000	
29–32	1	0001	32
33–36	0	0000	
37–40	0	0000	
41–44	0	0000	
45–48	0	0000	
49–52	1	0001	52
53–56	0	0000	
57–60	2	0010	59
61–64	2	0010	63
65–68	0	0000	
69–72	0	0000	
73–76	0	0000	
77–80	0	0000	
81–84	0	0000	

Data Elements	Bit Map Value	Hex Bit Values	Data Elements Present
85–88	0	0000	
89–92	0	0000	
93–96	0	0000	
97–100	0	0000	
101–104	4	0100	102
105–108	0	0000	
109–112	0	0000	
113–116	0	0000	
117–120	0	0000	
121–124	0	0000	
125–128	0	0000	

Data Element Structures

The following paragraphs describe how data elements in the BASE24 external message must be structured. These guidelines are followed by BASE24 Remote Banking and must be adhered to by RBSI-compliant devices sending messages to BASE24 Remote Banking.

Fixed-Length Data Elements

Data placed in numeric, fixed-length data elements must be right-justified, with leading zeros. Data placed in all other fixed-length data elements must be left-justified, with trailing blanks.

Variable-Length Data Elements

Data placed in variable-length data elements can vary in length from zero positions up to the individual maximum length stated for the data element.

The actual length of the data placed in a variable-length data element must always be specified in a fixed-length prefix immediately preceding the data. This fixed-length field is known as the *field length indicator*.

For variable-length data elements with a maximum length of less than 100 characters, a two-position field length indicator containing the length of the data in the field precedes the data element.

For variable-length data elements with a maximum length greater than 99 and less than 1,000 characters, a three-position field length indicator containing the length of the data in the field precedes the data element.

These prefixes must be right-justified and zero filled. For example, if a variable-length data element could be up to 200 characters, but only seven characters were actually to be loaded into the element, the required fixed-length prefix would be 007. In this case, if the seven characters were 1234567, the entire data element to be included in the external message would be 0071234567—ten positions in all.

Had the data element been limited to a maximum of less than 100 characters, the fixed-length prefix would be 07, and the entire data element would be 071234567—nine positions in all.

Documentation of Field Length Indicators

All variable-length data elements must be preceded by a field length indicator, and users must allow for this. Some variable-length data elements contain data structures while others contain a single field. In sections 5 and 6 of this manual, the field length indicator is not included in the format of the data elements.

Because the value shown with the FORMAT tag for the data element is the length of the data field only, the length of the field length indicator must be added to the value shown with the FORMAT tag for the data element and users must allow for the field length indicator preceding the data element. For example, the Additional Response Data element (P-44) is variable length up to 8 characters. In this case, the user must allow for a 2-position field length indicator preceding this 8-character element, for a total of up to 10 characters.

Zero-Length Data Elements

BASE24 Remote Banking can include zero-length data elements if an RBSI-compliant device has inadvertently made mandatory a data element that BASE24 Remote Banking does not support. If BASE24 Remote Banking has no data to place in a mandatory data element, it simply sets the data element prefix to zeros (00 or 000) and sends the prefix.

Private Use Data Elements

The ISO 8583 standard includes several private use data elements, which can be defined in any manner by an organization implementing the standard. BASE24 Remote Banking uses a number of these data elements to carry pieces of information that are required for processing by BASE24 Remote Banking but are not specifically defined by the ISO 8583 standard. Some of these private use data elements are defined differently depending on the BASE24 Remote Banking transaction to which a message applies. An example is data element P-48, which can carry information specific to BASE24-telebanking or BASE24-billpay. For BASE24-telebanking, this data element can carry the following information:

- New PINs for PIN change transactions
- The search direction for customer account list inquiry transactions
- The check number for check clearance inquiry transactions
- The high check number, low check number, amount, expiration date, description, check date, and the date and time the stop request was added for check stop payment add transactions
- The source type for last count of account by source inquiry transactions
- An optional amount for check clearance inquiry, last count of check transactions inquiry, and last count of debit or credit transactions inquiry transactions
- An optional start and end date for last count of debit/credit transactions inquiry transactions as well as for statement download and statement closing download transactions
- Positioning information for other *last count of* inquiry transactions
- Data specific to user-defined or ACI-use-only transactions

For BASE24-billpay, this data element can carry the following information:

- Optional vendor specific data for scheduled payment transactions
- The activation allowed flag for customer activate transactions
- The FIID for loan application transactions
- Search and key data for inquiry transactions
- Key data for scheduled payment or transfer updates and deletes
- Key data for customer, customer account, and customer vendor database maintenance transactions

Data Elements by Transaction Message

Detailed descriptions of the data elements used for each transaction message are listed in the ***BASE24 Remote Banking Standard Interface Message Specifications***. This manual lays out the message format for each message that can be exchanged between BASE24 Remote Banking and an RBSI-compliant device. This manual is intended for software engineers responsible for programming an RBSI-compliant device.

Section 5

Data Elements 1 Through 64

This section contains descriptions for data elements 1 through 64 of the BASE24-telebanking external message.

P-1 Secondary Bit Map

Format: AN 16

ISO Field Name: Bit Map, Secondary

Used By: BASE24-telebanking
BASE24-billpay

The secondary bit map identifies the presence or absence of data elements 65 through 128 in the BASE24-telebanking external message. It functions the same as the primary bit map, except that the primary bit map identifies the presence or absence of data elements 1 through 64 and the secondary bit map identifies the presence or absence of data elements 65 through 128. The secondary bit map field is sent as a series of 16 hexadecimal digits in which the binary representation of each digit corresponds to four fields.

The secondary bit map is required if any of data elements 65 through 128 are included in the message. Otherwise, it is not used.

The presence or absence of the secondary bit map is identified by bit position 1 in the primary bit map. Data elements 65 through 128 cannot be included in the message if the secondary bit map is not present.

P-2 Primary Account Number (PAN)

Format: AN ..19

ISO Field Name: Primary Account Number (PAN)

Used By: BASE24-telebanking
BASE24-billpay

The primary account number is a series of digits used to identify a customer account or relationship. The Primary Account Number (P-2) data element and the Primary Account Number, Extended (P-34) data element cannot both be present in the external message at the same time.

This data element contains the customer ID if the length of the customer ID is less than 20 digits. If the customer ID is 20 or more digits, the Primary Account Number, Extended (P-34) data element is used instead.

Some transactions can have a customer ID of zeros. The bank list inquiry, bank branch list inquiry, loan application, and customer inquiry transactions can have a customer ID of zeros. This indicates that the customer does not have to be a customer of the institution to perform the transaction. In these cases, the length of the customer ID must be at least one and less than twenty, with the data element containing the appropriate number of zeros.

When the customer ID is set to zeros on the initial request of a bank list inquiry or bank branch list inquiry transaction, subsequent requests must echo back the customer ID value returned in the initial response from BASE24. On the initial response, this data element will contain the customer ID of the default Customer Table (CSTT) row for the FIID. All subsequent requests must then use this value rather than a value of zeros.

Note: The data element definition is a variation of the ISO standard definition. The ISO standard defines the Primary Account Number as a numeric data element. BASE24 Remote Banking defines the data element as an alphanumeric data element.

On incoming messages, the PAN from this data element or the Primary Account Number, Extended (P-34) data element is placed in the ITD-PAN.NUM field of the Internal Transaction Data (ITD).

On outgoing messages, this value is taken from the ITD-PAN.NUM field in the ITD.

P-3 Processing Code

Format: AN 6

ISO Field Name: Processing Code

Used By: BASE24-telebanking
BASE24-billpay

The Processing Code data element contains a series of digits used to describe the effect of a transaction on the customer account and the accounts affected. It is logically divided into three 2-digit subfields whose possible values are defined in ISO 8583-1993 standard. All processing codes that begin with a 9 are reserved for private use.

Positions 1 and 2 comprise the transaction code subfield, which describes the specific transaction to be performed (e.g., debit, credit, inquiry, transfer). The following transaction codes are supported in BASE24 Remote Banking.

Position 1	Position 2	Description
3	0	Available funds inquiry
3	A	Check clearance inquiry
3	B	Last count of debit/credit transactions inquiry
3	C	Last count of account by source transactions inquiry
3	D	Last count of check transactions inquiry
3	E	Last count of debit transactions inquiry
3	F	Last count of credit transactions inquiry
3	G	Last count of account transfer transactions inquiry
3	H	Customer vendor list inquiry
3	J	Scheduled payments list inquiry
3	K	Scheduled transfers list inquiry
3	L	History inquiry – payments and transfers
3	M	Scheduled transactions list inquiry

Position 1	Position 2	Description
3	N	Customer account list inquiry
3	P	Multiple account balance inquiry
3	Q	Statement closing download
3	R	Statement download
4	0	Transfer – immediate
9	0	PIN change
9	1	PIN verify
9	A	Schedule payment – immediate
9	B	Schedule payment – future
9	C	Schedule payment – recurring
9	D	Schedule transfer – future
9	E	Schedule transfer – recurring
9	F	Scheduled payment delete
9	G	Scheduled transfer delete
9	H	Scheduled payment update
9	J	Scheduled transfer update
9	K	Customer add
9	L	Customer inquiry
9	M	Customer update
9	N	Customer vendor add
9	P	Customer vendor deactivate
9	Q	Customer vendor update
9	R	Master vendor list inquiry

Position 1	Position 2	Description
9	T	Customer vendor request mailing
9	V	Loan application
9	W	Miscellaneous request
9	X	History Inquiry – all transactions
A	0	Check stop payment add
B	1	Bank list inquiry
B	2	Bank branch list inquiry
B	4	Miscellaneous requests inquiry
B	5	Customer account add
B	6	Customer account activate
B	7	Customer account deactivate
B	8	Customer account delete
B	9	Customer account update
B	B	Customer activate
B	C	Customer deactivate
B	D	Customer vendor activate
B	E	Customer vendor mask verify list inquiry
B	F	Customer verify
B	G	Customer select
B	<i>x</i>	Reserved—BASE24-billpay, where <i>x</i> = H–Z
P	<i>x</i>	Reserved—custom changes, where <i>x</i> = 0–9, A–Z
Q	<i>x</i>	Reserved—custom changes, where <i>x</i> = 0–9, A–Z
R	<i>x</i>	Reserved—custom changes, where <i>x</i> = 0–9, A–Z

Position 1	Position 2	Description
U	<i>x</i>	User defined, request to financial institution, where <i>x</i> = 0–9, A–Z
W	<i>x</i>	Reserved—ACI-use-only, where <i>x</i> = 0–9, A–Z
X	<i>x</i>	Reserved for distributors, where <i>x</i> = 0–9, A–Z
Y	<i>x</i>	Reserved for distributors, where <i>x</i> = 0–9, A–Z
Z	<i>x</i>	Reserved for distributors, where <i>x</i> = 0–9, A–Z

Positions 3 and 4 describe the account type for account 1. Positions 5 and 6 describe the account type for account 2. The following account types are supported in BASE24 Remote Banking for account 1 and account 2:

Position		Description
3 or 5	4 or 6	
1	0	Savings account
2	0	DDA (checking) account
3	0	Credit account
3	8	Line of credit (private use)
5	8	Certificate (private use)
5	9	Retirement account (private use)
9	0	Interest-bearing checking account (private use)
9	A	Commercial loan (private use)
9	B	Installment loan (private use)
9	C	Mortgage loan (private use)

On incoming messages, the processing code from this data element is translated and placed in the ITD-PROC-CDE.TXN-CDE, ITD-PROC-CDE.ACCT1-TYP, and ITD-PROC-CDE.ACCT2-TYP fields of the Internal Transaction Data (ITD).

On outgoing messages, the processing code is translated from the ITD-PROC-CDE.TXN-CDE, ITD-PROC-CDE.ACCT1-TYP, and ITD-PROC-CDE.ACCT2-TYP fields of the ITD and placed in this data element.

P-4 Transaction Amount

Format: N 12

ISO Field Name: Amount, Transaction

Used By: BASE24-telebanking
BASE24-billpay

The Transaction Amount data element contains the amount of funds requested (either for debit or credit) in the local currency of the source location of the transaction, excluding any fees.

Decimalization of the amount is implied by the Transaction Currency Code (P-49) data element. For example, if the currency code indicates U.S. currency, 000000001000 would indicate \$10.00. However, if the currency code indicates Italian currency, the amount would be 1000 lire.

On incoming messages, the value from this data element is placed in the ITD-AMT-TXN.TXN.AMT field of the Internal Transaction Data (ITD).

On outgoing messages, this value is taken from the ITD-AMT-TXN.TXN.AMT field in the ITD.

P-11 System Trace Audit Number

Format: N 6

ISO Field Name: System Trace Audit Number

Used By: BASE24-telebanking

The System Trace Audit Number data element contains a number assigned by a message initiator that uniquely identifies a transaction within a 24-hour time period. The system trace audit number remains unchanged for all messages of the same class throughout the life of the transaction (e.g., it is the same for the request and response, but is different for a subsequent reversal). The system trace audit number must be the same for all multiple account select requests in a single transaction.

On incoming messages, the value from this data element is placed in the ITD-SEQ-NUM field of the Internal Transaction Data (ITD).

On outgoing messages, this value is taken from the ITD-SEQ-NUM field in the ITD.

P-12 Local Transaction Date/Time

Format: N 12 (YYMMDDhhmmss)

ISO Field Name: Date and Time, Local Transaction

Used By: BASE24-telebanking

The Local Transaction Date and Time data element contains the local date and time at which the transaction takes place.

On incoming messages, the value from this data element is placed in the ITD-LOCAL-TXN.DAT and ITD-LOCAL-TXN.TIM fields of the Internal Transaction Data (ITD).

On outgoing messages, this value is taken from the ITD-LOCAL-TXN.DAT and ITD-LOCAL-TXN.TIM fields in the ITD.

P-17 Capture Date

Format: N 4 (MMDD)

ISO Field Name: Date, Capture

Used By: BASE24-telebanking

The Capture Date data element contains the month and day on which the transaction data was processed by the acquirer.

On outgoing messages, this value is taken from the ITD-CAPTR-DAT fields in the Internal Transaction Data (ITD).

P-19 Acquirer Country Code

Format: N 3

ISO Field Name: Country Code, Acquirer Institution

Used By: BASE24-telebanking

The Acquirer Institution Country Code data element contains the code of the country where the acquiring institution is located. Country Codes are documented in the ISO 3166 standard, *Codes for the Representation of Names of Countries*.

On incoming messages, the value from this data element is placed in the ITD-ACQ.INST-CNTRY-CDE field of the Internal Transaction Data (ITD). If this data element is not provided in the incoming message, the value from the LGNT-COUNTRY-ABBR param in the LCONF is used.

On outgoing messages, this value is taken from the ITD-ACQ.INST-CNTRY-CDE field in the ITD.

P-22 Point of Service Data Code

Format: AN 12

ISO Field Name: Point of Service Data Code

Used By: BASE24-telebanking

The Point of Service Data Code data element contains a series of codes intended to identify terminal capability, terminal environment, and presentation security data. It is used to indicate specific conditions that were present at the time the transaction took place at the point of service.

If the PIN is not verified locally, the Point of Service Data Code data element should be set to 61050061510C. If the PIN is verified locally, the data element should be set to 61050061510S. The following table describes what each position of this data element means for BASE24-telebanking.

Position	Length	Code	Description
1	1	6	Card data input capability (key entered)
2	1	1	Cardholder authentication capability (PIN)
3	1	0	Card capture capability (none)
4	1	5	Operating environment (on premises of cardholder, unattended)
5	1	0	Cardholder present (cardholder present)
6	1	0	Card present (card not present)
7	1	6	Card data input mode (key entered)
8	1	1	Cardholder authentication method (PIN)
9	1	5	Cardholder authentication entity (other)
10	1	1	Card data output capability (none)
11	1	0	Terminal output capability (unknown)

Position	Length	Code	Description
12	1	C or S	PIN capture capability (private use). Valid values are as follows: C = PIN not verified locally S = PIN verified locally

In every request from the device, either the Personal Identification Number (PIN) (P-52) data element must be present or position 12 of the Point of Service Data Code data element must be set to a value of S.

Note: If either the Primary Account Number (P-2) data element, or the Primary Account Number, Extended (P-34) data element contains zeros, position 12 must be set to a value of S.

On incoming messages, the information from this data element is placed in the ITD-PT-SVC fields of the Internal Transaction Data (ITD). If this data element is not provided in the incoming message, the value from the DEFAULT POINT OF SVC DATA CODE field on VCD screen 18 is used.

On outgoing messages, this information is taken from the ITD-PT-SVC fields in the ITD.

P-24 Function Code

Format: N 3

ISO Field Name: Function Code

Used By: BASE24-telebanking

A code that indicates the specific purpose of the message within its message class. The following table lists the values used with the message type identifiers indicated.

Function Code	Description	Message Type Identifier
650	Administrative Message Parse Failure	Administrative Notification (1644)
811	System Security/Key Change	NMM Request/Repeat (1804/1805)
970	More Account 1 Accounts	Authorization/Financial Request/Repeat/Response (1100/1200, 1101/1201, 1110/1210) (Private Use)
971	More Account 2 Accounts	Authorization/Financial Request/Repeat/Response (1100/1200, 1101/1201, 1110/1210) (Private Use)
972	No More Accounts	Authorization/Financial Request/Repeat/Response (1100/1200, 1101/1201, 1110/1210) (Private Use)
973	More Customers	Authorization Request/Repeat/Response (1100, 1101, 1110)

Function codes 970–972 are private use codes and only pertain to multiple account selection. They must be echoed back to the BASE24 system in subsequent multiple account select requests. Function code 973 is also a private use code that is used when more customer IDs or personals IDs are available during customer

selection. This function code must be echoed back to the BASE24 system in subsequent customer select respond requests. Other function codes pertinent to a particular message type can be found in the ISO 8583-1993 document.

On incoming authorization and financial messages, the value from this data element is placed in the ITD-FNCT.CDE field of the Internal Transaction Data (ITD).

On outgoing authorization and financial messages, this value is taken from the ITD-FNCT.CDE field in the ITD.

P-25 Message Reason Code

Format: N 4

ISO Field Name: Message Reason Code

Used By: BASE24-telebanking
BASE24-billpay

This data element provides the receiver of a request, advice, or notification message with the reason, or purpose, of that message. For original authorizations and financial transactions, it identifies why the type of message was sent (e.g., why a request was sent instead of an advice). If the transaction is a reversal, this data element indicates the reason the reversal was generated.

Codes for requests (1100/1101/1200/1201) are in the range of 1500–1999. Codes for reversals (1420/1421) are in the range of 4000–4499. For specific message reason codes, refer to the ISO 8583-1993 document.

On incoming messages, the value from this data element is placed in the ITD-RSN-CDE-MSG field of the Internal Transaction Data (ITD).

On outgoing messages, this value is taken from the ITD-RSN-CDE-MSG field in the ITD.

P-30 Original Amounts

Format: N 24

ISO Field Name: Amounts, Original

Used By: BASE24-telebanking
BASE24-billpay

The Original Amounts data element contains the original transaction and reconciliation amounts if the transaction was denied. BASE24-telebanking and BASE24-billpay use only the transaction amount. The reconciliation amount is zeros in the ISO message. The format of this data element in the ISO message is as follows.

Position	Length	Field Description
1-12	N 12	Original Transaction Amount The original amount of the transaction, expressed in the currency of the remote banking endpoint or customer service representative terminal where the transaction was initiated. On outgoing messages, this value is taken from the ITD-AMT-TXN-ORIG.TERM field in the Internal Transaction Data (ITD).
13-24	N 12	Original Reconciliation Amount This field is not currently used and is zero filled.

P-32 Acquiring Institution ID Code

Format: N ..11

ISO Field Name: Acquiring Institution Identification Code

Used By: BASE24-telebanking
BASE24-billpay

The Acquiring Institution Identification Code data element identifies the acquiring institution for the transaction. If this data element is not included in the message from the device, the default value is obtained from the BASE24 database.

On incoming messages, the value from this data element is placed in the ITD-ACQ.ID-CDE field of the Internal Transaction Data (ITD). If this data element is not provided in the incoming message, the value from the DEFAULT ACQUIRER ID field on VCD screen 18 is used.

On outgoing messages, this value is taken from the ITD-ACQ.ID-CDE field in the ITD.

P-34 Primary Account Number, Extended

Format: N ..28

ISO Field Name: Acquiring Institution Identification Code

Used By: BASE24-telebanking
BASE24-billpay

The Primary Account Number, Extended data element contains a series of digits used to identify a customer account or relationship. The Primary Account Number, Extended data element and the Primary Account Number (P-2) data element cannot both be present in the external message at the same time.

This data element contains the customer ID if the length of the customer ID is 20 digits or more. If the customer ID is less than 20 digits, the Primary Account Number (P-2) data element is used instead.

The following transactions can have a customer ID of zeros: bank list inquiry, bank branch list inquiry, loan application, and customer inquiry. A customer does not have to be a customer of the institution to perform these transactions. In these cases the length of the customer ID must be at least 20 digits, with the data element containing the appropriate number of zeros.

Note: This data element format varies from the ISO standard. The ISO standard defines the Primary Account Number, Extended data element as a numeric data element. BASE24-telebanking and BASE24-billpay define the data element as an alphanumeric data element.

On incoming messages, the PAN from this data element or the Primary Account Number (P-2) data element is placed in the ITD-PAN.NUM field of the Internal Transaction Data (ITD).

On outgoing messages, this value is taken from the ITD-PAN.NUM field in the ITD.

P-38 Approval Code

Format: ANP 6

ISO Field Name: Approval Code

Used By: BASE24-telebanking

The Approval Code data element contains the code assigned by the authorizing institution to indicate approval. This data element is returned in approved response messages and if the approved transaction is reversed, this data element is echoed in the reversal message.

Note: BASE24 Remote Banking generates approval codes for transactions authorized on BASE24. If a transaction is sent to a host or a third-party service provider for authorization, it is the responsibility of the host or third-party service provider to provide an approval code in the response message.

On outgoing messages, this value is taken from the ITD-APPRV.CDE field in the ITD.

P-39 Action Code

Format: N 3

ISO Field Name: Action Code

Used By: BASE24-telebanking
BASE24-billpay

The Action Code data element contains a code that defines the disposition of a message, as defined in the ISO 8583-1993 standard. Action codes that can be used are documented in this standard. BASE24-telebanking and BASE24-billpay also define the following private use action codes. Detailed descriptions of these action codes are provided in the ***BASE24 Remote Banking Transaction Processing Manual***.

Action Code	Description	Message Type Identifier
080	Approved, Backup Account Used	Financial Request Response (1210)
081	Approved, Overdraft Used	Financial Request Response (1210)
082	Approved, Customer Reactivated	Authorization Request Response (1110)
180	Denied, Amount Not Found	Authorization Request Response (1110)
181	Denied, PIN Change Required	All Responses
182	Denied, New PIN Invalid	Authorization Request Response (1110)
183	Denied, Bank Not Found	Authorization Request Response (1110)
184	Denied, Bank Not Effective	Authorization Request Response (1110)
185	Denied, Customer Vendor Not Found	Authorization Request Response (1110)

Action Code	Description	Message Type Identifier
186	Denied, Customer Vendor Not Effective	Authorization Request Response (1110)
187	Denied, Customer Vendor Account Invalid	Authorization Request Response (1110)
188	Denied, Vendor Not Found	Authorization Request Response (1110)
189	Denied, Vendor Not Effective	Authorization Request Response (1110)
190	Denied, Vendor Data Invalid	Authorization Request Response (1110)
191	Denied, Payment Date Invalid	Authorization Request Response (1110)
192	Denied, Personal ID Not Found	Authorization Request Response (1110)
193	Denied, Scheduled Transactions Exist	Authorization Request Response (1110)
940	Denied, Database Error	All Responses
941	Denied, Currency Code Not Supported	All Responses
942	Denied, Format Error – Amount	Authorization Request Response (1110)
943	Denied, Format Error – Customer Vendor	Authorization Request Response (1110)
944	Denied, Format Error – Date	Authorization Request Response (1110)
945	Denied, Format Error – Name	Authorization Request Response (1110)
946	Denied, Format Error – Account	Authorization Request Response (1110)

Action Code	Description	Message Type Identifier
947	Denied, Recurring Data Error	Authorization Request Response (1110)
948	Denied, Update Not Allowed	Authorization Request Response (1110)

On outgoing response messages, this value is taken from the ITD-ACT.CDE field in the ITD.

P-44 Additional Response Data

Format: ANS ..8

ISO Field Name: Additional Response Data

Used By: BASE24-billpay

The Additional Response Data data element is used to provide additional data for certain BASE24-billpay transaction responses. BASE24-billpay uses this data element to carry the payment or transfer date in response messages for the following transactions:

- Schedule payment – immediate
- Schedule payment – future
- Schedule payment – recurring
- Schedule transfer – future
- Schedule transfer – recurring
- Scheduled payment update
- Scheduled transfer update

The payment date indicates the business date for immediate and future transactions. It indicates the start date for recurring transactions. The format of this data element in the ISO message is as follows.

Position	Length	Field Description
1–8	AN 8	Payment date

This field is in YYYYMMDD format.

On outgoing response messages, this value is taken from the ITD-PMNT-DAT field in the ITD.

P-48 Additional Data, Private

Format: ANS ...200

ISO Field Name: Additional Data, Private

Used By: BASE24-telebanking
BASE24-billpay

The Additional Data, Private data element is used to provide additional data for certain BASE24-telebanking and BASE24-billpay transaction requests and responses.

Note: The Additional Data data element must be even in length in all requests from the device.

On incoming messages, the information from this data element is placed in the ITD-ADNL-DATA.FDATA fields of the Internal Transaction Data (ITD).

On outgoing messages, this information is taken from the ITD-ADNL-DATA.FDATA fields in the ITD.

BASE24-telebanking

BASE24-telebanking uses this data element to carry the following information:

- A new PIN in a PIN change transaction
- The search direction in a customer account list inquiry transaction
- The check number in a check clearance inquiry transaction
- Optional start and end posting dates for statement download and statement closing download transactions
- The high check number, low check number, amount, expiration date, description, check date, and an optional date and time indicating when the stop request was added for check stop payment add transactions
- The source type in a last count of account by source transactions inquiry transaction
- An optional amount in check clearance inquiry, last count of check transactions inquiry, last count of credit transactions inquiry, and last count of debit transactions inquiry transactions

- An optional start date and end date in a last count of debit/credit transactions inquiry transaction
- Positioning information in other *last count of* inquiry transactions

In all other BASE24-telebanking transactions, this data element can be used to transport data between the device and the host for user-defined or ACI-use-only transactions.

PIN Change

The PIN Change transaction request requires the Additional Data data element to have the following data.

Position	Length	Field Description
1–16	AN 16	New PIN 1 The new PIN.
17–32	AN 16	New PIN 2 The new PIN repeated for confirmation.

The new PINs (New PIN 1 and New PIN 2) sent in a PIN change transaction request must have the same PIN block type and use the same verification method as the current PIN.

Check Clearance Inquiry and Last Count of Inquiries

The format of the Additional Data data element for the check clearance inquiry transaction and the *last count of* transaction inquiries (transaction codes 3A–3G) is as follows.

Position	Length	Field Description
1	AN 1	Request/Response Indicator A code that indicates whether this is the initial request, a continuation request, or a response. Valid values are as follows: 0 = Initial request 1 = Continuation request or response

Position	Length	Field Description
2	AN 1	Source Type A code that indicates the source of the transaction. Valid values are as follows: A = ATM B = Telebanking device P = POS device T = Teller = All of the above, where is a blank character For check clearance inquiry transactions, this field must contain a blank space.
3–14	AN 12	Check Number The number of the check for which the customer wants to determine clearance. Embedded spaces are not allowed. The number entered must be right-justified and padded with leading zeros. This is a required field for check clearance inquiry transactions.
15–32	AN 18	Amount The amount of the check or transaction in whole and fractional currency units (for example, U.S. dollars and cents) for which the customer wants to determine clearance or inquire upon. A decimal point can be entered if the amount includes fractional currency units. A negative amount cannot be entered. This is an optional field used to refine the search for a particular check or transaction.
33–40	AN 8	Start Date The starting date (YYYYMMDD) for which debit and credit transactions are to be reported for this account. Any debit or credit transactions that occurred prior to this date are not included in the transaction response. This is an optional field used to refine the search for last count of debit/credit transactions inquiry transactions for a specific date range. The date entered in this field must be less than the date entered in the End Date field.

Position	Length	Field Description
41–48	AN 8	End Date The ending date (YYYYMMDD) for which debit and credit transactions are to be reported for this account. Any debit or credit transactions that occurred after this date are not included in the transaction response. This is an optional field used to refine the search for last count of debit/credit transactions inquiry transactions to a specific date range. The date entered in this field must be greater than the date entered in the Start Date field.

Note: If a subfield does not pertain to the transaction, the subfield must be blank filled. If a check number is present, it must be right-justified and zero filled.

The Start Date and End Date fields are optional for the last count of debits/credit transactions inquiry transaction. A start date is required if an end date is specified. However, an end date is not required if a start date is specified.

The device formats the additional data on the initial request. The device must set the Request/Response Indicator field to a value of 0.

For transaction inquiry continuation requests, the additional data in the continuation request must be echoed back exactly as it was sent in the previous inquiry request response. The device must **not** reset any of the subfields of the Additional Data, Private data element that were returned in the response message from BASE24.

The response message from BASE24 contains additional data that BASE24 needs to process continuation requests. Therefore, the length of the Additional Data, Private data element will be greater than 48 bytes in the response message.

Check Stop Payment Add

The format of the Additional Data data element for the check stop payment add transaction (transaction code A0) is as follows.

Position	Length	Field Description
1–12	AN 12	High Check Number The highest check number in a range of checks for which the customer wants to stop payment. If the customer wants to stop payment on an individual check, this field contains the specific check number. Embedded spaces are not allowed. The number entered must be right-justified and padded with leading zeros. This is a required field.
13–24	AN 12	Low Check Number The lowest check number in a range of checks for which the customer wants to stop payment. If the customer wants to stop payment on an individual check, this field contains blanks. Embedded spaces are not allowed. The number entered must be right-justified and padded with leading zeros. This is an optional field.
25–42	AN 18	Amount The amount of the checks in whole and fractional currency units (for example, U.S. dollars and cents) for which the customer wants to stop payment. A decimal point can be entered if the amount includes fractional currency units. A negative amount cannot be entered. This is an optional field used to refine the search for checks for which payment is to be stopped.

Position	Length	Field Description
43–50	AN 8	Stop Date The date (YYYYMMDD) on which this stop payment order was entered. If no date is entered in this field, it defaults to the current date. For the request, either both the Stop Date and Stop Time fields must contain values or both fields must be left blank. If these fields are left blank, BASE24 sets them to the current date and time. On the response, this field may echo back the value entered in the request, be set by BASE24, or be overridden by the host.
51–58	AN 8	Stop Time The time (hhmmssst) at which this stop payment order was entered. If no time is entered in this field, it defaults to the current time. On the response, this field may echo back the value entered in the request, be set by BASE24, or be overridden by the host.
59–66	AN 8	Expiration Data The date (YYYYMMDD) on which this stop payment order is to expire. This is a required field.
67–101	AN 35	Description An optional information-only field for entering a message regarding the stop payment order. The information in this field has no effect on BASE24 processing.
102–109	AN 8	Check Date The date (YYYYMMDD) on the checks for which payment is to be stopped. This is an optional field used to refine the search for checks for which payment is to be stopped.
110–200	AN 91	Filler

Customer Account List Inquiry

The format of the Additional Data, Private data element for the customer account list inquiry transaction (transaction code 3N) is as follows.

Position	Length	Field Description
1	AN 1	Request/Response Indicator A code that indicates whether this is the initial request, a continuation request, or a response. Valid values are as follows: 0 = Initial request 1 = Continuation request or response
2–3	AN 2	Search Direction A code that indicates whether the search direction is ascending or descending. Valid values are as follows: AC = Ascending DC = Descending
4	AN 1	Reserved This field is not used.
5–200	ANS 196	Key Data (variable length)

Note: The device does not need to send this data element in the initial request message. The request/response indicator defaults to 0, the search direction defaults to AC, and the key data is always spaces for the initial request.

If the device does send this data element in the initial request, the Request/Response Indicator field must be set to 0, the Search Direction flag must be set to AC, and the Key Data field must be blank filled.

For transaction inquiry continuation requests, the Request/Response Indicator and Key Data fields must be echoed back exactly as they were sent in the previous inquiry request response. The device must **not** reset these two subfields of the Additional Data, Private data element that are returned in the response message from BASE24. The device can reset the Search Direction flag if the customer wants to change the direction of the search.

Statement Download and Statement Closing Download

The format of the Additional Data data element for the statement download and statement closing download transactions (transaction codes 3Q and 3R) is as follows.

Position	Length	Field Description
1	AN 1	Request/Response Indicator A code that indicates whether this is the initial request, a continuation request, or a response. Valid values are as follows: 0 = Initial request 1 = Continuation request or response
2	AN 1	Source Type This field is not currently used.
3–14	AN 12	Check Number This field is not currently used.
15–32	AN 18	Amount This field is not currently used.
33–40	AN 8	Start Posting Date The starting posting date (YYYYMMDD) for which statement information is to be reported for this account. Any transactions that were posted prior to this date are not included in the transaction response. This is an optional field used to refine the search for statement information. The date entered in this field must be less than the date entered in the End Posting Date field.

Position	Length	Field Description
41–48	AN 8	End Posting Date The ending posting date (YYYYMMDD) for which statement information is to be reported for this account. Any transactions that were posted after this date are not included in the transaction response. This is an optional field used to refine the search for statement information. The date entered in this field must be greater than the date entered in the Start Posting Date field.

BASE24-billpay

BASE24-billpay uses the Additional Data, Private data element for several transactions. The format of the data in this data element is specific to each transaction.

Customer Activate Data

This data element is mandatory for a customer activate transaction. The following is the format of the Additional Data, Private data element for customer activate transactions:

Position	Length	Field Description
1	AN 1	Reactivating Allowed by Client A code that indicates whether the client allows customer IDs to be reactivated. Valid values are as follows: Y = Yes, reactivation is allowed. N = No, reactivation is not allowed.
2–29	AN 28	Personal ID The identifier for this customer's personal information.
30	AN 1	Reserved This field is not used.

The Reactivating Allowed by Client flag must be present and the Personal ID field is optional. If the Personal ID field is blank, BASE24 Remote Banking uses the value of the customer ID for the personal ID.

Customer Add Data

This data element is mandatory for a customer add transaction. The following is the format of the Additional Data, Private data element for customer add transactions:

Position	Length	Field Description
1–4	AN 4	FIID The FIID of the financial institution for this customer. The FIID is an identifier that must be unique within the logical network. It is used throughout BASE24 products to identify each BASE24 institution.

Customer Deactivate Data

This data element is optional for a customer deactivate transaction. If this data element is not provided, BASE24 Remote Banking uses the value of the customer ID for the personal ID. The following is the format of the Additional Data, Private data element for customer deactivate transactions:

Position	Length	Field Description
1–28	AN 28	Personal ID The identifier for this customer's personal information.

Customer Inquiry Data

This data element is optional for a customer inquiry transaction. The following is the format of the Additional Data, Private data element for customer inquiry transactions:

Position	Length	Field Description
1–28	AN 28	Personal ID The identifier for this customer's personal information.
29–32	AN 4	FIID The FIID of the financial institution for this customer. The FIID is an identifier that must be unique within the logical network. It is used throughout BASE24 products to identify each BASE24 institution.

If the PAN in data element 2 (or data element 34 if using extended PAN numbers) is all zeros, the FIID field must be provided.

If the Personal ID is blanks, BASE24 Remote Banking uses the value of the customer ID for the personal ID. If data element 2 (PAN) is zeros, then the Personal ID field must be blank and the FIID field must be provided.

Customer Update Data

This data element is optional for a customer update transaction. If this data element is not provided, BASE24 Remote Banking uses the value of the customer ID for the personal ID. The following is the format of the Additional Data, Private data element for customer update transactions:

Position	Length	Field Description
1–28	AN 28	Personal ID The identifier for this customer's personal information.

Customer Account Data

This data element is mandatory for customer account activate, customer account add, customer account deactivate, customer account delete, and customer account update transactions. The following is the format of the Additional Data, Private data element for customer account transactions:

Position	Length	Field Description
1-19	AN 19	Account Number The customer application account number.

Position	Length	Field Description
20–21	AN 2	Account Type The ISO account type code for this application account. The BASE24 reserved account type values are as follows: 10 = Savings account 1A = Savings account 1 1B = Savings account 2 1C = Savings account 3 1D = Savings account 4 1E = Savings account 5 1F = Savings account 6 20 = Demand deposit account 2A = Demand deposit account 1 2B = Demand deposit account 2 2C = Demand deposit account 3 2D = Demand deposit account 4 2E = Demand deposit account 5 2F = Demand deposit account 6 2G = Demand deposit account 7 2H = Demand deposit account 8 30 = Credit account 3A = Credit account 1 3B = Credit account 2 3C = Credit account 3 3D = Credit account 4 3E = Credit account 5 3F = Credit account 6 3G = Credit account 7 38 = Line of credit 58 = Certificate 59 = Retirement account 90 = Interest-bearing checking account 9A = Commercial loan 9B = Installment loan 9C = Mortgage loan
22	AN 1	Reserved This field is not used.

Customer Vendor Data

This data element is mandatory for customer vendor activate, customer vendor deactivate, and customer vendor update transactions. The following is the format of the Additional Data, Private data element for customer vendor transactions:

Position	Length	Field Description
1–5	N 5	Customer Vendor Number The customer vendor number generated by the BASE24 system when this customer vendor was added. This number is unique among all customer vendors for this customer.
6	AN 1	Reserved This field is not used.

Loan Application Data

This data element is mandatory for a loan application transaction. The following is the format of the Additional Data, Private data element for loan application transactions:

Position	Length	Field Description
1–4	AN 4	FIID The FIID of the financial institution for this customer. The FIID is an identifier that must be unique within the logical network. It is used throughout BASE24 products to identify each BASE24 institution.

Schedule Payment Data

This data element is optional for schedule payment – immediate, scheduled payment – future, and scheduled payment – recurring transactions. If present, it will contain vendor (merchant) specific payment data. (The Vendor Specific Data Type field in the customer vendor list inquiry response indicates if vendor specific

payment data is required for a Schedule Payment transaction.) The following is the format of the Additional Data, Private data element for schedule payment transactions:

Position	Length	Field Description
1–200	ANS 200	Vendor Specific Payment Data (variable length) Vendor-specific data required by this vendor from a customer whenever a payment is made. A description of the data required is stored in the Vendor Table (VNDR).

BASE24-billpay supports a length of 30 bytes for the Vendor Specific Payment Data field. Third-party bill pay service providers can support other lengths for the data. However, the length of this field cannot exceed 200 bytes.

Scheduled Payment Delete Data

This data element is mandatory for scheduled payment delete transactions. The following is the format of the Additional Data, Private data element for scheduled payment delete transactions:

Position	Length	Field Description
1–6	N 6	Customer Reference Number A unique BASE24 system-generated number used to reference historical data for this transaction. This number is generated by the XPNET or Pathway Billpay Server process for identification purposes when it processes a transaction.

Scheduled Payment Update Data

This data element is mandatory for scheduled payment update transactions. The following is the format of the Additional Data, Private data element for scheduled payment update transactions:

Position	Length	Field Description
1–6	N 6	Customer Reference Number A unique BASE24 system-generated number used to reference historical data for this transaction. This number is generated by the XPNET or Pathway Billpay Server process for identification purposes when it processes a transaction.
7–25	AN 19	From Account Number The number of the customer's account to be debited for this scheduled payment.
26–27	AN 2	From Account Type The type of customer account (e.g., 20 for a demand deposit account) to be debited for this scheduled payment.

Position	Length	Field Description
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28–33	AN 6	From Account Type Description
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A text description of the account type.

The BASE24 reserved account type name values are as follows:

CD	= Certificate
CMRCL	= Commercial loan
CR	= Credit account
CR1	= Credit account
CR2	= Credit account
CR3	= Credit account
CR4	= Credit account
CR5	= Credit account
CR6	= Credit account
CR7	= Credit account
DDA	= Demand deposit account
DDA1	= Demand deposit account
DDA2	= Demand deposit account
DDA3	= Demand deposit account
DDA4	= Demand deposit account
DDA5	= Demand deposit account
DDA6	= Demand deposit account
DDA7	= Demand deposit account
DDA8	= Demand deposit account
INSTL	= Installment loan
IRA	= Retirement account
LOCR	= Line of credit
MRTGL	= Mortgage loan
NONE	= No account
NOW	= Interest-bearing checking account
SAV	= Savings account
SAV1	= Savings account
SAV2	= Savings account
SAV3	= Savings account
SAV4	= Savings account
SAV5	= Savings account
SAV6	= Savings account

Position	Length	Field Description
34–63	AN 30	Vendor Specific Data Vendor-specific data required by this vendor from a customer whenever a payment is made. A description of the data required is stored in the Vendor Table (VNDR).
64	AN 1	Reserved This field is not used.

The From Account Number field must contain either spaces or numeric digits. If the From Account Number field contains spaces, the From Account Type and From Account Type Description fields must also contain spaces.

If the From Account Number field does not contain spaces, either the From Account Type or the From Account Type Description field must be provided. If the From Account Type field contains spaces, the From Account Type Description field is used to look up the *from* account type.

Scheduled Transfer Delete Data

This data element is mandatory for scheduled transfer delete transactions. The following is the format of the Additional Data, Private data element for scheduled transfer delete transactions:

Position	Length	Field Description
1–6	N 6	Customer Reference Number A unique BASE24 system-generated number used to reference historical data for this transaction. This number is generated by the XPNET or Pathway Billpay Server process for identification purposes when it processes a transaction.

Scheduled Transfer Update Data

This data element is mandatory for scheduled transfer update transactions. The following is the format of the Additional Data, Private data element for scheduled transfer update transactions:

Position	Length	Field Description
1–6	N 6	Customer Reference Number A unique BASE24 system-generated number used to reference historical data for this transaction. This number is generated by the XPNET or Pathway Billpay Server process for identification purposes when it processes a transaction.
7–25	AN 19	From Account Number The number of the customer's account to be debited for this scheduled transfer.
26–27	AN 2	From Account Type An ISO code indicating the type of customer account (e.g., 20 for a demand deposit account) to be debited for this scheduled transfer. Refer to the Account Type field description provided earlier for a list of the BASE24 reserved account type values.
28–33	AN 6	From Account Type Description A text description of the <i>from</i> account type. Refer to the From Account Type Description field description provided earlier for a list of the BASE24 reserved account type name values.
34–52	AN 19	To Account Number The number of the customer's account to be credited for this scheduled transfer.

Position	Length	Field Description
53–54	AN 2	To Account Type An ISO code indicating the type of customer account (e.g., 20 for a demand deposit account) to be credited for this scheduled transfer. Refer to the Account Type field description provided earlier for a list of the BASE24 reserved account type values.
55–60	AN 6	To Account Type Description A text description of the <i>to</i> account type. Refer to the From Account Type Description field description provided earlier for a list of the BASE24 reserved account type name values.

The From Account Number field must contain spaces or numeric digits. If the From Account Number field contains spaces, the From Account Type and From Account Type Description fields must also contain spaces.

If the From Account Number field does not contain spaces, either the From Account Type or the From Account Type Description field must be provided. If the From Account Type field contains spaces, the From Account Type Description field is used to look up the *from* account type. The From Account information will be updated.

The To Account Number field must contain spaces or numeric digits. If the To Account Number field contains spaces, the To Account Type and To Account Type Description fields must also contain spaces.

If the To Account Number field does not contain spaces, either the To Account Type or the To Account Type Description field must be provided. If the To Account Type field contains spaces, the To Account Type Description field is used to look up the *to* account type.

Inquiry Transaction Data

The first five fields (first eight characters) are required for all requests listed below.

For subsequent requests, the client should always echo all fields received from the previous response, except for the Search Direction field. The Search Direction field can be modified to allow the client to change directions, unless noted otherwise.

Note: All reserved fields listed below must be space filled. Also, if an area is required, but not used (e.g., filter data), that area must also be space filled, including any unused portion of that area.

The Additional Data, Private data element contains data for the following inquiry transactions:

- Bank list inquiry
- Bank branch list inquiry
- Customer vendor list inquiry
- Customer vendor mask verify list inquiry
- History inquiry – payments and transfers
- History inquiry – all transactions
- Master vendor list inquiry
- Miscellaneous requests inquiry
- Scheduled payments list inquiry
- Scheduled transfers list inquiry
- Scheduled transactions list inquiry

Search Direction Flag. The following example describes how the Search Direction flag is used for inquiry transactions. Assume there are 30 records that match the criteria for the specified inquiry request (e.g., history inquiry – payments and transfers). Record 30 is the most current payment and record 1 is the oldest payment. Assume that five records can be returned in a single response to the device. The initial request always begins with the most current data so records 26–30 are returned in the initial response. If a subsequent request is received and the search direction is descending, records 21–25 are returned. If another request is received and the search direction has been changed to ascending, records 26–30 are returned again. Key data that is returned in response messages and echoed in subsequent requests is used by BASE24 with the Search Direction flag to determine what data is returned in the subsequent response message.

Bank List Inquiry Data. This data element is mandatory for bank list inquiry transactions. The following is the format of the Additional Data, Private data element for bank list inquiry transactions:

Position	Length	Field Description
1	AN 1	Request/Response Indicator A code that indicates whether this is the initial request, a continuation request, or a response. Valid values are as follows: 0 = Initial request 1 = Continuation request or response ␣ = Use defaults, where ␣ is a blank character
2–3	AN 2	Search Direction A code that indicates the direction of the search. Valid values are as follows: AC = Ascending (default) DC = Descending
4	AN 1	Key Data Type A code that indicates the type of key to be used for the search. Valid values are as follows: 0 = Bank Name (default) 1 = FIID
5–7	N 3	Cumulative Transaction Count A running count of the number of items (i.e., financial institutions) returned thus far for the bank list inquiry transaction.
8	AN 1	Reserved This field is not used.
9–38	AN 30	Filter Data—Not currently used This field must be space-filled if a Key Data field is present in the message.

Position	Length	Field Description
39–60	AN 22	Key Data: Bank Name The name of the financial institution to be used for the search.
or	or	or
39–42	AN 4	Key Data: FIID The FIID of the financial institution to be used for the search. The FIID is an identifier that must be unique within the logical network. It is used throughout BASE24 products to identify each BASE24 institution.

The first five fields must be provided (the first eight bytes). If the Request/Response Indicator field contains a space, then the following defaults are used:

- The search direction is ascending
- The key data type is 0 (Bank Name)
- The Bank Name value is spaces

If the Key Data field is present, the Filter Data field must be present. Since the Filter Data field is not currently used, it must contain spaces if key data is present.

If defaults are not used, the Key Data field cannot contain spaces.

Bank Branch List Inquiry Data. This data element is mandatory for bank branch list inquiry transactions. The following is the format of the Additional Data, Private data element for bank list inquiry transactions:

Position	Length	Field Description
1	AN 1	Request/Response Indicator A code that indicates whether this is the initial request, a continuation request, or a response. Valid values are as follows: 0 = Initial request 1 = Continuation request or response ␣ = Use defaults, where ␣ is a blank character

Position	Length	Field Description
2–3	AN 2	Search Direction A code that indicates the direction of the search. Valid values are as follows: AC = Ascending (default) DC = Descending
4	AN 1	Key Data Type A code that indicates the type of key to be used for the search. The only valid value is 0 (bank branch), which is also the default value.
5–7	N 3	Cumulative Transaction Count A running count of the number of items (i.e., financial institution branches) returned thus far for the bank branch list inquiry transaction.
8	AN 1	Reserved This field is not used.
9–38	AN 30	Filter Data—Not currently used This field must be space-filled if a Key Data field is present in the message.
39–42	AN 4	Bank Branch Key—FIID The FIID of the financial institution to be used for the search. The FIID is an identifier that must be unique within the logical network. It is used throughout BASE24 products to identify each BASE24 institution.
43–67	AN 25	Bank Branch Key—City The city in which bank branches are located to be used in the search.
68	AN 1	Reserved This field is not used.

The first five fields must be provided (the first eight bytes). If the Request/Response Indicator field contains a space then the following defaults are used:

- The search direction is ascending
- The key data type is 0 (Bank Branch)
- The Bank Branch Key (FIID and City) value is spaces

If not using defaults, the Bank Branch Key - FIID field must be provided. The Bank Branch Key - City field is optional.

If key data is present, the Filter Data field must be present. Since the Filter Data field is not currently used, it must contain spaces if key data is present.

Customer Vendor List Inquiry Data. This data element is mandatory for customer vendor list inquiry transactions. The following is the format of the Additional Data, Private data element for customer vendor list inquiry transactions:

Position	Length	Field Description
1	AN 1	Request/Response Indicator A code that indicates whether this is the initial request, a continuation request, or a response. Valid values are as follows: 0 = Initial request 1 = Continuation request or response ␣ = Use defaults, where ␣ is a blank character
2-3	AN 2	Search Direction A code that indicates the direction of the search. The only valid value is AC (ascending), which is also the default value.
4	AN 1	Key Data Type A code that indicates the type of key to be used for the search. Valid values are as follows: 0 = Vendor name (default) 1 = Customer vendor number

Position	Length	Field Description
5–7	N 3	Cumulative Transaction Count A running count of the number of items (i.e., customer vendors) returned thus far for the customer vendor list inquiry transaction.
8	AN 1	Reserved This field is not used.
9–38	AN 30	Filter Data—Not currently used This field must be space filled if a Key Data field is present in the message.
39–68	AN 30	Key Data: Vendor Name (default) A vendor name used as a starting point for searching forward through customer vendors by vendor name.
or	or	or
39–43	N 5	Key Data: Customer Vendor Number A customer vendor number used as a starting point for searching forward through customer vendors by customer vendor number. This is a unique BASE24 system-generated number used to reference historical data for this transaction. This number is generated by the XPNET or Pathway Billpay Server process for identification purposes when it processes a transaction.
44	AN 1	Key Data: Reserved This field is not used.

The first five fields must be provided (the first eight bytes). If the Request/Response Indicator field contains a space, then the following defaults are used:

- The search direction is ascending
- The key data type is 0 (Vendor Name)
- The Vendor Name value is spaces

Only ascending order is allowed for this transaction.

If key data is present, the Filter Data field must be present. Since the Filter Data field is not currently used, it must contain spaces if key data is present.

If defaults are not being used, the key data cannot be spaces.

Customer Vendor Mask Verify List Inquiry Data. This data element is mandatory for customer vendor mask verify list inquiry transactions. The following is the format of the Additional Data, Private data element for customer vendor mask verify list inquiry transactions:

Position	Length	Field Description
1	AN 1	Request/Response Indicator A code that indicates whether this is the initial request, a continuation request, or a response. Valid values are as follows: 0 = Initial request 1 = Continuation request or response
2-3	AN 2	Search Direction A code that indicates the direction of the search. Valid values are as follows: AC = Ascending DC = Descending
4	AN 1	Key Data Type A code that indicates the type of key to be used for the search. Valid values are as follows: 0 = Vendor branch by name 1 = Vendor branch by number
5-7	N 3	Cumulative Transaction Count A running count of the number of items (i.e., customer vendor masks) returned thus far for the customer vendor mask verify list inquiry transaction.
8	AN 1	Reserved This field is not used.

Position	Length	Field Description
9–36	AN 28	Customer Account with Vendor The customer's account number with this vendor. This is the number of the account to which payments from this customer are posted or credited. If a value is provided in this field, it is verified against each mask returned.
37–38	AN 2	Reserved This field is not used.

Position	Length	Field Description
39–68	AN 30	Key Data: Vendor Name A vendor name used as a starting point for searching through customer vendor masks.
69–98	AN 30	Key Data: Vendor Branch Name The name of a vendor branch used as a starting point for searching through customer vendor masks.
99–128	AN 30	Key Data: Address 1 The first line of address information of a vendor branch used as a starting point for searching through customer vendor masks.
129–158	AN 30	Key Data: Address 2 The second line of address information of a vendor branch used as a starting point for searching through customer vendor masks.
or	or	or
39–47	N 9	Key Data: Vendor Number A vendor number used as a starting point for searching through customer vendor masks.
48–51	N 4	Key Data: Vendor Branch Number A vendor branch number used as a starting point for searching through customer vendor masks.
52	AN 1	Key Data: Reserved This field is not used.

The first five fields must be provided (the first eight bytes). Defaults are not used with this transaction.

If a key data value is used, values must be provided for applicable key data fields. If you are using the vendor branch by name key data, values must be provided for the Vendor Name, Vendor Branch Number, Address 1, and Address 2 fields. If you are using the vendor branch by number key, values must be provided for the Vendor Number, Vendor Branch Number, and the Reserved fields (space filled).

If a customer account with vendor is supplied as filter data, it is verified against each mask returned.

If key data is present, the Customer Account With Vendor field must be present. Any unused positions must be space filled.

History Inquiry Data. This data element is mandatory for history inquiry – all transactions and history inquiry – payments and transfers transactions. The following is the format of the Additional Data, Private data element for history inquiry transactions:

Position	Length	Field Description
1	AN 1	Request/Response Indicator A code that indicates whether this is the initial request, a continuation request, or a response. Valid values are as follows: 0 = Initial request 1 = Continuation request or response = Use defaults, where is a blank character
2–3	AN 2	Search Direction A code that indicates the direction of the search. Valid values are as follows: AC = Ascending DC = Descending (default)
4	AN 1	Key Data Type A code that indicates the type of key to be used for the search. Valid values are as follows: 0 = Business date (default) 1 = Customer reference number
5–7	N 3	Cumulative Transaction Count A running count of the number of items (i.e., transactions) returned thus far for the customer vendor mask verify list inquiry transaction.
8	AN 1	Reserved

Position	Length	Field Description
9	AN 1	Filter Data A code that indicates the type of filter to be used for the search. Valid values are as follows: A = Approved transactions D = Denied transactions B = Both
10–38	AN 29	Reserved This field is not used.
39–46	N 8	Key Data: Business Date The business date (YYYYMMDD) to be used as a starting point for searching through previously logged transactions. For financial transactions, this the date on which funds were collected from the <i>from</i> account. For nonfinancial transactions, this is the date on which the transaction was processed.
or	or	or
39–44	N 6	Key Data: Customer Reference Number The customer reference number to be used as a starting point for searching through previously logged transactions.

The first five fields must be provided (the first eight bytes). If the Request/Response Indicator field contains a space, then the following defaults are used:

- The search direction is descending
- The key data type is 0 (Business Date)
- The Business Date field is set to the current date.

If key data is present, filter data must be present.

If defaults are not used, the key data field cannot contain spaces.

Vendor List Inquiry Data. This data element is mandatory for master vendor list inquiry and vendor summary inquiry transactions. The following is the format of the Additional Data, Private data element for master vendor list inquiry and vendor summary inquiry transactions:

Position	Length	Field Description
1	AN 1	Request/Response Indicator A code that indicates whether this is the initial request, a continuation request, or a response. Valid values are as follows: 0 = Initial request 1 = Continuation request or response = Use defaults, where is a blank character
2–3	AN 2	Search Direction A code that indicates the direction of the search. Valid values are as follows: AC = Ascending (default) DC = Descending
4	AN 1	Key Data Type A code that indicates the type of key to be used for the search. The only valid value is 0 (vendor name).
5–7	N 3	Cumulative Transaction Count A running count of the number of items (i.e., vendors) returned thus far for the master vendor list inquiry transaction.
8	AN 1	Reserved This field is not used.
9	AN 1	Filter Data: List All Branches A code that indicates whether the list all branches filter is used for the search. Valid values are as follows: Y = Yes N = No

Position	Length	Field Description
10	AN 1	Filter Data: List Current Vendors A code that indicates whether the list current vendors filter is used for the search. Valid values are as follows: Y = Yes N = No
11	AN 1	Filter Data: List Verified Vendors A code that indicates whether the list verified vendors filter is used for the search. Valid values are as follows: Y = Yes N = No
12	AN 1	Filter Data: List Public Vendors A code that indicates whether the list public vendors filter is used for the search. Valid values are as follows: Y = Yes N = No
13–38	AN 26	Reserved This field is not used.
39–68	AN 30	Key Data: Vendor Name The name of the vendor to be used as a starting point for searching through the vendor registry.

The first five fields must be provided (the first eight bytes). If the Request/Response Indicator field contains a space, then the following defaults are used:

- The search direction is ascending
- The key data type is 0 (Vendor Name)
- The Vendor Name field is set to spaces

If key data is present, filter data must be present. The unused portion of the filter data must be space filled.

If defaults are not used, the Key Data field cannot contain spaces.

Miscellaneous Requests Inquiry Data. This data element is mandatory for miscellaneous requests inquiry transactions. The following is the format of the Additional Data, Private data element for miscellaneous requests inquiry transactions:

Position	Length	Field Description
1	AN 1	Request/Response Indicator A code that indicates whether this is the initial request, a continuation request, or a response. Valid values are as follows: 0 = Initial request 1 = Continuation request or response = Use defaults, where is a blank character
2–3	AN 2	Search Direction A code that indicates the direction of the search. Valid values are as follows: AC = Ascending DC = Descending (default)
4	AN 1	Key Data Type A code that indicates the type of key to be used for the search. Valid values are as follows: 0 = Transaction source date
5–7	N 3	Cumulative Transaction Count A running count of the number of items (i.e., miscellaneous requests) returned thus far for the miscellaneous requests inquiry transaction.
8	AN 1	Reserved This field is not used.
9–38	AN 30	Filter Data—Not currently used This field must be space-filled if a Key Data field is present in the message.

Position	Length	Field Description
39–46	N 8	Key Data: Transaction Source Date The transaction source date (YYYYMMDD) to be used as a starting point for searching for miscellaneous requests. This is the date the request was entered.

The first five fields must be provided (the first eight bytes). If the Request/Response Indicator field contains a space, then the following defaults are used:

- The search direction is descending
- The key data type is 0 (Transaction Source Date)
- The Transaction Source Date field is set to the current date

If key data is present, filter data must be present. Since the Filter Data field is not currently used, it must contain spaces if key data is present.

If defaults are not used, the Key Data field cannot contain spaces.

Scheduled Payments List Inquiry Data. This data element is mandatory for scheduled payments list inquiry transactions. The following is the format of the Additional Data, Private data element for scheduled payments list inquiry transactions:

Position	Length	Field Description
1	AN 1	Request/Response Indicator A code that indicates whether this is the initial request, a continuation request, or a response. Valid values are as follows: 0 = Initial request 1 = Continuation request or response ␣ = Use defaults, where ␣ is a blank character
2–3	AN 2	Search Direction A code that indicates the direction of the search. Valid values are as follows: AC = Ascending (default) DC = Descending

Position	Length	Field Description
4	AN 1	Key Data Type A code that indicates the type of key to be used for the search. The only valid value is 0 (next business date).
5–7	N 3	Cumulative Transaction Count A running count of the number of items (i.e., scheduled payments) returned thus far for the scheduled payments list inquiry transaction.
8	AN 1	Reserved This field is not used.
9–13	N 5	Filter Data: Customer Vendor Number A customer vendor number used to refine the search for scheduled payments. This field is required if key data is present.
14–38	AN 25	Reserved This field is not used.
39–46	N 8	Key Data: Next Business Date The next business date to be used as a starting point for searching for scheduled payments. This is the date on which the next payment is to be made.

The first five fields must be provided (the first eight bytes). If the Request/Response Indicator field contains a space, then the following defaults are used:

- The search direction is ascending
- The key data type is 0 (Next Business Date)
- The Next Business Date field is set to all zeros

If key data is present, filter data must be present.

If defaults are not used, the Key Data field cannot contain spaces.

Scheduled Transfers List Inquiry Data. This data element is mandatory for scheduled transfers list inquiry transactions. The following is the format for Scheduled Transfers List Inquiry transactions:

Position	Length	Field Description
1	AN 1	Request/Response Indicator A code that indicates whether this is the initial request , a continuation request, or a response. Valid values are as follows: 0 = Initial request 1 = Continuation request or response ␣ = Use defaults, where ␣ is a blank character
2–3	AN 2	Search Direction A code that indicates the direction of the search. Valid values are as follows: AC = Ascending (default) DC = Descending
4	AN 1	Key Data Type A code that indicates the type of key to be used for the search. The only valid value is 0 (next business date).
5–7	N 3	Cumulative Transaction Count A running count of the number of items (i.e., scheduled transfers) returned thus far for the scheduled transfers list inquiry transaction.
8	AN 1	Reserved This field is not used.
9–38	AN 30	Filter Data—Not currently used This field must be space-filled if key data is present in the message.
39–46	N 8	Key Data: Next Business Date The next business date to be used as a starting point for searching for scheduled transfers. This is the date on which the next transfer is to be made.

The first five fields must be provided (the first eight bytes). If the Request/Response Indicator field contains a space, then the following defaults are used:

- The search direction is ascending
- The key data type of 0 (Next Business Date)
- The Next Business Date field is set to all zeros

If key data is present, filter data must be present. Since the Filter Data field is not currently used, it must contain spaces if key data is present.

If defaults are not used, the Key Data field cannot contain spaces.

Scheduled Transactions List Inquiry Data. This data element is mandatory for scheduled transactions list inquiry transactions. The following is the format of the Additional Data, Private data element for scheduled transactions list inquiry transactions:

Position	Length	Field Description
1	AN 1	Request/Response Indicator A code that indicates whether this is the initial request, a continuation request, or a response. Valid values are as follows: 0 = Initial request 1 = Continuation request or response ␣ = Use defaults, where ␣ is a blank character
2–3	AN 2	Search Direction A code that indicates the direction of the search. Valid values are as follows: AC = Ascending DC = Descending (default)
4	AN 1	Key Data Type A code that indicates the type of key to be used for the search. Valid values are as follows: 0 = Payment date 1 = Customer reference number

Position	Length	Field Description
5–7	N 3	Cumulative Transaction Count A running count of the number of items (i.e., scheduled transactions) returned thus far for the scheduled transactions list inquiry transaction.
8	AN 1	Reserved This field is not used.
9–13	N 5	Filter Data: Customer Vendor Number A customer vendor number used to refine the search for scheduled payments. This field is required if key data is present.
14–38	AN 25	Reserved This field is not used.
39–46	N 8	Key Data: Payment Date The payment date (YYYYMMDD) to be used as a starting point for searching for scheduled transactions. This is the date on which the next payment or transfer is to be made for this scheduled transaction.
or	or	or
39–44	N 6	Key Data: Customer Reference Number The customer reference number to be used as a starting point for searching through scheduled transactions.

The first five fields must be provided (the first eight bytes). If the Request/Response Indicator field contains a space, then the following defaults are used:

- The search direction is descending
- The key data type is 0 (Payment Date)
- The Payment Date field is set to all zeros

If key data is present, filter data must be present.

If defaults are not used, the Key Data field cannot contain spaces.

P-49 Transaction Currency Code

Format: N 3

ISO Field Name: Currency Code, Transaction

Used By: BASE24-telebanking
BASE24-billpay

The Transaction Currency Code data element contains a code that defines the local currency of the acquirer or source location of the transaction. This is the currency that applies to the Transaction Amount (P-4) data element.

On incoming messages, the value from this data element is placed in the ITD-AMT-TXN.TXN.CRNCY-CDE field of the Internal Transaction Data (ITD).

On outgoing messages, this value is taken from the ITD-AMT-TXN.TXN.CRNCY-CDE field of the ITD.

P-52 Personal Identification Number (PIN)

Format: AN 16

ISO Field Name: Personal Identification Number (PIN) Data

Used By: BASE24-telebanking

Personal Identification Number (PIN) data is assigned to a customer to uniquely identify that customer at the point of service. This element can contain a clear PIN or an encrypted PIN, and is sent as 16 hexadecimal digits. If a clear PIN is used, it is left justified and padded to the right with a pad character.

In every request from the device, the PIN data must be present or position 12 of the Point of Service Data Code (P-22) data element must be set to a value of S.

On incoming messages, the value from this data element is placed in the ITD-PIN-DATA.PIN field of the Internal Transaction Data (ITD).

P-54 Additional Amounts

Format: ANS ...120

ISO Field Name: Additional Amounts

Used By: BASE24-telebanking
BASE24-billpay

The Additional Amounts data element carries balance information for the accounts involved in the transaction and the remaining amount. This data element is used to return balance information for an available funds inquiry, transfer – immediate or schedule payment – immediate transaction. If BASE24 is configured to return balances in PIN verification responses and if a default account is defined for a customer, the balances associated with the default account are returned in the response in this data element.

The remaining amount is the maximum amount available for the customer to transfer when insufficient funds exist in the account or when the original amount requested for a transfer would cause the BASE24-telebanking transfer limits in the Positive Balance File (PBF) to be exceeded.

The following ISO definition can occur up to 6 times:

Position	Length	Field Description
1-2	N 2	Account Type An ISO code indicating the type of customer account (e.g., 20 for a demand deposit account) to which this balance information applies. Refer to the Account Type field description provided earlier in this section for a list of the BASE24 reserved account type values. On outgoing messages, the value for this field is taken from the ITD-PROC-CDE.ACCT1-TYP or ACCT2-TYP fields of the Internal Transaction Data (ITD).

Position	Length	Field Description
3–4	N 2	Amount Type A code indicating the type of balance amount provided for this account. Valid values are as follows: 01 = Amount 1 ledger balance 02 = Amount 1 available balance 03 = Amount owed 04 = Amount due 05 = Available credit 16 = Amount 2 ledger balance 17 = Amount 2 available balance 20 = Amount remaining On outgoing messages, the value for this field is taken from the ITD-BAL.INFO.AMT-TYP field of the Internal Transaction Data (ITD).
5–7	N 3	Currency Code A code indicating the currency in which the balance amount is provided. Valid currency codes are documented in the ISO 4217 standard, <i>Codes for the Representation of Currencies and Funds</i>. On outgoing messages, the currency code for the amount remaining is taken from the ITD-AMT-TXN.TXN.CRNCY-CDE field of the Internal Transaction Data (ITD). The currency code for all other balance amounts is taken from the ITD-BAL.INFO.CRNCY-CDE field of the ITD.
8–20	X + N 12	Amount The account balance in whole and fractional currency units. If the balance is negative, the amount is preceded by a minus sign (–). On outgoing messages, the amount remaining is taken from the ITD-AMT-REMAIN field of the Internal Transaction Data (ITD). All other balance amounts are taken from the ITD-BAL.INFO.AMT field of the ITD.

P-56 Original Data Elements

Format: N ..35

ISO Field Name: Original Data Elements

Used By: BASE24-telebanking
BASE24-billpay

The Original Data Elements data element is a group of five sub-elements contained in a reversal, intended to identify the original transaction being reversed.

This data element is also sent in the network management request/repeat messages that acknowledge the receipt of a new key. When present in the network management message, the original system trace audit number sub-element must contain the system trace audit number of the message that contained the new key.

On incoming messages, this information is placed in the ITD-ORIG-DATA fields of the Internal Transaction Data (ITD).

The following ISO format is used for this data element. Absence of data is indicated by zeros.

Position	Length	Field Description
1–4	N 4	Original Message Type Identifier The type of message that was reversed. Valid values are as follows: 1110 = Authorization request response from an Integrated Authorization Server process 1210 = Financial request response from an Integrated Authorization Server process
5–10	N 6	Original System Trace Audit Number The sequence number originally assigned by the message initiator to uniquely identify the message.
11–22	N 12	Original Local Transaction Date and Time The date and time, in YYMMDDhhmmss format, the original transaction was processed by the BASE24 system.

Position	Length	Field Description
23–24	N 2	Original Acquiring Institution ID Code Length The length of the original acquiring institution ID.
25–35	N 11	Original Acquiring Institution ID Code A code identifying the acquiring institution. In the United States, this is the routing and transit number.

P-59 Transport Data

Format: ANS ...30

ISO Field Name: Transport Data

Used By: BASE24-telebanking
BASE24-billpay

The Transport Data data element contains data from the originator of the message that must be returned unaltered in the response. For example, this data element can contain information that the device uses and requires to be present in response messages.

BASE24 Remote Banking does not use the data in this data element. The VRU Interface Object echoes back any information received in this data element in a request message in the corresponding response message.

On incoming messages, the information from this data element is placed in the ITD-ACQ-DPNT.FDATA field in the Internal Transaction Data (ITD).

On outgoing messages, the information from this data element is taken from the ITD-ACQ-DPNT.FDATA field in the Internal Transaction Data (ITD).

P-62 Recurring Transaction Data

Format: ANS ...7

ISO Field Name: BASE24-billpay Recurring Transaction Data

Used By: BASE24-billpay

The Recurring Transaction Data element contains additional information that is included in request or response messages for scheduled payment and transfer transactions.

On incoming messages, this information is placed in the ITD-RECUR-TXN-DATA fields of the Internal Transaction Data (ITD).

On outgoing messages, this information is taken from the ITD-RECUR-TXN-DATA fields of the ITD.

The format of this data element in the ISO message is as follows:

Position	Length	Field Description
1-2	AN 2	Period Type A code that indicates the period type for this transaction. Valid values are as follows: WK = Weekly 2W = Every two weeks MN = Monthly Q = Quarterly S = Semi-annually A = Annually
3-6	AN 4	Number of Periods / Periods Remaining The remaining number of times a recurring transaction is scheduled to be initiated.

Position	Length	Field Description
7	N 1	Skip Next Payment Indicator A code that indicates whether the next scheduled payment or transfer is to be skipped for this transaction. Valid values are as follows: 0 = Do not skip next payment or transfer Any other value = Skip next payment or transfer

The Period Type subfield indicates the type of period (e.g., monthly, etc.) for a recurring payment or transfer. This subfield must contain a valid value for schedule payment – recurring and schedule transfer – recurring transaction requests. The same value is echoed back in the response message.

The Period Type subfield must contain a valid value in scheduled payment update and scheduled transfer update transaction requests for recurring transactions if the period type is changed by the customer. If this data element is sent in a scheduled payment update or scheduled transfer update transaction for a recurring transaction and the period type is not changed, the Period Type field must be set to spaces.

The Number of Periods/Periods Remaining field contains the number of periods for a recurring payment or transfer. This field must contain a valid value in schedule payment – recurring and schedule transfer – recurring transaction requests. The value in the response message indicates the number of periods remaining.

The Number of Periods/Periods Remaining field must contain a valid value in scheduled payment update and scheduled transfer update transaction requests for recurring transactions if the number of periods is changed by the customer. If this data element is sent in a scheduled payment update or scheduled transfer update transaction request for a recurring transaction and the number of periods is not changed, the Number of Periods/Periods Remaining field must be set to spaces.

The Skip Next Payment Indicator field specifies whether the next scheduled recurring payment or transfer should be skipped. This field must contain a valid value in scheduled payment update and scheduled transfer update transaction requests for recurring transactions.

P-63 Source Code

Format: ANS ...2

ISO Field Name: Source Code

Used By: BASE24-telebanking
BASE24-billpay

The Source Code data element identifies the type of device that initiated the transaction.

On incoming messages, the value from this data element is placed in the ITD-SRC-CDE field of the Internal Transaction Data (ITD). If this data element is not provided in the incoming message, the value from the DEFAULT SOURCE CODE field on VCD screen 18 is used.

On outgoing messages, this value is taken from the ITD-SRC-CDE field of the ITD.

The format of this data element in the ISO message is as follows:

Position	Length	Field Description
1-2	AN 2	Source Code
		A code that indicates the type of device that initiated the transaction. Valid values are as follows:
		AD = Audio device
		BL = Billpay billing
		IB = Inbound CSR
		PC = Personal computer
		SP = Screen phone
		All other values are user defined.

The Source Code field values AD, BL, IB, PC and SP are BASE24-defined valid values. The value BL is generated by the BASE24-billpay Billing system. Any other alphanumeric value is accepted as a user-defined value.

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Section 6

Data Elements 65 Through 128

This section contains descriptions for data elements 65 through 128 of the BASE24-telebanking external message. It also includes descriptions for the external data block that carries data used in conjunction with data element 125 for certain transaction request and response messages.

S-72 Data Record

Format: ANS ...999

ISO Field Name: Data Record

Used By: BASE24-telebanking
BASE24-billpay

The Data Record data element is used when BASE24 Remote Banking or the device is unable to read a message. It contains a three-byte numeric field that indicates the data element in question followed by the original message that is being rejected.

If the message type indicator is invalid, the three-byte numeric field is set to a value of 200. If the primary bit map is invalid, the three-byte numeric field is set to a value of 201. This data element is sent in Administrative Notify (1644) messages only and is not mapped to or from the Internal Transaction Data (ITD).

S-73 Action Date

Format: N 6 (YYMMDD)

ISO Field Name: Action Date

Used By: BASE24-billpay

The Action Date data element contains the date of the payment or transfer specified by the customer in schedule payment – future, schedule payment – recurring, schedule transfer – future, and schedule transfer – recurring transaction requests.

The Action Date data element is optional in scheduled payment update and scheduled transfer update transaction requests for future or recurring transactions. This data element is included if the customer has specified a new date for the next payment or transfer.

On incoming messages, the value from this data element is placed in the ITD-ACT-DAT field of the Internal Transaction Data (ITD).

On outgoing messages, the value is taken from the ITD-ACT-DAT field of the ITD.

S-95 Card Issuer Reference Data

Format: ANS ..6

ISO Field Name: Card Issuer Reference Data

Used By: BASE24-billpay

The Card Issuer Reference Data element contains the customer reference number assigned by BASE24-billpay. The customer reference number is returned in all approved—and a subset of denied—BASE24-billpay response messages, and must be echoed in any subsequent reversal messages. The customer reference number is a value generated by the XPNET or Pathway Billpay Server process for identification purposes when it processes a transaction.

On incoming messages, the value from this data element is placed in the ITD-CUST-REF-NUM field of the Internal Transaction Data (ITD).

On outgoing messages, the value of this data element is taken from the ITD-CUST-REF-NUM field of the ITD.

Note: This field is only present in the first response of the following BASE24-billpay inquiry transactions that require multiple requests and responses:

- Customer vendor list inquiry
- Scheduled payments list inquiry
- Scheduled transfers list inquiry
- History inquiry – payments and transfers
- History inquiry – all transactions
- Bank list inquiry
- Bank branch list inquiry
- Customer vendor mask verify list inquiry
- Master vendor list inquiry
- Miscellaneous requests inquiry
- Scheduled transactions list inquiry

S-96 Key Management Data

Format: ANS ...114

ISO Field Name: Key Management Data

Used By: BASE24-telebanking

The ANSI X9.17-1985 standard, *Financial Institution Key Management (Wholesale)* establishes the standard for key management and defines the Cryptographic Service Message (CSM). The CSM is a free format ANSI X9.17 standard data element that is used for moving key management data between two entities when the keys are distributed automatically.

CSM information is returned in data element 96 in responses when a key change must occur. Returning the key information in this manner provides a way to initiate a key change without sending an unsolicited message to the device.

On incoming messages, the information from this data element is placed in the ITD-ACQ-DPNT.FDATA field of the Internal Transaction Data (ITD).

On outgoing messages, the value of this data element is taken from the ITD-ACQ-DPNT.FDATA field of the ITD.

Cryptographic Service Messages (CSM)

There are several classes of CSMs, three of which are used by BASE24 Remote Banking. These include the Error Service Message (ESM), Key Service Message (KSM), and Response Service Message (RSM). All other CSM classes are considered optional and are ignored if received from the device.

The following subsections show the mandatory fields used for the BASE24 Remote Banking implementation of the ANSI X9.17-1985 standard. The intent of these subsections is to illustrate the format and possible content of the CSM. These subsections are not intended to reiterate the ANSI X9.17-1985 standard.

The ANSI X9.17-1985 standard uses key tags as delimiters and field identifiers in the actual CSM Data area. An example of a key tag is MCL/. Message data fields are illustrated in lower case and indicate data that is inserted into these fields. Note that the CSM itself always begins with the literal CSM, and the message contents are always carried between a pair of parentheses.

For detailed descriptions of the fields used in CSMs, refer to the ***BASE24 Integrated Server Transaction Security Manual***.

Key Service Message (KSM)

A Key Service Message (KSM) is generated when a threshold has been surpassed or a BASE24 operator has requested a new key.

Refer to the ANSI X9.17-1985 standard for more information on the KSM.

The CSM data area is listed below. Note that the Length column includes the length of the key tag as well as the field value and represents the maximum length allowed for the field. A `␣` character denotes a blank space.

Key Tag	Field	Length	Description	Value
CSM(		4	CSM Tag and Start Parenthesis	
MCL/	mmm␣	8	Message Class	KSM
RCV/	rrrrrrrrrrrrrr␣	21	Message Receiver	
ORG/	ooooooooooooooooo␣	21	Message Originator	
KPTR/	kkkkkkkkkkkkkkkk␣	22	New Key	
CTR/	hhhhhhhhhhhhhh␣	19	Originator's Hexadecimal Key Count	
VVK/	vvvv	8	Key Verification Value	
)		1	End Parenthesis	

The ORG and RCV field values must be agreed upon by the two parties. The BASE24 originating ID must be the receiving ID for the device and vice versa.

The hexadecimal key counter has a range of 0 to FFFFFFFF. Leading zeros are suppressed.

Error Service Message (ESM)

An Error Service Message (ESM) is sent in response to the detection of one or more error conditions.

Note: BASE24 expects the device to always accept or adopt the count provided by BASE24 in the Key Service Message (KSM). Therefore, the device should never reject the new key because of a count synchronization error.

Refer to the ANSI X9.17-1985 standard for more information on the ESM.

The CSM data area is listed below. Note that the Length column includes the length of the key tag as well as the field value and represents the maximum length allowed for the field. A `␣` character denotes a blank space.

Key Tag	Field	Length	Description	Value
CSM(		4	CSM Tag and Start Parenthesis	
MCL/	mmm␣	8	Message Class	ESM
RCV/	rrrrrrrrrrrrrrrr␣	21	Message Receiver	
ORG/	oooooooooooooooooooo␣	21	Message Originator	
CTP/	hhhhhhhhhhhhhhhh␣	19	Receiver's Hexadecimal Key Count	
CTR/	hhhhhhhhhhhhhhhh␣	19	Originator's Hexadecimal Key Count	
ERF/	eeeeeeeeeeeeeeee␣	21	Error Field	
)		1	End Parenthesis	

The CTP field contains the count expected and the CTR contains the original count sent in the KSM message. Since BASE24 always expects the device to adopt the CTR count, the CTP field is never present. Refer to section 7.3 of the ANSI X9.17-1985 standard for a discussion of counters.

The CTR field is always present. It has a range of 0 to FFFFFFFF. Leading zeros are suppressed.

The ERF field contains a variable-length string containing one or more of the following characters depending on the errors detected:

Error Code	Definition
C	Cannot process (general error)
F	Format (syntax) error

Error Code	Definition
H	User defined (defined to mean Invalid ID)
P	CTP error

The value in the ORG field in the ESM must equal the value in the RCV field for the KSM, and the value in the RCV field in the ESM must equal the value in the ORG field for the KSM.

Response Service Message (RSM)

A Response Service Message (RSM) is generated following receipt of an acceptable KSM.

Refer to the ANSI X9.17-1985 standard for more information on the RSM.

The CSM data area is listed below. Note that the Length column includes the length of the key tag as well as the field value and represents the maximum length allowed for the field. A character denotes a blank space.

Key Tag	Field	Length	Description	Value
CSM(		4	CSM Tag and Start Parenthesis	
MCL/	mmm	8	Message Class	RSM
RCV/	rrrrrrrrrrrrrr	21	Message Receiver	
ORG/	oooooooooooooooo	21	Message Originator	
)		1	End Parenthesis	

The value in the ORG field in the RSM must equal the value in the RCV field for the KSM, and the value in the RCV field in the RSM must equal the value in the ORG field for the KSM.

S-98 Payee

Format: ANS 25

ISO Field Name: Payee

Used By: BASE24-billpay

The Payee data element contains the customer vendor number. This field must be included in schedule payment – immediate, schedule payment – future, and schedule payment – recurring requests, and is optional for scheduled payment update and scheduled payment delete requests. If this data element is provided in a scheduled payment delete or scheduled payment update transaction, it is informational only and is not used.

The value in this field is left justified and blank filled.

On incoming messages, the value from this data element is placed in the ITD-PAYEE.FLD field of the Internal Transaction Data (ITD).

On outgoing messages, the value for this data element is taken from the ITD-PAYEE.FLD field of the ITD.

S-102 Account Identification 1

Format: ANS ..28

ISO Field Name: Account Identification 1

Used By: BASE24-telebanking
BASE24-billpay

The Account Identification 1 data element contains a series of digits used to identify a customer account, usually some account that is related to the customer ID in the Primary Account Number (P-2) or Primary Account Number, Extended (P-34) data element. This is the account associated with the account type found in positions 3 and 4 of the Processing Code (P-3) data element and identifies the *from* account involved in a transfer transaction, the first account involved in a dual inquiry, or the single account involved in transactions other than a transfer.

This data element does not specify the account number used for customer account transactions (i.e., add, activate, delete, deactivate, and update).

If an account 1 qualifier is specified in the Account Qualifiers (S-123) data element, this field can be omitted.

Note: BASE24 Remote Banking currently supports only account numbers that are less than 20 digits in length. Also, although ISO defines this field as alphanumeric, BASE24 Remote Banking supports only numeric values in this field.

On incoming messages, the value from this data element is placed in the ITD-ACCT1-NUM field of the Internal Transaction Data (ITD).

On outgoing messages, the value for this data element is taken from the ITD-ACCT1-NUM field of the ITD.

S-103 Account Identification 2

Format: ANS ..28

ISO Field Name: Account Identification 2

Used By: BASE24-telebanking
BASE24-billpay

The Account Identification 2 data element contains a series of digits used to identify a customer account, usually some account that is related to the customer ID in the Primary Account Number (P-2) or Primary Account Number, Extended (P-34) data element. This is the account associated with the account type found in positions 5 and 6 of the Processing Code (P-3) data element and identifies the *to* account involved in a transfer transaction, the second account involved in a dual available funds inquiry transaction, or the second account involved in a user-defined transaction.

If an account 2 qualifier is specified in the Account Qualifiers (S-123) data element, this field can be omitted.

Note: BASE24 Remote Banking currently supports only account numbers that are less than 20 digits in length. Also, although ISO defines this field as alphanumeric, BASE24 Remote Banking supports only numeric values in this field.

On incoming messages, the value from this data element is placed in the ITD-ACCT2-NUM field of the Internal Transaction Data (ITD).

On outgoing messages, the value for this data element is taken from the ITD-ACCT2-NUM field of the ITD.

S-104 Transaction Description

Format: ANS ...60

ISO Field Name: Transaction Description

Used By: BASE24-billpay

The Transaction Description data element contains additional payment data. This data is returned in all approved, and a subset of denied, schedule payment – immediate, schedule payment – future, and schedule payment – recurring transaction response messages.

Note: Even though this data element carries only 60 bytes of data for BASE24 Remote Banking, the length field preceding the data still has three positions.

On incoming messages, the information from this data element is placed in the ITD-PAYEE-DESCR fields of the Internal Transaction Data (ITD).

On outgoing messages, the information for this data element is taken from the ITD-PAYEE-DESCR fields of the ITD.

The format of this field in the ISO message is as follows:

Position	Length	Field Description
1–28	AN 28	Customer’s Account Number with Vendor The customer account with the vendor to receive the customer payment.
29–60	AN 30	Vendor (Merchant) Name The name of the vendor to receive the customer payment.

S-123 Account Qualifiers

Format: AN ...8

ISO Field Name: BASE24-telebanking/BASE24-billpay Account Qualifiers

Used By: BASE24-telebanking
BASE24-billpay

The Account Qualifiers data element contains the account qualifiers used to determine the account 1 number and type and the account 2 number and type. Account qualifiers (sometimes referred to as short codes or account codes) can be assigned to account numbers. The customer can enter an account qualifier in place of the account number for a transaction. This means that data elements 102 and 103 can be omitted if data element 123 is specified. Data element 123 is available for both account 1 and account 2 qualifiers.

Note: Even though this data element carries only 8 bytes of data for BASE24 Remote Banking, the length field preceding the data still has three positions.

On incoming messages, the information from this data element is placed in the ITD-ACCT1-QUAL and ITD-ACCT2-QUAL fields of the Internal Transaction Data (ITD).

On outgoing messages, the information for this data element is taken from the ITD-ACCT1-QUAL and ITD-ACCT2-QUAL fields of the ITD.

The format of this field in the ISO message is as follows:

Position	Length	Field Description
1-4	AN 4	Account 1 Qualifier A code used in place of the account number and account type to identify account 1, which is the <i>from</i> account involved in a transfer transaction, the first account involved in a dual inquiry, user-defined transaction, or the single account involved in transactions other than a transfer.

Position	Length	Field Description
5-8	AN 4	Account 2 Qualifier A code used in place of the account number and account type to identify account 2, which is the <i>to</i> account involved in a transfer transaction, the second account involved in a dual available funds inquiry transaction, or the second account involved in a user-defined transaction.

S-124 Message Format Type ID

Format: AN ...8

ISO Field Name: Message Format Type ID

Used By: BASE24-telebanking
BASE24-billpay

This data element is reserved for future use. It is intended to be used when different message formats are required. The Product Type field is reserved for ACI line of business use only, while the Other Type field can be used as required for custom development.

On incoming messages, the information from this data element is placed in the ITD-FRMT-TYP-ID fields of the Internal Transaction Data (ITD).

On outgoing messages, the information for this data element is taken from the ITD-FRMT-TYP-ID fields of the ITD.

The format of this field in the ISO message is as follows:

Position	Length	Field Description
1–4	AN 4	Product Type This field is reserved for product use.
5–8	AN 4	Other Type This field can be used as required.

S-125 Transaction Data, Private

Format: ANS ...998

ISO Field Name: BASE24-telebanking Multiple Account Information Request Data
BASE24-telebanking Transaction History Response Data
BASE24-telebanking Account List Response Data
BASE24-telebanking Multiple Account Balance Response Data
BASE24-telebanking Customer Verification Data
BASE24-telebanking Statement Download Response Data
BASE24-telebanking Statement Closing Download Response Data
BASE24-billpay Request/Response Data

Used By: BASE24-telebanking
BASE24-billpay

Multiple Account Information Request Data

Multiple account information request data carries the information needed to begin processing a multiple account balance inquiry transaction. The Multiple Account Information structure can occur up to a maximum of 13 times. A two-byte numeric count field provided at the beginning of the data indicates how many occurrences exist for the Multiple Account Information structure in the message.

On incoming messages, the value from this data element is placed in the ITD-CUST-REF-NUM field of the Internal Transaction Data (ITD).

On outgoing messages, the value of this data element is taken from the ITD-CUST-REF-NUM field of the ITD.

The following format pertains to the multiple account balance inquiry transaction request:

Position	Length	Field Description
1-2	N 2	Count The number of accounts for which balance information is requested in this request message. Valid values are 0-13.
...		Multiple Account Information Structure The following fields occur up to 13 times. Each occurrence is 30 characters in length.
	AN 28	Account Number The number of a customer application account for which balance information is requested.

Position	Length	Field Description
	AN 2	Account Type
		The ISO account type code for this application account.
		The BASE24 reserved account type values are as follows:
		10 = Savings account
		1A = Savings account 1
		1B = Savings account 2
		1C = Savings account 3
		1D = Savings account 4
		1E = Savings account 5
		1F = Savings account 6
		20 = Demand deposit account
		2A = Demand deposit account 1
		2B = Demand deposit account 2
		2C = Demand deposit account 3
		2D = Demand deposit account 4
		2E = Demand deposit account 5
		2F = Demand deposit account 6
		2G = Demand deposit account 7
		2H = Demand deposit account 8
		30 = Credit account
		3A = Credit account 1
		3B = Credit account 2
		3C = Credit account 3
		3D = Credit account 4
		3E = Credit account 5
		3F = Credit account 6
		3G = Credit account 7
		38 = Line of credit
		58 = Certificate
		59 = Retirement account
		90 = Interest-bearing checking account
		9A = Commercial loan
		9B = Installment loan
		9C = Mortgage loan

Transaction History Response Data

Transaction history response data carries the results of all BASE24-telebanking last count of transactions inquiry requests. The Last Transaction structure can occur up to a maximum of 15 times. A two-byte numeric count field provided at the beginning of the data indicates how many occurrences exist for the Last Transaction structure in the message. The format defined below pertains to the following transactions:

- Check clearance inquiry
- Last count of debit/credit transactions inquiry
- Last count of account by source transactions inquiry
- Last count of check transactions inquiry
- Last count of debit transactions inquiry
- Last count of credit transactions inquiry
- Last count of account transfer transactions inquiry

On outgoing messages, the last transaction structure is taken from either an internal buffer maintained by the Authorization Combined Positive Customer object for transactions authorized offline, or from the DATA field in the Telebanking Standard Internal Message (BSTM) for transactions authorized online by a host. The LAST-X-OFST and the LAST-X-LGTH fields in the BSTM indicate the position of this transaction history information in the DATA field.

The internal buffer is structured according to the LAST-TXN.LAST-TXN-INFO definition described in the CDEFSGDS file on the core data definitions subvolume.

Transaction history response data is obtained from the Transaction History File (THF).

Position	Length	Field Description
1-2	N 2	Count
		The number of items (i.e., transactions) returned in this response message. Valid values are 0-15.

Position	Length	Field Description
3–5	AN 3	Currency Code A code that identifies the currency of the amount carried in the Amount field. Valid values are listed in the 1990 ISO 4217 standard, <i>Codes for the Representation of Currencies and Funds</i> .
6	AN 1	Reserved This field is not used.
...		Last Transaction Structure The following fields occur up to 15 times. Each occurrence is 56 characters in length.
	N 6	Date (YYMMDD) The date on which this transaction occurred.
	N 12	Amount The amount of the transaction in whole and fractional currency units.
	AN 12	Check Number The serial number of the check written for the transaction. The presence of a serial number identifies a check transaction.
	AN 1	Debit/Credit Indicator A code that identifies whether this transaction is a debit or credit. Valid values are as follows: C = Credit D = Debit

Position	Length	Field Description
AN 1		Source Type A code that identifies the source of a transaction. Any value can be used. However, values reserved for use by BASE24 products are as follows: A = ATM B = Telebanking device P = POS Device T = Teller
AN 1		Funds Indicator A code that identifies whether the account had sufficient funds to approve a debit transaction. Valid values are as follows: 0 = Sufficient funds 1 = Not sufficient funds
AN 1		Reversal Indicator A code that identifies whether the transaction had to be reversed. Valid values are as follows: 0 = Not reversed 1 = Reversed
AN 14		Reserved Transaction Flags This field is not currently used by BASE24 and should be zero filled.
AN 8		Host Transaction Code The code assigned to this transaction for processing on the host.

Account List Response Data

Account list response data carries the results of the BASE24-telebanking customer account list inquiry transaction request. The Account List structure can occur up to a maximum of 15 times. A two-byte numeric count field provided at the beginning of the data indicates how many occurrences exist for the Account List structure in the message. Customer information precedes the account information.

On outgoing messages, the account list structure is taken from either an internal buffer maintained by the Authorization Combined Positive Customer object for transactions authorized offline, or from the DATA field in the Telebanking Standard Internal Message (BSTM) for transactions authorized online by a host. The LAST-X-OFST and the LAST-X-LGTH fields in the BSTM indicate the position of this account information in the DATA field.

The internal buffer is structured according to the LAST-TXN.ACCT-LIST-INFO definition described in the CDEFSGDS file on the core data definitions subvolume.

Data for this response is obtained from the Customer Table (CSTT) and Customer/Account Relation Table (CACT).

The following format pertains to the account list inquiry transaction response:

Position	Length	Field Description
1-2	N 2	Count The number of items (i.e., accounts) returned in this response message. Valid values are 0-15.
3	AN 1	Additional Data Flag A code indicating whether more application accounts exist for this customer ID. Valid values are as follows: Y = Yes, more accounts exist N = No, no more accounts exist
4-5	AN 2	Customer Type A code identifying the type of customer associated with this customer ID. This is an information-only field that is not used in processing. Valid values are as follows: B = Business customer C = Individual customer All other values are user defined. All values are left justified.

Position	Length	Field Description
6–7	AN 2	Customer Verify Flag A code identifying the status of the customer relationship. Valid values are as follows: C = Closed I = Issued, not active V = Verified All other values are user defined. All user-defined codes are treated the same as closed accounts. Customer service representatives can add accounts for the customer only if this field contains a value of I or V. Customer service representatives can delete accounts for the customer only when this field contains a value of I. The customer ID cannot be used in processing unless this field contains a value of V. When a customer activate transaction is processed, the value of this field is changed from an I to a V. All values are left justified.
8–27	AN 20	Customer Information Buffer #1 User-defined customer verification information. This information is returned on customer verify transactions and can be used for customer verification purposes. For example, this field and the Customer Information Buffer #2 and #3 fields could be used to contain the customer's government ID, birth date, and mother's maiden name.
28–47	AN 20	Customer Information Buffer #2 See the Customer Information Buffer #1 field description above.
48–67	AN 20	Customer Information Buffer #3 See the Customer Information Buffer #1 field description above.

Position	Length	Field Description
68	AN 1	Reserved This field is not used.
...		Account List Structure The following fields occur up to 15 times. Each occurrence is 48 characters in length.
	AN 19	Account Number An application account number.
	AN 2	Account Type An ISO code indicating the type of application account. Refer to the Account Type field description provided in the Multiple Account Information request redefine for a list of the BASE24 reserved account type values.
	AN 4	Account Qualifier A user-defined code that can be used in place of the application account number and account type to identify and access this account.
	AN 4	FIID The FIID of the customer's financial institution for this application account.

Position	Length	Field Description
AN 1	Status	A code indicating the current status of the application account and the action to be taken by the Integrated Authorization Server process if this account status is encountered during transaction processing. Valid values are as follows: 0, A, B, C = Inactive (deny transactions) 1, D, E, F, G, H, I = Open (accept transactions) 2, J, K, L = Restricted to Deposits (accept all transactions except transfers and payments from this account) 3, M, N, O, P, Q, R = Open Primary (accept transactions) 4, S, T, U = Restricted Primary (accept all transactions except transfers and payments from this account) 9, V, W, X, Y, Z = Closed (deny transactions) BASE24 products do not distinguish between the alphabetic and numeric values grouped together above. The alphabetic values are intended to give institutions a wider range of values for assigning account statuses
AN 15	Account Description	A text description for this application account.
AN 1	Allowed: Debit	A code indicating whether the customer can perform transfer or payment transactions that debit this account. Valid values are as follows: Y = Yes, account can be debited for a transfer or payment N = No, account cannot be debited for a transfer or payment

Position	Length	Field Description
	AN 1	Action Allowed: Credit A code indicating whether the customer can perform credit transactions on this account. Credit transactions include transfers to this account. Valid values are as follows: Y = Account can be credited N = Account cannot be credited
	AN 1	Action Allowed: Inquiry A code indicating whether the customer can perform inquiry transactions (e.g., available funds inquiry and scheduled transfers list inquiry) on this account. Valid values are as follows: Y = Account can be inquired upon N = Account cannot be inquired upon

Multiple Account Balance Response Data

Multiple account balance response data carries the results of an approved BASE24-telebanking multiple account balance inquiry transaction request. The Multiple Account Balance structure can occur up to a maximum of 13 times. A two-byte numeric count field provided at the beginning of the data indicates how many occurrences exist for the Multiple Account Balance structure in the message.

Note: If the multiple account balance inquiry transaction is denied, the request data is returned in the message.

On outgoing messages, the multiple account balance structure is taken from either an internal buffer maintained by the Authorization Combined Positive Customer object for transactions authorized offline, or from the DATA field in the Telebanking Standard Internal Message (BSTM) for transactions authorized online by a host. The LAST-X-OFST and the LAST-X-LGTH fields in the BSTM indicate the position of this multiple account balance information in the DATA field.

The internal buffer is structured according to the LAST-TXN.ACCT-INFO definition described in the CDEFSGDS file on the core data definitions subvolume.

The following format pertains to the multiple account balance inquiry transaction response:

Position	Length	Field Description
1-2	N 2	Count The number of items (i.e., accounts) returned in this response message. Valid values are 0-13.
...		Multiple Account Balance Structure The following fields occur up to 13 times. Each occurrence is 59 characters in length.
AN 19		Account Number The number of the application account to which this balance information applies.
AN 2		Account Type An ISO code indicating the type of application account. Refer to the Account Type field description provided in the Multiple Account Information request redefine for a list of the BASE24 reserved account type values.
AN 3		Action Code The action code returned as a result of performing the inquiry. Refer to the <i>BASE24 Remote Banking Transaction Processing Manual</i> for a list of all valid action codes.
AN 8		Balance Date (YYYYMMDD) The date on which the balance amount in the Balance Amount field was determined.
AN 3		Currency Code A code identifying the currency of the balance amount in the Balance Amount fields. Reserved values are listed in the 1990 ISO 4217 standard, <i>Codes for the Representation of Currencies and Funds</i> .

Position	Length	Field Description
Balance Entry Information		
The following fields occur 2 times for each Multiple Account Balance Structure occurrence.		
AN 2		Balance Amount Type
A code indicating the type of balance information returned. Valid values are as follows:		
01 = Ledger balance of account 1 (noncredit account)		
02 = Available balance of account 1 (noncredit account)		
03 = Credit balance (credit account)		
04 = Amount due (not currently supported)		
05 = Available credit balance (credit account)		
16 = Ledger balance of account 2 (noncredit account)		
17 = Available balance of account 2 (noncredit account)		
AN 1		Balance Amount Sign
If the balance is negative, this field contains a minus sign (–). If the balance is positive, this field contains either a blank space or a plus sign (+).		
AN 12		Balance Amount
The balance for this account.		

Customer Verification Data

Customer verification data includes the initial customer verify transaction request, and a response of either a customer verify transaction response or customer select transaction request. The response depends on whether more than one customer exists that matches the customer verify request and whether a customer name is entered in the request. If a customer name is entered in the request, a customer select transaction request is always returned from BASE24. If a customer ID or account number is entered in the request and only one customer matches the customer verify request, a customer verify transaction response is returned from BASE24. If a customer ID or account number is entered in the request and a selection needs to be made at the client, a customer select request is returned from BASE24.

When a customer select request is returned to the client, the customer is prompted to select the occurrence of data that is appropriate. A customer select respond request is then sent from the client to BASE24.

There is no reversal capability for customer verification data since this is logic to identify a customer prior to an actual (financial or nonfinancial) transaction being performed.

Note: More than one set of customer select and customer select respond request messages can be sent depending on the customer ID and personal ID relationships. Also, a continuation of the customer select respond request can be sent from the client if more customers need to be provided.

Customer Verification Request Data

Customer verification request data carries customer verification information. The request data is sent from the client to BASE24 for customer verify transaction requests.

Customer verification request data from this data element is not carried in the Internal Transaction Data (ITD) or Telebanking Standard Internal Message (BSTM).

The following format pertains to the customer verify transaction request:

Position	Length	Field Description
1	AN 1	Search Indicator A code indicating whether BASE24 is to search by customer ID, account number, or customer name. This value is required. Valid values are as follows: A = Account number C = Customer ID N = Name
2	AN 1	Reserved This field is not used.

Position	Length	Field Description
3–30	AN 28	Search: PAN (Customer ID) The unique identification number assigned to this customer. An unlimited number of individual application accounts and persons can be associated with one customer ID, and one customer can be associated with several customer IDs. The customer ID entered in this field must match a customer ID defined in the BASE24 system. This field is required if the Search Indicator field is set to a value of C.
31–58	AN 28	Search: Account Number An application account number for this customer. The account number entered in this field must match an application account number configured to the system for the customer. This field is required if the Search Indicator field is set to a value of A.
59–78	AN 20	Search: Family Name The family name or last name of the customer. This field is required if the Search Indicator field is set to a value of N.
79–88	AN 10	Search: Given Name The given name or first name of the customer. If a value is provided in this field, a value must also be provided in the Search: Family Name field.
89	AN 1	Search: Middle Initial The customer's middle initial. If a value is provided in this field, values must also be provided in the Search: Family Name and Given Name fields.
90	AN 1	Reserved This field is not used.

Either the customer ID, account number, or customer name must be provided. If the search is by name, the family name must be entered at a minimum. If the given name is entered, the family name must also be entered. If the middle initial is entered, the given and family names must also be entered. If the account number is entered, it must be numeric.

Customer Verification Response Data

Customer verification response data carries customer verification information. The response data is sent from BASE24 to the client for customer verify transaction responses.

Data for this response is obtained from the Customer Table (CSTT) and Personal Information Table (PIT).

Customer verification response data from this data element is not carried in the Internal Transaction Data (ITD) or Telebanking Standard Internal Message (BSTM).

The following format pertains to the customer verify transaction response:

Position	Length	Field Description
1–22	AN 22	FIID Name The name of the financial institution associated with this customer. Spaces that precede the name or are embedded in the name are considered part of the name. Spaces following the name are not considered part of the name.
23–50	AN 28	Search: PAN (Customer ID) Echoes the value provided in the request.
51–70	AN 20	Search: Family Name Echoes the value provided in the request.
71–80	AN 10	Search: Given Name Echoes the value provided in the request.
81	AN 1	Search: Middle Initial Echoes the value provided in the request.
82	AN 1	Reserved This field is not used.

Position	Length	Field Description
83–102	AN 20	Customer Information 1 User-defined customer verification information. For example, this field and the Customer Information 2 and 3 fields could be used to contain the customer's government ID, birth date, and mother's maiden name.
103–122	AN 20	Customer Information 2 See the Customer Information 1 field description above.
122–142	AN 20	Customer Information 3 See the Customer Information 1 field description above.
143–170	AN 28	Personal ID The identifier of a particular person for this customer ID. For example, this field could contain an employee number for a business customer. For an individual customer, this field could contain the individual's social security number or the customer ID.
171–176	AN 6	Title The social title (e.g., Mr., Mrs., Ms., Dr., Rev.), civil and military title (e.g., Gen., Gov., Prof.), or any other title used to address this customer.
177–196	AN 20	Name: Family The family name or last name of the customer.
197–206	AN 10	Name: Given The given name or first name of the customer.
207	AN 1	Name: Middle Initial The customer's middle initial.
208–222	AN 15	Government ID The government ID (e.g., U.S. social security number) assigned to this customer. The government ID can be used by customer service representatives to verify the identity of a calling customer.

Position	Length	Field Description
223–230	N 8	Date of Birth The date of birth (YYYYMMDD) for this customer.
231–260	AN 30	Address 1 The first line of the street or mailing address for this customer.
261–290	AN 30	Address 2 The second line of the street or mailing address for this customer.
291–320	AN 30	Address 3 The third line of the street or mailing address for this customer.
321–345	AN 25	City The name of the city for this customer's residence or mailing address.
346–347	AN 2	State The two-letter U.S. state or Canadian province abbreviation for this customer's residence or mailing address. State codes are based on the 1988 ANSI X3.38 standard, <i>Identification of the States, the District of Columbia, and the Outlying and Associated Areas of the United States for Information Interchange</i> .
348–350	AN 3	Country Code The country code associated with this customer's residence or mailing address. Country codes are based on the 1993 ISO 3166 standard, <i>Codes for the Representation of Names of Countries</i> .
351–360	AN 10	Postal Code The postal code of this customer's residence or mailing address.
361–380	AN 20	Phone: Home The home telephone number for this customer.

Position	Length	Field Description
381–400	AN 20	Phone: Work The work telephone number for this customer.
401–420	AN 20	Phone Other 1 Any other telephone number (e.g., wireless, paging, or fax) that can be used to reach this customer.
421–430	AN 10	Phone Other 1 Description A description (for example, WIRELESS, PAGER, or FAX) of the telephone number provided in the Phone Other 1 field.
431–450	AN 20	Phone Other 2 Any other telephone number (e.g., wireless, paging, or fax) that can be used to reach this customer.
451–460	AN 10	Phone Other 2 Description A description (for example, WIRELESS, PAGER, or FAX) of the telephone number provided in the Phone Other 2 field.
461–480	AN 20	Mother's Maiden Name The maiden name of this customer's mother. This name can be used by customer service representatives to verify the identity of a calling customer.
481–482	AN 2	Language Indicator A user-defined code indicating the primary language spoken by this customer. Valid values are 00–99.

Position	Length	Field Description
483	AN 1	PIN Required at CSR A code identifying whether a PIN is required with a transaction that is entered by a customer service representative. Valid values are as follows: Y = Yes, a PIN is required. N = No, a PIN is not required. O = The customer service representative can override a transaction that is entered without a PIN. However, a PIN is preferred. This means that the institution normally requires a PIN, but the customer service representative can still authorize a customer transaction without a PIN if he or she is confident of the customer's identity through other means.
484	AN 1	PIN Change Allowed at CSR A code identifying whether a PIN Change transaction can be performed by a customer service representative. Valid values are as follows: Y = Yes, PIN Change transactions are allowed. N = No, PIN Change transactions are not allowed.

Position	Length	Field Description
485	AN 1	PIN Change Required A flag that indicates whether a customer needs to initiate a PIN change transaction. If the institution wants the customer to initiate a PIN change transaction to change the current PIN, this field must be set to the value Y. If the institution wants to define a customer before the initial PIN has been selected, this field must be set to the value S. When this field is set to the value S or Y, the only transaction the customer can perform is a PIN change transaction. This field is set to the value N after a successful PIN change transaction is performed. Valid values are as follows: Y = Yes, a PIN change is required. The customer must change the current PIN; the only transaction allowed is the PIN change transaction. N = No, a PIN change is not required; the current PIN can be used. S = Customer select. The customer must select the initial PIN; the only transaction allowed is the PIN change transaction.
486	AN 1	Reserved This field is not used.
487–488	AN 2	Customer Type A code identifying the type of customer associated with this customer ID. This is an information-only field that is not used in processing. Valid values are as follows: B = Business customer C = Individual customer All other values are user defined.

Position	Length	Field Description
489–490	AN 2	Verify Flag A code identifying the status of the customer relationship. Valid values are as follows: C = Closed I = Issued, but not active V = Verified Customer service representatives can add accounts for the customer only if this field contains a value of I or V. Customer service representatives can delete accounts for the customer only when this field displays a value of I. This customer ID cannot be used in financial transaction processing unless this field contains a value of V. When a customer activate transaction is processed, the value of this field is changed from an I to a V. All other values are user defined. User-defined codes are treated the same as closed accounts.
491	N 1	MAS Display Options A code that identifies which information the institution prefers to have displayed on customer service representative (CSR) terminal screens when a multiple account selection situation is encountered. Valid values are as follows: 1 = Display account descriptions only 2 = Display account numbers only 3 = Display account descriptions and numbers; account descriptions are preferred 4 = Display account descriptions and numbers; account numbers are preferred
492	AN 1	Reserved This field is not used.

Position	Length	Field Description
493–500	AN 8	Route Profile An identifier grouping institutions together for transaction routing purposes. The routing profile is part of the key used by the BASE24-telebanking and BASE24-billpay products to select the Institution Routing Configuration File (IRCF) and Processing Code Definition File (PCDF) records for customers of this institution.
501–504	AN 4	FIID The FIID of the customer's financial institution. The FIID is an identifier that must be unique within the logical network. It is used throughout BASE24 products to identify each BASE24 institution.

Customer Select Request Data

Customer select request data carries customer select information. The request data is sent from BASE24 to the client for customer select transaction requests.

Both Private Data fields must be saved and echoed back in the customer select respond request.

If the More Indicator field indicates that more customer IDs or personal IDs can be retrieved, the Select Index field can be returned with a space. Otherwise, a value must be placed in the Select Index field.

Data for this request is obtained from the Customer Table (CSTT) and Personal Information Table (PIT).

On outgoing messages, the customer select request data is taken from an internal buffer maintained by the Voice Response Unit Interface object. This internal buffer is structured according to the CUST-SELCT definition described in the CDEFSGDS file on the core data definitions subvolume and private data definitions used by the object.

The following format pertains to the customer select transaction request:

Position	Length	Field Description
1–2	AN 2	Select Index The only valid value is a blank space.
3–120	AN 118	Private Data 1 Private data used by BASE24 Remote Banking for positioning if a subsequent customer select respond request is received.
121–122	N 2	Customer Count The number of customer items contained in this message. Valid values are 1–6.
123–125	AN 3	More Indicator A code indicating whether more customer IDs or personal IDs exist. Valid values are as follows: 972 = No more customer IDs or personal IDs 973 = More customer IDs or personal IDs
126	AN 1	Reserved This field is not used.
127–146	AN 20	Private Data 2 Private data used by BASE24 Remote Banking when a customer requests to view more customer information.
...		Customer Information Structure The following fields occur up to 6 times. Each occurrence is 140 characters in length.
	AN 28	Customer ID The unique identification number assigned to a customer. An unlimited number of individual application accounts and persons can be associated with one customer ID, and one customer can be associated with several customer IDs.

Position	Length	Field Description
AN 28	Personal ID	The identifier of a particular person for a customer ID. For example, this field could contain an employee number for a business customer. For an individual customer, this field could contain the individual's social security number or the customer ID.
AN 20	Name: Family	The family name or last name of a customer.
AN 10	Name: Given	The given name or first name of a customer.
AN 1	Name: Middle Initial	The middle initial of a customer name.
AN 2	ID Type	The only valid value is CU for the Customer Table (CSTT).
AN 30	Address 1	The first line of the street or mailing address for a customer.
AN 20	Home Phone	The home telephone number for a customer.
AN 1	Reserved	This field is not used.

Customer Select Respond Request Data

Customer select respond request data carries customer select information. The request data is sent from the client to BASE24 for customer select transactions responses. A **␣** denotes a blank space.

The Customer Information structure must be echoed back whether or not a customer is selected.

If the More Indicator field has a value of 972 (there are no more customer IDs or personal IDs to retrieve), the Select Index field must contain a value from 1 through 6.

Data for this request is obtained from the Customer Table (CSTT) and Personal Information Table (PIT).

On incoming messages, the customer select respond request data is placed in an internal buffer maintained by the Authorization Combined Positive Customer object. This internal buffer is structured according to the CUST-SELECT definition described in the CDEFSGDS file on the core data definitions subvolume.

The following format pertains to the customer select transaction response:

Position	Length	Field Description
1–2	AN 2	Select Index If a customer was selected, this field contains the index value of the customer information structure array for that customer. Otherwise, this field should contain a blank character. Valid values are 1–6.
3–120	AN 118	Private Data 1 Private data used by BASE24 Remote Banking for positioning if a subsequent customer select respond request is received.
121–122	N 2	Customer Count The number of customer items contained in this message. Valid values are 1–6.
123–125	AN 3	More Indicator A code indicating whether more customer IDs or personal IDs exist. Valid values are as follows: 972 = No more customer IDs or personal IDs 973 = More customer IDs or personal IDs
126	AN 1	Reserved This field is not used.

Position	Length	Field Description
127–146	AN 20	Private Data 2 Private data used by BASE24 Remote Banking when a customer requests to view more customer information.
...		Customer Information Structure The following fields occur up to 6 times. Each occurrence is 140 characters in length.
	AN 28	Customer ID The unique identification number assigned to a customer. An unlimited number of individual application accounts and persons can be associated with one customer ID, and one customer can be associated with several customer IDs.
	AN 28	Personal ID The identifier of a particular person for a customer ID. For example, this field could contain an employee number for a business customer. For an individual customer, this field could contain the individual's social security number or the customer ID.
	AN 20	Name: Family The family name or last name of a customer.
	AN 10	Name: Given The given name or first name of a customer.
	AN 1	Name: Middle Initial The middle initial of a customer name.
	AN 2	ID Type The only valid value is CU for the Customer Table (CSTT).
	AN 30	Address 1 The first line of the street or mailing address for a customer.

Position	Length	Field Description
	AN 20	Home Phone The home telephone number for a customer.
	AN 1	Reserved This field is not used.

Statement Download Response Data

The Transaction Data, Private data element carries response data for approved statement download transactions (transaction code 3R). The response data is sent from BASE24 to the client for statement download responses.

The following format pertains to the statement download transaction response:

Position	Length	Field Description
1–2	N 2	Count The number of transactions returned in this response message. Valid values are 0–3.
3–5	AN 3	Currency Code A code that identifies the currency of the amount carried in the Amount field. Valid values are listed in the 1990 ISO 4217 standard, <i>Codes for the Representation of Currencies and Funds</i> .
6	AN 1	Additional Data Flag A code indicating whether more transactions exist for this customer account. Valid values are as follows: Y = Yes, more data possible. N = No, no more data.
...		Statement Download Structure The following fields occur up to 3 times. Each occurrence is 250 characters in length.

Position	Length	Field Description
N 6	Date	The date (YYMMDD) on which this transaction was posted.
N 12	Amount	The amount of this transaction in whole and fractional currency units (e.g., U.S. dollars and cents).
AN 12	Check Number	The serial number of the check written for this transaction. The presence of a serial number identifies a check transaction.
AN 1	Debit/Credit Indicator	A code that identifies whether this transaction is a debit or credit. Valid values are as follows: C = Credit D = Debit
AN 1	Source Type	A code that identifies the source of this transaction. Any value can be used. However, values reserved for use by BASE24 products are as follows: A = ATM B = Telebanking device P = POS Device T = Teller
AN 1	Funds Indicator	A code that identifies whether the account had sufficient funds to approve a debit transaction. Valid values are as follows: 0 = Sufficient funds 1 = Not sufficient funds

Position	Length	Field Description
AN 1		Reversal Indicator A code that identifies whether the transaction had to be reversed. Valid values are as follows: 0 = Not reversed 1 = Reversed
AN 14		Reserved Transaction Flags This field is not currently used by BASE24 and should be zero filled.
AN 8		Host Transaction Code The code assigned to this transaction for processing on the host.
AN 32		FI Transaction ID A unique, nonexpiring financial institution identifier for this account.
AN 1		Special Data Type A code indicating whether payment data, transfer data, or no special data is present for this transaction. Valid values are as follows: P = Payment data present T = Transfer data present ␣ = No special data present, where ␣ is a blank space
AN 1		Reserved

Position	Length	Field Description
AN 160		Special Data
		The following fields contain data relating to a payment or transfer transaction. If the transaction is not a payment or transfer, these fields are blank.
		The PMNT fields identify the payee (vendor) in a payment transaction. The XFER fields identify the other account involved in a transfer transaction, that is the account that is not identified in data element S-102 or S-123.
AN 30		PMNT – Payee Name
AN 30		PMNT – Payee Address 1
AN 30		PMNT – Payee Address 2
AN 25		PMNT – Payee City
AN 2		PMNT – Payee State
AN 10		PMNT – Payee Zip
AN 20		PMNT – Payee Phone
		or
AN 19		XFER – Other Account
AN 2		XFER – Other Account Type
AN 139		XFER – Reserved

Statement Closing Download Response Data

The Transaction Data, Private data element carries response data for approved statement closing download transactions (transaction code 3Q). The response data is sent from BASE24 to the client for statement closing download responses.

The following format pertains to the statement closing download transaction response:

Position	Length	Field Description
1–2	N 2	Count
		The number of transactions returned in this response message. Valid values are 0–16.

Position	Length	Field Description
3–5	AN 3	Currency Code A code that identifies the currency of the amount carried in the Closing Balance field. Valid values are listed in the 1990 ISO 4217 standard, <i>Codes for the Representation of Currencies and Funds</i> .
6	AN 1	Additional Data Flag A code indicating whether more transactions exist for this customer account. Valid values are as follows: Y = Yes, more data possible. N = No, no more data.
...		Statement Closing Download Structure The following fields occur up to 16 times. Each occurrence is 62 characters in length.
	AN 32	FI Transaction ID A unique, nonexpiring financial institution identifier for this account.
	AN 6	Closing Date The closing date (YYMMDD) for this statement.
	N 12	Closing Balance The closing balance for this account in whole and fractional currency units (e.g., U.S. dollars and cents).
	AN 6	Begin Posting Date The beginning posting date (YYMMDD) for this statement.
	AN 6	End Posting Date The ending posting date (YYMMDD) for this statement.

Billpay Request and Response Data

Data element 125 is used to indicate whether an external data block is appended to the end of the ISO message. The external data block carries a block of data for the BASE24-billpay transaction requests and responses listed on the following page. See the “External Data Block” discussion later in this section for the format of the actual data in the external data block.

Data element 125 uses the following definition for the transactions listed below to indicate whether the external block exists:

Position	Length	Field Description
1	AN 1	External Data Block Exists The only valid value is Y to indicate that the external data block exists.

Requests

The following transactions carry request data in the external data block. For these transactions data element 125 should be defined with a length of 001 and the value defined above:

Customer add
Customer update
Customer account add
Customer account update
Customer vendor add
Customer vendor update
Loan application
Miscellaneous request

Responses

The following transactions carry response data in the external data block. For these transactions, data element 125 should be checked if it exists, the length of data element 125 should be set to 001, and the value should be set as defined above:

- Bank list inquiry
- Bank branch list inquiry
- Customer activate
- Customer add
- Customer deactivate
- Customer inquiry
- Customer update
- Customer account activate
- Customer account add
- Customer account deactivate
- Customer account delete
- Customer account update
- Customer vendor activate
- Customer vendor add
- Customer vendor deactivate
- Customer vendor list inquiry
- Customer vendor mask verify list inquiry
- Customer vendor update
- History inquiry – all transactions
- History inquiry – payments and transfers
- Master vendor list inquiry
- Miscellaneous request
- Miscellaneous requests inquiry
- Scheduled payments list inquiry
- Scheduled transfers list inquiry
- Scheduled transactions list inquiry

S-126 Multiple Account Selection Data

Format: ANS ...757
 ANS ...004

ISO Field Name: BASE24-telebanking/BASE24-billpay Multiple Account Selection Data

Used By: BASE24-telebanking
 BASE24-billpay

The Multiple Account Selection Data (S-126) element is used to send multiple accounts and account descriptions from BASE24 to the device to allow the customer to select an account for the current transaction. The Multiple Account Selection Data element is also used to return the account types of the selected accounts from the device to BASE24.

Multiple Accounts in Response to Device

The Multiple Account Selection Data element is used to carry the multiple accounts and account descriptions for multiple account selection. The account structure can occur up to a maximum of 15 times in the data element. The Account 1 Count field defines the number of occurrences for account 1 (i.e., how many *from* accounts are provided in the account structure). The Account 2 Count field defines the number of occurrences for account 2 (i.e., how many *to* accounts are provided in the account structure).

Note: The total value from the Account 1 Count and Account 2 Count fields cannot exceed the maximum value of 15.

The More Indicator field is not part of the account structure.

Data for this response is obtained from the Customer/Account Relation Table (CACT).

On outgoing messages, the multiple account selection data is taken from either an internal buffer maintained by the Authorization Combined Positive Customer object for transactions authorized offline, or from the DATA field in the Telebanking Standard Internal Message (BSTM) for transactions authorized online by a host. The LAST-X-OFST and the LAST-X-LGTH fields in the BSTM indicate the position of this multiple account selection data information in the DATA field.

The internal buffer is structured according to the MAS definition described in the CDEFSGDS file on the core data definition subvolume.

The table below contains the layout of the Multiple Account Selection Data element when Multiple Account Selection Data is returned by BASE24 to the device.

Position	Length	Field Description
1–2	N 2	Account 1 Count The number of <i>from</i> accounts returned in this response message. Valid values are 0–15.
3–4	N 2	Account 2 Count The number of <i>to</i> accounts returned in this response message. Valid values are 0–15.
...		Account Structure The following fields occur up to 15 times. Each occurrence is 50 characters in length.
	AN 28	Account Number An application account number for this customer.
	AN 2	Account Type An ISO code indicating the type of application account. Refer to the Account Type field description provided in the Transaction Data, Private (S-125) data element for a list of the BASE24 reserved account type values.
	AN 15	Account Description A text description for this application account.
	AN 1	Account Indicator A code indicating whether the accounts returned in this response message are associated with a <i>from</i> account (account 1) or a <i>to</i> account (account 2). Valid values are as follows: 1 = Associated with account 1 2 = Associated with account 2

Position	Length	Field Description
	AN 4	Account Qualifier A user-defined code that can be used in place of the application account number and account type to identify and access this account.
Varies	AN 3	More Indicator A code indicating whether more <i>from</i> accounts (account 1), <i>to</i> accounts (account 2), or no more accounts exist for this customer. Valid values are as follows: 970 = More account 1 accounts 971 = More account 2 accounts 972 = No more accounts available

Save Account Types in Request to BASE24

When a customer selects an account during multiple account selection, the selected account number is returned in the appropriate data element (S-102 for account 1 and S-103 for account 2) in the subsequent message to BASE24. The corresponding account type from the multiple account selection data must also be returned to BASE24. This account type value can differ from the account type value in the processing code and BASE24 requires both account types for transaction processing. The account type from the Multiple Account Selection Data element is referred to as the save account type. The device must return the appropriate save account type value in the subsequent message to BASE24.

On incoming messages, the value from this data element is placed in either the ITD-ACCT1.TYP or ITD-ACCT2.TYP field of the Internal Transaction Data (ITD).

The table below contains the layout of data element S-126 when save account type data is returned by the device to BASE24 during multiple account selection processing.

Position	Length	Description
1-2	AN 2	Save Account 1 Type The account type for the <i>from</i> account (account 1) selected from the multiple account selection data. Refer to the Account Type field description provided in the Transaction Data, Private (S-125) data element for a list of the BASE24 reserved account type values.
3-4	AN 2	Save Account 2 Type The account type for the <i>to</i> account (account 2) selected from the multiple account selection data. Refer to the Account Type field description provided in the Transaction Data, Private (S-125) data element for a list of the BASE24 reserved account type values.

External Data Block

Format: ANS3936

ISO Field Name: Not applicable

Used By: BASE24-billpay

The external data block carries a block of data for the transactions listed below. The external data block is appended to the end of the ISO message and is used in conjunction with data element 125.

On incoming messages, the length and address of the external data block is placed in the ITD-LAST-X-AREA field of the Internal Transaction Data (ITD). The external data block itself is placed in the DATA field of the BSTM when BASE24-telebanking sends the message to BASE24-billpay.

On outgoing messages, BASE24-telebanking takes the external data block from the DATA field of the BSTM.

The following transactions use the external data block for requests:

- Customer add
- Customer update
- Customer account add
- Customer account update
- Customer vendor add
- Customer vendor update
- Loan application
- Miscellaneous request

The following transactions use the external data block for responses:

- Bank list inquiry
- Bank branch list inquiry
- Customer activate
- Customer add
- Customer deactivate
- Customer inquiry
- Customer update
- Customer account activate
- Customer account add
- Customer account deactivate
- Customer account delete

Customer account update
Customer vendor activate
Customer vendor add
Customer vendor deactivate
Customer vendor list inquiry
Customer vendor mask verify list inquiry
Customer vendor update
History inquiry – all transactions
History inquiry – payments and transfers
Master vendor list inquiry
Miscellaneous request
Miscellaneous requests inquiry
Scheduled payments list inquiry
Scheduled transfers list inquiry
Scheduled transactions list inquiry

BASE24-billpay Request Data

The external data block contains request data for several BASE24-billpay transactions. The format of the data in this field is specific to each transaction.

Customer Add and Update Request Data

The external data block carries customer information for customer add and customer update transaction requests.

Data for this request is used to add or update rows in the Customer Table (CSTT) and Personal Information Table (PIT).

The format of this data in the external data block is as follows:

Position	Length	Field Description
1-30	AN 30	Alternate Contact The name and telephone number of the alternate contact associated with this customer ID. The primary contact is defined using one or more personal IDs associated with this customer ID. This is an optional field.

Position	Length	Field Description
31–50	AN 20	Customer Information: Line 1 User-defined customer verification information. For example, this field and the Customer Information 2 and 3 fields could be used to contain the customer's government ID, birth date, and mother's maiden name. This is an optional field.
51–70	AN 20	Customer Information: Line 2 See the Customer Information: Line 1 field description above.
71–90	AN 20	Customer Information: Line 3 See the Customer Information: Line 1 field description above.
91–92	AN 2	Customer Type A code identifying the type of customer associated with this customer ID. This is an information-only field that is not used in processing. Valid values are as follows: B = Business customer C = Individual customer All other values are user defined. This is a required field.
93–96	N 4	Branch Number The number for the main branch of the financial institution where this customer account is serviced. This is the branch customers are referred to by customer service representatives for additional information on this customer account. This is an optional field.
97–115	N 19	Default Account Number The number of the account for which balance information is returned to the customer with PIN verify transactions. This is an optional field.

Position	Length	Field Description
116–117	AN 2	Default Account Type An ISO code identifying the default account type for which balance information is returned to the customer with PIN verify transactions. This is an optional field. Refer to the Account Type field description provided in the Transaction Data, Private (S-125) data element for a list of the BASE24 reserved account type values. Note: For the Default Account Type and Service Fee Account Type fields, if an account number is not assigned, the account type fields should be filled in with a value of 00 (i.e., an account type description of “NONE”). Also, if a default account number or service fee account number is assigned, either the account type or the account description can be provided for the account. If the account type is provided, the account description is not used.

Position	Length	Field Description
118–123	AN 6	Default Account Type Description A text description of the account type. This is an optional field. The BASE24 reserved account type name values are as follows: CD = Certificate CMRCL = Commercial loan CR = Credit account CR1 = Credit account CR2 = Credit account CR3 = Credit account CR4 = Credit account CR5 = Credit account CR6 = Credit account CR7 = Credit account DDA = Demand deposit account DDA1 = Demand deposit account DDA2 = Demand deposit account DDA3 = Demand deposit account DDA4 = Demand deposit account DDA5 = Demand deposit account DDA6 = Demand deposit account DDA7 = Demand deposit account DDA8 = Demand deposit account INSTL = Installment loan IRA = Retirement account LOCR = Line of credit MRTGL = Mortgage loan NONE = No account NOW = Interest-bearing checking account SAV = Savings account SAV1 = Savings account SAV2 = Savings account SAV3 = Savings account SAV4 = Savings account SAV5 = Savings account SAV6 = Savings account (See the note for the Default Account Type field description above.)

Position	Length	Field Description
124–142	N 19	Service Fee: Account Number The account number charged by the customer's financial institution using the automatic debit feature for the billpay service. This is an optional field.
143–144	AN 2	Service Fee: Account Type The account type associated with the service fee account number for this customer ID. This field identifies the type of account charged by the customer's financial institution using the automatic debit feature for the billpay service. Note: For the Default Account Type and Service Fee Account Type fields, if an account number is not assigned, the account type fields should be filled in with a value of 00 (i.e., an account type description of "NONE"). Also, if a default account number or service fee account number is assigned, either the account type or the account description can be provided for the account. If the account type is provided, the account description is not used.
145–150	AN 6	Service Fee: Account Type Description A text description of the service fee account type. This is an optional field. Refer to the Default Account Type Description field for a list of BASE24 reserved values. (See the note for the Service Fee: Account Type field description above.)
151–158	AN 8	Customer Profile An identifier used to group primary customer IDs together for transaction processing. This is a required field.

Position	Length	Field Description
159–178	AN 20	Family Name The family name or last name of this customer. This field accepts special characters required for languages other than U.S. English. This is a required field.
179–188	AN 10	Given Name The given name or first name of this customer. This field accepts special characters required for languages other than U.S. English. This is a required field.
189	AN 1	Middle Initial The initial of this customer's middle name. This is an optional field.
190–195	AN 6	Title The social title (e.g., Mr., Mrs., Ms., Dr., Rev.), civil and military title (e.g., Gen., Gov., Prof.), or any other title used to address this customer. This is an optional field.
196–210	AN 15	Government ID The government ID (for example, U.S. social security number) assigned to this customer. The government ID can be used by customer service representatives to verify the identity of a calling customer. This is an optional field.
211–218	N 8	Birth Date The date of birth (YYYYMMDD) for this customer. The date of birth can be used by customer service representatives to verify the identity of a calling customer. This is an optional field.
219–248	AN 30	Address 1 The first line of the street or mailing address for this customer. This is an optional field.

Position	Length	Field Description
249–278	AN 30	Address 2 The second line of the street or mailing address for this customer. This is an optional field.
279–308	AN 30	Address 3 The third line of the street or mailing address for this customer. This is an optional field.
309–333	AN 25	City The name of the city for this customer's residence or mailing address. This is an optional field.
334–335	AN 2	State The two-letter U.S. state or Canadian province abbreviation for this customer's residence or mailing address. State codes are based on the 1988 ANSI X3.38 standard, <i>Identification of the States, the District of Columbia, and the Outlying and Associated Areas of the United States for Information Interchange</i> . This is an optional field.
336–338	AN 3	Country Code The country code associated with this customer's residence or mailing address. Country codes are based on the 1993 ISO 3166 standard, <i>Codes for the Representation of Names of Countries</i> . This is an optional field.
339–348	AN 10	Postal Code The postal code of this customer's residence or mailing address. This is an optional field.
349–350	N 2	Language Indicator A user-defined code indicating the primary language spoken by this customer. Valid values are 00–99. This is an optional field.

Position	Length	Field Description
351–370	AN 20	Phone: Home The home telephone number for this customer. This is an optional field.
371–390	AN 20	Phone: Work The work telephone number for this customer. This is an optional field.
391–410	AN 20	Phone: Other 1 Any other telephone number (e.g., wireless, paging, or fax) that can be used to reach this customer. This is an optional field.
411–420	AN 10	Phone: Other 1 Description A description (e.g., WIRELESS, PAGER, or FAX) of the telephone number provided in the Phone: Other 1 field. This is an optional field.
421–440	AN 20	Phone: Other 2 Any other telephone number (e.g., wireless, paging, or fax) that can be used to reach this customer. This is an optional field.
441–450	AN 10	Phone: Other 2 Description A description (e.g., WIRELESS, PAGER, or FAX) of the telephone number provided in the Phone: Other 2 field. This is an optional field.
451–470	AN 20	Mother's Maiden Name The maiden name of this customer's mother. This name can be used by customer service representatives to verify the identity of a calling customer. This is an optional field.

Customer Account Add Request Data

The external data block carries customer account information for customer account add transaction requests.

Data for this request is used to add rows in the Customer/Account Relation Table (CACT).

The format of this data in the external data block is as follows:

Position	Length	Field Description
1–6	AN 6	Account Type Description A text description of the application account type. If the ISO account type value is provided in the Additional Data, Private (P-48) data element, this field is not used. If the ISO account type is not provided in the P-48 data element, this field is required. Refer to the Default Account Type Description field for a list of BASE24 reserved values.
7–10	AN 4	Account Qualifier A user-defined code that can be used in place of the account number and account type to identify this account. Account qualifiers must be unique across all accounts assigned to the customer. For example, if the customer has two checking accounts, DDA and DDA1 can be used as qualifiers for the two accounts. The same account qualifier cannot be used for two different accounts. This is an optional field.
11–25	AN 15	Account Description A text description of the account. This is an optional field.

Position	Length	Field Description
26	AN 1	Allowed: Debit A code indicating whether the customer can perform transfer or payment transactions that debit this account. Valid values are as follows: Y = Yes, account can be debited for a transfer or payment N = No, account cannot be debited for a transfer or payment
27	A 1	Allowed: Credit A code indicating whether the customer can perform credit transactions on the account. Credit transactions include transfers to this account. Valid values are as follows: Y = Yes N = No This is a required field.
28	A 1	Allowed: Inquiry A code indicating whether the customer can perform inquiry transactions (e.g., available funds inquiry and scheduled transfers list inquiry) on the account. Valid values are as follows: Y = Yes N = No This is a required field.

Customer Account Update Request Data

The external data block carries customer account information for customer account update transaction requests.

Data for this request is used to update rows in the Customer/Account Relation Table (CACT).

The format of this data in the external data block is as follows:

Position	Length	Field Description
1–4	AN 4	Account Qualifier A user-defined code that can be used in place of the account number and account type to identify this account. Account qualifiers must be unique across all accounts assigned to the customer. For example, if the customer has two checking accounts, DDA and DDA1 can be used as qualifiers for the two accounts. The same account qualifier cannot be used for two different accounts. This is an optional field.
5–19	AN 15	Account Description A text description of the account. This is an optional field.
20	AN 1	Allowed: Debit A code indicating whether the customer can perform transfer or payment transactions that debit this account. Valid values are as follows: Y = Yes, account can be debited for a transfer or payment. N = No, account cannot be debited for a transfer or payment.

Position	Length	Field Description
21	A 1	Allowed: Credit A code indicating whether the customer can perform credit transactions on the account. Credit transactions include transfers to this account. Valid values are as follows: Y = Yes N = No This is a required field.
22	A 1	Allowed: Inquiry A code indicating whether the customer can perform inquiry transactions (e.g., available funds inquiry and scheduled transfers list inquiry) on the account. Valid values are as follows: Y = Yes N = No This is a required field.

Customer Vendor Add Request Data

The external data block carries customer vendor information for customer vendor add transaction requests.

Data for this request is used to add rows in the Customer Vendor Table (CVND). This request can also cause rows to be added to the Vendor Table (VNDR) and Vendor Branch Table (VNDB).

Note: Either the Vendor Name or Vendor Number fields should be present, but not both. If the Vendor Number field contains a non-zero value, it is considered present.

Either the Vendor Branch Number or the Vendor Branch Name fields (Name, Address 1, Address 2, City, State, Postal Code, Phone Number, Customer Account With Vendor) should be present. If the Vendor Branch Number field contains a nonzero value, it is considered present.

If the Vendor Branch Name fields are present, the Vendor Branch Table (VNDB) is read by the Vendor Branch Name, Address 1, and Address 2 key. If a vendor branch is not found with that key data, a vendor branch is added.

If the Vendor Branch Name fields are present, the Vendor Address 1, City, State, Postal Code, and Customer Account with Vendor fields must be non-blank.

The format of this data in the external data block is as follows:

Position	Length	Field Description
1–30	AN 30	Vendor Name The name of the vendor to whom this customer makes payments. (See the note preceding the table.)
31–39	N 9	Vendor Number An internal number generated by the system for this vendor. This number is unique among all registered vendors and is automatically generated when the vendor is added to the system. (See the note preceding the table.)
40–59	AN 20	Vendor User Remit Information 1 Specific information about the remittance that is required by the vendor. The information must be included in the remittance file produced for the vendor's remittance type. This is an optional field.
60–79	AN 20	Vendor User Remit Information 2 Specific information about the remittance that is required by the vendor. The information must be included in the remittance file produced for the vendor's remittance type. This is an optional field.
80–109	AN 30	Vendor Branch Name The name of the vendor branch used by the customer. (See the note preceding the table.)

Position	Length	Field Description
110–113	N 4	Vendor Branch Number A number that identifies the branch. This value must be unique for the vendor. (See the note preceding the table.)
114–143	AN 30	Address 1 The first line of the mailing address or street address for the vendor branch. (See the note preceding the table.)
144–173	AN 30	Address 2 The second line of the mailing address or street address for the vendor branch. (See the note preceding the table.)
174–198	AN 25	City The city in which the vendor branch is located. (See the note preceding the table.)
199–200	AN 2	State A code for the state in which the vendor branch is located. The U.S. state codes are based on standards described in the 1988 ANSI X3.38 standard, <i>Identification of the States, the District of Columbia, and the Outlying and Associated Areas of the United States for Information Interchange</i> . This field can also contain a value other than an ANSI code. (See the note preceding the table.)
201–210	AN 10	Postal Code The postal code of the vendor branch. (See the note preceding the table.)

Position	Length	Field Description
211–230	AN 20	Phone Number The business telephone number for the vendor branch. This is an optional field.
231–250	AN 20	Vendor Branch User Remit Information 1 Specific information about the remittance that is required by the vendor. The information is included in the remittance file produced for the vendor's remittance type. This is an optional field.
251–270	AN 20	Vendor Branch User Remit Information 2 Specific information about the remittance that is required by the vendor. The information is included in the remittance file produced for the vendor's remittance type. This is an optional field.
271–298	AN 28	Customer Account with Vendor The customer account with the vendor that is credited with any customer payments. Any value entered here must conform to the vendor masks configured for this vendor and vendor branch combination. (See the note preceding the table.)
299–308	AN 10	Customer Vendor Short Name The free-format customer short name for the vendor. This field is for information only and is not required.
309–328	AN 20	Customer Vendor User Information 1 This field contains customer-specific information about the customer vendor. This field is for information only and is not required.
329–348	AN 20	Customer Vendor User Information 2 This field contains customer-specific information about the customer vendor. This field is for information only and is not required.

Position	Length	Field Description
349–352	AN 4	Budget Category A code identifying the budget category to which this vendor is assigned. This is an optional field.

Customer Vendor Update Request Data

The external data block carries customer vendor information for customer vendor update transaction requests.

Data for this request is used to update rows in the Customer Vendor Table (CVND). This request can also cause rows to be updated in the Vendor Table (VNDR) and Vendor Branch Table (VNDB).

Note: Either the Vendor Branch Number or the Vendor Branch Name fields (Name, Address 1, Address 2, City, State, Postal Code, Phone Number, Customer Account With Vendor) should be present. If the Vendor Branch Number field contains a non-zero value, it is considered present.

If the Vendor Branch Name fields are present, the Vendor Branch Table (VNDB) is read by Vendor Branch Name, Address 1, and Address 2 key. If the VNDB row is found, it is updated with the fields from the request. If a VNDB row is not found with that key data, a VNDB row is added and the Customer Vendor Table (CVND) row is updated with the vendor branch number. The Address 1, City, State, Postal Code, and Customer Account with Vendor fields must be non-blank for this situation.

If the Vendor Branch Number field is present and the value equals the vendor branch number in the CVND row, the VNDB row is read using the vendor branch number. The VNDB row is updated with the fields from the request. If the VNDB row is not present, the update request is denied.

If the Vendor Branch Number field is present and does not equal the vendor branch number in the CVND row, the VNDB row is read using the vendor branch number from the update request. The VNDB row is updated with the fields from the update request. In addition, the CVND row is updated with the vendor branch number from the update request. If the vendor branch number is present in the request but the VNDB row is not found, the update is denied.

The format of this data in the external data block is as follows:

Position	Length	Field Description
1–30	AN 30	Vendor Name The name of the vendor to whom this customer makes payments.
31	AN 1	Update Vendor User Information A code indicating whether to update vendor user information in the Vendor Table (VNDR) with this request. Valid values are as follows: Y = Yes, update vendor user information. N = No, do not update vendor user information.
32–51	AN 20	Vendor User Remit Information 1 Specific information about the remittance that is required by the vendor. The information must be included in the remittance file produced for the vendor's remittance type. This is an optional field.
52–71	AN 20	Vendor User Remit Information 2 Specific information about the remittance that is required by the vendor. The information must be included in the remittance file produced for the vendor's remittance type. This is an optional field.
72–101	AN 30	Vendor Branch Name The name of the vendor branch used by the customer. (See the note preceding the table.)
102–105	N 4	Vendor Branch Number A number that identifies the branch. This value must be unique for the vendor. (See the note preceding the table.)

Position	Length	Field Description
106–135	AN 30	Address 1 The first line of the mailing address or street address for the vendor branch. (See the note preceding the table.)
136–165	AN 30	Address 2 The second line of the mailing address or street address for the vendor branch. (See the note preceding the table.)
166–190	AN 25	City The city in which the vendor branch is located. (See the note preceding the table.)
191–192	AN 2	State A code for the state in which the vendor branch is located. The U.S. state codes are based on standards described in the 1988 ANSI X3.38 standard, <i>Identification of the States, the District of Columbia, and the Outlying and Associated Areas of the United States for Information Interchange</i> . This field can also contain a value other than an ANSI code. (See the note preceding the table.)
193–202	AN 10	Postal Code The postal code of the vendor branch. (See the note preceding the table.)
203–222	AN 20	Phone Number The business telephone number for the vendor branch. This is an optional field.

Position	Length	Field Description
223	AN 1	Update Vendor Branch User Information A code indicating whether to update user information in the Vendor Branch Table (VNDB) with this request. Valid values are as follows: Y = Yes, update user information. N = No, do not update user information.
224–243	AN 20	Vendor Branch User Remit Information 1 Specific information about the remittance that is required by the vendor. The information is included in the remittance file produced for the vendor's remittance type. This is an optional field.
244–263	AN 20	Vendor Branch User Remit Information 2 Specific information about the remittance that is required by the vendor. The information is included in the remittance file produced for the vendor's remittance type. This is an optional field.
264–291	AN 28	Customer Account with Vendor The customer account with the vendor that is credited with any customer payments. Any value entered here must conform to the vendor masks configured for this vendor and vendor branch combination. (See the note preceding the table.)
292–301	AN 10	Customer Vendor Short Name The free-format customer short name for the vendor. This field is for information only and is not required.
302–321	AN 20	Customer Vendor User Information 1 This field contains customer-specific information about the customer vendor. This field is for information only and is not required.

Position	Length	Field Description
322–341	AN 20	Customer Vendor User Information 2 This field contains customer-specific information about the customer vendor. This field is for information only and is not required.
342–345	AN 4	Budget Category A code identifying the budget category to which this vendor is assigned. This is an optional field.
346	AN 1	Reserved This field is not used.

Loan Application Request Data

The external data block carries loan application information for loan application transaction requests.

Data in this request is used to add rows in the Loan Table (LOAN).

Note: All amount fields accept numbers with leading or trailing spaces, but do not accept decimal points.

The format of this data in the external data block is as follows:

Position	Length	Field Description
1–20	AN 20	Family Name The family name or last name of the loan applicant. This field accepts special characters required for languages other than U.S. English. This field is required.

Position	Length	Field Description
21–30	AN 10	Given Name The given name or first name of the loan applicant. This field accepts special characters required for languages other than U.S. English. This is a required field.
31	AN 1	Middle Initial The middle initial of the loan applicant. This is an optional field.
32–51	N 20	Transaction Source Last Changed The date and time, in SQL timestamp format (e.g., YYYYMMDDhhmmssmmmmmm), this loan application was entered. This is an optional field.
52–55	AN 4	FIID The FIID of the financial institution from which the customer is requesting a loan. The FIID is an identifier that must be unique within the logical network. It is used throughout BASE24 products to identify each BASE24 institution. This is a required field.
56	AN 1	Loan Type A user-defined code indicating the type of loan requested. This is an optional field.
57–66	AN 10	Referred by: Given Name The given name or first name of the person who referred this customer, if applicable. This field accepts special characters required for languages other than U.S. English. This is an optional field.

Position	Length	Field Description
67–86	AN 20	Referred by: Family Name The family name or last name of the person who referred this customer, if applicable. This field accepts special characters required for languages other than U.S. English. This is an optional field.
87–98	N 12	Amount Requested The amount of the requested loan in whole and fractional currency units. A decimal point can be entered if the amount requested includes fractional currency units. A negative amount cannot be entered. This is a required field.
99–123	AN 25	Collateral Free-format text describing the collateral offered for this loan. This is an optional field.
124–183	AN 60	Purpose of Loan Free-format text describing the purpose for this loan. This is an optional field.
184–185	N 2	Number of Dependents The number of dependents for the loan applicant. This is a required field. This is a required field.
186–215	AN 30	Ages of Dependents Free-form text identifying the ages of each dependent for the loan applicant. This is an optional field.
216–223	N 8	Date of Birth The birth date (YYYYMMDD) of the loan applicant. This is an optional field.
224–238	AN 15	Government ID The government ID (for example, U.S. social security number) assigned to the loan applicant. This is an optional field.

Position	Length	Field Description
239–258	AN 20	Home Phone Number The home telephone number for the loan applicant. This is an optional field.
259	AN 1	Married A flag indicating whether the loan applicant is married. Valid values are as follows: Y = Yes, married. N = No, not married. (Default) This is a required field.
260–289	AN 30	Current Address: Line 1 The first line of the street or mailing address for the loan applicant's current residence. This is an optional field.
290–319	AN 30	Current Address: Line 2 The second line of the street or mailing address for the loan applicant's current residence. This is an optional field.
320–344	AN 25	Current Address: City The name of the city where the loan applicant currently resides. This is an optional field.
345–346	AN 2	Current Address: State The two-letter U.S. state or Canadian province abbreviation for the loan applicant's current residence. Reserved state codes are defined in the 1988 ANSI X3.38 standard, <i>Identification of the States, the District of Columbia, and the Outlying and Associated Areas of the United States for Information Interchange</i> . This is an optional field.
347–356	AN 10	Current Address: Postal Code The postal code of the loan applicant's current residence. This is an optional field.

Position	Length	Field Description
357–358	N 2	Current Address: Years at Address The number of years the loan applicant has resided at his or her current address. This is a required field.
359	AN 1	Current Address: Own A flag indicating whether the loan applicant owns his or her current residence. Valid values are as follows: Y = Yes, own. N = No, do not own.
360–371	N 12	Current Address: Monthly Payment The amount, in whole and fractional currency units, of the monthly mortgage or rent payment for the loan applicant's current residence. A decimal point can be entered if the payment includes fractional currency units.
372–393	AN 22	Current Address: Name Rent Payments are made to The company or person to which the loan applicant makes rent payments, if the applicant rents his or her current residence.
394–423	AN 30	Previous Address: Line 1 The first line of the street or mailing address for the loan applicant's previous residence.
424–453	AN 30	Previous Address: Line 2 The second line of the street or mailing address for the loan applicant's previous residence.
454–478	AN 25	Previous Address: City The name of the city for the loan applicant's previous residence.

Position	Length	Field Description
479–480	AN 2	Previous Address: State The two-letter U.S. state or Canadian province abbreviation for the loan applicant's previous residence. Reserved state codes are defined in the 1988 ANSI X3.38 standard, <i>Identification of the States, the District of Columbia, and the Outlying and Associated Areas of the United States for Information Interchange</i> .
481–490	AN 10	Previous Address: Postal Code The postal code of the loan applicant's previous residence.
491–492	N 2	Previous Address: Years at Address The number of years the loan applicant resided at his or her previous address.
493	AN 1	Previous Address: Own A flag indicating whether the loan applicant owned his or her previous residence. Valid values are as follows: Y = Yes, owned. N = No, did not own.
494–513	AN 20	Occupation A description of the loan applicant's current occupation.
514–535	AN 22	Current Employer: Name The name of the loan applicant's current employer.
536–565	AN 30	Current Employer: Address, Line 1 The first line of the street or mailing address for the loan applicant's current employer.
566–595	AN 30	Current Employer: Address, Line 2 The second line of the street or mailing address for the loan applicant's current employer.

Position	Length	Field Description
596–620	AN 25	Current Employer: City The name of the city for the loan applicant's current employer.
621–622	AN 2	Current Employer: State The two-letter U.S. state or Canadian province abbreviation for the loan applicant's current employer. Reserved state codes are defined in the 1988 ANSI X3.38 standard, <i>Identification of the States, the District of Columbia, and the Outlying and Associated Areas of the United States for Information Interchange</i> .
623–632	AN 10	Current Employer: Postal Code The postal code for the loan applicant's current employer.
633–634	N 2	Current Employer: Years At The number of years this loan applicant has worked for his or her current employer.
635–646	N 12	Current Employer: Monthly Income The amount, in whole and fractional currency units, the loan applicant makes per month in his or her current job. This field cannot contain a negative amount.
647–668	AN 22	Previous Employer: Name The name of the loan applicant's previous employer.
669–698	AN 30	Previous Employer: Address, Line 1 The first line of the street or mailing address for the loan applicant's previous employer.
699–728	AN 30	Previous Employer: Address, Line 2 The second line of the street or mailing address for the loan applicant's previous employer.
729–753	AN 25	Previous Employer: City The name of the city for the loan applicant's previous employer.

Position	Length	Field Description
754–755	AN 2	Previous Employer: State The two-letter U.S. state or Canadian province abbreviation for the loan applicant's previous employer. Reserved state codes are defined in the 1988 ANSI X3.38 standard, <i>Identification of the States, the District of Columbia, and the Outlying and Associated Areas of the United States for Information Interchange</i> .
756–765	AN 10	Previous Employer: Postal Code The postal code for the loan applicant's previous employer.
766–767	N 2	Previous Employer: Years At The number of years the loan applicant worked for his or her previous employer.
768–779	N 12	Previous Employer: Monthly Income The amount, in whole and fractional currency units, the loan applicant made per month in his or her previous job.
780–801	AN 22	Other Source of Income Any other source from which the loan applicant currently derives income besides his or her current job.
802–813	N 12	Other Source of Income: Amount The amount, in whole and fractional currency units, the loan applicant currently derives per month from a source other than his or her current job.
814	AN 1	Judgements A flag indicating whether the loan applicant has ever had a legal ruling or judgement against him or her. Valid values are as follows: Y = Yes, the loan applicant has had a legal ruling or judgement against him or her. N = No, the loan applicant has never had a legal ruling against him or her.

Position	Length	Field Description
815	AN 1	Bankruptcy A flag indicating whether the loan applicant has ever declared a personal bankruptcy. Valid values are as follows: Y = Yes, the loan applicant has declared a personal bankruptcy. N = No, the loan applicant has never declared a personal bankruptcy.
816	AN 1	Back Taxes A flag indicating whether the loan applicant has ever owed any back taxes. Valid values are as follows: Y = Yes, the loan applicant has owed back taxes. N = No, the loan applicant has never owed any back taxes.
817	AN 1	Income Reduction A flag indicating whether the loan applicant has ever had any income reductions (e.g., automatic deductions for child support or alimony payments) as a result of a legal ruling or judgement. Valid values are as follows: Y = Yes, the loan applicant has had an income reduction. N = No, the loan applicant has never had an income reduction.
818–827	AN 10	Coapplicant: Given Name The given name or first name of the coapplicant for this loan. This field accepts special characters required for languages other than U.S. English.
828	AN 1	Coapplicant: Middle Initial The middle initial of the coapplicant for this loan. This field accepts special characters required for languages other than U.S. English.

Position	Length	Field Description
829–848	AN 20	Coapplicant: Family Name The family name or last name of the coapplicant for this loan. This field accepts special characters required for languages other than U.S. English.
849–856	N 8	Coapplicant: Date of Birth The birth date (YYYYMMDD) of the coapplicant for this loan.
857–871	AN 15	Coapplicant: Government ID The government ID (e.g., U.S. social security number) assigned to the coapplicant for this loan.
872–891	AN 20	Coapplicant: Home Phone The home telephone number of the coapplicant for this loan.
892–901	AN 10	Coapplicant: Relationship The relationship of the coapplicant to the customer requesting the loan.
902–921	AN 20	Coapplicant: Occupation A description of the coapplicant's current occupation.
922–943	AN 22	Coapplicant: Current Employer, Name The name of the coapplicant's current employer.
944–973	AN 30	Coapplicant: Current Employer, Address, Line 1 The first line of the street or mailing address for the coapplicant's current employer.
974–1003	AN 30	Coapplicant: Current Employer, Address, Line 2 The second line of the street or mailing address for the coapplicant's current employer.
1004–1028	AN 25	Coapplicant: Current Employer, City The name of the city for the coapplicant's current employer.

Position	Length	Field Description
1029–1030	AN 2	Coapplicant: Current Employer, State The two-letter U.S. state or Canadian province abbreviation for the coapplicant's current employer. Reserved state codes are defined in the 1988 ANSI X3.38 standard, <i>Identification of the States, the District of Columbia, and the Outlying and Associated Areas of the United States for Information Interchange</i> .
1031–1040	AN 10	Coapplicant: Current Employer, Postal Code The postal code for the coapplicant's current employer.
1041–1042	N 2	Coapplicant: Current Employer, Years At The number of years the coapplicant has worked for his or her current employer.
1043–1054	N 12	Coapplicant: Current Employer, Monthly Income The amount, in whole and fractional currency units, the coapplicant makes per month in his or her current job. A negative amount cannot be entered in this field.
1055–1076	AN 22	Coapplicant: Previous Employer, Name The name of the coapplicant's previous employer.
1077–1106	AN 30	Coapplicant: Previous Employer, Address, Line 1 The first line of the street or mailing address for the coapplicant's previous employer.
1107–1136	AN 30	Coapplicant: Previous Employer, Address, Line 2 The second line of the street or mailing address for the coapplicant's previous employer.
1137–1161	AN 25	Coapplicant: Previous Employer, City The name of the city for the coapplicant's previous employer.

Position	Length	Field Description
1162–1163	AN 2	Coapplicant: Previous Employer, State The two-letter U.S. state or Canadian province abbreviation for the coapplicant's previous employer. Reserved state codes are defined in the 1988 ANSI X3.38 standard, <i>Identification of the States, the District of Columbia, and the Outlying and Associated Areas of the United States for Information Interchange</i> .
1164–1173	AN 10	Coapplicant: Previous Employer, Postal Code The postal code for the coapplicant's previous employer.
1174–1175	N 2	Coapplicant: Previous Employer, Years At The number of years this coapplicant worked for his or her previous employer.
1176–1187	N 12	Coapplicant: Previous Employer, Monthly Income The amount, in whole and fractional currency units, the coapplicant made per month in his or her previous job. This field cannot contain a negative amount.
1188–1209	AN 22	Coapplicant: Other Source of Income Any other source from which the coapplicant currently derives income besides his or her current job.
1210–1221	N 12	Coapplicant: Other Source of Income Amount The amount, in whole and fractional currency units, the coapplicant currently derives per month from a source other than his or her current job. This field cannot contain a negative amount.
1222–1243	AN 22	Institution Info (1): Institution, Name The name of a financial institution where the loan applicant maintains an account (e.g., savings or demand deposit) considered to be an asset.
1244–1265	AN 22	Institution Info (2): Institution, Name The name of a second financial institution where the loan applicant maintains an account (e.g., savings or demand deposit) considered to be an asset.

Position	Length	Field Description
1266–1287	AN 22	Institution Info (3): Institution, Name The name of a third financial institution where the loan applicant maintains an account (e.g., savings or demand deposit) considered to be an asset.
1288–1306	AN 19	Institution Info (1): Institution, Account Number The number of an account considered to be an asset at the first financial institution.
1307–1325	AN 19	Institution Info (2): Institution, Account Number The number of an account considered to be an asset at the second financial institution.
1326–1344	AN 19	Institution Info (3): Institution, Account Number The number of an account considered to be an asset at the third financial institution.
1345–1346	AN 2	Institution Info (1): Institution, Account Type The type of the account considered to be an asset at the first financial institution.
1347–1348	AN 2	Institution Info (2): Institution, Account Type The type of the account considered to be an asset at the second financial institution.
1349–1350	AN 2	Institution Info (3): Institution, Account Type The type of the account considered to be an asset at the third financial institution.
1351–1362	N 12	Institution Info (1): Institution, Account Value The estimated current value, in whole and fractional currency units, of the account considered to be an asset at the first institution. This field cannot contain a negative amount.

Position	Length	Field Description
1363–1374	N 12	Institution Info (2): Institution, Account Value The estimated current value, in whole and fractional currency units, of the account considered to be an asset at the second institution. This field cannot contain a negative amount.
1375–1386	N 12	Institution Info (3): Institution, Account Value The estimated current value, in whole and fractional currency units, of the account considered to be an asset at the third institution. This field cannot contain a negative amount.
1387–1416	AN 30	Institution Info (1): Institution, Address, Line 1 The first line of the street or mailing address for the first financial institution maintaining an asset account.
1417–1446	AN 30	Institution Info (2): Institution, Address, Line 1 The first line of the street or mailing address for the second financial institution maintaining an asset account.
1447–1476	AN 30	Institution Info (3): Institution, Address, Line 1 The first line of the street or mailing address for the third financial institution maintaining an asset account.
1477–1506	AN 30	Institution Info (1): Institution, Address, Line 2 The second line of the street or mailing address for the first financial institution maintaining an asset account.
1507–1536	AN 30	Institution Info (2): Institution, Address, Line 2 The second line of the street or mailing address for the second financial institution maintaining an asset account.
1537–1566	AN 30	Institution Info (3): Institution, Address, Line 2 The second line of the street or mailing address for the third financial institution maintaining an asset account.
1567–1591	AN 25	Institution Info (1): Institution, City The name of the city where the first financial institution maintaining an asset account is located.

Position	Length	Field Description
1592–1616	AN 25	Institution Info (2): Institution, City The name of the city where the second financial institution maintaining an asset account is located.
1617–1641	AN 25	Institution Info (3): Institution, City The name of the city where the third financial institution maintaining an asset account is located.
1642–1643	AN 2	Institution Info (1): Institution, State The two-letter U.S. state or Canadian province abbreviation for the state where the first financial institution maintaining an asset account is located. Reserved state codes are defined in the 1988 ANSI X3.38 standard, <i>Identification of the States, the District of Columbia, and the Outlying and Associated Areas of the United States for Information Interchange</i> .
1644–1645	AN 2	Institution Info (2): Institution, State The two-letter U.S. state or Canadian province abbreviation for the state where the second financial institution maintaining an asset account is located.
1646–1647	AN 2	Institution Info (3): Institution, State The two-letter U.S. state or Canadian province abbreviation for the state where the third financial institution maintaining an asset account is located.
1648–1657	AN 10	Institution Info (1): Institution, Postal Code The postal code for the first financial institution maintaining an asset account.
1658–1667	AN 10	Institution Info (2): Institution, Postal Code The postal code for the second financial institution maintaining an asset account.
1668–1677	AN 10	Institution Info (3): Institution, Postal Code The postal code for the third financial institution maintaining an asset account.

Position	Length	Field Description
1678–1681	N 4	Auto (1): Year (YYYY) The model year of an automobile owned by the applicant.
1682–1685	N 4	Auto (2): Year (YYYY) The model year of a second automobile owned by the applicant.
1686–1705	AN 20	Auto (1): Make The make, or manufacturer, of the first automobile owned by the applicant.
1706–1725	AN 20	Auto (2): Make The make, or manufacturer, of the second automobile owned by the applicant.
1726–1745	AN 20	Auto (1): Model The model name of the first automobile owned by the applicant.
1746–1765	AN 20	Auto (2): Model The model name of the second automobile owned by the applicant.
1766–1777	N 12	Residence Purchase Price If the applicant requesting the loan owns his or her current residence, this is the amount, in whole and fractional currency units, the applicant paid for the property. This field cannot contain a negative amount.
1778–1789	N 12	Down Payment The amount, in whole and fractional currency units, the applicant paid as a down payment for this residence. This field cannot contain a negative amount.
1790–1801	N 12	Life Insurance Value The value, in whole and fractional currency units, of life insurance policies owned by the loan applicant. This field cannot contain a negative amount.

Position	Length	Field Description
1802	AN 1	Other Property A flag indicating whether the loan applicant owns any property other than his or her current residence. Valid values are as follows: Y = Yes, the loan applicant owns additional property. N = No, the loan applicant does not own any additional property.
1803–1832	AN 30	Other Property Description A description of property owned by the loan applicant other than his or her current residence.
1833	AN 1	Other Assets A flag indicating whether the loan applicant has any assets other than those already described. Valid values are as follows: Y = Yes, the loan applicant has other assets. N = No, the loan applicant does not have any other assets.
1834–1863	AN 30	Other Assets Description A description of assets owned by the loan applicant other than those already described.
1864–1883	AN 20	Liability (1): Type User-defined text describing a loan applicant liability. Suggested liability types include the following: Personal, Credit Cards, Auto, Installment, Mortgage, Charge Cards, Cosignatory, Alimony, Child Support.
1884–1903	AN 20	Liability (2): Type User-defined text describing a second loan applicant liability.
1904–1923	AN 20	Liability (3): Type User-defined text describing a third loan applicant liability.

Position	Length	Field Description
1924–1943	AN 20	Liability (4): Type User-defined text describing a fourth loan applicant liability.
1944–1963	AN 20	Liability (5): Type User-defined text describing a fifth loan applicant liability.
1964–1983	AN 20	Liability (6): Type User-defined text describing a sixth loan applicant liability.
1984–2003	AN 20	Liability (7): Type User-defined text describing a seventh loan applicant liability.
2004–2023	AN 20	Liability (8): Type User-defined text describing an eighth loan applicant liability.
2024–2043	AN 20	Liability (9): Type User-defined text describing a ninth loan applicant liability.
2044–2063	AN 20	Liability (10): Type User-defined text describing a tenth loan applicant liability.
2063–2083	AN 20	Liability (11): Type User-defined text describing an eleventh loan applicant liability.
2084–2105	AN 22	Liability (1): Institution Name User-defined text containing the name of the creditor for the first liability.

Position	Length	Field Description
2106–2127	AN 22	Liability (2): Institution Name User-defined text containing the name of the creditor for the second liability.
2128–2149	AN 22	Liability (3): Institution Name User-defined text containing the name of the creditor for the third liability.
2150–2171	AN 22	Liability (4): Institution Name User-defined text containing the name of the creditor for the fourth liability.
2172–2193	AN 22	Liability (5): Institution Name User-defined text containing the name of the creditor for the fifth liability.
2194–2215	AN 22	Liability (6): Institution Name User-defined text containing the name of the creditor for the sixth liability.
2216–2237	AN 22	Liability (7): Institution Name User-defined text containing the name of the creditor for the seventh liability.
2238–2259	AN 22	Liability (8): Institution Name User-defined text containing the name of the creditor for the eighth liability.
2260–2281	AN 22	Liability (9): Institution Name User-defined text containing the name of the creditor for the ninth liability.
2282–2303	AN 22	Liability (10): Institution Name User-defined text containing the name of the creditor for the tenth liability.

Position	Length	Field Description
2304–2325	AN 22	Liability (11): Institution Name User-defined text containing the name of the creditor for the eleventh liability.
2326–2337	N 12	Liability (1): Amount Owed The current amount, in whole and fractional currency units, of the first liability. This amount should include any interest on the debt for which the loan applicant is liable. This field cannot contain a negative amount.
2338–2349	N 12	Liability (2): Amount Owed The current amount, in whole and fractional currency units, of the second liability. This amount should include any interest on the debt for which the loan applicant is liable. This field cannot contain a negative amount.
2350–2361	N 12	Liability (3): Amount Owed The current amount, in whole and fractional currency units, of the third liability. This amount should include any interest on the debt for which the loan applicant is liable. This field cannot contain a negative amount.
2362–2373	N 12	Liability (4): Amount Owed The current amount, in whole and fractional currency units, of the fourth liability. This amount should include any interest on the debt for which the loan applicant is liable. This field cannot contain a negative amount.
2374–2385	N 12	Liability (5): Amount Owed The current amount, in whole and fractional currency units, of the fifth liability. This amount should include any interest on the debt for which the loan applicant is liable. This field cannot contain a negative amount.
2386–2397	N 12	Liability (6): Amount Owed The current amount, in whole and fractional currency units, of the sixth liability. This amount should include any interest on the debt for which the loan applicant is liable. This field cannot contain a negative amount.

Position	Length	Field Description
2398–2409	N 12	Liability (7): Amount Owed The current amount, in whole and fractional currency units, of the seventh liability. This amount should include any interest on the debt for which the loan applicant is liable. This field cannot contain a negative amount.
2410–2421	N 12	Liability (8): Amount Owed The current amount, in whole and fractional currency units, of the eighth liability. This amount should include any interest on the debt for which the loan applicant is liable. This field cannot contain a negative amount.
2422–2433	N 12	Liability (9): Amount Owed The current amount, in whole and fractional currency units, of the ninth liability. This amount should include any interest on the debt for which the loan applicant is liable. This field cannot contain a negative amount.
2434–2445	N 12	Liability (10): Amount Owed The current amount, in whole and fractional currency units, of the tenth liability. This amount should include any interest on the debt for which the loan applicant is liable. This field cannot contain a negative amount.
2446–2457	N 12	Liability (11): Amount Owed The current amount, in whole and fractional currency units, of the eleventh liability. This amount should include any interest on the debt for which the loan applicant is liable. This field cannot contain a negative amount.
2458–2469	N 12	Liability (1): Monthly Payment The amount, in whole and fractional currency units, the loan applicant makes in monthly payments, if applicable, toward absolving the debt on the first liability. This field cannot contain a negative amount

Position	Length	Field Description
2470–2481	N 12	Liability (2): Monthly Payment The amount, in whole and fractional currency units, the loan applicant makes in monthly payments, if applicable, toward absolving the debt on the second liability. This field cannot contain a negative amount
2482–2493	N 12	Liability (3): Monthly Payment The amount, in whole and fractional currency units, the loan applicant makes in monthly payments, if applicable, toward absolving the debt on the third liability. This field cannot contain a negative amount
2494–2505	N 12	Liability (4): Monthly Payment The amount, in whole and fractional currency units, the loan applicant makes in monthly payments, if applicable, toward absolving the debt on the fourth liability. This field cannot contain a negative amount
2506–2517	N 12	Liability (5): Monthly Payment The amount, in whole and fractional currency units, the loan applicant makes in monthly payments, if applicable, toward absolving the debt on the fifth liability. This field cannot contain a negative amount
2518–2529	N 12	Liability (6): Monthly Payment The amount, in whole and fractional currency units, the loan applicant makes in monthly payments, if applicable, toward absolving the debt on the sixth liability. This field cannot contain a negative amount
2530–2541	N 12	Liability (7): Monthly Payment The amount, in whole and fractional currency units, the loan applicant makes in monthly payments, if applicable, toward absolving the debt on the seventh liability. This field cannot contain a negative amount

Position	Length	Field Description
2542–2553	N 12	Liability (8): Monthly Payment The amount, in whole and fractional currency units, the loan applicant makes in monthly payments, if applicable, toward absolving the debt on the eighth liability. This field cannot contain a negative amount
2554–2565	N 12	Liability (9): Monthly Payment The amount, in whole and fractional currency units, the loan applicant makes in monthly payments, if applicable, toward absolving the debt on the ninth liability. This field cannot contain a negative amount
2566–2577	N 12	Liability (10): Monthly Payment The amount, in whole and fractional currency units, the loan applicant makes in monthly payments, if applicable, toward absolving the debt on the tenth liability. This field cannot contain a negative amount
2578–2589	N 12	Liability (11): Monthly Payment The amount, in whole and fractional currency units, the loan applicant makes in monthly payments, if applicable, toward absolving the debt on the eleventh liability. This field cannot contain a negative amount
2590	AN 1	Reserved This field is not used.

Miscellaneous Request Data

The external data block carries miscellaneous message information for miscellaneous request transaction requests.

Data in this request is used to add rows in the Miscellaneous Request Table (MISC).

The format of this data in the external data block is as follows:

Position	Length	Field Description
1–20	AN 20	Family Name The family name or last name of this customer. This field accepts special characters required for languages other than U.S. English.
21–30	AN 10	Given Name The given name or first name of this customer. This field accepts special characters required for languages other than U.S. English.
31–50	AN 20	Phone Number The telephone number of the contact person for this request.
51	A 1	Contact Type A code indicating the type of contact for this request. Reserved values are as follows: B = Bank (financial institution) C = Customer O = Other V = Vendor This field can also contain a user-defined contact type.
52	N 1	Priority The priority assigned to this request. Requests with a priority of 1 are considered the highest priority while requests with a priority of 5 are considered the lowest priority. Valid values are 1–5.

Position	Length	Field Description
53	A 1	Request Type A code indicating the type of this request. Valid values are as follows: A = Account maintenance C = Complaint Q = Question or query User-defined values can be used for other requests.
54–75	AN 22	Department User-defined text identifying the financial institution or service provider department to manage this request.
76–699	AN 624	Contact Details User-defined text explaining the details of this request.
700	A 1	Reserved This field is not used.

BASE24-billpay Response Data

The external data block contains response data for several BASE24-billpay transactions. The format of the data in this field is specific to each transaction.

Bank List Inquiry Response Data

The external data block carries bank information when a bank list inquiry transaction is performed.

Data in this response is obtained from the Bank Table (BANK).

The format of this data in the external data block is as follows:

Position	Length	Field Description
1–2	N 2	Count The number of items (i.e., banks) returned in this response message. Valid values are 0–19.
3	AN 1	Additional Data Flag A code indicating whether more financial institution entries exist. Valid values are as follows: Y = Yes, more data possible. N = No, no more data.
4	AN 1	Reserved This field is not used.
...		Bank Information The following fields occur up to 19 times. Each occurrence is 212 characters in length.
	AN 4	FIID The FIID of a financial institution. The FIID is an identifier that must be unique within the logical network. It is used throughout BASE24 products to identify each BASE24 institution.
	AN 11	Institution ID The financial institution's routing and transit number or issuer identification number.
	AN 22	Bank Name The name of the financial institution associated with this record. Spaces that precede the name or are embedded in the name are considered part of the name.
	AN 4	Main Branch Number The main branch number for this financial institution. This is the branch to which customers for this financial institution are referred for additional information on their billpay or enhanced telebanking service.

Position	Length	Field Description
	N 8	Begin Date The date (YYYYMMDD) on which billpay or enhanced telebanking service will become available for use by this financial institution. No payments or transfers can be processed, or scheduled to occur, prior to this date for any customers of this financial institution.
	N 8	End Date The date (YYYYMMDD) on which remote banking service will no longer be available for use by this financial institution. No payments or transfers can be processed, or scheduled to occur, later than this date for any customers of this financial institution.
	N 12	Payment High Limit The maximum amount, in whole currency units, a customer for this institution can make in an individual payment transaction. Valid values range from 0 to 999999999999999999. If the payment high limit in a Customer Table (CSTT) row for a customer of this institution is set to zero and this field is set to a nonzero value, then the limit for the financial institution is checked. In this case, the payment transaction amount must be equal to or less than the amount specified in this field. If the payment high limit in a CSTT row for a customer of this institution is set to a nonzero value, then this field is not checked. If the payment high limit in the CSTT and this field are both set to zero, then no limit is imposed on the payment transaction amount for a customer of this institution.

Position	Length	Field Description
AN 2		Subsequent Transaction Source A code indicating the transaction source for subsequent scheduled or recurring transactions performed on behalf of this institution's customers. A transaction source is only provided with an initial scheduled or recurring transaction because all subsequent scheduled and recurring transactions originate from the BASE24billpay Scheduled Transaction Initiator process. A transaction source must be associated with every transaction for billing purposes. All two-character values, including all blanks, are valid. A user-defined value other than one of the reserved codes indicated below should be used. A value of ST is suggested to represent the Scheduled Transaction Initiator process. The following reserved codes should not be used: AD = Audio device BL = Billing IB = Inbound customer service representative (CSR) PC = Personal computer SP = Screen phone
AN 4		Billing Group The billing group assigned to this financial institution for bank billing purposes. The billing group and billing type are used to access the appropriate rows in the billing tables (i.e., Billing Group Table, Billing Type Table, Billing Rate Table, Rate Table, and Rate Group Table). This information is used by the system owner to bill this financial institution for billpay or enhanced telebanking service. This field is not used to bill customers of this financial institution. This field is only used if the billing subsystem is being used.

Position	Length	Field Description
AN 2		Billing Type The billing type assigned to this financial institution for bank billing purposes. The billing type and billing group are used to access the appropriate rows in the billing tables (i.e., Billing Group Table, Billing Type Table, Billing Rate Table, Rate Table, and Rate Group Table). This information is used by the system owner for billing this financial institution for billpay or enhanced telebanking service. This field is used only if the billing subsystem is being used.
A 1		Customer Vender Initial Verify A code indicating the initial customer vendor verification status used when a customer vendor is added for this institution's BASE24-billpay customers. This code indicates the status of the customer's account with the vendor. A customer cannot direct payments to a vendor unless the customer's account with the vendor has been verified. All values except V are considered to indicate an unverified status and require that some type of action be taken to verify the customer's account with the vendor. All values other than the reserved values are user defined. A blank space is considered valid and represents a user-defined value. The following values are reserved: F = Failed. The customer's account with the vendor failed verification. The customer vendor is considered to be unverified. I = Initial. The customer's account with the vendor is considered to be unverified. V = Verified. The customer's account with the vendor has been verified. Payments can be sent to the customer vendor.

Position	Length	Field Description
N 9		Vendor Number The vendor number associated with this financial institution for online customer billing purposes if the financial institution supports online extraction of customer fees for billpay or enhanced telebanking service. Fees are extracted using a payment transaction from the account specified in the SERVICE FEE ACCOUNT NUMBER field of the customer's Customer Table (CSTT) row. The financial institution is considered the vendor in this transaction.
AN 28		Default Customer ID The customer ID for the default Customer Table (CSTT) row for this financial institution. A default row in the Customer/Personal ID Relation Table (CPIT) and Personal Information Table (PIT) are also accessed using this customer ID. If customers were not preloaded for this financial institution (see the Customer Database Preload field below), data from these default rows is displayed on the customer service representative (CSR) Customer Signup screen for this institution's customers or returned in a customer add response when they are initially signed up for billpay or enhanced telebanking service. The value from this field is also used when logging a transaction that does not contain a customer ID.
AN 19		Default Account Number The account number for the default Customer/Account Relation Table (CACT) row for this financial institution. If customers were not preloaded for this financial institution (see the Customer Database Preload field below), data from this default row is displayed on the customer service representative (CSR) Customer Account List screen for this institution's customers when they are initially signed up for billpay or enhanced telebanking service.

Position	Length	Field Description
A	1	Customer Database Preload A code indicating whether the customer rows for this institution have been automatically preloaded from an existing database or tape. If customer rows for this institution were preloaded, the appropriate Customer Table (CSTT), Customer/Account Relation Table (CACT), Customer/Personal ID Relation Table (CPIT), and Personal Information Table (PIT) rows were automatically created when the data was preloaded. The verification status assigned to the CSTT rows is determined by the institution. If the verification status was set to the value V (verified), then the customer is already considered to be activated and the Customer Signup screen cannot be accessed. If the verification status was set to the value I (Initial), then customer service representatives can activate the customer simply by pressing a function key on the Customer Signup screen when a customer for this institution calls to sign up for billpay or enhanced telebanking service. If customer rows for this institution were not preloaded for this institution, the appropriate default Customer Table (CSTT), Customer/Account Relation (CACT), Customer/Personal ID Relation (CPIT), and Personal Information Table (PIT) rows are used to display default data on the Customer Signup and Customer Signup Accounts screens when signing up new customers for this financial institution (see the Default Customer ID and Default Account Number field descriptions above). In this case, customer service representatives must fill in all required fields on the Customer Signup and Customer Signup Accounts screens before pressing a function key to add the appropriate table rows on behalf of the customer. The customer service representative must then press another function key to activate the customer rows. Valid values are as follows: Y = Yes, customer rows for this institution were automatically preloaded. N = No, customer rows for this institution were not automatically preloaded.

Position	Length	Field Description
A 1		Customer ID Generation Allowed A flag indicating whether the system is to generate customer ID numbers. Valid values are as follows: Y = Yes, generate customer ID numbers. N = No, do not generate customer ID numbers.
A 1		Reactivation Allowed A flag indicating whether the system is to allow a customer to be reactivated. Valid values are as follows: Y = Yes, customer reactivation allowed N = No, customer reactivation not allowed
A 1		Reserved This field is not used.
AN 75		File Maintenance Tag See the “File Maintenance Tag Data” discussion at the end of this subsection for a description of this field.

Bank Branch List Response Data

The external data block carries branch information when a bank branch list inquiry transaction is performed.

Data in this response is obtained from the Bank Branch Table (BKBR).

The format of this data in the external data block is as follows:

Position	Length	Field Description
1–2	N 2	Count The number of items (i.e., bank branches) returned in this response message. Valid values are 0–16.

Position	Length	Field Description
3	AN 1	Additional Data Flag A code indicating whether more financial institution branches entries exist. Valid values are as follows: Y = Yes, more data possible. N = No, no more data.
4	AN 1	Reserved This field is not used.
...		Bank Branch Information The following fields occur up to 16 times. Each occurrence is 243 characters in length.
	AN 4	FIID The financial institution identifier. The value in this field links the branch in this row to the financial institution.
	AN 4	Branch Number The number that uniquely identifies this branch to the financial institution.
	AN 30	Address Line 1 The first line of the mailing or street address for this branch.
	AN 30	Address Line 2 The second line of the mailing or street address for this branch.
	AN 25	City The city in which this branch is located.

Position	Length	Field Description
AN 2		State An ANSI code for the state in which the branch is located. The U.S. state codes are based on standards described in the 1988 ANSI X3.38 standard, <i>Identification of the States, the District of Columbia, and the Outlying and Associated Areas of the United States for Information Interchange</i> . This field can contain a value other than an ANSI code.
AN 10		Postal Code The postal code of this branch.
AN 20		Phone Number The business telephone number for this branch.
AN 22		Contact Name The name of the contact person at this branch. This is the person to which customers can be referred by customer service representatives for additional information on their billpay or enhanced telebanking service.
AN 20		Fax The fax telephone number for this branch.
AN 1		Reserved This field is not used.
AN 75		File Maintenance Tag See the “File Maintenance Tag Data” discussion at the end of this subsection for a description of this field.

Customer Response Data

The external data block carries customer information for customer activate, customer add, customer deactivate, customer inquiry, and customer update transactions.

Data in this response is obtained from the Customer Table (CSTT), Customer Billpay Table (CBPT), and Personal Information Table (PIT).

The format of this data in the external data block is as follows:

Position	Length	Field Description
1-4	N 4	Version The version number of this CSTT row. This field is reserved for future use.
5-8	AN 4	FIID The FIID of the financial institution for this CSTT row. The FIID is an identifier that must be unique within the logical network. It is used throughout BASE24 products to identify each BASE24 institution and to secure file record and table row access by financial institution.
9-16	AN 8	Customer Profile An identifier used to group primary customer IDs together for transaction processing. The value in this field must match a value in the CUSTOMER PROFILE field on Customer Allowed Transaction Table (CATT) screen 1.
17-18	AN 2	Customer Type A code identifying the type of customer associated with this customer ID. This is an information-only field that is not used in processing. Valid values are as follows: B = Business customer C = Individual customer This field can also contain a user-defined type code.

Position	Length	Field Description
19–22	N 4	Bad PIN Count The number of times an incorrect PIN has been entered for this CSTT row during the current usage accumulation period. Each institution defines the maximum number of incorrect PIN tries allowed for the customer in the MAX PIN TRIES field on Institution Definition File (IDF) screen 2. When this number of incorrect PIN tries is exceeded, subsequent transactions for the customer are rejected according to the setting in the BAD PIN ACTION field on IDF screen 2.
23–24	AN 2	Customer Verify Flag A code identifying the status of the customer relationship. Valid values are as follows: C = Closed I = Issued but inactive V = Verified Customer service representatives can add accounts for the customer only if this field contains a value of I or V. Customer service representatives can delete accounts for the customer only when this field displays a value of I. This CSTT row cannot be used in financial transaction processing unless this field displays a value of V. When a customer activate transaction is processed, the value of this field is changed from an I to a V. Additional user-defined codes can be entered; however, all user-defined codes are treated the same as closed accounts.

Position	Length	Field Description
25–40	AN 16	PIN Offset The PIN offset value, PIN verification number (PVN), or the PIN verification value (PVV) used in PIN verification processing. The information in this field depends on the PIN verification parameters specified in the PIN CHECK TYPE and POFST/PVV LOC fields on Institution Definition File (IDF) screen 2.
41–60	N 20	Last Change: Customer Verify Status The date and time (YYYYMMDDhhmmssmmmmmm) the Customer Verify Flag field was last updated for this CSTT row.
61–80	N 20	Last Change: PIN Offset The date and time (YYYYMMDDhhmmssmmmmmm) the PIN Offset field was last updated for this CSTT row.
81	AN 1	PVK Index Identifies the location of the PIN verification value (PVV) when the Visa PVV PIN verification method is specified in the PIN CHECK TYPE field on Institution Definition File (IDF) screen 2. Valid values are 1 through 6 for the Visa PVV verification method and blank for other PIN verification methods.
82–100	N 19	Default Account Number The number of the account for which balance information is returned to the customer with PIN verify transactions.
101–102	AN 2	Default Account Type An ISO code identifying the default account type for which balance information is returned to the customer with PIN verify transactions.

Position	Length	Field Description
103–106	N 4	Max Online History Records The maximum number of Transaction History File (THF) records that can be returned in historical inquiry transactions to a remote banking endpoint device for this customer. This field specifies the total number of items that can be returned for the transaction, not just for each response message included in the transaction. This maximum does not apply to historical inquiries requested by a customer service representative using the BASE24 CSR terminal screens. Valid values are as follows: 0 = No maximum is enforced. 1–9999 = Maximum number of THF records that can be returned.
107–126	AN 20	Customer Information: Line 1 User-defined customer verification information. For example, this field and the Customer Information: Line 2 and Line 3 fields could be used to contain the customer's government ID, birth date, and mother's maiden name.
127–146	AN 20	Customer Information: Line 2 See the Customer Information: Line 1 field description above.
147–166	AN 20	Customer Information: Line 3 See the Customer Information: Line 1 field description above.
167–196	AN 30	Alternate Contact The name and telephone number of the alternate contact associated with this customer ID. The primary contact is defined using one or more personal IDs associated with this customer ID.
197–204	N 8	Begin Date The first date (YYYYMMDD) this customer ID row can be used in financial transaction processing.

Position	Length	Field Description
205–212	N 8	End Date The last date (YYYYMMDD) this row can be used in financial transaction processing.
213–216	N 4	Branch Number The identification number for the main branch of the financial institution where this CSTT row is serviced. This is the branch to which customers are referred by customer service representatives for additional information on this CSTT row.
217	AN 1	PIN Change Required A flag that indicates whether a customer needs to initiate a PIN change transaction. If the institution wants the customer to initiate a PIN change transaction to change the current PIN, this field must be set to the value Y. If the institution wants to define a customer before the initial PIN has been selected, this field must be set to the value S. When this field is set to the value S or Y, the only transaction the customer can perform is a PIN change transaction. This field is set to the value N after a successful PIN change transaction is performed. Valid values are as follows: Y = Yes, a PIN change is required. The customer must change the current PIN; the only transaction allowed is the PIN change transaction. N = No, a PIN change is not required; the current PIN can be used. S = Customer select. The customer must select the initial PIN; the only transaction allowed is the PIN change transaction.
218–219	AN 2	Billing Type A code that identifies a group of customers for billing purposes throughout the BASE24-billpay database. This code also defines the service level and fee structure for the billpay or enhanced telebanking service.

Position	Length	Field Description
220	AN 1	Packet Status A user-defined code indicating whether an informational packet explaining the billpay or enhanced telebanking service has been sent for this customer ID.
221–240	N 20	Billing Type Last Change The date and time (YYYYMMDDhhmmssmmmmmm) the Billing Type field was last updated.
241–260	N 20	Packet Status Last Change The date and time (YYYYMMDDhhmmssmmmmmm) the Packet Status field was last updated.
261–264	AN 4	Billpay Billing Group A code identifying the billing group, which is an arbitrary value used to assign BASE24-billpay or Enhanced Telebanking customers to a particular date for customer billing.
265–276	N 12	Payment High Limit The maximum amount, in whole currency units, that can be requested in an individual BASE24-billpay schedule payment transaction for this CSTT row. If this field is set to a nonzero value, this limit overrides the customer's financial institution payment limit specified in the Bank Table (BANK).
277–351	AN 75	Customer File Maintenance Tag See the "File Maintenance Tag Data" discussion at the end of this subsection for a description of this field.
352–371	N 20	Last Reset Timestamp The date and time (YYYYMMDDhhmmssmmmmmm) the Bad PIN Count field was last cleared. The Bad PIN Count field is cleared when the customer performs the first transaction of a business day or enters a correct PIN, based on the value in the PIN TRIES RESET OPTION field on Institution Definition File (IDF) screen 2.

Position	Length	Field Description
372–390	N 19	Service Fee: Account Number The account number charged by the customer’s financial institution using the automatic debit feature for billpay or enhanced telebanking service. The Service Fee: Account Type field indicates the type of account associated with this account number.
391–392	AN 2	Service Fee: Account Type The ISO account type associated with the service fee account number for this customer ID. This field identifies the type of account charged by the customer’s financial institution using the automatic debit feature for billpay or enhanced telebanking service.
393	AN 1	Reserved This field is not used.
394–398	N 5	Number of Cycles The number of billing cycles that have been processed for this customer using the current billing type. If the BASE24-billpay Billing Application is not being used, this field will contain zero.
399–404	N 6	Next Customer Reference Number The reference number to use for this customer’s next billpay or enhanced telebanking transaction. The Billpay Server process and the Scheduled Transaction Initiator process manage the value in this field.
405–408	AN 4	Billing Group The billing group to which this customer currently belongs. If the BASE24-billpay Billing Application is not being used, this field can be blank.

Position	Length	Field Description
409–413	N 5	Last Customer Vendor Number The last customer vendor number used for this customer. The next vendor number for this customer is the value in this field plus 1. The Billpay Server process and the Billpay Table Maintenance Server process manage the value in this field.
414–421	N 8	Last Cycle End Date The end date (YYYYMMDD) of the last billing cycle for this customer. If the BASE24-billpay Billing Application is not being used, this field will contain zero.
422–423	AN 2	Original Billing Type The billing type that existed in the associated Customer Table (CSTT) row when this row was originally created in the CBPT. If the BASE24-billpay Billing Application is not being used, this field can be blank.
424–429	N 6	Last Billing Reference Number The last reference number processed by the BASE24-billpay Billing Application, if used. The Customer Billing process manages the value in this field, which identifies the most recent History Table (HIST) row used for fee calculation purposes.
430–431	AN 2	Current Billing Type The billing type that is currently being used. If the BASE24-billpay Billing Application is not being used, this field can be blank.
432–507	AN 75	Customer Billpay File Maintenance Tag See the “File Maintenance Tag Data” discussion at the end of this subsection for a description of this field.

Position	Length	Field Description
508–535	AN 28	Personal ID The identifier for this PIT row. The identifier can be prescribed by the service provider or chosen by the customer, but must be unique within the PIT. For example, this field could contain an employee number for a business customer. For an individual customer, this field could contain the individual's social security number or the customer ID. When a new customer is added using the customer add transaction from the CSR Customer Signup screen or an RBSI-compliant device, the personal ID is always set to the value of the customer ID.
536–539	N 4	Version The version number of this PIT row. This field is reserved for future use.
540–543	AN 4	FIID The FIID of the financial institution associated with this personal ID.
544–563	AN 20	Family Name The family name or last name of this customer. This field accepts special characters required for languages other than U.S. English.
564–573	AN 10	Given Name The given name or first name of this customer. This field accepts special characters required for languages other than U.S. English.
574	AN 1	Middle Initial The initial of this customer's middle name.
575–590	AN 16	Family Name Token The tokenized family name that is derived using a Soundex algorithm.

Position	Length	Field Description
591–596	AN 6	Given Name Token The tokenized given name that is derived using a Soundex algorithm.
597–602	AN 6	Title The social title (e.g., Mr., Mrs., Ms., Dr., Rev.), civil and military title (e.g., Gen., Gov., Prof.), or any other title used to address this customer.
603–617	AN 15	Government ID The government ID (e.g., U.S. social security number) assigned to this customer. The government ID can be used by customer service representatives to verify the identity of a calling customer.
618–625	N 8	Birth Date The date of birth (YYYYMMDD) for this customer.
626–655	AN 30	Address 1 The first line of the street or mailing address for this customer.
656–685	AN 30	Address 2 The second line of the street or mailing address for this customer.
686–715	AN 30	Address 3 The third line of the street or mailing address for this customer.
716–740	AN 25	City The name of the city for this customer's residence or mailing address

Position	Length	Field Description
741–742	AN 2	State The two-letter U.S. state or Canadian province abbreviation for this customer’s residence or mailing address. State codes are based on the 1988 ANSI X3.38 standard, <i>Identification of the States, the District of Columbia, and the Outlying and Associated Areas of the United States for Information Interchange</i> .
743–745	AN 3	Country The country code associated with this customer’s residence or mailing address. Country codes are based on the 1993 ISO 3166 standard, <i>Codes for the Representation of Names of Countries</i> .
746–757	AN 10	Postal Code The postal code of this customer’s residence or mailing address.
758–759	N 2	Language Indicator A user-defined code indicating the primary language spoken by this customer. Valid values are 00–99.
760–777	AN 20	Phone: Home The home telephone number for this customer.
778–797	AN 20	Phone: Work The work telephone number for this customer.
798–817	AN 20	Phone: Other 1 Any other telephone number (e.g., wireless, paging, or fax) that can be used to reach this customer.
818–827	AN 10	Phone: Other 1 Description A description (e.g, WIRELESS, PAGER, or FAX) of the telephone number provided in the PHONE: Other 1 field.

Position	Length	Field Description
828–847	AN 20	Phone: Other 2 Any other telephone number (e.g., wireless, paging, or fax) that can be used to reach this customer.
848–857	AN 10	Phone: Other 2 Description A description (e.g, WIRELESS, PAGER, or FAX) of the telephone number provided in the PHONE: Other 2 field.
858–877	AN 20	Mother's Maiden Name The maiden name of this customer's mother. This name can be used by customer service representatives to verify the identity of a calling customer. This field accepts special characters required for languages other than U.S. English.
878–952	AN 75	Personal File Maintenance Tag See the "File Maintenance Tag Data" discussion at the end of this subsection for a description of this field.
953–958	AN 6	Default Account Type Description The account type name for the ISO account type code provided in the Default Account Type field. This is the name configured for the code in the Account Type Table File (ATT). Refer to the Default Account Type Description field provided in the Customer Add and Update Request Data redefine for a list of BASE24 reserved values.
959–964	AN 6	Service Fee Account Type Description The account type name for the ISO account type code provided in the Service Fee: Account Type field. This is the name configured for the code in the Account Type Table File (ATT). Refer to the Default Account Type Description field provided in the Customer Add and Update Request Data redefine for a list of BASE24 reserved values.

Customer Account Response Data

The external data block carries customer account information for customer account activate, customer account add, customer account deactivate, customer account delete, and customer account update transactions.

Data in this response is obtained from the Customer/Account Relation Table (CACT). Detailed field descriptions for the CACT are provided in the ***BASE24 Core Files and Tables Maintenance Manual***.

The format of this data in the external data block is as follows:

Position	Length	Field Description
1–19	N 19	Account Number The application account number.
20–21	AN 2	Account Type The ISO account type code for this application account. Refer to the Account Type field description provided in the Transaction Data, Private (S-125) data element for a list of the BASE24 reserved account type values.
22–25	AN 4	Account Qualifier A user-defined code that can be used in place of the account number and account type to identify this account. Account qualifiers must be unique across all accounts assigned to the customer. For example, if the customer has two checking accounts, DDA and DDA1 can be used as qualifiers for the two accounts. The same account qualifier cannot be used for two different accounts.
26–29	AN 4	FIID The FIID of the customer's financial institution for this application account.

Position	Length	Field Description
30	AN 1	Status A code indicating the current status of the application account and the action to be taken by the Integrated Authorization Server process if this account status is encountered during transaction processing. Valid values are as follows: 0, A, B, C = Inactive (deny transactions) 1, D, E, F, G, H, I = Open (accept transactions) 2, J, K, L = Restricted to Deposits (accept all transactions except transfers and payments from this account) 3, M, N, O, P, Q, R = Open Primary (accept transactions) 4, S, T, U = Restricted Primary (accept all transactions except transfers and payments from this account) 9, V, W, X, Y, Z = Closed (deny transactions) BASE24 products do not distinguish between the alphabetic and numeric values grouped together above. The alphabetic values are intended to give institutions a wider range of values for assigning account statuses
31–34	N 4	Version The version number of this CACT row. This field is reserved for future use.
35–49	AN 15	Description A text description for this application account.
50	AN 1	Allowed: Debit A code indicating whether the customer can perform transfer or payment transactions that debit this account. Valid values are as follows: Y = Yes, account can be debited for a transfer or payment. N = No, account cannot be debited for a transfer or payment.

Position	Length	Field Description
51	A 1	Allowed: Credit A code indicating whether the customer can perform credit transactions on this account. Credit transactions include transfers to this account. Valid values are as follows: Y = Yes, account can be credited. N = No, account cannot be credited.
52	A 1	Allowed: Inquiry A code indicating whether the customer can perform inquiry transactions (e.g., available funds inquiry and scheduled transfers list inquiry) on this account. Valid values are as follows: Y = Yes, account can be inquired upon. N = No, account cannot be inquired upon.
53–59	N 8	Begin Date The first date (YYYYMMDD) this application account can be used in financial transaction processing.
60–67	N 8	End Date The last date (YYYYMMDD) this application account can be used in financial transaction processing.
68–142	AN 75	File Maintenance Tag See the “File Maintenance Tag Data” discussion at the end of this subsection for a description of this field.
143–148	AN 6	Account Type Description A text description of the service fee account type. Refer to the Default Account Type Description field provided in the Customer Add and Update Request Data redefine for a list of BASE24 reserved values.

Position	Length	Field Description
149–156	AN 8	Account Status Description The current status of the account indicating the action to be taken by the Integrated Authorization Server process if this account status is encountered during transaction processing. Valid status codes and the actions taken for each code are as follows: CLOSED = Closed. Deny transactions. DEPONLY = Restricted to deposits only. Accept all transactions except transfers from this account. INACTV = Inactive. Deny transactions. OPEN = Open. Accept transactions. OPENPRI = Open primary account. Accept transactions. RSTOPRI = Primary account restricted to deposits. Accept all transactions except transfers from this account. UNKNOWN = Unknown.

Customer Vendor Response Data

The external data block carries customer vendor information for customer vendor activate, customer vendor add, customer vendor deactivate, and customer vendor update transactions.

Note: If a *limited* vendor is added on a Customer Vendor Add response, the Vendor Number and Vendor Branch fields will contain the numbers to the new vendor and vendor branch rows that were inserted. If the vendor and vendor branch previously existed, the data from the existing values for the City, State, Postal Code, and Phone Number fields are returned. If any of these values differed from what was supplied in the request, the transaction is approved, but an action code of 002 is returned, indicating that not all the data was accepted.

For a customer vendor update transaction, two groups of fields may or may not be updated, depending on the following values: vendor verify status, vendor type, and account with vendor status. The two groups of fields are as follows:

Group 1 includes the following Customer Vendor Table (CVND) fields that can be updated:

- Account with Vendor
- Short Name
- Customer Vendor User Info 1
- Customer Vendor User Info 2.

Group 2 includes the following Vendor Table (VNDR) fields that can be updated:

- Vendor Name
- Vendor Branch Name
- Address Line 1
- Address Line 2
- City
- State
- Postal Code
- Vendor Phone Number
- Vendor User Remit Info 1
- Vendor User Remit Info 2
- Vendor Branch User Remit Info 1
- Vendor Branch User Remit Info 2.

The following table defines the circumstances for which each group of fields can be updated.

Vendor Verify[*]	Account With Vendor[†]	Vendor Type[‡]	Group 1 Fields	Group 2 Fields
Verified	Verified	Public	All	None
Verified	Initial	Public	All	None
Initial	Initial	Public	All	All
Initial	Verified	Public	All	All

Vendor Verify[*]	Account With Vendor[†]	Vendor Type[‡]	Group 1 Fields	Group 2 Fields
Verified	Verified	Limited	All	All
Verified	Initial	Limited	All	All
Initial	Initial	Limited	All	All
Initial	Verified	Limited	All	All
Not Verified or Initial	Not Verified or Initial	-----	None	None

^{*} The vendor verify value is from the VRFY column of the Vendor Table (VNDR).

[†] The account with vendor status value is from the ACCT_VRFY column of the Customer Vendor Table (CVND).

[‡] The vendor type value is from the VEND_TYP column of the Vendor Table (VNDR).

Data for this response is obtained from the Customer Vendor Table (CVND), Vendor Table (VNDR), Vendor Branch Table (VNDB), and Vendor Budget Category Table (VBUD). Detailed field descriptions for the CVND, VNDR, and VNDB are provided in the *BASE24-billpay Tables Maintenance Manual*.

The format of customer vendor response data in the external data block is as follows:

Position	Length	Field Description
1-5	N 5	Customer Vendor Number
		A value that uniquely identifies the relationship between the customer and vendor. The Billpay Table Maintenance Server process and Billpay Server process use the value in the Last Customer Vendor Number field on Customer Billpay Table (CBPT) screen 1 when adding a new customer vendor row, then update the Last Customer Vendor Number field by adding 1 to it.

Position	Length	Field Description
6–33	AN 28	Account with Vendor The customer account with the vendor that is credited with any customer payments.
34	AN 1	Account Verify A code indicating the result of the last attempt to verify this customer vendor information. Values can be reserved or user defined; however, any value other than V is considered to be unverified. The reserved values are as follows: F = Failed. Failed verification. I = Initial. Customer vendor information has been entered in the table but has not been verified. V = Verified. Customer vendor information has been verified and can be used for transactions.
35–54	N 20	Account Verify Last Change Date The date and time (YYYYMMDDhhmmssmmmmmm) the Account Verify field was initially set or last modified.
55–62	N 8	Customer Vendor Begin Date The first date (YYYYMMDD) the associated customer account with the vendor can be used to initiate a payment.
63–70	N 8	Customer Vendor End Date The last date (YYYYMMDD) the associated customer account with the vendor can be used to initiate a payment.
71–74	AN 4	Budget Category A code identifying the budget category to which this vendor is assigned. Any code used in this field must first be defined in the Vendor Budget Category Table (VBUD).

Position	Length	Field Description
75–84	AN 10	Customer Vendor Short Name The free-format customer short name for the vendor. This field is for information only.
85–104	AN 20	User Information 1 This field contains customer-specific information about the customer vendor. This field is for information only.
105–124	AN 20	User Information 2 This field contains customer-specific information about the customer vendor. This field is for information only.
125–200	AN 75	Customer Vendor File Maintenance Tag See the “File Maintenance Tag Data” discussion at the end of this subsection for a description of this field.
201–230	AN 30	Vendor Name The name of the vendor.
231–234	N 4	Main Branch Number The number of the vendor branch considered to be the primary location for this vendor.
235–238	N 4	Remit Branch Number The number of the vendor branch considered to be the location where remittance is expected to be received.
239–260	AN 22	Remit Person The name of the person responsible for remittance for the vendor. This person is normally at the remit branch.

Position	Length	Field Description
261	AN 1	Verify Status A code identifying the current verification status of a vendor. Reserved values are as follows: A = Awaiting ACH prenotification process to begin. This value is automatically set if the Remit Type is ACH so the ACH remittance processing software can identify this vendor as needing prenotification. No transactions can be paid to this vendor. F = Failed. Failed verification; no transactions can be paid to this vendor. I = Initial. This value is entered when the vendor row is inserted into the database. No transactions can be paid to this vendor. P = Pending ACH. Pending ACH prenotification to complete; this value is used after the vendor's prenotification has been submitted for ACH processing. No transactions can be paid to this vendor. V = Verified. Transactions can be paid to this vendor. The values normally used for table maintenance are I, F, and V. ACH processing uses the value P during prenotification processing of vendors who are paid through ACH processing. Other values can be used; however, only the value V allows transactions to be initiated by a customer.
262–271	AN 10	Verify Status Description A text description of the code contained in the Verify Status field. Text descriptions for reserved codes are as follows: A = WAIT ACH F = FAILED I = INITIAL P = PEND ACH V = VERIFIED A text description of USER-DFN is used for all other values.

Position	Length	Field Description
272–291	N 20	Verify Status Last Change The date and time (YYYYMMDDhhmmssmmmmmm) the value in the Verify Status field was last changed.
292–299	N 8	Vendor Begin Date The first date (YYYYMMDD) this vendor can be used in financial transaction processing.
300–307	N 8	Vendor End Date The last date (YYYYMMDD) this vendor can be used in financial transaction processing.
308–322	AN 15	Government ID The government identifier for this vendor. It is normally used as a tax identifier.
323–333	AN 11	Institution ID The ID of the financial institution owning the vendor's account to be credited for payments.
334–352	AN 19	Account Number The account number credited for payments made to this vendor. This account is located at the financial institution identified in the Institution ID field.
353–354	AN 2	Account Type An ISO code identifying the type of account credited for payments made to this vendor. Refer to the Account Type field description in the Multiple Account Information request redefine for a list of the BASE24 reserved account type values.

Position	Length	Field Description
355–357	AN 3	Remit Type A code indicating the form of payment the vendor wants to receive. Valid values are as follows: ACH = The vendor receives payment in the form of an automated clearing house (ACH) transaction. CGP = The vendor receives payment in the form of a group-item check. A group-item check is one check for multiple payments. The Billpay Table Maintenance Server process uses this value if no value is entered before adding a row. CIT = The vendor receives payment in the form of a single-item check. A single-item check is one check for each payment. User-defined values can be used for other forms of remittance. The Remittance Generator process uses this code to create various remit files. Specifically, the first character of the code is used as the first letter of the name of the remit file. For check processing, the full remit type is used to determine whether a group-item check or a single-item check is to be produced; however, all check remittances go into the same check remit file.
358–359	AN 2	Remit Report Type A user-defined code identifying how the vendor is notified about the customer detail accompanying the actual remittance. Suggested values are as follows: AA = ACH addenda DK = Diskette FX = Facsimile HD = Hard copy printout This field is for information only and is not currently used.

Position	Length	Field Description
360–363	AN 4	Default Budget Type A code identifying the budget category to which this vendor is assigned. Any code used in this field must first be defined in the Vendor Budget Category Table (VBUD).
364–383	AN 20	Audio Token The value given to a client device to retrieve data that is presented to the customer. The data retrieved is normally the vendor's name. This optional token is intended to allow a remote banking endpoint device to speak a vendor's name, but can be used by other input client devices. The PIT-NAM-ALGO param in the Logical Network Configuration File (LCONF) must be set to the value Y before the BASE24-billpay product uses the value from this field to generate an audio token.
384–403	N 20	Warning Payment Limit Amount The maximum amount, in whole and fractional currency units, of a customer payment to this vendor. If a customer payment to this vendor exceeds this amount, the online BASE24-billpay software outputs a warning event message. A zero value means there is no limit.
404–407	AN 4	Vendor Mask Group The name of the mask used to verify the account numbers customers have with this vendor. If this field and the Account Mask Group field on Vendor Branch Table (VNDB) screen 1 contain values, the account mask identified in the Account Mask Group field on VNDB screen 1 is used to verify the account numbers of all customers using the vendor branch identified by that VNDB row. If this field contains a value and the Account Mask Group field on VNDB screen 1 does not contain a value, the account mask identified in this field is used to verify the account numbers of all customers using the vendor branch identified by that VNDB row.

Position	Length	Field Description
408	AN 1	Vendor Type A code identifying the type of vendor. Valid values are as follows: L = Limited P = Public A public vendor is a vendor available for all customers and uses. A limited vendor is a vendor available for a limited number of customers and uses. A limited vendor is typically added when a customer wants to pay a person who is not a recognized business. This field is for information purposes only. It has no impact on transaction processing.
409–415	AN 7	Vendor Type Description A text description of the code contained in the Vendor Type field. Text descriptions for valid values are as follows: P = Public L = Limited
416–423	N 8	Prenote Date The date (YYYYMMDD) when a prenotification for this vendor was sent for ACH processing. This date is meaningful only for vendors with a remit type of ACH.
424–425	AN 2	Vendor Specific Data Type A user-defined code that identifies the type of vendor-specific data required.
426–455	AN 30	Vendor Specific Data Description Text that describes vendor-specific data required by this vendor for payment transactions. The presence of an entry in this field controls whether vendor-specific data is required from a remote banking endpoint device.

Position	Length	Field Description
456–475	AN 20	Vendor User Remit Information 1 Specific information about the remittance that is required by the vendor. The information must be included in the remittance file produced for the vendor's remittance type.
476–495	AN 20	Vendor User Remit Information 2 See the Vendor User Remit Information 1 field description above.
496–499	AN 4	Category Code A code used to classify the industry in which the vendor is involved.
500–574	AN 75	Vendor File Maintenance Tag See the “File Maintenance Tag Data” discussion at the end of this subsection for a description of this field.
575	AN 1	Reserved This field is not used.
576–584	N 9	Vendor Number The number that identifies the vendor. This value is generated by the system and is unique.
585–588	N 4	Vendor Branch Number The number of the vendor branch considered to be the location where remittance is expected to be received.
589–618	AN 30	Vendor Branch Name The name of the vendor remittance branch.
619–648	AN 30	Address 1 The first line of the mailing address or street address of the vendor remittance branch.

Position	Length	Field Description
649–678	AN 30	Address 2 The second line of the mailing address or street address of the vendor remittance branch.
679–703	AN 25	City The city in which the vendor remittance branch is located.
704–705	AN 2	State A code for the state in which the vendor remittance branch is located. The U.S. state codes are based on standards described in the 1988 ANSI X3.38 standard, <i>Identification of the States, the District of Columbia, and the Outlying and Associated Areas of the United States for Information Interchange</i> . This field can contain a value other than an ANSI code.
706–715	AN 10	Postal Code The postal code of the vendor remittance branch.
716–718	AN 3	County A code indicating the county in which the vendor remittance branch is located. The U.S. county and county equivalent codes are based on standards described in the 1988 ANSI X3.31 standard, <i>Structure for the Identification of the Counties and County Equivalents of the United States and its Outlying and Associated Areas for Information Interchange</i> . This field can also contain a value other than an ANSI code.

Position	Length	Field Description
719–721	AN 3	Country A code for the country in which the vendor remittance branch is located. The country codes are available in the 1993 ISO 3166 standard, <i>Codes for the Representation of Names of Countries</i> . This field can also contain a value other than an ISO code.
722–741	AN 20	Phone Number The business telephone number for the vendor remittance branch.
742–763	AN 22	Contact Person The name of the contact person at the vendor remittance branch.
764–783	AN 20	Fax Phone The fax telephone number for the vendor remittance branch.
784–803	AN 20	Data Communications Phone Number The data communication telephone number for data transmission with the vendor remittance branch, if applicable.
804–807	AN 4	Mask Group The name of the mask used to verify the account numbers customers have with this vendor branch. If this field contains a value, the account mask identified in this field is used to verify the account numbers of all customers using this vendor branch. If this field does not contain a value, the account mask identified in the Account Mask Group field on VNDR screen 1 is used to verify the account numbers of all customers using this vendor branch.

Position	Length	Field Description
808–827	AN 20	Vendor Branch User Remit Information 1 Specific information about the remittance that is required by the vendor. This information is included in the remittance file produced for the vendor’s remittance type.
828–847	AN 20	Vendor Branch User Remit Information 2 See the Vendor Branch User Remit Information 1 field description above.
848–922	AN 75	Vendor Branch File Maintenance Tag See the “File Maintenance Tag Data” discussion at the end of this subsection for a description of this field.
923–952	AN 30	Customer Vendor Budget Category Description A free-format text description of the budget category for this customer vendor in the Budget Category field.
953–882	AN 30	Vendor Default Budget Category Description A free-format text description of the default budget category to which this vendor is assigned in the Default Budget Category field.

Customer Vendor List Inquiry Response Data

The external data block carries customer vendor information when a customer vendor list inquiry transaction is performed. The format of this data in the external data block is as follows:

Note: The response data for the customer vendor list inquiry is the same as for the customer vendor response data list above with the exception of an additional four bytes at the beginning of the customer vendor list inquiry response data. The additional four bytes are listed below.

Position	Length	Field Description
1–2	N 2	Count The number of items (i.e., customer vendors) returned in this response message. Valid values are 0–4.
3	AN 1	Additional Data Flag: A code indicating whether more customer vendors exist for this customer ID. Valid values are as follows: Y = Yes, more data possible. N = No, no more data.
4	AN 1	Reserved This field is not used.
...		Customer Vendor Information Occurs up to a maximum of 4 times. See the Customer Vendor Response Data format above.

Note: The Additional Data Flag field value Y means that more data is possible. If the maximum number of occurrences are returned, the value is set to Y. It is possible that the last occurrence is the last CVND row for the customer and the next request will not contain any CVND inquiry occurrences. The Additional Data Flag is only set to N if the number of occurrences returned is less than the maximum possible number of occurrences that can be returned.

Customer Vendor Mask Verify Response Data

The external data block carries customer vendor mask information for customer vendor mask verify list inquiry transactions.

Data for this response is obtained from the Vendor Mask Table (VNDM). Detailed field descriptions for the VNDM are provided in the ***BASE24-billpay Tables Maintenance Manual***.

The format of this data in the external data block is as follows:

Position	Length	Description
1-2	N 2	Count The number of items (i.e., vendor masks) returned in this response message. Valid values are 0-8.
3	AN 1	Additional Items A code indicating whether more vendor masks exist for this customer vendor. Valid values are as follows: Y = Yes, more data possible. N = No, no more data.
4	AN 1	Reserved This field is not used.
...		Vendor Mask Information The following fields occur up to a maximum of 8 times. Each occurrence is 476 characters in length.
	AN 4	Mask Group The name of the mask that is used to verify customer account numbers with this vendor.

Position	Length	Description
AN 70	Account Mask	An edit mask that is used to analyze the account number a customer has with a vendor. The edit that uses this value is performed after account lengths are checked and before a user-defined algorithm is checked. The edit mask may contain combinations of the following characters to represent the valid format of an account number: A = Matches any alphabetic character (A through Z). X = Matches any alphanumeric character (A through Z, 0 through 9). 9 = Matches any digit (0 through 9). ? = Matches any single character. * = Matches any number of characters (including none). “<i>text</i>” = Matches <i>text</i>, where <i>text</i> is the group of characters being matched. Since single quotation marks indicate where the text being matched begins and ends, two quotation marks together represent a quotation mark that appears in the text being matched.

Position	Length	Description
	N 5	User Check Number A number that identifies a user-defined algorithm used to analyze the account number a customer has with a vendor. The edit that uses this value is performed after account number lengths or account masking are checked. Valid values are as follows: 0 = The account number is considered valid without being analyzed (default value). 1–50 = The account number is analyzed with the user-defined algorithm identified by this value. The value in this field represents the numeric value in a CASE statement in the software into which the user-defined algorithm is inserted. The standard BASE24-billpay product does not have any account number algorithms. 51–9999 = The account number is considered invalid without being analyzed. Note: Numbers between 51 and 9999 can be used if the CASE statement is changed to recognize them.
	N 2	Minimum Length The smallest number of characters the account number can contain. This value cannot be zero and must be less than or equal to the value in the Maximum Length field. The edit that uses this value is performed before the account masking or user-defined algorithms are checked.
	N 2	Maximum Length The greatest number of characters the account number can contain. This value cannot be zero and must be greater than or equal to the value in the Minimum Length field. The edit that uses this value is performed before the account masking or user-defined algorithms are checked.

Position	Length	Description
AN 312		Help Text A free-format, information-only field that can be used to provide additional information.
AN 78		Comment An information-only field that can be used to describe the purpose of the account mask.
N 2		Verification Failed Indicator A code indicating whether the customer account with vendor provided in the inquiry request passed or failed verification for this vendor mask. Valid values are as follows: 00 = Accepted 01 = No account 02 = Failed minimum 03 = Failed maximum 04 = Failed mask 05 = Failed user check
AN 1		Reserved This field is not used.

History Inquiry Response Data

The external data block carries historical data for history inquiry – payments and transfers transactions and history inquiry – all transactions.

Data for this response is obtained from the History Table (HIST). Detailed field descriptions for the HIST are provided in the ***BASE24-billpay Tables Maintenance Manual***.

The format of this data in the external data block is as follows:

Position	Length	Description
1	N 2	Count The number of items (i.e., transactions) returned in this response message. Valid values are 0–7.
3	AN 1	Additional Data Flag: A code indicating whether more transactions exist for this customer ID. Valid values are as follows: Y = Yes, more data possible. N = No, no more data.
4	AN 1	Reserved This field is not used.
...		Inquiry Response Structure The following fields occur up to a maximum of 7 times. Each occurrence is 523 characters in length.
	N 5	Partition Number A number indicating the History Table (HIST) partition in which this transaction is stored. The HIST is partitioned for performance reasons.
	N 20	GMT The date and time (YYYYMMDDhhmmssmmmmmm) on which this transaction was written to the History Table (HIST).
	N 6	Customer Reference Number A unique system-generated number used to reference historical data for this transaction for this customer. Every transaction processed by the system for this customer is assigned a unique reference number.
	N 5	Customer Vendor Number The customer vendor number identifying the customer vendor for a payment transaction

Position	Length	Description
N 9	Vendor Number	The number that identifies the vendor for a payment transaction. This value is generated by the system and is unique.
AN 28	Customer Account with Vendor	The customer's account number with this customer vendor. This is the number of the account to which a payment transaction is posted or credited.
N 20	Billpay GMT	The date and time (YYYYMMDDhhmmssmmmmmm) this transaction was entered into the system (i.e., was received by the Billpay Server process).
AN 2	Transaction Type	A code indicating the type of transaction processed. Refer to the Processing Code (P-3) data element description for a list of the reserved transaction type codes.
AN 4	From FIID	The FIID of the financial institution owning the <i>from</i> account for this transaction. The FIID is used throughout BASE24 products to identify each BASE24 institution.
AN 19	From Account Number	The number of the customer account debited in a payment or transfer transaction. This is the account from which a payment or transfer is made.
AN 2	From Account Type	An ISO code identifying the type of <i>from</i> account used in this transaction. Refer to the Account Type field description provided in the Transaction Data, Private (S-125) data element for a list of the BASE24 reserved account type values.

Position	Length	Description
AN 4	To FIID	The FIID of the financial institution owning the <i>to</i> account for this transaction. The FIID is used throughout BASE24 products to identify each BASE24 institution.
AN 19	To Account Number	The number of the customer account credited in a transfer transaction. This is the account to which a transfer is made.
AN 2	To Account Type	An ISO code identifying the type of <i>to</i> account used in this transaction. Refer to the Account Type field description provided in the Transaction Data, Private (S-125) data element for a list of the BASE24 reserved account type values.
N 12	Amount	The amount of the payment or transfer in whole and fractional currency units (e.g., U.S. dollars and cents).
AN 2	Transaction Source	A code indicating the type of device or program used to enter this transaction. User-defined values other than the following reserved values are also valid: AD = Audio device (e.g., an interactive voice response system) BL = BASE24-billpay Billing Server process IB = Inbound customer service representative PC = Personal computer SP = Screen phone
N 6	Transaction Source Number	A unique number assigned by the transaction source device or software to this transaction.

Position	Length	Description
N 20		Transaction Source Local Civil Time The date and time (YYYYMMDDhhmmssmmmmmm) this transaction was entered into the system at the transaction source or device. This time is provided in the local civil time of the transaction source.
AN 30		Vendor Specific Data Any vendor-specific data that was entered for a payment transaction.
N 8		Payment Date The date (YYYYMMDD) the customer requested this transaction to be performed.
AN 1		Schedule Type A flag indicating whether this is an immediate, future, or recurring scheduled financial transaction or a nonfinancial transaction. Valid values are as follows: I = Immediate F = Future R = Recurring ␣ = N/A (nonfinancial), where ␣ is a blank space
AN 28		Bill Text Any billing text associated with this transaction. This is typically the text that will appear on the customer's billing statement from the customer's financial institution.
N 6		Original Customer Reference Number The original reference number for this transaction, if it is different from the current reference number. If the original reference number is the same as the current reference number, this field contains all blanks. For transfers and payments that were scheduled, this is the reference number of the original scheduled transaction.

Position	Length	Description
AN 3		Response Code A code indicating the processing outcome of this transaction. Refer to the <i>BASE24 Remote Banking Transaction Processing Manual</i> for a list of all possible action codes.
AN 4		Reversal Reason Code A code used by the transaction authorizer to indicate the reason this transaction is reversed. If the transaction is not reversed, this field is blank.
AN 3		Auth Action Code See the Response Code field description above.
AN 9		Auth Approve Code The approval code assigned to the transaction by the transaction authorizer. An approval code is a random number generated for an approved transaction. For denied transactions, this field is blank.
AN 1		Collection Status The status for the collection of funds from the <i>from</i> account for this transaction, if applicable. User-defined values other than the following reserved values also are valid. C = Complete. Collection of customer funds for this transaction is complete. F = Failed. The collection of customer funds for this transaction failed. I = Initial. The collection of customer funds for this transaction has not yet begun. P = Pending. The collection of customer funds for this transaction is pending. ␣ = Unknown, where ␣ is a blank space. Displayed for all nonfinancial transactions. A value of all blanks is considered to be user defined.

Position	Length	Description
N 8		Business Date The date (YYYYMMDD) on which funds were collected from the <i>from</i> account for financial transactions. For nonfinancial transactions, this is the date on which the transaction was processed.
N 8		Remit Business Date The date (YYYYMMDD) on which remittance was initiated to a vendor for a payment.

Position	Length	Description
A	1	Remit Status The current status of remittance processing for a payment transaction. Valid values are as follows: C = The Billpay Server process has received acknowledgment that the remittance was successfully sent to the vendor. When this status is displayed, the Remit Ack fields will contain values. F = The Billpay Server process was unable to generate remittance for this payment transaction. This value can also be used to indicate that the vendor acknowledged that they failed to receive the payment or that a third-party processor acknowledged that generation of the required remittance failed. In either case, the Remit Ack Received GMT field indicates the date the acknowledgment was received. I = Remittance is waiting to be sent out by the Remittance Generator process. L = The automated clearinghouse (ACH) transaction or check used to pay the vendor has cleared the account from which the payment was issued. This implies that the vendor has been credited with funds for the payment. S = The Remittance Generator process has processed this transaction and made an entry in the appropriate Remit File. ␣ = Displayed for all nonpayment transactions (where ␣ is a blank space). A value of all blanks is considered to be user defined. For all other values, the remittance status is considered to be unknown. For a payment remitted by ACH processing, the typical sequence of status values for the life of the payment transaction are as follows: I, S, and L; or I, S, and F. For a payment remitted by check, the typical sequence of status values for the life of the payment transaction are as follows: I, S, C, and L. The exact sequence of remittance status values depends on the acknowledgments that can be returned from a third-party processor or the vendor.

Position	Length	Description
A 3	Remit Type	A code indicating the type of remittance generated for a payment transaction. Valid values include the following: ACH = Automated Clearinghouse (ACH) payment CGP = Group-item check CIT = Single-item check = Displayed for all nonpayment transactions, where is a blank space
N 20	Remit Ack GMT	The date and time (YYYYMMDDhhmmssmmmmmm) on which the vendor was actually paid, rather than the date on which the Remittance Generator process generated the remittance for a payment transaction.
AN 10	Remit Ack Number	A number used on the remittance generated for a payment transaction, such as a check number.
N 20	Remit Ack Received GMT	The date and time (YYYYMMDDhhmmssmmmmmm) on which an outside remitter (e.g., an automated clearing house) notified the BASE24-billpay system of the date the vendor was actually paid (Remit Ack GMT field) and the number used on the remittance (Remit Ack Number field).
AN 75	File Maintenance Tag	See the “File Maintenance Tag Data” discussion at the end of this subsection for a description of this field.

Position	Length	Description
AN 6		From Account Type Description The account type name for the ISO account type code provided in the From Account Type field. This is the name configured for the code in the Account Type Table File (ATT). Refer to the Default Account Type Description field provided in the Customer Add and Update Request Data redefine for a list of BASE24 reserved values.
AN 6		To Account Type Description The account type name for the ISO account type code provided in the To Account Type field. This is the name configured for the code in the Account Type Table File (ATT). Refer to the Default Account Type Description field provided in the Customer Add and Update Request Data redefine for a list of BASE24 reserved values.
AN 30		Response Code Description User-defined text describing the response code provided in the Response Code field. This is the description configured for the code in the Response Text Table (RTXT).
AN 30		Vendor Name The name of the vendor identified by the number in the Vendor Number field.

Position	Length	Description
AN 20		Vendor Token The value given to a client device to retrieve data that is presented to the customer. The data retrieved is normally the vendor's name. This optional token is intended to allow a remote banking endpoint device to speak a vendor's name, but can be used by other input client devices. The PIT-NAM-ALGO param in the Logical Network Configuration File (LCONF) must be set to the value Y before the BASE24-billpay product uses the value from this field to generate an audio token.
N 3		Currency Code A code indicating the primary type of currency the logical network uses (e.g. 840) for transaction amounts stored in the BASE24 database. For the currency code values, refer to the ISO 4217-1990 standard, <i>Codes for the Representation of Currencies and Funds</i>. The currency code in this field can be used to determine the number of fractional currency unit positions to account for in the Amount field.
AN 1		Reserved This field is not used.
AN 4		Transaction Type Financial Description A text description for financial transactions. Valid values for reserved financial transaction type codes are as follows: 9A, 9B, 9C, 5A, 5B = BLPY 9D, 9E, 4A, 4B = TRFR All other values = ¯ (where ¯ is a blank space)

The Customer Vendor Number, Vendor Token, Vendor Name and Customer Account with Vendor fields are blank for transfer transactions. The To Account Number and To Account Type fields are blank for payment transactions.

Note: The Additional Data Flag field value of Y means that more data is possible. If the maximum number of occurrences are returned, the value is set to Y. It is possible that the last occurrence is the last payment or transfer for the customer and the next request will not contain any payment or transfer occurrences. The Additional Data Flag field is only set to a value of N if the number of occurrences returned is less than the maximum possible number of occurrences that can be returned.

Master Vendor List Inquiry Response Data

Master vendor list inquiry response data carries vendor information. The response data is returned for master vendor list transactions.

Data for this response is obtained from the Vendor Table (VNDR), Vendor Branch Table (VNDB), and Vendor Budget Category Table (VBUD). Detailed field descriptions for the VNDR, VNDB, and VBUD are provided in the *BASE24-billpay Tables Maintenance Manual*.

The format of this data in the external data block is as follows:

Position	Length	Description
1-2	N 2	Count The number of items (i.e., vendors) returned in this response message. Valid values are 0-5.
3	AN 1	Additional Data Flag A code indicating whether more vendors exist in the vendor registry. Valid values are as follows: Y = Yes, more data possible. N = No, no more data.
4	AN 1	Reserved This field is not used.
...		Vendor Information The following fields occur up to a maximum of 5 times. Each occurrence is 782 characters in length.

Position	Length	Description
AN 30	Vendor Name	The name of the vendor.
N 4	Main Branch Number	The identification number of the vendor branch considered to be the primary location for this vendor.
N 4	Remit Branch Number	The identification number of the vendor branch considered to be the location where remittance is expected to be received.
AN 22	Remit Person	The name of the person responsible for remittance for the vendor. This person is normally at the remit branch.

Position	Length	Description
AN 1		Verify Status A code identifying the current verification status of a vendor. Reserved values are as follows: A = Awaiting ACH prenotification process to begin. This value is automatically set if the Remit Type is ACH so the ACH remittance processing software can identify this vendor as needing prenotification. No transactions can be paid to this vendor. F = Failed. Failed verification; no transactions can be paid to this vendor. I = Initial. This value is entered when the vendor row is inserted into the database. No transactions can be paid to this vendor. P = Pending ACH. Pending ACH prenotification to complete; this value is used after the vendor's prenotification has been submitted for ACH processing. No transactions can be paid to this vendor. V = Verified. Transactions can be paid to this vendor. The values normally used for table maintenance are I, F, and V. ACH processing uses the value P during prenotification processing of vendors who are paid through ACH processing. Other values can be used; however, only the value V allows transactions to be initiated by a customer.
AN 10		Verify Status Description A text description of the code contained in the Verify Status field. Text descriptions for reserved codes are as follows: A = WAIT ACH F = FAILED I = INITIAL P = PEND ACH V = VERIFIED A text description of USER-DFN is used for all other values.

Position	Length	Description
N 20		Verify Status Last Change The date and time (YYYYMMDDhhmmssmmmmmm) the value in the Verify Status field was last changed.
N 8		Vendor Begin Date The first date (YYYYMMDD) this vendor can be used in financial transaction processing.
N 8		Vendor End Date The last date (YYYYMMDD) this vendor can be used in financial transaction processing.
AN 15		Government ID The government identifier for this vendor. It is normally used as a tax identifier.
AN 11		Institution ID The ID of the financial institution owning the vendor's account to be credited for payments.
AN 19		Account Number The account number credited for payments made to this vendor. This account is located at the financial institution identified in the Institution ID field.
AN 2		Account Type An ISO code identifying the type of account credited for payments made to this vendor. Refer to the Account Type field description provided in the Transaction Data, Private (S-125) data element for a list of the BASE24 reserved account type values.

Position	Length	Description
AN 3		Remit Type A code indicating the form of payment the vendor wants to receive. Valid values are as follows: ACH = The vendor receives payment in the form of an automated clearing house (ACH) transaction. CGP = The vendor receives payment in the form of a group-item check. A group-item check is one check for multiple payments. The Billpay Table Maintenance Server process uses this value if no value is entered before adding a row. CIT = The vendor receives payment in the form of a single-item check. A single-item check is one check for each payment. User-defined values can be used for other forms of remittance. The Remittance Generator process uses this code to create various remit files. Specifically, the first character of the code is used as the first letter of the name of the remit file. For check processing, the full remit type is used to determine whether a group-item check or a single-item check is to be produced; however, all check remittances go into the same check remit file.
AN 2		Remit Report Type A user-defined code identifying how the vendor is notified about the customer detail accompanying the actual remittance. Suggested values are as follows: AA = ACH addenda DK = Diskette FX = Facsimile HD = Hard copy printout This field is for information only and is not currently used.

Position	Length	Description
AN 4		Default Budget Type A code identifying the budget category to which this vendor is assigned. Any code used in this field must first be defined in the Vendor Budget Category Table (VBUD).
AN 20		Audio Token The value given to a client device to retrieve data that is presented to the customer. The data retrieved is normally the vendor's name. This optional token is intended to allow a remote banking endpoint device to speak a vendor's name, but can be used by other input client devices. The PIT-NAM-ALGO param in the Logical Network Configuration File (LCONF) must be set to the value Y before the BASE24-billpay product uses the value from this field to generate an audio token.
N 20		Warning Payment Limit Amount The maximum amount, in whole and fractional currency units, of a customer payment to this vendor. If a customer payment to this vendor exceeds this amount, the online BASE24-billpay software outputs a warning event message. A zero value means there is no limit.
AN 4		Vendor Mask Group The name of the mask used to verify the account numbers customers have with this vendor. If this field and the Account Mask Group field on Vendor Branch Table (VNDB) screen 1 contain values, the account mask identified in the Account Mask Group field on VNDB screen 1 is used to verify the account numbers of all customers using the vendor branch identified by that VNDB row. If this field contains a value and the Account Mask Group field on VNDB screen 1 does not contain a value, the account mask identified in this field is used to verify the account numbers of all customers using the vendor branch identified by that VNDB row.

Position	Length	Description
AN 1		Vendor Type A code identifying the type of vendor. Valid values are as follows: L = Limited P = Public A public vendor is a vendor available for all customers and uses. A limited vendor is a vendor available for a limited number of customers and uses. A limited vendor is typically added when a customer wants to pay a person who is not a recognized business. This field is for information purposes only. It has no impact on transaction processing.
AN 7		Vendor Type Description A text description of the code contained in the Vendor Type field. Text descriptions for valid values are as follows: P = Public L = Limited
N 8		Prenote Date The date (YYYYMMDD) when a prenotification for this vendor was sent for ACH processing. This date is meaningful only for vendors with a remit type of ACH.
AN 2		Vendor Specific Data Type A user-defined code that identifies the type of vendor-specific data required.
AN 30		Vendor Specific Data Description Text that describes vendor-specific data required by this vendor for payment transactions. The presence of an entry in this field controls whether vendor-specific data is required from a remote banking endpoint device.

Position	Length	Description
AN 20		Vendor User Remit Information 1 Specific information about the remittance that is required by the vendor. The information must be included in the remittance file produced for the vendor's remittance type.
AN 20		Vendor User Remit Information 2 See the Vendor User Remit Information 1 field description above.
AN 4		Category Code A code used to classify the industry in which the vendor is involved.
AN 75		Vendor File Maintenance Tag See the "File Maintenance Tag Data" discussion at the end of this subsection for a description of this field.
AN 1		Reserved This field is not used.
N 9		Vendor Number The number that identifies the vendor. This value is generated by the system and is unique.
N 4		Vendor Branch Number The number of the vendor branch considered to be the location where remittance is expected to be received.
AN 30		Vendor Branch Name The name of the vendor remittance branch.
AN 30		Address 1 The first line of the mailing address or street address of the vendor remittance branch.

Position	Length	Description
AN 30	Address 2	The second line of the mailing address or street address of the vendor remittance branch.
AN 25	City	The city in which the vendor remittance branch is located.
AN 2	State	A code for the state in which the vendor remittance branch is located. The U.S. state codes are based on standards described in the 1988 ANSI X3.38 standard, <i>Identification of the States, the District of Columbia, and the Outlying and Associated Areas of the United States for Information Interchange</i>. This field can contain a value other than an ANSI code.
AN 10	Postal Code	The postal code of the vendor remittance branch.
AN 3	County	A code indicating the county in which the vendor remittance branch is located. The U.S. county and county equivalent codes are based on standards described in the 1988 ANSI X3.31 standard, <i>Structure for the Identification of the Counties and County Equivalents of the United States and its Outlying and Associated Areas for Information Interchange</i>. This field can also contain a value other than an ANSI code.

Position	Length	Description
AN 3		Country A code for the country in which the vendor remittance branch is located. The country codes are available in the 1993 ISO 3166 standard, <i>Codes for the Representation of Names of Countries</i> . This field can also contain a value other than an ISO code.
AN 20		Phone Number The business telephone number for the vendor remittance branch.
AN 22		Contact Person The name of the contact person at the vendor remittance branch.
AN 20		Fax Phone The fax telephone number for the vendor remittance branch.
AN 20		Data Communications Phone Number The data communication telephone number for data transmission with the vendor remittance branch, if applicable.
AN 4		Mask Group The name of the mask used to verify the account numbers customers have with this vendor branch. If this field contains a value, the account mask identified in this field is used to verify the account numbers of all customers using this vendor branch. If this field does not contain a value, the account mask identified in the Account Mask Group field on VNDR screen 1 is used to verify the account numbers of all customers using this vendor branch.

Position	Length	Description
AN 20		Vendor Branch User Remit Information 1 Specific information about the remittance that is required by the vendor. This information is included in the remittance file produced for the vendor's remittance type.
AN 20		Vendor Branch User Remit Information 2 See the Vendor Branch User Remit Information 1 field description above.
AN 75		Vendor Branch File Maintenance Tag See the "File Maintenance Tag Data" discussion at the end of this subsection for a description of this field.
AN 30		Customer Vendor Budget Category Description A free-format text description of the budget category for this customer vendor in the Budget Category field.
AN 30		Vendor Default Budget Category Description A free-format text description of the default budget category to which this vendor is assigned in the Default Budget Category field.

Miscellaneous Response Data

The external data block carries miscellaneous message information for miscellaneous request response transactions.

Data for this response is obtained from the Miscellaneous Request Table (MISC). Detailed field descriptions for the MISC are provided in the ***BASE24 Remote Banking Customer Service Support Manual***.

The format of this data in the external data block is as follows:

Position	Length	Description
1–20	N 20	Transaction Source Local Civil Time The date and time (YYYYMMDDhhmmssmmmmmm) this transaction was entered into the system at the transaction source or device. This time is provided in the local civil time of the transaction source.
21–40	AN 20	Family Name The family name or last name of this customer.
41–50	AN 10	Given Name The given name or first name of this customer.
51–70	AN 20	Phone Number The telephone number of the contact person for this request.
71–74	AN 4	FIID The FIID of the customer's financial institution. The FIID is an identifier that must be unique within the logical network. It is used throughout BASE24 products to identify each BASE24 institution.
75	A 1	Contact Type A code indicating the type of contact for this request. Reserved values are as follows: B = Bank (financial institution) C = Customer O = Other V = Vendor This field can also contain a user-defined contact type.
76	N 1	Priority The priority assigned to this request. Requests with a priority of 1 are considered the highest priority while requests with a priority of 5 are considered the lowest priority. Valid values are 1–5.

Position	Length	Description
77	A 1	Request Type A code indicating the type of this request. Valid values are as follows: A = Account maintenance C = Complaint Q = Question or query User-defined values can be used for other requests.
78–99	AN 22	Department User-defined text identifying the financial institution or service provider department to manage this request.
100–723	AN 624	Contact Details User-defined text explaining the details of this request.
723–799	AN 75	File Maintenance Tag See the “File Maintenance Tag Data” discussion at the end of this subsection for a description of this field.
800	A 1	Reserved This field is not used.

Miscellaneous Requests Inquiry Response Data

The external data block carries miscellaneous message information for miscellaneous requests inquiry response transactions.

Data for this response is obtained from the Miscellaneous Request Table (MISC). Detailed field descriptions for the MISC are provided in the ***BASE24 Remote Banking Customer Service Support Manual***.

The format of this data in the external data block is as follows:

Position	Length	Description
1–2	N 2	Count The number of items (i.e., miscellaneous requests) returned in this response message. Valid values are 0–5.
3	AN 1	Additional Data Flag: A code indicating whether more miscellaneous requests exist for this search. Valid values are as follows: Y = Yes, more data possible. N = No, no more data.
4	AN 1	Reserved This field is not used.
...		Miscellaneous Information The Miscellaneous Information fields occur up to a maximum of 5 times. Each occurrence is 800 characters in length. See the Miscellaneous Response Data format above for descriptions of these fields.

Scheduled Payments List Inquiry Response Data

The external data block carries scheduled payment information for scheduled payments list inquiry transaction responses.

Data for this response is obtained from the Future Table (FUTR) and Recurring Table (RCUR). Detailed field descriptions for the FUTR and RCUR are provided in the *BASE24 Remote Banking Customer Service Support Manual*.

The format of this data in the external data block is as follows:

Position	Length	Description
1–2	N 2	Count The number of items (i.e., scheduled payments) returned in this response message. Valid values are 0–9.

Position	Length	Description
3	AN 1	Additional Data Flag A code indicating whether more scheduled payments exist for this customer. Valid values are as follows: Y = Yes, more data possible N = No, no more data Note: The Additional Data Flag field value of Y means that more data is possible. If the maximum number of occurrences are returned, the value is set to Y. It is possible that the last occurrence is the last scheduled payment for the customer and the next request will not contain any scheduled payment occurrences. The Additional Data Flag field is only set to a value of N if the number of occurrences returned is less than the maximum possible number of occurrences that can be returned.
4	AN 1	Reserved This field is not used.
...		Inquiry Response Structure The following fields occur up to a maximum of 9 times. Each occurrence is 449 characters in length.
	N 4	Payments Made The total number of payments already made.
	N 4	Total Payments The total number of payments to make.
	N 8	Initial Payment Date The date (YYYYMMDD) on which the initial payment is to be made.

Position	Length	Description
AN 2	Recur Period	A code indicating the payment frequency. Valid values are as follows: WK = Weekly 2W = Every two weeks MN = Monthly Q = Quarterly S = Semi-annually A = Annually
N 5	STI Partition Number	A number indicating the Future Table (FUTR) partition in which this transaction is stored. Each Scheduled Transaction Initiator process handles a separate FUTR partition.
N 8	STI Payment Date	The date (YYYYMMDD) on which the Scheduled Transaction Initiator process will next attempt to make a payment for this scheduled transaction. This date may be different from the date displayed in the Payment Date field if the process was unable to complete the transaction successfully on the specified next payment date.
N 20	Retry Processing GMT	If the transaction is currently in a retry cycle, this is the date and time (YYYYMMDDhhmmssmmmmmm) the system will next attempt to process the transaction.
N 6	Customer Reference Number	A unique system-generated number used to reference historical data for this scheduled payment transaction.
N 5	Customer Vendor Number	The number of the customer vendor to receive this payment.

Position	Length	Description
N 9	Vendor Number	The number of the vendor to receive this payment. The vendor number is an internal number generated by the system for this vendor. This number is unique among all registered vendors and is automatically generated when the vendor is added to the system.
AN 28	Customer's Account with Vendor	The customer's account number with the vendor to which this payment is posted or credited.
N 20	Billpay GMT	The date and time (YYYYMMDDhhmmssmmmmmm) this scheduled payment transaction was entered into the system (i.e., was received by the BASE24-billpay Billpay Server process).
AN 2	Transaction Type	A code indicating the specific type of scheduled payment transaction. Valid values are as follows: 9B = Schedule payment – future 9C = Schedule payment – recurring
AN 4	From FIID	The FIID of the customer's financial institution for the <i>from</i> account. The FIID is an identifier that must be unique within the logical network. It is used throughout BASE24 products to identify each BASE24 institution.
AN 19	From Account Number	The customer's account number to be debited for this payment.
AN 2	From Account Type	An ISO code identifying the type of account debited for this payment. Refer to the Account Type field description in the Transaction Data, Private (S-125) data element for a list of the BASE24 reserved account type values.

Position	Length	Description
AN 4	To FIID	This field should be blank. <i>To</i> accounts do not apply for scheduled payments.
AN 19	To Account Number	This field should be blank. <i>To</i> accounts do not apply for scheduled payments.
AN 2	To Account Type	This field should be blank. <i>To</i> accounts do not apply for scheduled payments.
N 12	Amount	The amount of the payment in whole and fractional currency units (e.g., U.S. dollars and cents).
AN 2	Transaction Source	A code indicating the originating source of this scheduled payment transaction. User-defined values other than the following reserved values are also valid: AD = Audio device (e.g., an interactive voice response system) BL = BASE24-billpay Billing Server process IB = Inbound customer service representative PC = Personal computer SP = Screen phone
N 6	Transaction Source Number	A unique number assigned by the transaction source device or software to this scheduled payment transaction.
N 20	Transaction Source Local Civil Time	The date and time (YYYYMMDDhhmmssmmmmmm) this scheduled payment transaction was entered into the system at the transaction source or device. This time is provided in the local civil time of the transaction source.

Position	Length	Description
AN 30		Vendor Specific Data Any vendor-specific data that was entered for this scheduled payment transaction.
N 8		Payment Date The date (YYYYMMDD) on which the next payment is to be made. For future scheduled payments, this field is blank or contains the same date as the Initial Payment Date field.
A 1		Schedule Type A flag indicating the type of scheduled payment. Valid values are as follows: F = Future payment R = Recurring payment
AN 28		Billing Text The billing text, if any, associated with this transaction. This is the text normally displayed on the customer's account statement for the transaction.

Position	Length	Description
AN 2	Processing Status	A code indicating the current processing status of this scheduled payment transaction. Valid values are as follows: = Not started, where is a blank space PA = Pending authorization RP = Retry processing is pending RW = Waiting to retry All others = Unknown Note: Customers can only update transaction data, delete transactions, or skip a payment period if the Scheduled Transaction Initiator process is not currently processing the transaction. If the status indicates the transaction is pending authorization or that retry processing is pending, the Scheduled Transaction Initiator process is currently processing the transaction and it cannot be altered. If the status indicates that processing has not started, is waiting to be retried, or is unknown, then the transaction can be altered.
N 20	STI Request GMT	The date and time (YYYYMMDDhhmmssmmmmmm) when the Scheduled Transaction Initiator process requested a funds authorization for this payment transaction. A value is contained in this field only during the Scheduled Transaction Initiator process authorization cycle.
N 4	Retry Count Transient	The number of times this payment transaction has been retried due to a transient error condition. A transient error condition specifies a condition that typically exists for only a brief period of time and that often can be rectified without any operator intervention.

Position	Length	Description
N 4		Retry Count Nontransient The number of times the transaction has been retried due to a nontransient error condition. A nontransient error condition specifies a condition that may exist for an extended period of time (e.g., insufficient funds) or may require operator intervention to rectify the condition.
AN 3		Retry Last Response Code A code indicating the processing outcome of the last retry attempt to process this payment transaction. Refer to the <i>BASE24 Remote Banking Transaction Processing Manual</i> for descriptions of all possible action codes.
AN 3		Remit Type A code that indicates the from of payment that the vendor is to receive. Valid values are as follows: ACH = The vendor is to receive payment in the form of Automated Clearing House (ACH) transactions. CGP = The vendor is to receive a check that includes a group of payments. CIT = The vendor is to receive individual checks for each payment.
AN 75		File Maintenance Tag See the “File Maintenance Tag Data” discussion at the end of this subsection for a description of this field.
AN 30		Vendor Name The name of the vendor to receive this scheduled payment.

Position	Length	Description
AN 20		Vendor Token The value given to a client device to retrieve data that is presented to the customer. The data retrieved is normally the vendor's name. This optional token is intended to allow a remote banking endpoint device to speak a vendor's name, but can be used by other input client devices. The PIT-NAM-ALGO param in the Logical Network Configuration File (LCONF) must be set to the value Y before the BASE24-billpay product uses the value from this field to generate an audio token.
N 3		Currency Code A code indicating the primary type of currency the logical network uses (e.g. 840) for transaction amounts stored in the BASE24 database. For the currency code values, refer to the ISO 4217-1990 standard, <i>Codes for the Representation of Currencies and Funds</i>. The currency code in this field can be used to determine the number of fractional currency unit positions to account for in the Amount field.
AN 6		From Account Type Description The account type name for the ISO account type code provided in the From Account Type field. This is the name configured for the code in the Account Type Table File (ATT). Refer to the Default Account Type Description field provided in the Customer Add and Update Request Data redefine for a list of BASE24 reserved values.
AN 1		Reserved This field is not used.

Scheduled Transfers List Inquiry Response Data

The external data block carries scheduled transfer information for scheduled transfers list inquiry transaction responses.

Data for this response is obtained from the Future Table (FUTR) and Recurring Table (RCUR). Detailed field descriptions for the FUTR and RCUR are provided in the *BASE24 Remote Banking Customer Service Support Manual*.

The format of this data in the external data block is as follows:

Position	Length	Description
1-2	N 2	Count The number of items (i.e., scheduled transfers) returned in this response message. Valid values are 0-10.
3	AN 1	Additional Data Flag A code indicating whether more scheduled transfers exist for this customer. Valid values are as follows: Y = Yes, more data possible. N = No, no more data. Note: The Additional Data Flag field value Y means that more data is possible. If the maximum number of occurrences are returned, the value is set to Y. It is possible that the last occurrence is the last scheduled transfer for the customer and the next request will not contain any scheduled transfer occurrences. The Additional Data Flag field is only set to a value of N if the number of occurrences returned is less than the maximum possible number of occurrences that can be returned.
4	AN 1	Reserved This field is not used.
...		Inquiry Response Structure The following fields occur up to a maximum of 10 times. Each occurrence is 405 characters in length.

Position	Length	Description
N 4	Transfers Made	The total number of transfers already made.
N 4	Total Transfers	The total number of transfers to make.
N 8	Initial Transfer Date	The date (YYYYMMDD) on which the initial transfer is to be made.
AN 2	Recur Period	A code indicating the transfer frequency. Valid values are as follows: WK = Weekly 2W = Every two weeks MN = Monthly Q = Quarterly S = Semi-annually A = Annually
N 5	STI Partition Number	A number indicating the Future Table (FUTR) partition in which this transaction is stored. Each Scheduled Transaction Initiator process handles a separate FUTR partition.
N 8	STI Transfer Date	The date (YYYYMMDD) on which the Scheduled Transaction Initiator process will next attempt to make a transfer for this scheduled transaction. This date may be different from the date displayed in the Transfer Date field if the process was unable to complete the transaction successfully on the specified next transfer date.
N 20	Retry Processing GMT	The date and time (YYYYMMDDhhmmssmmmmmm) the system will next attempt to process the transaction, if the transaction is currently in a retry cycle.

Position	Length	Description
N 6		Customer Reference Number A unique system-generated number used to reference historical data for this scheduled transfer transaction.
N 5		Customer Vendor Number This field should be blank. Customer vendors do not apply for scheduled transfers.
N 9		Vendor Number This field should be blank. Vendors do not apply for scheduled transfers.
AN 28		Customer Account with Vendor This field should be blank. Customer accounts with vendors do not apply for scheduled transfers.
N 20		Billpay GMT The date and time (YYYYMMDDhhmmssmmmmmm) this scheduled transfer transaction was entered into the system (i.e., was received by the BASE24-billpay Billpay Server process).
AN 2		Transaction Type A code indicating the specific type of scheduled transfer transaction. Valid values are as follows: 9D = Schedule transfer – future 9E = Schedule transfer – recurring
AN 4		From FIID The FIID of the customer's financial institution for the <i>from</i> account. The FIID is an identifier that must be unique within the logical network. It is used throughout BASE24 products to identify each BASE24 institution.
AN 19		From Account Number The customer's account number to be debited for this transfer.

Position	Length	Description
AN 2		From Account Type An ISO code identifying the type of account debited for this transfer. Refer to the Account Type field description provided in the Transaction Data, Private (S-125) data element for a list of the BASE24 reserved account type values.
AN 4		To FIID The FIID of the customer's financial institution for the <i>to</i> account. The FIID is an identifier that must be unique within the logical network. It is used throughout BASE24 products to identify each BASE24 institution.
AN 19		To Account Number The customer's account number to be credited for this transfer.
AN 2		To Account Type An ISO code identifying the type of account credited for this transfer. Refer to the Account Type field description provided in the Transaction Data, Private (S-125) data element for a list of the BASE24 reserved account type values.
N 12		Amount The amount of the transfer in whole and fractional currency units (e.g., U.S. dollars and cents).
AN 2		Transaction Source A code indicating the originating source of this scheduled transfer transaction. User-defined values other than the following reserved values are also valid: AD = Audio device (e.g., an interactive voice response system) BL = BASE24-billpay Billing Server process IB = Inbound customer service representative PC = Personal computer SP = Screen phone

Position	Length	Description
N 6		Transaction Source Number A unique number assigned by the transaction source device or software to this scheduled transfer transaction.
N 20		Transaction Source LCT The date and time (YYYYMMDDhhmmssmmmmmm) this scheduled transfer transaction was entered into the system at the transaction source or device. This time is provided in the local civil time of the transaction source.
AN 30		Vendor Specific Data This field should be blank. Vendors do not apply for scheduled transfers.
N 8		Transfer Date The date on which the next transfer is to be made. For future scheduled transfers, this field is blank or contains the same date as the Initial Transfer Date field.
A 1		Schedule Type A flag indicating the type of scheduled transfer. Valid values are as follows: F = Future transfer R = Recurring transfer
AN 28		Billing Text The billing text, if any, associated with this transaction. This is the text normally displayed on the customer's account statement for the transaction.

Position	Length	Description
AN 2		Processing Status A code indicating the current processing status of this scheduled transfer transaction. Valid values are as follows: ␣ = Not started, where ␣ is a blank space PA = Pending authorization RP = Retry processing is pending RW = Waiting to retry All others = Unknown Note: Customers can only update transaction data, delete transactions, or skip a transfer period if the Scheduled Transaction Initiator process is not currently processing the transaction. If the status indicates the transaction is pending authorization or that retry processing is pending, the Scheduled Transaction Initiator process is currently processing the transaction and it cannot be altered. If the status indicates that processing has not started, is waiting to be retried, or is unknown, then the transaction can be altered.
N 20		STI Request GMT The date and time (YYYYMMDDhhmmssmmmmmm) when the Scheduled Transaction Initiator process requested a funds authorization for this transfer transaction. A value is contained in this field only during the Scheduled Transaction Initiator process authorization cycle.
N 4		Retry Count Transient The number of times this transfer transaction has been retried due to a transient error condition. A transient error condition specifies a condition that typically exists for only a brief period of time and that often can be rectified without any operator intervention.

Position	Length	Description
N 4		Retry Count Nontransient The number of times the transaction has been retried due to a nontransient error condition. A nontransient error condition specifies a condition that may exist for an extended period of time (e.g., insufficient funds) or may require operator intervention to rectify the condition.
AN 3		Retry Last Response Code A code indicating the processing outcome of the last retry attempt to process this transfer transaction. Refer to the <i>BASE24 Remote Banking Transaction Processing Manual</i> for descriptions of all possible action codes.
AN 3		Remit Type This field should be blank. Remittances are not generated for scheduled transfer transactions.
AN 75		File Maintenance Tag See the “File Maintenance Tag Data” discussion at the end of this subsection for a description of this field.
N 3		Currency Code A code indicating the primary type of currency the logical network uses (e.g. 840) for transaction amounts stored in the BASE24 database. For the currency code values, refer to the ISO 4217-1990 standard, <i>Codes for the Representation of Currencies and Funds</i> . The currency code in this field can be used to determine the number of fractional currency unit positions to account for in the Amount field.

Position	Length	Description
AN 6		From Account Type Description The account type name for the ISO account type code provided in the From Account Type field. This is the name configured for the code in the Account Type Table File (ATT). Refer to the Default Account Type Description field provided in the Customer Add and Update Request Data redefine for a list of BASE24 reserved values.
AN 6		To Account Type Description The account type name for the ISO account type code provided in the To Account Type field. This is the name configured for the code in the Account Type Table File (ATT). Refer to the Default Account Type Description field provided in the Customer Add and Update Request Data redefine for a list of BASE24 reserved values.
AN 1		Reserved This field is not used.

Scheduled Transactions List Inquiry Response Data

The external data block carries scheduled payment and transfer information for scheduled transactions list inquiry transaction responses.

Data for this response is obtained from the Future Table (FUTR) and Recurring Table (RCUR). Detailed field descriptions for the FUTR and RCUR are provided in the *BASE24 Remote Banking Customer Service Support Manual*.

The format of this data in the external data block is as follows:

Position	Length	Description
1-2	N 2	Count The number of items (i.e., scheduled payments and transfers) returned in this response message. Valid values are 0-8.
3	AN 1	Additional Data Flag A code indicating whether more scheduled payments and transfers exist for this customer. Valid values are as follows: Y = Yes, more data possible. N = No, no more data. Note: The Additional Data Flag field value of Y means that more data is possible. If the maximum number of occurrences are returned, the value is set to Y. It is possible that the last occurrence is the last scheduled payment for the customer and the next request will not contain any scheduled payment occurrences. The Additional Data Flag field is only set to a value of N if the number of occurrences returned is less than the maximum possible number of occurrences that can be returned.
4	AN 1	Reserved This field is not used.
...		Inquiry Response Structure The following fields occur up to a maximum of 8 times. Each occurrence is 455 characters in length.

Position	Length	Description
N 4		Payments Made The total number of payments or transfers already made for this scheduled transaction.
N 4		Total Payments The total number of payments or transfers to make for this scheduled transaction.
N 8		Initial Payment Date The date (YYYYMMDD) on which the initial payment or transfer is to be made.
AN 2		Recur Period A code indicating the payment or transfer frequency. Valid values are as follows: WK = Weekly 2W = Every two weeks MN = Monthly Q = Quarterly S = Semi-annually A = Annually
N 5		STI Partition Number A number indicating the Future Table (FUTR) partition in which this scheduled transaction is stored. Each Scheduled Transaction Initiator process handles a separate FUTR partition.
N 8		STI Payment Date The date (YYYYMMDD) on which the Scheduled Transaction Initiator process will next attempt to make a payment or transfer for this scheduled transaction. This date may be different from the date displayed in the Payment Date field if the process was unable to complete the transaction successfully on the specified next transfer date.

Position	Length	Description
	N 20	Retry Processing GMT If the transaction is currently in a retry cycle, this is the date and time (YYYYMMDDhhmmssmmmmmm) the system will next attempt to process the transaction.
	N 6	Customer Reference Number A unique system-generated number used to reference historical data for this scheduled payment or transfer transaction.
	N 5	Customer Vendor Number For scheduled payment transactions, this field contains the number of the customer vendor to receive this payment. For scheduled transfer transactions, this field should be blank. Customer vendors do not apply for scheduled transfers.
	N 9	Vendor Number For scheduled payment transactions, this field contains the number of the vendor to receive this payment. The vendor number is an internal number generated by the system for this vendor. This number is unique among all registered vendors and is automatically generated when the vendor is added to the system. For scheduled transfer transactions, this field should be blank. Vendors do not apply for scheduled transfers.
	AN 28	Customer's Account with Vendor For scheduled payment transactions, this field contains the customer's account number with the vendor to which this payment is posted or credited. For scheduled transfer transactions, this field should be blank. Customer accounts with vendors do not apply for scheduled transfers.

Position	Length	Description
N 20	Billpay GMT	The date and time (YYYYMMDDhhmmssmmmmmm) this scheduled payment or transfer transaction was entered into the system (i.e., was received by the BASE24-billpay Billpay Server process).
AN 2	Transaction Type	A code indicating the specific type of scheduled payment transaction. Valid values are as follows: 9B = Schedule payment – future 9C = Schedule payment – recurring 9D = Schedule transfer – future 9E = Schedule transfer – recurring
AN 4	From FIID	The FIID of the customer's financial institution for the <i>from</i> account. The FIID is an identifier that must be unique within the logical network. It is used throughout BASE24 products to identify each BASE24 institution.
AN 19	From Account Number	The customer's account number to be debited for this payment or transfer.
AN 2	From Account Type	An ISO code identifying the type of account debited for this payment or transfer. Refer to the Account Type field description provided in the Transaction Data, Private (S-125) data element for a list of the BASE24 reserved account type values.
AN 4	To FIID	The FIID of the customer's financial institution for the <i>to</i> account. The FIID is an identifier that must be unique within the logical network. It is used throughout BASE24 products to identify each BASE24 institution. For scheduled payment transactions, this field should be blank. <i>To</i> accounts do not apply for scheduled payments.

Position	Length	Description
AN 19		To Account Number The customer's account number to be credited for this transfer. For scheduled payment transactions, this field should be blank. <i>To</i> accounts do not apply for scheduled payments.
AN 2		To Account Type An ISO code identifying the type of account credited for this transfer. Refer to the Account Type field description provided in the Transaction Data, Private (S-125) data element for a list of the BASE24 reserved account type values. For scheduled payment transactions, this field should be blank. <i>To</i> accounts do not apply for scheduled payments.
N 12		Amount The amount of the payment or transfer in whole and fractional currency units (e.g., U.S. dollars and cents).
AN 2		Transaction Source A code indicating the originating source of this scheduled payment or transfer transaction. User-defined values other than the following reserved values are also valid: AD = Audio device (e.g., an interactive voice response system) BL = BASE24-billpay Billing Server process IB = Inbound customer service representative PC = Personal computer SP = Screen phone
N 6		Transaction Source Number A unique number assigned by the transaction source device or software to this scheduled payment or transfer transaction.

Position	Length	Description
N 20		Transaction Source LCT The date and time (YYYYMMDDhhmmssmmmmmm) this scheduled payment or transfer transaction was entered into the system at the transaction source or device. This time is provided in the local civil time of the transaction source.
AN 30		Vendor Specific Data Any vendor-specific data that was entered for this scheduled payment transaction. For scheduled transfer transactions, this field should be blank. Vendors do not apply for scheduled transfers.
N 8		Payment Date The date on which the next payment or transfer is to be made. For future scheduled payments or transfers, this field is blank or contains the same date as the Initial Payment Date field.
A 1		Schedule Type A flag indicating the type of scheduled payment or transfer. Valid values are as follows: F = Future payment or transfer R = Recurring payment or transfer
AN 28		Billing Text The billing text, if any, associated with this transaction. This is the text normally displayed on the customer's account statement for the transaction.

Position	Length	Description
AN 2		Processing Status A code indicating the current processing status of this scheduled payment or transfer transaction. Valid values are as follows: ␣ = Not started, where ␣ is a blank space PA = Pending authorization RP = Retry processing is pending RW = Waiting to retry All others = Unknown Note: Customers can only update transaction data, delete transactions, or skip a payment or transfer period if the Scheduled Transaction Initiator process is not currently processing the transaction. If the status indicates the transaction is pending authorization or that retry processing is pending, the Scheduled Transaction Initiator process is currently processing the transaction and it cannot be altered. If the status indicates that processing has not started, is waiting to be retried, or is unknown, then the transaction can be altered.
N 20		STI Request GMT The date and time (YYYYMMDDhhmmssmmmmmm) when the Scheduled Transaction Initiator process requested a funds authorization for this payment or transfer transaction. A value is contained in this field only during the Scheduled Transaction Initiator process authorization cycle.
N 4		Retry Count Transient The number of times this payment or transfer transaction has been retried due to a transient error condition. A transient error condition specifies a condition that typically exists for only a brief period of time and that often can be rectified without any operator intervention.

Position	Length	Description
N 4		Retry Count Nontransient The number of times the transaction has been retried due to a nontransient error condition. A nontransient error condition specifies a condition that may exist for an extended period of time (e.g., insufficient funds) or may require operator intervention to rectify the condition.
AN 3		Retry Last Response Code A code indicating the processing outcome of the last retry attempt to process this payment or transfer transaction. Refer to the <i>BASE24 Remote Banking Transaction Processing Manual</i> for descriptions of all possible action codes.
AN 3		Remit Type For scheduled payment transactions, this field contains a code that indicates the from of payment that the vendor is to receive. Valid values are as follows: ACH = The vendor is to receive payment in the form of Automated Clearing House (ACH) transactions. CGP = The vendor is to receive a check that includes a group of payments. CIT = The vendor is to receive individual checks for each payment. For scheduled transfer transactions, this field should be blank. Remittances are not generated for scheduled transfer transactions.
AN 75		File Maintenance Tag See the “File Maintenance Tag Data” discussion at the end of this subsection for a description of this field.

Position	Length	Description
AN 30		Vendor Name The name of the vendor to receive this scheduled payment. For scheduled transfer transactions, this field should be blank. Vendors do not apply for scheduled transfer transactions.
AN 20		Vendor Token For scheduled payment transactions, this field contains the value given to a client device to retrieve data that is presented to the customer. The data retrieved is normally the vendor's name. This optional token is intended to allow a remote banking endpoint device to speak a vendor's name, but can be used by other input client devices. The PIT-NAM-ALGO param in the Logical Network Configuration File (LCONF) must be set to the value Y before the BASE24-billpay product uses the value from this field to generate an audio token. For scheduled transfer transactions, this field should be blank. Vendor tokens do not apply for scheduled transfer transactions.
N 3		Currency Code A code indicating the primary type of currency the logical network uses (e.g. 840) for transaction amounts stored in the BASE24 database. For the currency code values, refer to the ISO 4217-1990 standard, <i>Codes for the Representation of Currencies and Funds</i> . The currency code in this field can be used to determine the number of fractional currency unit positions to account for in the Amount field.

Position	Length	Description
	AN 6	From Account Type Description
		The account type name for the ISO account type code provided in the From Account Type field. This is the name configured for the code in the Account Type Table File (ATT).
		Refer to the Default Account Type Description field provided in the Customer Add and Update Request Data redefine for a list of BASE24 reserved values.
	AN 6	To Account Type Description
		The account type name for the ISO account type code provided in the To Account Type field. This is the name configured for the code in the Account Type Table File (ATT).
		Refer to the Default Account Type Description field provided in the Customer Add and Update Request Data redefine for a list of BASE24 reserved values.
	AN 1	Reserved
		This field is not used.

File Maintenance Tag Data

The File Maintenance Tag Data carries file maintenance information. This format is used throughout the response data formats for BASE24-billpay transactions in the external data block. The format of this data in the external data block is as follows:

Position	Length	Description
	N 5	Group Number
		The group number of the operator who last changed the record or row in a BASE24 file or table. This value is taken from the information provided when the operator last logged on to BASE24.

Position	Length	Description
	N 5	User Number The user number of the operator who last changed the record or row in a BASE24 file or table. This value is taken from the information provided when the operator last logged on to BASE24.
	AN 24	Terminal Name The name of the CRT terminal from which an operator last changed the record or row in a BASE24 file or table.
	N 20	AFT Timestamp The date and time (YYYYMMDDhhmmssmmmmmm) the record or row was last changed by an operator for file or table maintenance.
	N 20	OLTP Timestamp The date and time (YYYYMMDDhhmmssmmmmmm) the record or row was last changed due to online transaction processing.
	AN 1	Reserved This field is not used.

Appendix A

ISO Conversion Tables

This appendix contains conversion tables for BASE24 Remote Banking processing codes that must be converted to and from ISO standard codes by the Integrated Authorization Server process.

Processing Codes

Processing codes are carried in the Processing Code (P-3) data element of the BASE24 Remote Banking Standard Interface external message.

In BASE24-telebanking, processing codes are carried internally in three two-byte fields of the Internal Transaction Data (ITD):

- PROC-CDE.TXN-CDE (transaction code)
- PROC-CDE.ACCT1-TYP (account 1 type)
- PROC-CDE.ACCT2-TYP (account 2 type)

Data element P-3 is divided into three two-byte subfields with approximately the same meanings as those carried in the BASE24-telebanking ITD.

The following tables are used to show how BASE24-telebanking processing codes map to and from the ISO standard processing codes.

Mapping Remote Banking Processing Codes to ISO

Since ISO standards were used to develop the BASE24-telebanking and BASE24-billpay internal processing codes, the same codes are used when mapping internal processing codes used in the Telebanking Standard Internal Message (BSTM) to the ISO processing codes carried in the ISO external message. No translation takes place. The same codes are used both internally and externally. The BSTM contains the ITD as well as other information.

Mapping the Transaction Code

The following table shows the possible values for the first subfield of the Processing Code data element (positions 1–2) and what these values mean when mapped to the ISO message. These codes are mapped directly from the PROC-CDE.TXN-CDE field in the ITD.

Processing Codes	BASE24 Remote Banking	ISO
30	Available funds inquiry	Available funds inquiry
3A	Check clearance inquiry	Inquiry services

Processing Codes	BASE24 Remote Banking	ISO
3B	Last count of debit/credit transactions inquiry	Inquiry services
3C	Last count of account by source transactions inquiry	Inquiry services
3D	Last count of check transactions inquiry	Inquiry services
3E	Last count of debit transactions inquiry	Inquiry services
3F	Last count of credit transactions inquiry	Inquiry services
3G	Last count of account transfer transactions inquiry	Inquiry services
3H	Customer vendor list inquiry	Inquiry services
3J	Scheduled payments list inquiry	Inquiry services
3K	Scheduled transfers list inquiry	Inquiry services
3L	History inquiry – payments and transfers	Inquiry services
3M	Scheduled transactions list inquiry	Inquiry services
3N	Customer account list inquiry	Inquiry services
3P	Multiple account balance inquiry	Inquiry services
3Q	Statement closing download	Inquiry services
3R	Statement download	Inquiry services
40	Transfer – immediate	Cardholder accounts transfer
4A	Transfer – future	Transfer services

Processing Codes	BASE24 Remote Banking	ISO
4B	Transfer – recurring	Transfer services
50	Payment – immediate	Payment
5A	Payment – future	Payment services
5B	Payment – recurring	Payment services
90	PIN change	Reserved for private use
91	PIN verify	Reserved for private use
9A	Schedule payment – immediate	Reserved for private use
9B	Schedule payment – future	Reserved for private use
9C	Schedule payment – recurring	Reserved for private use
9D	Schedule transfer – future	Reserved for private use
9E	Schedule transfer – recurring	Reserved for private use
9F	Scheduled payment delete	Reserved for private use
9G	Scheduled transfer delete	Reserved for private use
9H	Scheduled payment update	Reserved for private use
9J	Scheduled transfer update	Reserved for private use
9K	Customer add	Reserved for private use
9L	Customer inquiry	Reserved for private use
9M	Customer update	Reserved for private use
9N	Customer vendor add	Reserved for private use
9P	Customer vendor deactivate	Reserved for private use
9Q	Customer vendor update	Reserved for private use
9R	Master vendor list inquiry	Reserved for private use

Processing Codes	BASE24 Remote Banking	ISO
9T	Customer vendor request mailing	Reserved for private use
9V	Loan application	Reserved for private use
9W	Miscellaneous request	Reserved for private use
9X	History inquiry – all transactions	Reserved for private use
A0	Check stop payment add	Reserved for private use
B1	Bank list inquiry	Reserved for private use
B2	Bank branch list inquiry	Reserved for private use
B4	Miscellaneous requests inquiry	Reserved for private use
B5	Customer account add	Reserved for private use
B6	Customer account activate	Reserved for private use
B7	Customer account deactivate	Reserved for private use
B8	Customer account delete	Reserved for private use
B9	Customer account update	Reserved for private use
BB	Customer activate	Reserved for private use
BC	Customer deactivate	Reserved for private use
BD	Customer vendor activate	Reserved for private use
BE	Customer vendor mask verify inquiry	Reserved for private use
BF	Customer verify	Reserved for private use
BG	Customer select	Reserved for private use

Mapping the Account Types

The following table shows the possible values of the PROC-CDE.ACCT1-TYP field and the PROC-CDE.ACCT2-TYP field in the ITD and what these values mean when mapped to the ISO message. These codes are mapped directly to the second and third subfields of data element P-3, respectively

Processing Codes	BASE24 Remote Banking	ISO
00	No account type	No account type
10	Savings account	Savings account
20	Checking account	Checking account
30	Credit account	Credit account
38	Line of credit	Credit account—reserved for private use
58	Certificate	Investment account—reserved for private use
59	Retirement account	Investment account—reserved for private use
90	Interest-bearing checking account	Reserved for private use
9A	Commercial loan	Reserved for private use
9B	Installment loan	Reserved for private use
9C	Mortgage loan	Reserved for private use

Mapping ISO Processing Codes to BASE24 Remote Banking

The Processing Code (P-3) data element in the BASE24 external message is carried as three subfields. The following topics describe how the ISO processing code values in the BASE24 external message are mapped to BASE24-telebanking and BASE24-billpay processing codes for carrying in the ITD. No translation takes place. The same codes are used both internally and externally.

Mapping the Transaction Code

The following table shows the possible values for the first subfield of data element P-3 in the BASE24 external message. The same value is placed in the PROC-CDE.TXN-CDE field in the ITD.

Processing Codes	ISO	BASE24 Remote Banking
30	Available funds inquiry	Available funds inquiry
3A	Inquiry services	Check clearance inquiry
3B	Inquiry services	Last count of debit/credit transactions inquiry
3C	Inquiry services	Last count of account by source transactions inquiry
3D	Inquiry services	Last count of check transactions inquiry
3E	Inquiry services	Last count of debit transactions inquiry
3F	Inquiry services	Last count of credit transactions inquiry
3G	Inquiry services	Last count of account transfer transactions inquiry
3H	Inquiry services	Customer vendor list inquiry
3J	Inquiry services	Scheduled payments list inquiry
3K	Inquiry services	Scheduled transfers list inquiry
3L	Inquiry services	History inquiry – payments and transfers
3N	Inquiry services	Customer account list inquiry
3P	Inquiry services	Multiple account balance inquiry
3Q	Inquiry services	Statement closing download

Processing Codes	ISO	BASE24 Remote Banking
3R	Inquiry services	Statement download
40	Cardholder accounts transfer	Transfer – immediate
4A	Transfer services	Transfer – future
4B	Transfer services	Transfer – recurring
50	Payment	Payment – immediate
5A	Payment services	Payment – future
5B	Payment services	Payment – recurring
90	Reserved for private use	PIN change
91	Reserved for private use	PIN verify
9A	Reserved for private use	Schedule payment – immediate
9B	Reserved for private use	Schedule payment – future
9C	Reserved for private use	Schedule payment – recurring
9D	Reserved for private use	Schedule transfer – future
9E	Reserved for private use	Schedule transfer – recurring
9F	Reserved for private use	Scheduled payment delete
9G	Reserved for private use	Scheduled transfer delete
9H	Reserved for private use	Scheduled payment update
9J	Reserved for private use	Scheduled transfer update
9K	Reserved for private use	Customer add
9L	Reserved for private use	Customer inquiry
9M	Reserved for private use	Customer update
9N	Reserved for private use	Customer vendor add

Processing Codes	ISO	BASE24 Remote Banking
9P	Reserved for private use	Customer vendor deactivate
9Q	Reserved for private use	Customer vendor update
9R	Reserved for private use	Master vendor list inquiry
9T	Reserved for private use	Customer vendor request mailing
9V	Reserved for private use	Loan application
9W	Reserved for private use	Miscellaneous request
9X	Reserved for private use	History inquiry – all transactions
A0	Reserved for private use	Check stop payment add
B1	Reserved for private use	Bank list inquiry
B2	Reserved for private use	Bank branch list inquiry
B4	Reserved for private use	Miscellaneous requests inquiry
B5	Reserved for private use	Customer account add
B6	Reserved for private use	Customer account activate
B7	Reserved for private use	Customer account deactivate
B8	Reserved for private use	Customer account delete
B9	Reserved for private use	Customer account update
BB	Reserved for private use	Customer activate
BC	Reserved for private use	Customer deactivate
BD	Reserved for private use	Customer vendor activate
BE	Reserved for private use	Customer vendor mask verify inquiry
BF	Reserved for private use	Customer verify

Processing Codes	ISO	BASE24 Remote Banking
BG	Reserved for private use	Customer select

The Integrated Authorization Server process also supports two-character codes beginning with the letters P, Q, R, U, X, Y, and Z. Processing codes beginning with the letter U identify user-defined transactions and codes beginning with the letter X, Y, or Z are reserved for distributor-specific code modifications. Processing codes beginning with letters P, Q, and R are reserved for customer-specific code modifications.

Any other codes are not supported by the Integrated Authorization Server process, and cause message rejection.

Mapping the Account Types

The following table shows the possible values for the second and third subfields of data element P-3 in the BASE24 external message. The same value is placed in the PROC-CDE.ACCT1-TYP field and PROC-CDE.ACCT2-TYP field in the ITD, respectively.

Processing Codes	ISO	BASE24 Remote Banking
00	No account specified	No account type
10	Savings account	Savings account
20	Checking account	Checking account
30	Credit account	Credit account
38	Credit account—reserved for private use	Line of credit
58	Investment account—reserved for private use	Certificate
59	Investment account—reserved for private use	Retirement account

Processing Codes	ISO	BASE24 Remote Banking
90	Reserved for private use	Interest-bearing checking account
9A	Reserved for private use	Commercial loan
9B	Reserved for private use	Installment loan
9C	Reserved for private use	Mortgage loan

ACI Worldwide Inc.

Appendix B

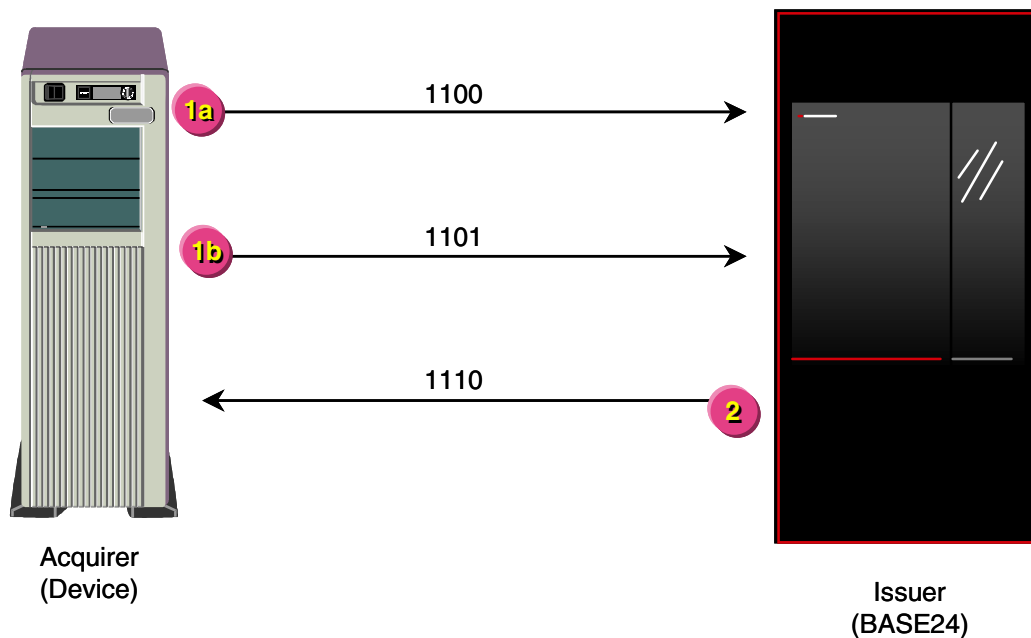
Transaction Flows

This appendix depicts the transaction processing performed between an RBSI-compliant device and BASE24 for various transaction flows.

In the transaction flow diagrams presented in this appendix, optional flow processing is depicted with broken lines.

Authorization Request/Response Flow

The following diagram details the transaction flow for an authorization request/repeat request and authorization request response. For this flow, it is assumed that the customer has entered or already selected an account number (i.e., multiple accounts have already been presented to the customer) and the customer is not performing a continuation of a transaction inquiry.



Processing Steps

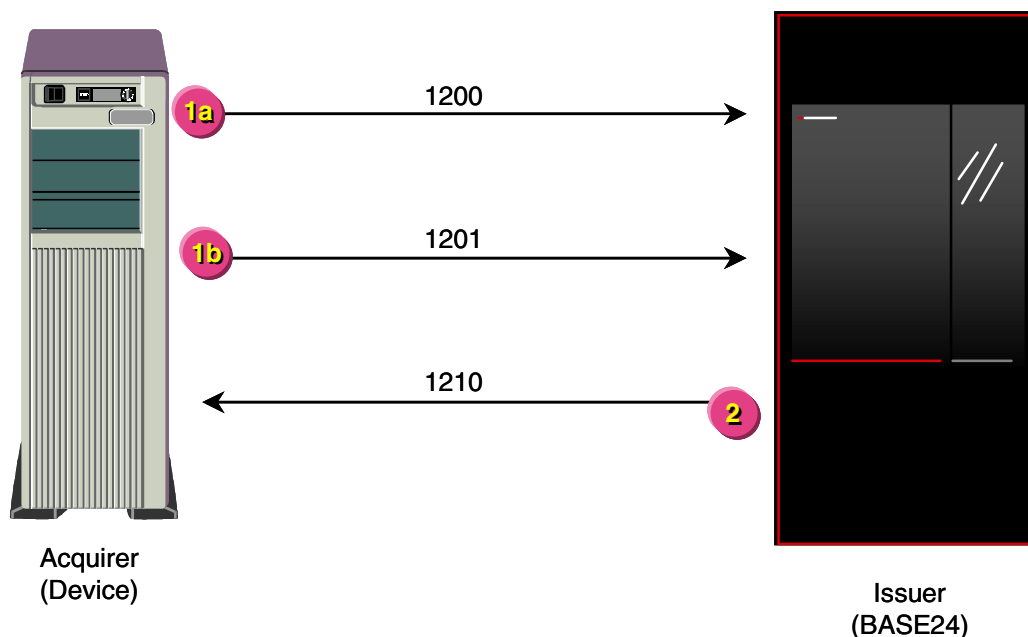
1. The device formats an authorization request or authorization repeat request and sends it to BASE24.
 - a. The device formats the authorization request (1100) for the appropriate transaction type that the customer has selected. The PIN can be verified locally or by BASE24. If the PIN is verified locally, the last position of the Point of Service Data Code (P-22) data element is set to a value of S. The device then sends the transaction to BASE24 and waits for a response.
 - b. If BASE24 does not respond to the device within a time limit defined by the device, the device formats an authorization repeat request (1101) for the appropriate transaction type. The device then resends the transaction to BASE24 and waits for a response.

2. BASE24 authorizes the transaction appropriately—either by authorizing the transaction itself or forwarding it on to the host if necessary.

BASE24 formats the authorization request response (1110) for the appropriate transaction type. Note that every time a customer has requested transaction inquiry information and more information is available, data element 48 is used to contain information for continuation of the inquiry on the next request. BASE24 sends the response to the device. The device responds appropriately to the customer.

Financial Request/Response Flow

The following diagram details the transaction flow for a financial request/repeat request and financial request response. For this flow, it is assumed that the customer has entered or already selected an account number (i.e., multiple accounts have already been presented to the customer).



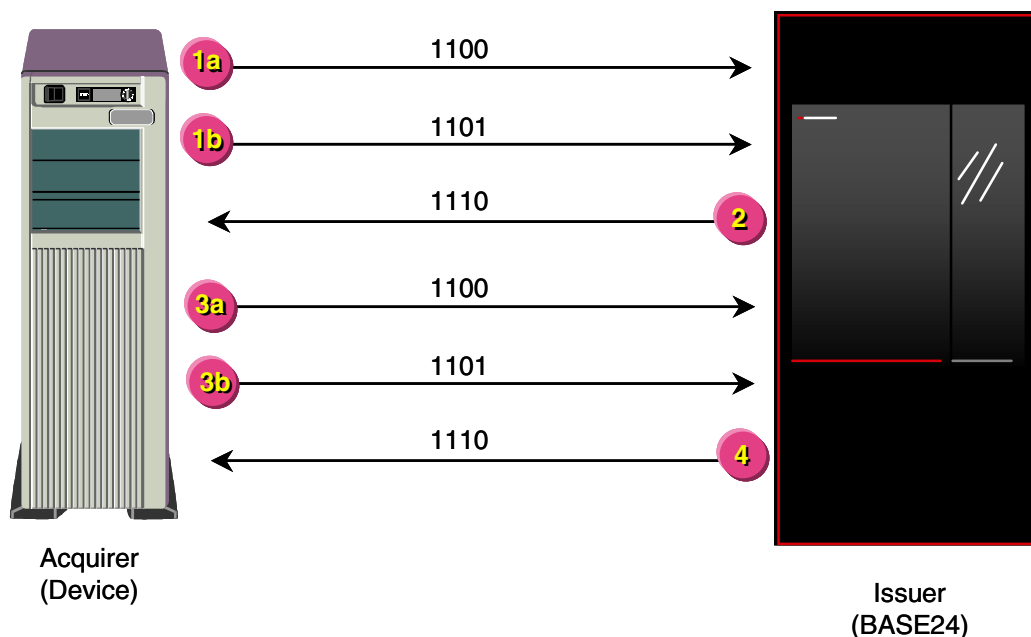
Processing Steps

1. The device formats a financial request or financial repeat request and sends it to BASE24.
 - a. The device formats the financial request (1200) for the appropriate transaction type (e.g., transfer) that the customer has selected. The PIN can be verified locally or by BASE24. If the PIN is verified locally, the last position of the Point of Service Data Code (P-22) data element is set to a value of S. The device then sends the transaction to BASE24 and waits for a response.
 - b. If BASE24 does not respond to the device within a time limit defined by the device, the device formats a financial repeat request (1201) for the appropriate transaction type. The device then resends the transaction to BASE24 and waits for a response.

2. BASE24 authorizes the transaction appropriately—either by authorizing the transaction itself or forwarding it on to the host if necessary.
 - BASE24 formats the financial request response (1210) for the appropriate transaction type. BASE24 sends the response to the device.
 - The device responds appropriately to the customer.

Transaction Inquiry Continuation Flow

The following diagram details the transaction flow for a transaction inquiry continuation. For this flow, it is assumed that the customer has entered or already selected an account number (i.e., multiple accounts have already been presented to the customer).



Processing Steps

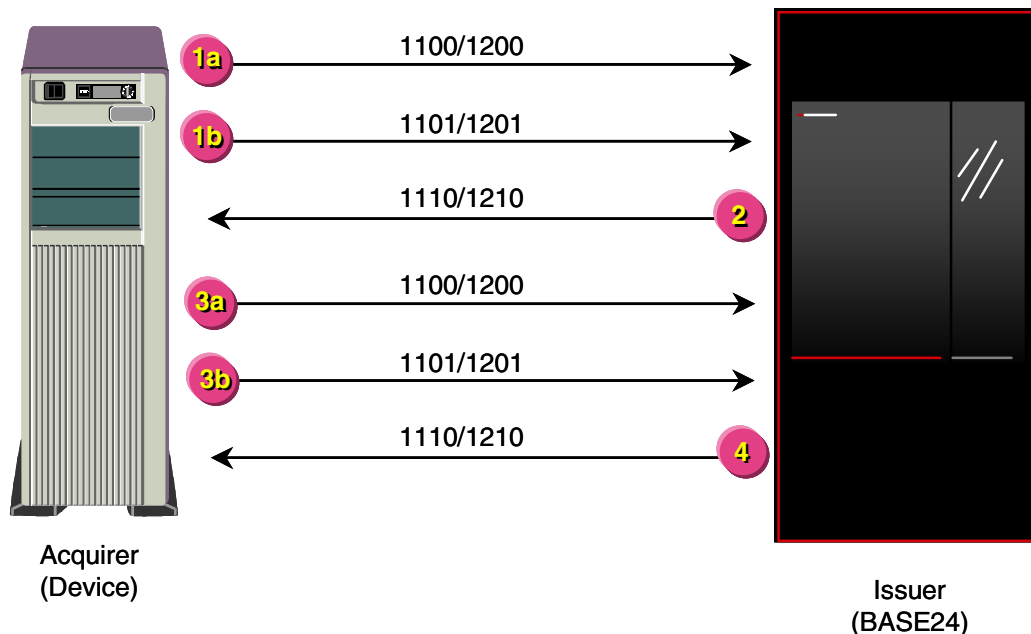
1. The device formats an authorization request or authorization repeat request and sends it to BASE24.
 - a. The device formats the authorization request (1100) for the transaction inquiry that the customer has selected. The PIN can be verified locally or by BASE24. If the PIN is verified locally, the last position of the Point of Service Data Code (P-22) data element is set to a value of S. The device then sends the transaction to BASE24 and waits for a response.
 - b. If BASE24 does not respond to the device within a time limit defined by the device, the device formats an authorization repeat request (1101) for the appropriate transaction type. The device then resends the transaction to BASE24 and waits for a response.

2. BASE24 authorizes the transaction appropriately—either by authorizing the transaction itself or forwarding it on to the host if necessary. BASE24 formats the authorization request response (1110) for the appropriate transaction type. Note that every time that a customer has requested transaction inquiry information and more information is available, data element 48 is used to contain information for continuation of the inquiry on the next request. BASE24 sends the response to the device.
3. The device responds with the transaction inquiry information to the customer contained in data element 125 or in the external data block. The customer is allowed to continue to retrieve information because data element 125 or the external data block indicates a non-zero count and data element 48 contains the last transaction key information used for positioning on the continuation request.
 - a. The customer elects to continue retrieving transaction inquiry information. The device formats the authorization request (1100) for the transaction inquiry that the customer has selected and includes data element 48 exactly as it was received in step 2, with the following exception. The device may update the Search Direction subfield of data element 48 for certain inquiry transactions to indicate whether the search is to continue in ascending or descending order. The device then sends the transaction to BASE24 and waits for a response.
 - b. If BASE24 does not respond to the device within a time limit defined by the device, the device formats an authorization repeat request (1101) for the appropriate transaction type. The device then resends the transaction to BASE24 and waits for a response.
4. BASE24 authorizes the transaction appropriately—either by authorizing the transaction itself or forwarding it on to the host if necessary. BASE24 formats the authorization request response (1110) for the appropriate transaction type. BASE24 sends the response to the device.

Steps 3 and 4 can be repeated until data element 125 is no longer present in the request message, which indicates that there is no more transaction information for the particular type of transaction requested.

Multiple Account Selection Flow

The following diagram details the transaction flow for multiple account selection.



Processing Steps

1. The device formats a request or repeat request and sends it to BASE24.
 - a. The device formats the authorization or financial request (1100 or 1200), depending upon the type of transaction that the customer has selected. The PIN can be verified locally or by BASE24. If the PIN is verified locally, the last position of the Point of Service Data Code (P-22) data element is set to a value of S. The device then sends the transaction to BASE24 and waits for a response.
 - b. If BASE24 does not respond to the device within a time limit defined by the device, then the device formats an authorization or financial repeat request (1101 or 1201) for the appropriate transaction type. The device then resends the transaction to BASE24 and waits for a response.
2. BASE24 determines that the customer is allowed to select from the valid accounts that he or she has. BASE24 formats the authorization or financial request response (1110 or 1210) for the appropriate transaction type with data element 126, which contains the list of accounts and data element 24, which contains a code that indicates whether more accounts exist. BASE24 sends the response to the device.

3. The device presents the multiple account information to the customer. The customer can then select from the list of accounts. If the function code in data element 24 indicates that more accounts exist for either account 1 (function code 970) or account 2 (function code 971), the customer can continue requesting more accounts. In this case, steps 3 and 4 are repeated until the customer selects an account or quits the transaction. If there are no more accounts for a customer to select, the function code in data element 24 is set to 972. Note that for a transaction requiring two accounts, such as a transfer, the customer can choose both accounts. The *from* accounts are presented first, however, if there is room in the message, some or all of the *to* accounts are presented in the same message, with an indicator of whether or not more *to* accounts are available. Up to 15 accounts can be carried in each message.

If the customer chooses the accounts, the device formats the authorization or financial request/repeat request (1100/1101 or 1200/1201) with the account number in the Account Identification 1 (S-102) or Account Identification 2 (S-103) data element. The device must also return the corresponding account type from the multiple account selection data in the Multiple Account Select Data (S-126) element. This account type value can differ from the account type value in the processing code and BASE24 requires both account types for transaction processing. The device then sends the transaction to BASE24 and waits for a response.

Note: When two accounts are required (such as for a transfer transaction) and a customer selects the first account (account 1), the device must move that to the Account Identification 1 data element and continue processing for the second account (account 2).

The customer elects to continue retrieving his or her list of accounts.

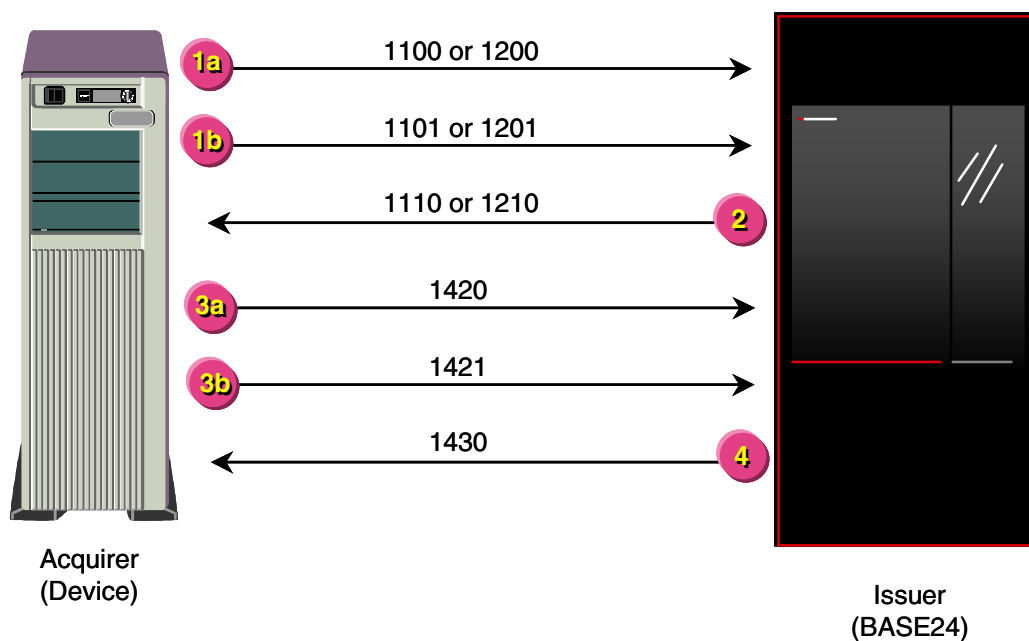
- a. The device formats an authorization or financial request (1100/1200). The device then sends the transaction to BASE24 and waits for a response.
- b. If BASE24 does not respond to the device within a time limit defined by the device, the device formats an authorization or financial repeat request (1101/1201) for the appropriate transaction type. The device then resends the transaction to BASE24 and waits for a response.

4. BASE24 authorizes the transaction appropriately—either by authorizing the transaction itself or forwarding it on to the host if necessary. BASE24 formats the authorization or financial request response (1110 or 1210) for the appropriate transaction type with data element 126, which contains the list of accounts and their corresponding account types, and data element 24, which contains a code indicating whether more accounts exist. BASE24 sends the response to the device.

Steps 3 and 4 can be repeated until the customer selects his or her accounts or cancels the transaction.

Reversal Flow

The following diagram details the transaction flow for a reversal of an authorization or financial transaction.



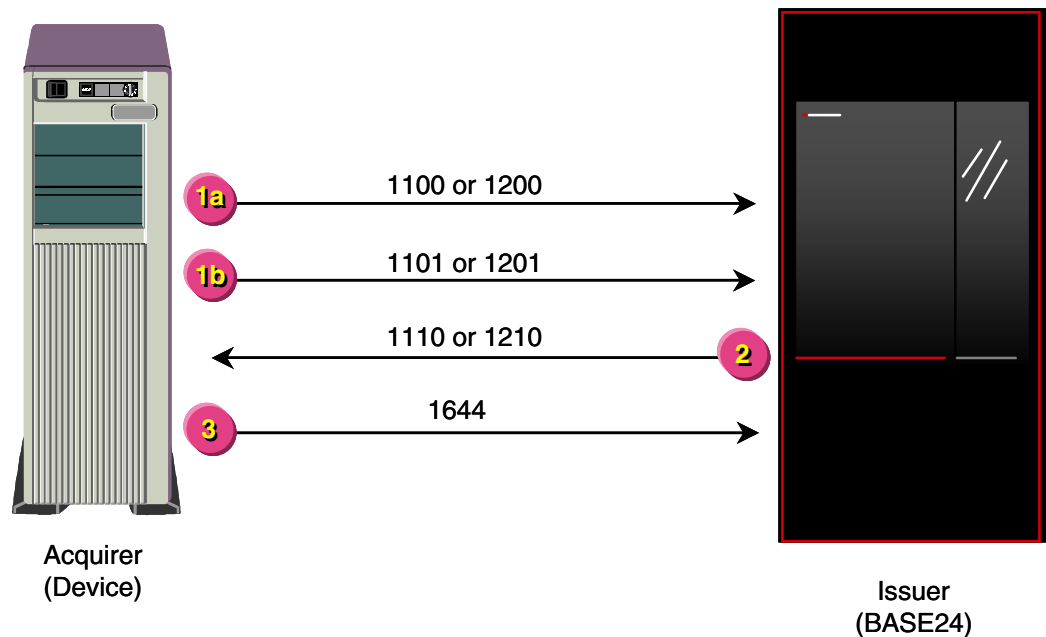
Processing Steps

1. The device formats a request or repeat request and sends it to BASE24.
 - a. The device formats an authorization or financial request (1100 or 1200) for the appropriate transaction type (e.g., transfer) that the customer has selected. The PIN can be verified locally or by BASE24. If the PIN is verified locally, the last position of the Point of Service Data Code (P-22) data element is set to a value of S. The device then sends the transaction to BASE24 and waits for a response.
 - b. If BASE24 does not respond to the device within a time limit defined by the device, the device formats an authorization or financial repeat request (1101 or 1201) for the appropriate transaction type. The device then resends the transaction to BASE24 and waits for a response.
2. BASE24 receives the transaction and authorizes it appropriately—either authorizing the transaction itself or forwarding it on to the host if necessary. BASE24 formats the authorization or financial request response (1110 or 1210) for the appropriate transaction type. BASE24 sends the response to the device.

3. The device determines that the transaction has not completed correctly.
 - a. The device formats a reversal advice request (1420) to send to BASE24. The original data is sent in data element 56 of the request.
 - b. If BASE24 does not respond to the device within a time limit defined to the device, the device formats a reversal advice repeat request (1421) for the appropriate transaction type. The device then resends the transaction to BASE24 and waits for a response.
4. BASE24 receives the transaction, performs the reversal, formats the reversal advice response (1430), and sends the response to the device.

Message Rejected by Device Flow

The following diagram details the transaction flow for a message rejected by an RBSI-compliant device.



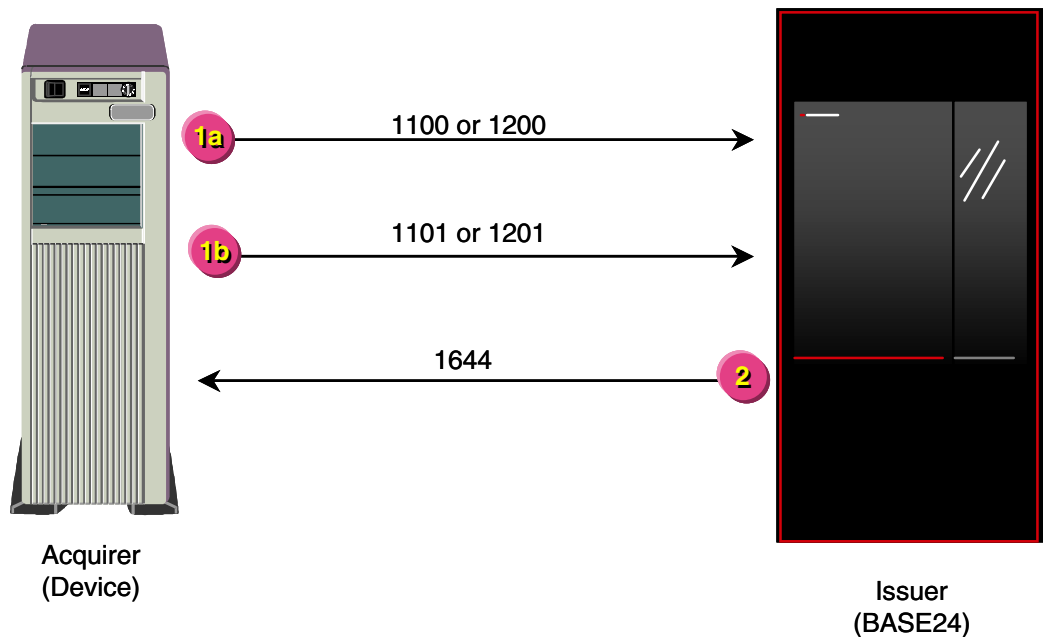
Processing Steps

1. The device formats a request or repeat request and sends it to BASE24.
 - a. The device formats an authorization or financial request (1100 or 1200) for the appropriate transaction type that the customer has selected. The PIN can be verified locally or by BASE24. If the PIN is verified locally, the last position of the Point of Service Data Code (P-22) data element is set to a value of S. The device then sends the transaction to BASE24 and waits for a response.
 - b. If BASE24 does not respond to the device within a time limit defined by the device, the device formats an authorization or financial repeat request (1101/1201) for the appropriate transaction type. The device then resends the transaction to BASE24 and waits for a response.
2. BASE24 authorizes the transaction appropriately—either by authorizing the transaction itself or forwarding it on to the host if necessary. BASE24 formats the authorization or financial request response (1110 or 1210) for the appropriate transaction type and sends the response to the device.

3. The device is unable to read the response.
 - The device formats an administrative notification (1644) to send to BASE24 with the number of the data element in error and the original message that is being rejected contained in data element 72.
 - BASE24 generates a reversal, if necessary.

Message Rejected by BASE24 Flow

The following diagram details the transaction flow for a message rejected by BASE24.



Processing Steps

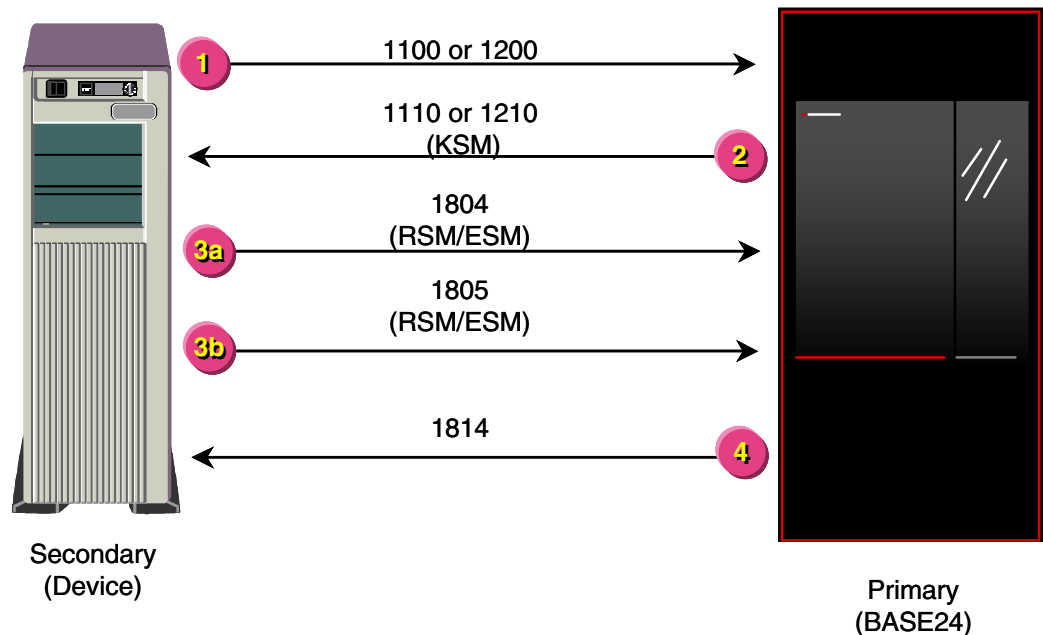
1. The device formats a request or repeat request and sends it to BASE24.
 - a. The device formats an authorization or financial request (1100 or 1200) for the appropriate transaction type that the customer has selected. The PIN can be verified locally or by BASE24. If the PIN is verified locally, the last position of the Point of Service Data Code (P-22) data element is set to a value of S. The device then sends the transaction to BASE24 and waits for a response.
 - b. If BASE24 does not respond to the device within a time limit defined by the device, then the device formats an authorization/financial repeat request (1101/1201) for the appropriate transaction type. The device then resends the transaction to BASE24 and waits for a response.
2. BASE24 rejects the transaction because of an invalid message type, unsupported message type, invalid value in a data element, or a required data element that is not present in the message. BASE24 formats and sends an

administrative notification (1644) to the device with the number of the data element in error and the original message that is being rejected contained in data element 72.

Note: This same transaction flow applies to network management request/repeat (1804/1805) as well as reversal advice/repeat (1420/1421) messages.

New Key Message Processing

A new key can be generated due to threshold processing, an expired timer, or a command received from the BASE24 network. The following illustration depicts the processing flow for a new key request.



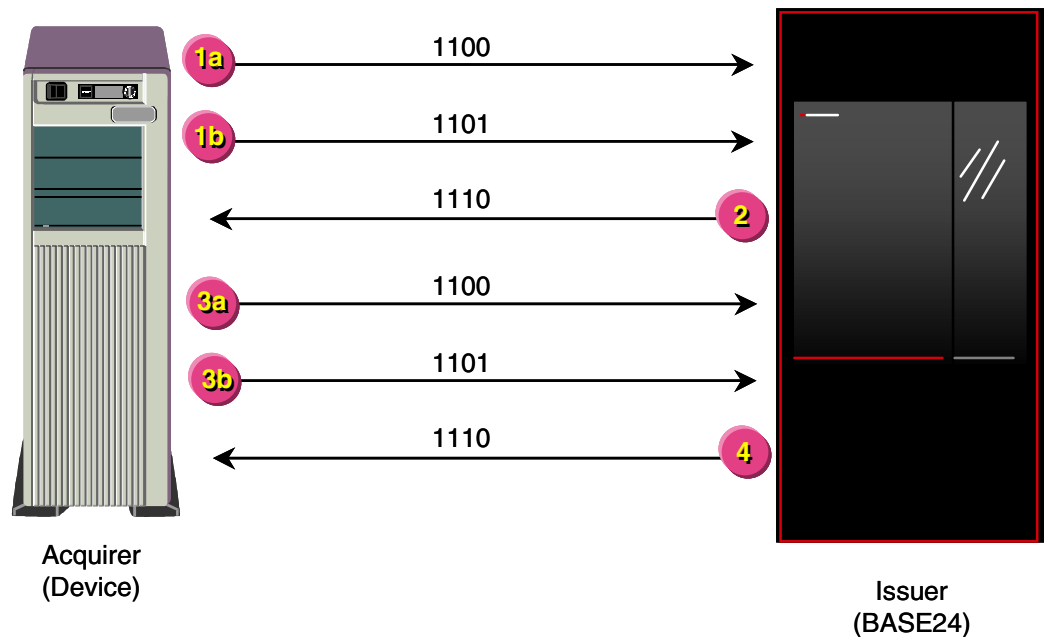
Processing Steps

1. The device formats an authorization/financial request or authorization/financial repeat request with a PIN and sends it to BASE24.
2. BASE24 determines a new key must be generated. This is due to either a command sent from the network or an exceeded threshold. In the response with a message type identifier of 1110/1210, BASE24 sends a key change indicator and the CSM of Key Service Message in data element 96. BASE24 also overwrites its oldest key with the key it just generated.
3. The device validates the new key and performs the following.
 - a. The device sends a network management request (1804) that contains an Error Service Message (ESM) or a Response Service Message (RSM) in data element 96. The function code in data element 24 is set to 811. Data element 56 must contain the system trace audit number from the authorization/financial request response (1110/1210) that contained the new key. The device begins using the new key when it sends the network management request (1804/1805) with an RSM.

- b. If BASE24 does not respond to the device within a time limit defined by the device, the device formats a network management request repeat (1805). The device then resends the transaction to BASE24 and waits for a response.
4. BASE24 processes the network management request or network management repeat request from the device. If the network management request indicates acceptance of the new key, BASE24 marks this latest key as current and the previous current key as old. If the network management request indicates that the device rejected the new key, BASE24 continues to treat this latest key as the old key. The last key accepted by the device remains the current key. BASE24 sends a network management request response (1814) to the device. BASE24 echoes back the 811 function code in data element 24.

Customer Verification and Selection Flow

The following diagram details the transaction flow for customer verification and selection.



Processing Steps

1. The device formats a request or repeat request and sends it to BASE24.
 - a. The device formats an authorization request (1100) for the customer verify transaction type. The device then sends the transaction to BASE24 and waits for a response.
 - b. If BASE24 does not respond to the device within a time limit defined by the device, then the device formats an authorization repeat request (1101) for the customer verify transaction type. The device then resends the transaction to BASE24 and waits for a response.
2. BASE24 authorizes the request and formats an 1110 authorization request response for the customer verify transaction and sends it to the acquiring device.
 - If a single customer could be determined from the information provided in the request and a customer name was not specified in the request, BASE24 returns verification information for the customer in data element 125 of the customer verify response.

- If a single customer could not be determined from the information provided in the request or if a customer name was specified in the request, BASE24 returns customer select data in data element 125 for one or more (up to six) customers that matched, or was similar to, the customer identification information provided in the customer select request.
3. If a single customer could not be determined from the customer information provided in the customer verify request, the RBSI-compliant device sends an 1100 authorization request for a customer select transaction to BASE24. The request contains customer select respond select data in data element 125.

If a single customer was determined, the customer is verified by the acquirer. For the sake of this transaction flow, assume that a single customer was not determined.
 - a. The device formats an authorization request (1100) for the customer select transaction type. The device then sends the transaction to BASE24 and waits for a response.
 - b. If BASE24 does not respond to the device within a time limit defined by the device, then the device formats an authorization repeat request (1101) for the customer select transaction type. The device then resends the transaction to BASE24 and waits for a response.
 4. BASE24 authorizes the request and formats an 1110 authorization request response for the customer select transaction and sends it to the acquiring device.
 - If a single customer could be determined from the information provided in the request, BASE24 returns verification information for the customer in data element 125 of the customer verify response.
 - If a single customer could not be determined from the information provided in the request, BASE24 returns customer select request data in data element 125 for one or more (up to six) customers that matched, or were similar to, the customer identification information provided in the customer select request.

Steps 3 and 4 are repeated until a single customer is verified.

Appendix C

Transaction Capabilities and Restrictions

This appendix details the various capabilities, requirements, and restrictions for transactions supported by the Remote Banking Standard Interface.

FIID Requirements

The device is responsible for providing a valid FIID, customer ID, or bank branch for any transaction that requires an FIID or bank branch. For applications that allow customers to be added to the system, the device must also know which FIIDs support preloaded customer data, and which do not. Because of these requirements, you may want to maintain a local financial institution database to maintain this information.

For devices that support the customer add transaction, knowledge of which FIIDs support preloaded customer data is required to properly script for customer add, customer activate, customer inquiry, and customer update transactions. The customer account add transaction is allowed whether or not the FIID supports preloaded customer data.

All BASE24 Remote Banking transactions require a valid FIID or customer ID, except for the bank list inquiry transaction. If the FIID or customer ID is not provided for this transaction, BASE24 uses the value of the BP-ACQ-FIID param in the LCONF to obtain the FIID.

Bank and Bank Branch List Inquiry Transactions

The bank list inquiry and bank branch list inquiry transactions can be performed in conjunction with other transactions as required, or they can be used to refresh a local database of financial institution information. When these transactions are used to refresh a local database, the customer ID should be set to all zeros in the transaction. These transactions can be sent periodically on an automated basis (e.g., once a day) or entered manually as needed to ensure that the local financial institution database is current with the BASE24 database.

Both the bank list inquiry and bank branch list inquiry transactions can be sent to BASE24 Remote Banking with a customer ID of all zeros.

Reactivation Flag

On customer activate transactions, the device is responsible for sending a flag indicating whether the device allows a customer ID to be reactivated in data element 48. Both the device and the customer's financial institution must allow a customer ID to be reactivated before BASE24 will allow it. A reactivation flag is maintained in the Bank Table (BANK) for each financial institution.

Customer ID and Customer Account Requirements

When a customer ID or account number is not required for a transaction, BASE24 Remote Banking expects to receive a value of all zeros for the customer ID from the device. When BASE24 Remote Banking receives a transaction with a customer ID of all zeros, it does not perform PIN processing or check the Customer Allowed Transaction Table (CATT) to determine whether the customer is allowed to perform the transaction. The device should prevent a PIN from being entered when a customer ID or account number is not part of the transaction.

Transactions for Which Customer Identification is not Required

The following BASE24 Remote Banking transactions do not require a customer ID or account number (i.e., the customer ID can use a value of all zeros):

- Bank list inquiry
- Bank branch list inquiry
- Customer inquiry (to retrieve default CSTT and PIT data)
- Loan application

Transactions for Which the Customer Verify Status is not Checked

The following transactions can be performed for a customer, regardless of the current status of the customer ID (i.e., closed, issued but inactive, or verified) or its begin and end dates:

- Customer account add
- Customer account activate
- Customer account delete
- Customer activate
- Customer add
- Customer deactivate
- Customer inquiry

- Customer update
- Customer vendor activate
- Customer vendor add
- Customer vendor deactivate
- Customer vendor list inquiry
- Customer vendor mask verify inquiry
- Customer vendor update

Transactions for Which the Customer Account Status is not Checked

The following transactions can be performed for a customer, regardless of the current status of the customer account (i.e., inactive, open, restricted to deposits, open primary account, primary account restricted to deposits, closed) and its begin and end dates:

- Customer account deactivate
- Customer account delete
- Customer account update

The customer account activate transaction is allowed if the account status is open primary, open, restricted, or restricted primary. The begin and end dates for the account are not checked for this transaction.

Customer Maximum History Records Restriction

If the MAX HISTORY RECORDS field in the CSTT is set to a nonzero value, the history inquiry – payments and transfers transaction and history inquiry – all transactions can only be performed in descending order for the customer. If this field is set to zero, these transactions can be performed in ascending or descending order.

Logging and Routing Restrictions

The customer verify and customer select transactions are neither routed or logged to the ITS Transaction Log File (ITLF).

The following transactions must always be routed to a host:

- Check stop payment add
- Statement download
- Statement closing download

Customer Context

The following three identifiers are used to identify a specific customer throughout a multiple transactions made in a single call for a particular customer. These identifiers, and related data (e.g., customer name, customer account number, etc.), are known as customer context.

- FIID
- Customer ID
- Personal ID

The customer context is established at the beginning of a transaction session for a customer, is retained throughout the session, and is cleared at the end of the session. It is the responsibility of the RBSI-compliant device to clear the context between sessions for different customers.

Customer Account and Customer Vendor Transaction Limits

Only one customer account or customer vendor can be processed in each of the following transactions. The device can bundle multiple transactions for transparent presentation to a user if required.

- Customer account add
- Customer account activate
- Customer account deactivate
- Customer account delete
- Customer account update
- Customer vendor add
- Customer vendor activate

- Customer vendor deactivate
- Customer vendor update

Account With Vendor Checking

When adding or updating a customer account with vendor, BASE24 checks any account masks defined for the vendor or vendor branch. The customer vendor mask list inquiry transaction allows devices to retrieve a list of all account masks defined for a vendor and vendor branch or to verify that an account number meets all account mask requirements.

Multiple Account Selection

To enable multiple account selection for the Remote Banking Standard Interface, the MULTIPLE ACCOUNT SELECT timer field and MULTIPLE ACCOUNT SELECT MAX COUNT field on the VRU Configuration Data (VCD) screen 18 must both be set to a nonzero value for the Remote Banking Standard Interface instance.

Refer to the *BASE24 Core Files and Tables Maintenance Manual* for more information on the VCD.

PIN Processing Requirements

PIN verification can be performed locally or by BASE24. If the device indicates a source of IB (inbound CSR) in the transaction, BASE24 also checks the Institution Definition File (IDF) to determine whether customer service representatives are required to perform PIN verify transactions or allowed to perform PIN change transactions on behalf of a customer. These flags are set in the PIN REQUIRED and PIN CHANGE ALLOWED fields on IDF screen 40, respectively.

BASE24 checks the PIN processing information in the CSTT for all transactions that provide a valid customer ID or account number, except for customer verify and customer select transactions. Also, BASE24 does not check whether a PIN change is required for customer add and customer activate transactions.

When a PIN has been successfully verified for a customer, the Point of Service Data Code in data element 52 should be set to a value of S in all subsequent request messages to avoid having the PIN re-verified for each request.

Duplicate Transaction Processing

When BASE24 detects a duplicate transaction while logging one of the following transactions, only the data retrieved from the ITS Transaction Log File (ITLF) can be returned in the transaction. The additional response data, which would have been placed in the external data block, is not logged to the ITLF and therefore cannot be retrieved and returned to the device when a duplicate transaction is detected.

- Customer add
- Customer deactivate
- Customer update
- Customer account activate
- Customer account add
- Customer account deactivate
- Customer account delete
- Customer account update
- Customer vendor activate
- Customer vendor add
- Customer vendor deactivate
- Customer vendor update

Appendix D

Recommended Transaction Sequences

This appendix provides recommended transaction sequences for performing BASE24 Remote Banking functions. Each sequence describes the recommended order of transactions and the data required for each transaction. Transaction codes are provided in parentheses following transaction names throughout this appendix.

This appendix is intended to be used as an aid in scripting software applications that use the Remote Banking Standard Interface.

Caution: Due to the flexible nature of the Remote Banking Standard Interface, the information presented in this appendix may or may not be appropriate to your application. Careful consideration should be made to the business requirements of your application before adopting any of these recommendations. Some transactions are intended for use by customer service representatives or branch employees only.

Recommended transaction sequences are provided for the following BASE24 Remote Banking functions:

- Verifying a customer
 - Identifying and verifying the customer
 - Verifying the customer's PIN
 - Changing the customer's PIN
- Entering a loan application
- Entering miscellaneous requests
- Retrieving miscellaneous requests

- Retrieving financial institution information
 - Retrieving financial institution information
 - Retrieving financial institution branch information
- Managing customer information
 - Adding a new customer
 - Retrieving customer information
 - Updating customer information
 - Activating a customer ID
 - Deactivating a customer ID
 - Reactivating a customer ID
- Managing customer accounts
 - Adding a customer account
 - Updating a customer account
 - Activating a customer account
 - Deactivating a customer account
 - Deleting a customer account
- Retrieving account information
 - Listing all configured customer accounts
 - Retrieving the balance of a single account
 - Retrieving the balance of two accounts simultaneously
 - Retrieving the balance of all (multiple) customer accounts
- Entering and managing payments
 - Entering a scheduled payment
 - Changing a scheduled payment
 - Deleting a scheduled payment
- Entering and managing transfers
 - Entering an immediate transfer
 - Entering a scheduled transfer
 - Changing a scheduled transfer
 - Deleting a scheduled transfer

- Retrieving currently scheduled transactions
 - Retrieving scheduled payments
 - Retrieving scheduled transfers
 - Retrieving scheduled transactions
- Entering a user-defined transaction
- Managing vendors
 - Listing all customer vendors available to a customer
 - Request mailing of a printed Customer Vendor List
 - Adding a customer vendor
 - Deactivating a customer vendor
 - Activating a customer vendor
 - Updating customer vendor information
- Retrieving previous transactions
 - Determining whether a particular check has cleared
 - Retrieving debit transactions
 - Retrieving credit transactions
 - Retrieving debit and credit transactions for a specified time period
 - Retrieving check transactions
 - Retrieving transactions entered from a particular source (e.g., a remote banking device)
 - Retrieving unscheduled transfer transactions
 - Downloading statement information
 - Retrieving scheduled payments and transfers
 - Retrieving BASE24-billpay transaction history
- Stopping payment on a check or range of checks

Verifying a Customer

This subsection provides transaction sequences for the following functions:

- Identifying and verifying the customer
- Verifying the customer's PIN
- Changing the customer's PIN

Identifying and Verifying the Customer

The following is a recommended transaction sequence for identifying and verifying a customer.

Assumptions: None.

1. Send a customer verify (BF) transaction request to BASE24. A customer ID, customer account number, or customer name is required.
2. Process the customer verification response data or customer select request data returned from BASE24.
 - a. If the response includes customer select request data and indicates that no more customer IDs or personal IDs exist, select the customer to be verified (using the Select Index field), and proceed with step 3.
 - b. If the response includes customer select request data and indicates that more customer IDs or personal IDs exist, but includes the customer to be verified, select the customer (using the Select Index field), and proceed with step 3.
 - c. If the response includes customer select request data and indicates that more customer IDs or personal IDs exist, and does not include the customer to be verified, proceed with step 3.
 - d. If the response contains customer verification response data for a single customer, use the information as required to verify the authenticity of the customer.
 - If the response indicates that the customer PIN needs to be changed, send a PIN change (90) transaction to BASE24. See “Changing the Customer's PIN” elsewhere in this appendix.
 - If the response indicates the customer PIN is required, either send a PIN verify (91) transaction to BASE24 or include the PIN in the next transaction sent to BASE24 for this customer. Once the PIN is

verified successfully, the Point of Service Data Code data element must be set to indicate that the PIN has been verified locally in any subsequent messages sent to BASE24. See “Verifying the Customer’s PIN” elsewhere in this appendix.

- If the customer is successfully verified, capture the customer context information for use with other transactions included in the same customer session.
3. Send the customer select respond request data to BASE24.
 4. Process the customer verification response data or customer select request data returned from BASE24.
 - a. If the response includes customer select request data and indicates that no more customer IDs or personal IDs exist, select the customer to be verified (using the Select Index field), and repeat step 3.
 - b. If the response includes customer select request data and indicates that more customer IDs or personal IDs exist, but includes the customer to be verified, select the customer (using the Select Index field), and repeat step 3.
 - c. If the response includes customer select request data and indicates that more customer IDs or personal IDs exist, and does not include the customer to be verified, repeat step 3.
 - d. If the response contains customer verification response data for a single customer, use the information as required to verify the authenticity of the customer.
 - If the response indicates that the customer PIN needs to be changed, send a PIN change (90) transaction to BASE24. See “Changing the Customer’s PIN” elsewhere in this appendix.
 - If the response indicates the customer PIN is required, either send a PIN verify (91) transaction to BASE24 or include the PIN in the next transaction sent to BASE24 for this customer. Once the PIN is verified successfully, the Point of Service Data Code data element must be set to indicate that the PIN has been verified locally in any subsequent messages sent to BASE24. See “Verifying the Customer’s PIN” elsewhere in this appendix.
 - If a PIN is not required and the customer is successfully verified, capture the customer context information for use with other transactions included in the same customer session.

Verifying the Customer's PIN

The following is a recommended transaction sequence for verifying a customer's PIN.

Note: This procedure uses the PIN verify (91) transaction to verify a customer's PIN. PINs can also be verified when processing any Remote Banking transaction that provides a PIN in the request message.

Assumptions: The customer is allowed to perform the PIN verify (91) transaction.

1. Verify the customer, if required. See “Verifying the Customer” elsewhere in this appendix. If the customer has already been successfully verified in the current call to BASE24 or if customer verification is not required, proceed with the next step.
2. Send a PIN verify (91) transaction request to BASE24. A customer ID and PIN are required.
3. Process the PIN verify response returned from BASE24 and convey the results of the transaction to the customer.
4. If this is the last transaction for this customer, clear the customer context information.

Changing the Customer's PIN

The following is a recommended transaction sequence for changing a customer's PIN.

Assumptions: The customer is allowed to perform the PIN change (90) transaction.

1. Verify the customer, if required. See “Verifying the Customer” elsewhere in this appendix. If the customer has already been successfully verified in the current call to BASE24 or if customer verification is not required, proceed with the next step.
2. Send a PIN change (90) transaction request to BASE24. A customer ID, the old PIN, and the new PIN are required.
3. Process the PIN change response returned from BASE24 and convey the results of the transaction to the customer.
4. If this is the last transaction for this customer, clear the customer context information.

Entering a Loan Application

The following is a recommended transaction sequence for entering a loan application.

Assumptions: If a customer ID or account number are provided, the customer is allowed to perform the loan application (9V) transaction.

1. Send a loan application (9V) transaction request to BASE24.
 - An FIID and full customer name (family name, given name, and middle initial) are required.
 - A customer ID and account number are optional. If they are not provided, the default customer ID and account numbers from the Bank Table (BANK) are used.
 - Because the institution accounts provided in the request are for information only, account type descriptions are not required.
 - If a customer ID or account number are provided, include the customer PIN if PIN verification is required and the PIN has not yet been verified in the current customer session.
2. Process the loan application response returned from BASE24 and convey the results of the transaction to the customer.
3. If this is the last transaction for this customer, clear the customer context information.

Entering Miscellaneous Information

The following is a recommended transaction sequence for entering miscellaneous information.

Assumptions: The customer is allowed to perform miscellaneous request (9W) transactions.

1. Verify the customer, if required. See “Verifying the Customer” elsewhere in this appendix. If the customer has already been successfully verified in the current call to BASE24 or if customer verification is not required, proceed with the next step.
2. Send a miscellaneous request (9W) transaction to BASE24.
 - A customer ID or account number is required.
 - Include the customer PIN if PIN verification is required and the PIN has not yet been verified in the current customer session.
3. Process the miscellaneous request response returned from BASE24 and convey the results of the transaction to the customer.
4. If this is the last transaction for this customer, clear the customer context information.

Retrieving Miscellaneous Requests

The following is a recommended transaction sequence for retrieving miscellaneous requests.

Assumptions: The customer is allowed to perform miscellaneous requests inquiry (B4) transactions.

1. Verify the customer, if required. See “Verifying the Customer” elsewhere in this appendix. If the customer has already been successfully verified in the current call to BASE24 or if customer verification is not required, proceed with the next step.
2. Send the miscellaneous requests inquiry (B4) transaction request to BASE24. The default search direction is descending by transaction source date.
 - A customer ID or account number is required.
 - Include the customer PIN if PIN verification is required and the PIN has not yet been verified in the current customer session.
3. Process the miscellaneous requests inquiry response returned from BASE24.
 - a. If no more miscellaneous requests exist, present the miscellaneous request information from the response to the customer and proceed with step 4.
 - b. If more miscellaneous requests exist, present the miscellaneous request information from the current response to the customer, capture the continuation key returned in the response, and repeat step 2. Refer to the topic “Transaction Inquiry Continuation” in section 1 for more information on continuing inquiry transactions.
4. If this is the last transaction for this customer, clear the customer context information.

Retrieving Financial Institution Information

This subsection provides transaction sequences for the following functions:

- Retrieving financial institution information
- Retrieving financial institution branch information

Retrieving Financial Institution Information

The following is a recommended transaction sequence for retrieving (listing) financial institutions.

Assumptions: If a customer ID or account number is provided, the customer is allowed to perform a bank list inquiry (B1) transaction.

1. Send the bank list inquiry (B1) transaction request to BASE24. The default search direction is ascending by bank name.
 - A bank name or an FIID is required to process a bank list inquiry (B1) transaction. If neither are provided, the FIID from the BP-ACQ-FIID param in the LCONF is used for logging the transaction to the ITLF.
 - A customer ID or account number are not required to process a bank list inquiry (B1) transaction. If the customer ID is not provided, the default customer ID configured for the financial institution in the Bank Table (BANK) is used for logging the transaction to the ITLF.
 - If a customer ID or account number is provided, include the customer PIN if PIN verification is required and the PIN has not yet been verified in the current customer session.
 - The inquiry transaction data in data element 48 is required for all BASE24-billpay list inquiry transactions.
2. Process the bank list inquiry response data returned from BASE24. Capture or select from the financial institution item information that is returned.
 - If the response indicates that no more financial institution items exist, proceed with step 3.
 - If the response indicates that more financial institution items exist, capture the continuation key returned in the response, determine whether the search is to continue in ascending or descending order, and repeat step 1. Refer to the topic “Transaction Inquiry Continuation” in section 1 for more information on continuing inquiry transactions.

3. If this is the last transaction for a customer, clear the customer context information.

Retrieving Financial Institution Branch Information

The following is a recommended transaction sequence for retrieving (listing) financial institution branch information.

Assumptions: If a customer ID or account number is provided, the customer is allowed to perform a bank branch list inquiry (B2) transaction.

1. Send the bank branch list inquiry (B2) transaction request to BASE24. The default search direction is ascending by bank name.
 - An FIID is required to process a bank branch list inquiry transaction.
 - A customer ID or account number are not required to process a bank branch list inquiry (B2) transaction. If the customer ID is not provided, the default customer ID configured for the financial institution in the Bank Table (BANK) is used for logging the transaction to the ITLF.
 - If a customer ID or account number is provided, include the customer PIN if PIN verification is required and the PIN has not yet been verified in the current customer session.
 - The inquiry transaction data in data element 48 is required for all BASE24-billpay list inquiry transactions.
2. Process the bank branch list inquiry response data returned from BASE24. Capture or select from the financial institution branch item information that is returned.
 - If the response indicates that no more financial institution branch items exist, proceed with step 3.
 - If the response indicates that more financial institution branch items exist, capture the continuation key returned in the response, determine whether the search is to continue in ascending or descending order, and repeat step 1. Refer to the topic “Transaction Inquiry Continuation” in section 1 for more information on continuing inquiry transactions.
3. If this is the last transaction for a customer, clear the customer context information.

Managing Customer Information

This subsection contains transaction sequences for the following functions:

- Adding a new customer
- Retrieving customer information
- Updating customer information
- Activating or reactivating a customer ID
- Deactivating a customer ID

Adding a New Customer

The following is a recommended transaction sequence for adding a new customer.

Assumptions: The RBSI-compliant device has knowledge of all FIIDs configured for the system and whether each financial institution uses preloaded customer data.

The customer or customer service representative is allowed to perform bank list inquiry (B1), bank branch list inquiry (B2), customer inquiry(9L), customer add (9K), customer activate (BB), customer account list inquiry (3N), and customer account add (B5) transactions.

1. If necessary, send the bank list inquiry (B1) transaction request to BASE24 to select the financial institution for the new customer. See “Retrieving Financial Institution Information” provided elsewhere in this appendix.
2. If the customer’s financial institution uses preloaded customer data (as indicated in the bank list inquiry response), verify the customer and proceed to step 4. See “Verifying the Customer” elsewhere in this appendix.
3. If necessary, send the bank branch list inquiry (B2) request to BASE24 to select the financial institution branch for the new customer. See “Retrieving Financial Institution Branch Information” provided elsewhere in this appendix.
4. Send the customer inquiry (9L) transaction request to BASE24 to retrieve the preloaded or default customer information.
 - If the customer’s financial institution uses preloaded customer information (as indicated in the bank list inquiry response), provide the FIID, customer ID or account number, and personal ID (retrieved during

customer verification) in the request. The preloaded Customer Table (CSTT) and Personal Information Table (PIT) information as well as Customer Billpay Table (CBPT) information is returned in the response.

- If the customer's financial institution does not use preloaded customer information (as indicated in the Bank List Inquiry response), provide the FIID only in the request. The default CSTT and PIT information for the customer's financial institution as well as CBPT information is returned in the response.
 - If the customer's data was preloaded, include the customer PIN if PIN verification is required and the PIN has not yet been verified in the current customer session.
5. Process the customer inquiry response returned from BASE24.
 - a. If the customer's information was preloaded, verify it as required by your organization and proceed to step 8. If any data needs to be modified, update the customer information. See "Updating Customer Information" elsewhere in this appendix.
 - b. If the customer's information was not preloaded, capture the customer information returned and modify it as needed to build the customer add request.
 6. For customers that were not preloaded, send the customer add (9K) transaction request to BASE24 to add the new customer.
 - If the customer's financial institution does not support automatically generated customer IDs, the customer ID must be provided in the request.
 - The Default Account Number and Service Fee Account Number fields must contain all numeric characters.
 - If the Default Account Type field is blank, the Default Account Type Description field must be provided.
 - If the Service Fee Account Type field is blank, the Service Fee Account Type Description field must be provided.
 - The customer's personal ID is set to the value of the customer ID, whether it is provided in the request or generated by BASE24.

Note: For financial institutions not using preloaded customer data, it is possible to add the same customer more than once. Therefore, appropriate procedures should be in place to avoid unnecessary duplication.

7. Process the customer response data returned from BASE24. Verify that the information is correct. If necessary, use the customer update request and response to change any data. See “Updating Customer Information” elsewhere in this appendix.
8. Verify and update the customer’s accounts or add them, depending on whether the customer’s data was preloaded.
 - a. If the customer’s accounts were preloaded, verify the customer’s accounts as required by your organization. See “Listing all Configured Customer Accounts” elsewhere in this appendix.
 - b. If the customer’s accounts were not preloaded, send the customer account add (B5) transaction request for each account to be added. Process the customer account response data for each account that is added.
 - If the account type is not provided in data element 48, the Account Type Description field must be provided.
 - The Account Qualifier field is optional.
 - If the Allowed - Debit, Allowed - Credit, and Allowed - Inquiry field values are not provided, the default account information configured for the customer’s financial institution are used.

Depending upon the policies of your organization, any accounts added may need to be verified by back-office personnel before they can be used in processing using Customer Table (CSTT) screen 2. In which case, the back-office personnel will update the account status appropriately after the account is successfully verified.
9. If appropriate, send the customer activate (BB) transaction request to BASE24 to initiate service for the customer.
10. Process the customer activate response returned from BASE24 and convey the results of the transaction to the customer.
11. If this is the last transaction for this customer, clear the customer context information.

Retrieving Customer Information

The following is a recommended transaction sequence for retrieving customer information.

Assumptions: The customer is allowed to perform a customer inquiry (9L) transaction.

1. Verify the customer, if required. See “Verifying the Customer” elsewhere in this appendix. If the customer has already been successfully verified in the current call to BASE24 or if customer verification is not required, proceed with the next step.
2. Send a customer inquiry (9L) transaction request to BASE24 to retrieve the customer information.
 - To retrieve information for a specific customer, provide the customer ID and, optionally, a personal ID in the request.
 - To retrieve default customer information for a financial institution, provide the FIID and a customer ID of all zeros in the request. This option allows users to retrieve default information when adding a customer whose data was not preloaded.
 - Include the customer PIN if PIN verification is required and the PIN has not yet been verified in the current customer session.
3. Process the customer response data returned from BASE24 and convey the results of the transaction to the customer.
4. If this is the last transaction for this customer, clear the customer context information.

Updating Customer Information

The following is a recommended transaction sequence for updating customer information.

Assumptions: The customer is allowed to perform customer inquiry (9L) and customer update (9M) transactions.

1. Verify the customer, if required. See “Verifying the Customer” elsewhere in this appendix. If the customer has already been successfully verified in the current call to BASE24 or if customer verification is not required, proceed with the next step.

2. Send a customer inquiry (9L) transaction request to BASE24 to retrieve the customer information.
 - To retrieve information for a specific customer, provide the customer ID and, optionally, a personal ID in the request.
 - To retrieve default customer information for a financial institution, provide the FIID only in the request.
 - Include the customer PIN if PIN verification is required and the PIN has not yet been verified in the current customer session.
3. Process the customer response data returned from BASE24. Capture the customer information returned in the response and use it as basis for building a customer update request.
4. Send a customer update (9M) transaction request to BASE24. Use the information retrieved from the previous step to build the request.
 - The Default Account Number and Service Fee Account Number fields must contain all numeric characters.
 - If the Default Account Type field is blank, the Default Account Type Description field must be provided.
 - If the Service Fee Account Type field is blank, the Service Fee Account Type Description field must be provided.
5. Process the customer response data returned from BASE24 and convey the results of the transaction to the customer.
6. If this is the last transaction for this customer, clear the customer context information.

Activating or Reactivating a Customer ID

The following is a recommended transaction sequence for activating or reactivating a customer ID.

Assumptions: The customer is allowed to perform customer inquiry (9L) and customer activate (BB) transactions.

If a customer is to be reactivated, the customer's financial institution allows reactivation.

1. Verify the customer, if required. See "Verifying the Customer" elsewhere in this appendix. If the customer has already been successfully verified in the current call to BASE24 or if customer verification is not required, proceed with the next step.
2. Send a customer inquiry (9L) transaction request to BASE24 to retrieve the customer information.
 - Include the customer ID or account number, and, optionally, a personal ID in the request.
 - Include the customer PIN if PIN verification is required and the PIN has not yet been verified in the current customer session.
3. Process the customer response data returned from BASE24. Verify that the customer information is correct. Update any customer information if necessary. See "Updating Customer Information" elsewhere in this appendix.
4. Send a customer activate (BB) transaction request to BASE24. The customer ID or account number, and reactivation flag are required. The personal ID is optional.

If the verify status of the customer is verified, both the reactivate flag in the request and the reactivate allowed field in the Bank Table (BANK) for the customer's financial institution must be set to allow activation.

5. Process the customer response data returned from BASE24 and convey the results of the transaction to the customer.
6. If this is the last transaction for this customer, clear the customer context information.

Deactivating a Customer ID

The following is a recommended transaction sequence for deactivating a customer ID.

Assumptions: The customer is allowed to perform customer inquiry (9L) and customer deactivate (BC) transactions.

1. Verify the customer, if required. See “Verifying the Customer” elsewhere in this appendix. If the customer has already been successfully verified in the current call to BASE24 or if customer verification is not required, proceed with the next step.
2. Send a customer inquiry (9L) transaction request to BASE24 to retrieve the customer information.
 - Include the customer ID or account number, and, optionally, a personal ID in the request.
 - Include the customer PIN if PIN verification is required and the PIN has not yet been verified in the current customer session.
3. Process the customer response data returned from BASE24. Verify that the customer information is correct.
4. Send a customer deactivate (BC) transaction request to BASE24. The customer ID or account number is required.
5. Process the customer response data returned from BASE24 and convey the results of the transaction to the customer.

Note: If the customer has any outstanding scheduled future or recurring transactions, deactivation is not allowed.

6. If this is the last transaction for this customer, clear the customer context information.

Managing Customer Accounts

This subsection contains recommended transaction sequences for the following functions:

- Adding a customer account
- Updating a customer account
- Activating a customer account
- Deactivating a customer account
- Deleting a customer account

Adding a Customer Account

The following is a recommended transaction sequence for adding a customer account.

Note: When a customer account is added successfully, it is automatically activated.

Assumptions: The customer is allowed to perform a customer add (9K) transaction.

1. Verify the customer, if required. See “Verifying the Customer” elsewhere in this appendix. If the customer has already been successfully verified in the current call to BASE24 or if customer verification is not required, proceed with the next step.
2. Send a customer account add (9K) transaction request to BASE24.
 - A customer ID (PAN), account number, and account type (or account type description) are required. If the account type is not provided, the account type description is required.
 - Include the customer PIN if PIN verification is required and the PIN has not yet been verified in the current customer session.
3. Process the customer account response data returned from BASE24 and convey the results of the transaction to the customer. Verify that the account information was added correctly. If necessary, update the account information. See “Updating a Customer Account” elsewhere in this appendix.

Default account information for the customer's financial institution is used for any fields not provided in the request.

4. If this is the last transaction for this customer, clear the customer context information.

Updating a Customer Account

The following is a recommended transaction sequence for updating a customer account.

Assumptions: The customer is allowed to perform customer account list inquiry (3N) and customer account update (B9) transactions.

1. Verify the customer, if required. See "Verifying the Customer" elsewhere in this appendix. If the customer has already been successfully verified in the current call to BASE24 or if customer verification is not required, proceed with the next step.
2. Send a customer account list inquiry (3N) transaction request to BASE24 to retrieve the information for the account to be updated. The search direction is ascending.
 - A customer ID or account number is required for this request.
 - Include the customer PIN if PIN verification is required and the PIN has not yet been verified in the current customer session.
3. Process the customer account list inquiry response returned from BASE24.
 - a. If the customer account to be updated is included in the response, select it and proceed with step 4.
 - b. If the customer account to be updated is not included in the response and the response indicates that additional customer account items exist, capture the continuation key returned in the response, determine whether the search is to continue in ascending or descending order, and repeat step 2. Refer to the topic "Transaction Inquiry Continuation" in section 1 for more information on continuing inquiry transactions.

4. Send a customer account update (B9) transaction request to BASE24. Use the account information selected in steps 2 and 3 as the basis for the request. The following fields can be updated in this request:
 - Account Qualifier
 - Debit Allowed Flag
 - Credit Allowed Flag
 - Inquiry Allowed Flag
 - Account Description
5. Process the customer account response data returned from BASE24 and convey the results of the transaction to the customer.
6. If this is the last transaction for this customer, clear the customer context information.

Activating a Customer Account

The following is a recommended transaction sequence for activating a customer account.

Assumptions: The customer is allowed to perform customer account list inquiry (3N) and customer account activate (B6) transactions.

1. Verify the customer, if required. See “Verifying the Customer” elsewhere in this appendix. If the customer has already been successfully verified in the current call to BASE24 or if customer verification is not required, proceed with the next step.
2. Send a customer account list inquiry (3N) transaction request to BASE24 to retrieve the account to be activated. The default search direction is ascending.
 - A customer ID or account number is required for this request.
 - Include the customer PIN if PIN verification is required and the PIN has not yet been verified in the current customer session.
3. Process the account list response data returned from BASE24.
 - a. If the customer account to be activated is included in the response, select it and proceed with step 4.
 - b. If the customer account to be activated is not included in the response and the response indicates that additional customer account items exist, capture the continuation key returned in the response, determine whether

the search is to continue in ascending or descending order, and repeat step 2. Refer to the topic “Transaction Inquiry Continuation” in section 1 for more information on continuing inquiry transactions.

4. Send a customer account activate (B6) transaction request to BASE24. Use the account information selected in steps 2 and 3 as the basis for the request. The account number and account type are required in data element 48.
5. Process the customer account response data returned from BASE24 and convey the results of the transaction to the customer.
6. If this is the last transaction for this customer, clear the customer context information.

Deactivating a Customer Account

The following is a recommended transaction sequence for deactivating a customer account.

Assumptions: The customer is allowed to perform customer account list inquiry (3N) and customer account deactivate (B7) transactions.

1. Verify the customer, if required. See “Verifying the Customer” elsewhere in this appendix. If the customer has already been successfully verified in the current call to BASE24 or if customer verification is not required, proceed with the next step.
2. Send a customer account list inquiry (3N) transaction request to BASE24 to retrieve the account to be deactivated. The default search direction is ascending.
 - A customer ID or account number is required for this request.
 - Include the customer PIN if PIN verification is required and the PIN has not yet been verified in the current customer session.
3. Process the account list response data returned from BASE24.
 - a. If the customer account to be deactivated is included in the response, select it and proceed with step 4.
 - b. If the customer account to be deactivated is not included in the response and the response indicates that additional customer account items exist, capture the continuation key returned in the response, determine whether the search is to continue in ascending or descending order, and repeat step 2. Refer to the topic “Transaction Inquiry Continuation” in section 1 for more information on continuing inquiry transactions.

4. Send a customer account deactivate (B7) transaction request to BASE24. Use the account information selected in steps 2 and 3 as the basis for the request. The account number and account type are required in data element 48.
5. Process the customer account response data returned from BASE24 and convey the results of the transaction to the customer.
6. If this is the last transaction for this customer, clear the customer context information.

Deleting a Customer Account

The following is a recommended transaction sequence for deleting a customer account.

Assumptions: The customer is allowed to perform customer account list inquiry (3N) and customer account delete (B8) transactions.

 The customer's verify status must be set to a value of I (issued but inactive).

1. Verify the customer, if required. See "Verifying the Customer" elsewhere in this appendix. If the customer has already been successfully verified in the current call to BASE24 or if customer verification is not required, proceed with the next step.
2. Send a customer account list inquiry (3N) transaction request to BASE24 to retrieve the account to be deleted. The default search direction is ascending.
 - A customer ID or account number is required for this request.
 - Include the customer PIN if PIN verification is required and the PIN has not yet been verified in the current customer session.
3. Process the account list response data returned from BASE24.
 - a. If the customer account to be deleted is included in the response, select it and proceed with step 4.
 - b. If the customer account to be deleted is not included in the response and the response indicates that additional customer account items exist, capture the continuation key returned in the response, determine whether the search is to continue in ascending or descending order, and repeat step 2. Refer to the topic "Transaction Inquiry Continuation" in section 1 for more information on continuing inquiry transactions.

4. Send a customer account delete (B8) transaction request to BASE24. Use the account information selected in steps 2 and 3 as the basis for the request. The account number and account type are required in data element 48.
5. Process the customer account response data returned from BASE24 and convey the results of the transaction to the customer.
6. If this is the last transaction for this customer, clear the customer context information.

Retrieving Account Information

This subsection contains transaction sequences for the following functions:

- Listing all customer accounts
- Retrieving the balance of one or two accounts
- Retrieving the balance of all (multiple) customer accounts

Listing All Customer Accounts

The following is a recommended transaction sequence for listing all customer accounts.

Assumptions: The customer is allowed to perform customer account list inquiry (3N) transactions.

1. Verify the customer, if required. See “Verifying the Customer” elsewhere in this appendix. If the customer has already been successfully verified in the current call to BASE24 or if customer verification is not required, proceed with the next step.
2. Send a customer account list inquiry (3N) transaction request to BASE24 to list all configured customer accounts. The default search direction is ascending.
 - A customer ID or account number is required for this request.
 - Include the customer PIN if PIN verification is required and the PIN has not yet been verified in the current customer session.
3. Process the account list response data returned from BASE24 and convey the account information returned in the response to the customer.
 - a. If the response indicates that there are no more customer accounts, proceed to step 4.
 - b. If the response indicates that additional customer account items exist, capture the continuation key returned in the response, determine whether the search is to continue in ascending or descending order, and repeat step 2. Refer to the topic “Transaction Inquiry Continuation” in section 1 for more information on continuing inquiry transactions.
4. If this is the last transaction for this customer, clear the customer context information.

Retrieving the Balance of One or Two Customer Accounts

The following is a recommended transaction sequence for retrieving the balance of one or two customer accounts. To retrieve more than two customer accounts, see “Retrieving the Balance of Multiple Customer Accounts” elsewhere in this subsection.

Assumptions: The customer is allowed to perform customer account list inquiry (3N) and available funds inquiry (30) transactions.

1. Verify the customer, if required. See “Verifying the Customer” elsewhere in this appendix. If the customer has already been successfully verified in the current call to BASE24 or if customer verification is not required, proceed with the next step.
2. Send an available funds inquiry (30) transaction request to BASE24 to retrieve the balance of one or two customer accounts.
 - A customer ID, and an account number, account type, or account qualifier for each account, are required for this request.
 - Include the customer PIN if PIN verification is required and the PIN has not yet been verified in the current customer session.
3. Process the available funds inquiry response returned from BASE24.
 - a. If the response includes multiple account select information and indicates that no more accounts exist, select the account and repeat steps 2 and 3.
 - b. If the response includes multiple account select information, indicates that more accounts exist, and does not contain the account to be used, capture the continuation key and repeat steps 2 and 3 until a particular account is selected. Refer to the topic “Transaction Inquiry Continuation” in section 1 for more information on continuing inquiry transactions.
 - c. If the response does not include multiple account select information, convey the balance information to the customer and proceed with step 4.
4. If this is the last transaction for this customer, clear the customer context information.

Retrieving the Balance of Multiple Customer Accounts

The following is a recommended transaction sequence for retrieving the balance of multiple customer accounts.

Assumptions: The customer is allowed to perform customer account list inquiry (3N) and multiple account balance inquiry (3P) transactions.

1. Verify the customer, if required. See “Verifying the Customer” elsewhere in this appendix. If the customer has already been successfully verified in the current call to BASE24 or if customer verification is not required, proceed with the next step.
2. Send a customer account list inquiry (3N) transaction request to BASE24 to list all configured customer accounts. The default search direction is ascending.
 - A customer ID or account number is required for this request.
 - Include the customer PIN if PIN verification is required and the PIN has not yet been verified in the current customer session.
3. Process the account list response data returned from BASE24 and capture the account numbers for which balances are requested.
 - a. If the response indicates that there are no more customer accounts, proceed to step 4.
 - b. If the response indicates that additional customer account items exist, capture the continuation key returned in the response, determine whether the search is to continue in ascending or descending order, and repeat step 2. Refer to the topic “Transaction Inquiry Continuation” in section 1 for more information on continuing inquiry transactions.
4. Send a multiple account balance inquiry (3P) transaction request to BASE24 to retrieve the balance of all specified accounts for a customer.
 - A customer ID or account number is required for this request.
 - Include the customer PIN if PIN verification is required and the PIN has not yet been verified in the current customer session.
 - The account number for each account balance to be listed is required.

5. Process the multiple account balance response data returned from BASE24 and convey the information returned in the response to the customer.
 - If the response indicates that no more account balance items exist, proceed with step 6.
 - If the response indicates that more account balance items exist, capture the continuation key returned in the response, determine whether the search is to continue in ascending or descending order, and repeat step 4. Refer to the topic “Transaction Inquiry Continuation” in section 1 for more information on continuing inquiry transactions.
6. If this is the last transaction for this customer, clear the customer context information.

Entering and Managing Payments

This subsection contains transaction sequences for the following functions:

- Entering a scheduled payment
- Changing a scheduled payment
- Deleting a scheduled payment

Entering a Scheduled Payment

The following is a recommended transaction sequence for entering a scheduled payment.

Assumptions:

- The customer vendor has been configured and is active.
- The customer *from* account has been configured, is active, and allows debit transactions.
- The customer is allowed to perform customer account list inquiry (3N) and customer vendor list inquiry (3H) transactions.
- The customer is allowed to perform a schedule payment – immediate (9A), schedule payment – future (9B), or schedule payment – recurring (9C) transaction.

1. Verify the customer, if required. See “Verifying the Customer” elsewhere in this appendix. If the customer has already been successfully verified in the current call to BASE24 or if customer verification is not required, proceed with the next step.
2. Identify the customer account to be debited in the payment (a valid account number or account qualifier is needed to identify the account). Send a customer account list inquiry (3N) transaction request for the customer if needed to identify the number or qualifier of the *from* account for the payment.
 - A customer ID or account number is required for this request.
 - Include the customer PIN if PIN verification is required and the PIN has not yet been verified in the current customer session.
3. Process the account list response data returned from BASE24.
 - a. If the account is identified in the list returned, select it and proceed with step 4.
 - b. If the account is not returned in the list and the response indicates that additional accounts exist, capture the continuation key returned in the response, determine whether the search is to continue in ascending or descending order, and repeat step 2. Refer to the topic “Transaction Inquiry Continuation” in section 1 for more information on continuing inquiry transactions.

4. Identify the customer vendor to be paid in the transaction (a valid customer vendor number or customer vendor short name is required to identify the customer vendor). Send a customer vendor list inquiry (3H) transaction request for the customer if needed to identify the customer vendor for the payment.
5. Process the customer vendor list inquiry response data returned from BASE24.
 - a. If the customer vendor is identified in the list returned, select it and proceed with step 6.
 - b. If the customer vendor is not returned in the list and the response indicates that additional customer vendors exist, capture the continuation key returned in the response, determine whether the search is to continue in ascending or descending order, and repeat step 4. Refer to the topic “Transaction Inquiry Continuation” in section 1 for more information on continuing inquiry transactions.
6. Send a schedule payment – immediate (9A), schedule payment – future (9B), or schedule payment – recurring (9C) transaction request identifying the *from* account, customer vendor, payment amount, payment type, initial payment date, vendor specific data (if required for the vendor), total number of recurring periods (for recurring payments only), and recurring period type (for recurring payments only).
7. Process the schedule payment response returned from BASE24 and convey the results of the transaction to the customer.
8. If this is the last transaction for this customer, clear the customer context information.

Changing a Scheduled Payment

The following is a recommended transaction sequence for changing a scheduled payment.

Assumptions:

- The scheduled payment to be updated has already been entered and has not yet expired.
- The customer is allowed to perform scheduled payments list inquiry (3J) or scheduled transactions list inquiry (3M) transactions.
- The customer is allowed to perform scheduled payment update (9H) transactions.

1. Verify the customer, if required. See “Verifying the Customer” elsewhere in this appendix. If the customer has already been successfully verified in the current call to BASE24 or if customer verification is not required, proceed with the next step.
2. Identify the scheduled payment to be changed. Send a scheduled payments list inquiry (3J) request or scheduled transactions list inquiry (3M) request to BASE24 to list scheduled payments for the customer.
 - A customer ID or account number is required.
 - Data element 48 is required for all BASE24-billpay list inquiry transactions. The scheduled payment list inquiry request can also include a customer vendor number to further define the search.
 - Include the customer PIN if PIN verification is required and the PIN has not yet been verified in the current customer session.
3. Process the scheduled payments list inquiry response data or scheduled transactions list inquiry response data returned from BASE24.
 - a. If the transaction item is identified in the list returned, select it and proceed with step 4.
 - b. If the transaction item is not returned in the list and the response indicates that additional items exist, capture the continuation key returned in the response, determine whether the search is to continue in ascending or descending order, and repeat step 2. Refer to the topic “Transaction Inquiry Continuation” in section 1 for more information on continuing inquiry transactions.

4. Send a scheduled payment update (9H) transaction request identifying the transaction (it must include the customer ID or account number and customer reference number) and the data (*from* account number and type, payment amount, initial payment date, vendor specific data, total number of recurring periods, or recurring period) to be changed.
5. Process the scheduled payment update response returned from BASE24.
 - a. If the response includes multiple account select information and indicates that no more accounts exist, select the account and repeat steps 4 and 5.
 - b. If the response includes multiple account select information, indicates that more accounts exist, and does not contain the account to be used, capture the continuation key and repeat steps 4 and 5 until a particular account is selected. Refer to the topic “Transaction Inquiry Continuation” in section 1 for more information on continuing inquiry transactions.
 - c. If the response does not include multiple account select information, convey the results of the transaction to the customer and proceed to step 6.
6. If this is the last transaction for this customer, clear the customer context information.

Deleting a Scheduled Payment

The following is a recommended transaction sequence for deleting a scheduled payment.

Assumptions: The scheduled payment to be deleted has already been entered and has not yet completed.

The customer is allowed to perform scheduled payments list inquiry (3J) or scheduled transactions list inquiry (3M) transactions.

The customer is allowed to perform scheduled payment delete (9F) transactions.

1. Verify the customer, if required. See “Verifying the Customer” elsewhere in this appendix. If the customer has already been successfully verified in the current call to BASE24 or if customer verification is not required, proceed with the next step.

2. Identify the scheduled payment to be deleted. Send a scheduled payments list inquiry (3J) request or scheduled transactions list inquiry (3M) request to BASE24 to list scheduled payments for the customer.
 - A customer ID or account number is required.
 - Data element 48 is required for all BASE24-billpay list inquiry transactions. The scheduled payment list inquiry request can also include a customer vendor number to further define the search.
 - Include the customer PIN if PIN verification is required and the PIN has not yet been verified in the current customer session.
3. Process the scheduled payments list inquiry response data or scheduled transactions list inquiry response data returned from BASE24.
 - a. If the transaction item is identified in the list returned, select it and proceed with step 4.
 - b. If the transaction item is not returned in the list and the response indicates that additional items exist, capture the continuation key returned in the response, determine whether the search is to continue in ascending or descending order, and repeat step 2. Refer to the topic “Transaction Inquiry Continuation” in section 1 for more information on continuing inquiry transactions.
4. Send a scheduled payment delete (9F) transaction request identifying the transaction (it must include the customer ID or account number and customer reference number) to be deleted.
5. Process the scheduled payment delete response returned from BASE24 and convey the transaction results to the customer.
6. If this is the last transaction for this customer, clear the customer context information.

Entering and Managing Transfers

This subsection contains transaction sequences for the following functions:

- Entering an immediate transfer
- Entering a scheduled transfer
- Changing a scheduled transfer
- Deleting a scheduled transfer

Entering an Immediate Transfer

The following is a recommended transaction sequence for entering an immediate transfer.

Assumptions: The customer *from* and *to* accounts have been configured and are active. The *from* account allows debit transactions and the *to* account allows credit transactions.

 The customer is allowed to perform a customer account list inquiry (3N) transaction.

 The customer is allowed to perform a transfer – immediate (40) transaction.

1. Verify the customer, if required. See “Verifying the Customer” elsewhere in this appendix. If the customer has already been successfully verified in the current call to BASE24 or if customer verification is not required, proceed with the next step.
2. Identify the customer *from* account to be debited in the transfer (a valid account number or account qualifier is needed to identify the account). Send a customer account list inquiry (3N) transaction request for the customer if needed to identify the number or qualifier of the *from* account for the transfer.
 - A customer ID or account number is required.
 - Include the customer PIN if PIN verification is required and the PIN has not yet been verified in the current customer session.
3. Process the account list response data returned from BASE24.
 - a. If the *from* account is identified in the list returned, select it and proceed with step 4.
 - b. If the *from* account is not returned in the list and the response indicates that additional accounts exist, capture the continuation key returned in the response, determine whether the search is to continue in ascending or descending order, and repeat step 2. Refer to the topic “Transaction Inquiry Continuation” in section 1 for more information on continuing inquiry transactions.
4. Identify the customer *to* account to be credited in the transfer (a valid account number or account qualifier is needed to identify the account). Send a customer account list inquiry (3N) request for the customer if needed to identify the number or qualifier of the *to* account for the transfer.

5. Process the account list response data returned from BASE24.
 - a. If the *to* account is identified in the list returned, select it and proceed with step 6.
 - b. If the *to* account is not returned in the list and the response indicates that additional accounts exist, capture the continuation key returned in the response, determine whether the search is to continue in ascending or descending order, and repeat step 4. Refer to the topic “Transaction Inquiry Continuation” in section 1 for more information on continuing inquiry transactions.
6. Send a transfer – immediate (40) transaction request identifying the *from* account, *to* account, transfer amount, and currency code.
7. Process the transfer – immediate response returned from BASE24 and convey the results of the transaction to the customer.
8. If this is the last transaction for this customer, clear the customer context information.

Entering a Scheduled Transfer

The following is a recommended transaction sequence for entering a scheduled transfer.

Assumptions: The customer *from* and *to* accounts have been configured and are active. The *from* account allows debit transactions and the *to* account allows credit transactions.

 The customer is allowed to perform a customer account list inquiry (3N) transaction.

 The customer is allowed to perform a schedule transfer – future (9D), or schedule transfer – recurring (9E) transaction.

1. Verify the customer, if required. See “Verifying the Customer” elsewhere in this appendix. If the customer has already been successfully verified in the current call to BASE24 or if customer verification is not required, proceed with the next step.
2. Identify the customer *from* account to be debited in the transfer (a valid account number or account qualifier is needed to identify the account). Send a customer account list inquiry (3N) transaction request for the customer if needed to identify the number or qualifier of the *from* account for the transfer.
 - A customer ID or account number is required.
 - Include the customer PIN if PIN verification is required and the PIN has not yet been verified in the current customer session.
3. Process the account list response data returned from BASE24.
 - a. If the *from* account is identified in the list returned, select it and proceed with step 4.
 - b. If the *from* account is not returned in the list and the response indicates that additional accounts exist, capture the continuation key returned in the response, determine whether the search is to continue in ascending or descending order, and repeat step 2. Refer to the topic “Transaction Inquiry Continuation” in section 1 for more information on continuing inquiry transactions.
4. Identify the customer *to* account to be credited in the transfer (a valid account number or account qualifier is needed to identify the account). Send a customer account list inquiry (3N) transaction request for the customer if needed to identify the number or qualifier of the *to* account for the transfer.

5. Process the account list response data returned from BASE24.
 - a. If the *to* account is identified in the list returned, select it and proceed with step 6.
 - b. If the *to* account is not returned in the list and the response indicates that additional accounts exist, capture the continuation key returned in the response, determine whether the search is to continue in ascending or descending order, and repeat step 4. Refer to the topic “Transaction Inquiry Continuation” in section 1 for more information on continuing inquiry transactions.
6. Send a schedule transfer – future (9D) or schedule transfer – recurring (9E) request identifying the *from* account, *to* account, transfer amount, transfer type, initial transfer date, total number of recurring periods (for recurring transfers only), and recurring period type (for recurring transfers only).
7. Process the schedule transfer response returned from BASE24 and convey the results of the transaction to the customer.
8. If this is the last transaction for this customer, clear the customer context information.

Changing a Scheduled Transfer

The following is a recommended transaction sequence for changing a scheduled transfer.

Assumptions: The scheduled transfer to be updated has already been entered and has not yet expired.

The customer is allowed to perform scheduled transfers list inquiry (3K) or scheduled transactions list inquiry (3M) transactions.

The customer is allowed to perform scheduled transfer update (9J) transactions.

1. Verify the customer, if required. See “Verifying the Customer” elsewhere in this appendix. If the customer has already been successfully verified in the current call to BASE24 or if customer verification is not required, proceed with the next step.
2. Identify the scheduled transfer to be changed. Send a scheduled transfers list inquiry (3K) transaction request or scheduled transactions list inquiry (3M) transaction request to BASE24 to list scheduled transfers for the customer.
 - A customer ID or account number is required.
 - Data element 48 is required for all BASE24-billpay list inquiry transactions.
 - Include the customer PIN if PIN verification is required and the PIN has not yet been verified in the current customer session.
3. Process the scheduled transfers list inquiry response data or scheduled transactions list inquiry response data returned from BASE24.
 - a. If the transaction item is identified in the list returned, select it and proceed with step 4.
 - b. If the transaction item is not returned in the list and the response indicates that additional items exist, capture the continuation key returned in the response, determine whether the search is to continue in ascending or descending order, and repeat step 2. Refer to the topic “Transaction Inquiry Continuation” in section 1 for more information on continuing inquiry transactions.

4. Send a scheduled transfer update (9J) transaction request identifying the transaction (it must include the customer ID or account number and customer reference number) and the data (*from* account number and type, *to* account number and type, transfer amount, initial transfer date, total number of recurring periods, or recurring period) to be changed.
5. Process the scheduled transfer update response returned from BASE24.
 - a. If the response includes multiple account select information for an updated account number and indicates that no more accounts exist, select the account and repeat steps 4 and 5.
 - b. If the response includes multiple account select information for an updated account number, indicates that more accounts exist, and does not contain the account to be used, capture the continuation key and repeat steps 4 and 5 until a particular account is selected. Refer to the topic “Transaction Inquiry Continuation” in section 1 for more information on continuing inquiry transactions.
 - c. If the response does not include multiple account select information, convey the results of the transaction to the customer and proceed with step 6.
6. If this is the last transaction for this customer, clear the customer context information.

Deleting a Scheduled Transfer

The following is a recommended transaction sequence for deleting a scheduled transfer.

Assumptions: The scheduled transfer to be deleted has already been entered and has not yet completed.

 The customer is allowed to perform scheduled transfers list inquiry (3K) or scheduled transactions list inquiry (3M) transactions.

 The customer is allowed to perform scheduled transfer delete (9G) transactions.

1. Verify the customer, if required. See “Verifying the Customer” elsewhere in this appendix. If the customer has already been successfully verified in the current call to BASE24 or if customer verification is not required, proceed with the next step.

2. Identify the scheduled transfer to be deleted. Send a scheduled transfers list inquiry (3K) transaction request or scheduled transactions list inquiry (3M) transaction request to BASE24 to list scheduled transfers for the customer.
 - A customer ID or account number is required.
 - Data element 48 is required for all BASE24-billpay list inquiry transactions.
 - Include the customer PIN if PIN verification is required and the PIN has not yet been verified in the current customer session.
3. Process the scheduled transfers list inquiry response data or scheduled transactions list inquiry response data returned from BASE24.
 - a. If the transaction item is identified in the list returned, select it and proceed with step 4.
 - b. If the transaction item is not returned in the list and the response indicates that additional items exist, capture the continuation key returned in the response, determine whether the search is to continue in ascending or descending order, and repeat step 2. Refer to the topic “Transaction Inquiry Continuation” in section 1 for more information on continuing inquiry transactions.
4. Send a scheduled transfer delete (9G) transaction request identifying the transaction (it must include the customer ID or account number and customer reference number) to be deleted.
5. Process the scheduled transfer delete response returned from BASE24 and convey the results of the transaction to the customer.
6. If this is the last transaction for this customer, clear the customer context information.

Retrieving Currently Scheduled Transactions

This subsection contains transaction sequences for the following functions:

- Retrieving scheduled payments
- Retrieving scheduled transfers
- Retrieving scheduled transactions

Once the withdrawal or transfer of funds for a scheduled payment or transfer is made or attempted (including all retry processing), the payment or transfer is considered to be complete. The customer can retrieve a completed payment or transfer by performing the “Retrieving Completed Payments and Transfers” procedure provided elsewhere in this section.

Retrieving Currently Scheduled Payments

The following is a recommended transaction sequence for retrieving currently scheduled payments.

Assumptions: The customer is allowed to perform a scheduled payments list inquiry (3J) or scheduled transactions list inquiry (3M) transaction.

1. Verify the customer, if required. See “Verifying the Customer” elsewhere in this appendix. If the customer has already been successfully verified in the current call to BASE24 or if customer verification is not required, proceed with the next step.
2. Send a scheduled payments list inquiry (3J) transaction request or scheduled transactions list inquiry (3M) transaction request to BASE24 to list scheduled payments for the customer.
 - A customer ID or account number is required.
 - Data element 48 is required for all BASE24-billpay list inquiry transactions. The scheduled payments list inquiry request can also include a customer vendor number to further refine the search.
 - Include the customer PIN if PIN verification is required and the PIN has not yet been verified in the current customer session.
3. Process the scheduled payments list inquiry response data or scheduled transactions list inquiry response data returned from BASE24 and convey the scheduled payment information returned in the response to the customer.
 - a. If the response indicates that no more scheduled payment items exist, proceed with step 4.
 - b. If the response indicates that additional scheduled payment items exist, capture the continuation key returned in the response, determine whether the search is to continue in ascending or descending order, and repeat step 2. Refer to the topic “Transaction Inquiry Continuation” in section 1 for more information on continuing inquiry transactions.
4. If this is the last transaction for this customer, clear the customer context information.

Retrieving Currently Scheduled Transfers

The following is a recommended transaction sequence for retrieving currently scheduled transfers.

Assumptions: The customer is allowed to perform a scheduled transfers list inquiry (3K) or scheduled transactions list inquiry (3M) transaction.

1. Verify the customer, if required. See “Verifying the Customer” elsewhere in this appendix. If the customer has already been successfully verified in the current call to BASE24 or if customer verification is not required, proceed with the next step.
2. Send a scheduled transfers list inquiry (3K) transaction request or scheduled transactions list inquiry (3M) transaction request to BASE24 to list scheduled transfers for the customer.
 - A customer ID or account number is required.
 - Data element 48 is required for all BASE24-billpay list inquiry transactions.
 - Include the customer PIN if PIN verification is required and the PIN has not yet been verified in the current customer session.
3. Process the scheduled transfers list inquiry response data or scheduled transactions list inquiry response data returned from BASE24 and convey the scheduled transfer information returned in the response to the customer.
 - a. If the response indicates that no more scheduled transfer items exist, proceed with step 4.
 - b. If the response indicates that additional scheduled transfer items exist, capture the continuation key returned in the response, determine whether the search is to continue in ascending or descending order, and repeat step 2. Refer to the topic “Transaction Inquiry Continuation” in section 1 for more information on continuing inquiry transactions.
4. If this is the last transaction for this customer, clear the customer context information.

Retrieving Currently Scheduled Transactions

The following is a recommended transaction sequence for retrieving currently scheduled transactions.

Assumptions: The customer is allowed to perform a scheduled transactions list inquiry (3M) transaction.

1. Verify the customer, if required. See “Verifying the Customer” elsewhere in this appendix. If the customer has already been successfully verified in the current call to BASE24 or if customer verification is not required, proceed with the next step.
2. Send a scheduled transactions list inquiry (3M) transaction request to BASE24 to list scheduled payments and transfers for the customer.
 - A customer ID or account number is required.
 - Data element 48 is required for all BASE24-billpay list inquiry transactions.
 - Include the customer PIN if PIN verification is required and the PIN has not yet been verified in the current customer session.
3. Process the scheduled transactions list inquiry response data returned from BASE24 and convey the scheduled transaction information returned in the response to the customer.
 - a. If the response indicates that no more scheduled payment or transfer items exist, proceed with step 4.
 - b. If the response indicates that additional scheduled payment or transfer items exist, capture the continuation key returned in the response, determine whether the search is to continue in ascending or descending order, and repeat step 2. Refer to the topic “Transaction Inquiry Continuation” in section 1 for more information on continuing inquiry transactions.
4. If this is the last transaction for this customer, clear the customer context information.

Entering a User-Defined Transaction

The following is a recommended transaction sequence for entering a user-defined transaction.

Assumptions: The customer is allowed to perform the user-defined (Ux) transaction.

1. Verify the customer, if required. See “Verifying the Customer” elsewhere in this appendix. If the customer has already been successfully verified in the current call to BASE24 or if customer verification is not required, proceed with the next step.
2. Send the user-defined (Ux) transaction request to BASE24.
 - A customer ID or account number is required.
 - Include the customer PIN if PIN verification is required and the PIN has not yet been verified in the current customer session.
3. Process the user-defined transaction response returned from BASE24.
 - a. If the response includes multiple account select information and indicates that no more accounts exist, select the account and repeat steps 2 and 3.
 - b. If the response includes multiple account select information, indicates that more accounts exist, and does not contain the account to be used for the transaction, capture the continuation key, and repeat steps 2 and 3 until a particular account is selected. Refer to the topic “Transaction Inquiry Continuation” in section 1 for more information on continuing inquiry transactions.
 - c. If the response does not include multiple account select information, convey the results of the transaction to the customer and proceed to step 4.
4. If this is the last transaction for this customer, clear the customer context information.

Vendor Management

This subsection contains transaction sequences for the following functions:

- Listing all customer vendors available to the customer
- Requesting a printed Customer Vendor List to be mailed to the customer
- Adding a customer vendor
- Activating or reactivating a customer vendor
- Deactivating a customer vendor
- Updating customer vendor information

Listing Customer Vendors

The following is a recommended transaction sequence for listing customer vendors.

Assumptions: The customer is allowed to perform a customer vendor list inquiry (3H) transaction.

1. Verify the customer, if required. See “Verifying the Customer” elsewhere in this appendix. If the customer has already been successfully verified in the current call to BASE24 or if customer verification is not required, proceed with the next step.
2. Send a customer vendor list inquiry (3H) transaction request to BASE24 to list all customer vendors for the customer.
 - A customer ID or account number is required.
 - Data element 48 is required for all BASE24-billpay list inquiry transactions.
 - Include the customer PIN if PIN verification is required and the PIN has not yet been verified in the current customer session.
3. Process the customer vendor list inquiry response data returned from BASE24 and convey the customer vendor information returned in the response to the customer.
 - a. If the response indicates that no more customer vendor items exist, proceed with step 4.
 - b. If the response indicates that additional customer vendor items exist, capture the continuation key returned in the response, determine whether the search is to continue in ascending or descending order, and repeat step 2. Refer to the topic “Transaction Inquiry Continuation” in section 1 for more information on continuing inquiry transactions.
4. If this is the last transaction for this customer, clear the customer context information.

Requesting a Printed Customer Vendor List

The following is a recommended transaction sequence for requesting a printed customer vendor list.

Assumptions: The customer is allowed to perform a customer vendor request mailing (9T) transaction.

1. Verify the customer, if required. See “Verifying the Customer” elsewhere in this appendix. If the customer has already been successfully verified in the current call to BASE24 or if customer verification is not required, proceed with the next step.
2. Send a customer vendor request mailing (9T) transaction request to BASE24 to print a Customer Vendor List to be mailed to the customer.
 - A customer ID or account number is required.
 - Include the customer PIN if PIN verification is required and the PIN has not yet been verified in the current customer session.
3. Process the customer vendor request mailing response returned from BASE24 and convey the results of the transaction to the customer.
4. If this is the last transaction for this customer, clear the customer context information.

Adding a Customer Vendor

The following is a recommended transaction sequence for adding a customer vendor.

Assumptions: The customer is allowed to perform master vendor list inquiry (9R), customer vendor mask verify list inquiry (BE), and customer vendor add (9N) transactions.

1. Verify the customer, if required. See “Verifying the Customer” elsewhere in this appendix. If the customer has already been successfully verified in the current call to BASE24 or if customer verification is not required, proceed with the next step.
2. Send a master vendor list inquiry (9R) transaction request to BASE24 to list registered vendors for the customer. A vendor name is optional and the search direction defaults to ascending. The vendor name is the key for the search.
 - Data element 48 is required for all BASE24-billpay list inquiry transactions. The master vendor list inquiry request should also include the following filter data indicators to further define the search if defaults are not used:
 - List all vendor branches
 - List current vendors
 - List verified vendors
 - List public vendors
 - Include the customer PIN if PIN verification is required and the PIN has not yet been verified in the current customer session.
3. Process the master vendor list inquiry response data returned from BASE24.
 - a. If the vendor to be added is identified in the list returned, select it and proceed with step 4.
 - b. If the vendor to be added is not returned in the list and the response indicates that additional vendors exist, capture the continuation key returned in the response, determine whether the search is to continue in ascending or descending order, and repeat step 2. The vendor name can be changed if needed to optimize the search. Refer to the topic “Transaction Inquiry Continuation” in section 1 for more information on continuing inquiry transactions.

- c. If it is determined that the customer vendor to be added is not a registered vendor and the customer is allowed to add unregistered vendors, proceed with step 6. The BP-MASTER-VEND-ADD-ALLOWED param in the LCONF determines whether the customer is allowed to add unregistered vendors.
4. Send a customer vendor mask verify list inquiry (BE) transaction request to retrieve a list of all vendor masks for the vendor and vendor branch identified in steps 3 and 4. The customer ID or account number and the vendor branch by name or vendor branch by number keys are required. This inquiry request can be performed in ascending order only.

Data element 48 is required for all BASE24-billpay list inquiry transactions. A customer account with vendor is optional with this request. If included, the account number is validated against each of the vendor masks returned.

5. Process the customer vendor mask verify list inquiry response data returned from BASE24. If a customer account with vendor was provided with the request, each vendor mask item indicates whether the account number was valid for that mask.
 - a. If no more vendor masks exist, present the mask information to the customer and proceed with step 6.
 - b. If more vendor masks exist, present the mask information from the current response to the customer, capture the continuation key returned in the response, and repeat step 4. Refer to the topic “Transaction Inquiry Continuation” in section 1 for more information on continuing inquiry transactions.
 6. Send a customer vendor add (9N) transaction request to BASE24 using the vendor information retrieved from steps 2 through 5. The values for the city, state, postal code, and phone number cannot be changed for registered vendors.

Include a customer vendor account number if a specific account exists with the vendor for this customer. Use the information from steps 5 and 6 to assist in entering a valid account number. If the customer does not have an account number with the vendor, the customer name is typically used as the account with vendor (e.g., JOHNQCUSTOMER). (A value must be placed in this field, even if the customer does not have an actual account with the vendor.)

When the customer adds the customer vendor, the account number is edited against a vendor’s branch account mask if one exists. Otherwise, it is edited against the vendor’s account mask. Edit masks ensure that the account is of the correct format, contains the correct number of digits, begins with a particular combination of digits, etc.

Optionally, enter or change the budget category. Any value the customer enters must already be defined to the system.

Optionally, include a customer vendor short name.

7. Process the customer vendor response data returned from BASE24 and convey the results of the transaction to the customer.
 - When a customer vendor is added, it is assigned a customer vendor number that is one greater than the last customer vendor number used.
 - The account with vendor status field is set to the initial value configured for the customer's financial institution. The account with vendor may need to be verified by back-office personnel before it can be used.
8. If this is the last transaction for this customer, clear the customer context information.

Activating or Reactivating a Customer Vendor

The following is a recommended transaction sequence for activating or reactivating a customer vendor.

Assumptions: The customer is allowed to perform a customer vendor list inquiry (3H) and customer vendor activate (BD) transaction.
If a customer vendor is to be reactivated, the customer's financial institution allows reactivation.

1. Verify the customer, if required. See “Verifying the Customer” elsewhere in this appendix. If the customer has already been successfully verified in the current call to BASE24 or if customer verification is not required, proceed with the next step.
2. If the customer vendor number for the customer vendor to be activated or reactivated is unknown, send a customer vendor list inquiry (3H) transaction request to BASE24 to list all customer vendors for the customer. Otherwise, continue with step 4.
 - Data element 48 is required for all BASE24-billpay list inquiry transactions.
 - Include the customer PIN if PIN verification is required and the PIN has not yet been verified in the current customer session.
3. Process the customer vendor list inquiry response data returned from BASE24.
 - a. If the customer vendor to be activated or reactivated is included in the response, select it and proceed with step 4.
 - b. If the customer vendor to be activated or reactivated is not included in the response and the response indicates that additional customer vendor items exist, capture the continuation key returned in the response, determine whether the search is to continue in ascending or descending order, and repeat step 2. Refer to the topic “Transaction Inquiry Continuation” in section 1 for more information on continuing inquiry transactions.
4. Send a customer vendor activate (BD) transaction request to BASE24. The customer ID or account number and customer vendor number are required.
5. Process the customer vendor activate response data returned from BASE24 and convey the results of the transaction to the customer.
6. If this is the last transaction for this customer, clear the customer context information.

Deactivating a Customer Vendor

The following is a recommended transaction sequence for deactivating a customer vendor.

Assumptions: The customer is allowed to perform a customer vendor list inquiry (3H) and customer vendor deactivate (9P) transaction.

1. Verify the customer, if required. See “Verifying the Customer” elsewhere in this appendix. If the customer has already been successfully verified in the current call to BASE24 or if customer verification is not required, proceed with the next step.
2. If the customer vendor number for the customer vendor to be deactivated is unknown, send a customer vendor list inquiry (3H) transaction request to BASE24 to list all customer vendors for the customer. Otherwise, continue with step 4.
 - Data element 48 is required for all BASE24-billpay list inquiry transactions.
 - Include the customer PIN if PIN verification is required and the PIN has not yet been verified in the current customer session.
3. Process the customer vendor list inquiry response data returned from BASE24.
 - a. If the customer vendor to be deactivated is included in the response, select it and proceed with step 4.
 - b. If the customer vendor to be deactivated is not included in the response and the response indicates that additional customer vendor items exist, capture the continuation key returned in the response, determine whether the search is to continue in ascending or descending order, and repeat step 2. Refer to the topic “Transaction Inquiry Continuation” in section 1 for more information on continuing inquiry transactions.
4. Send a customer vendor deactivate (9P) request to BASE24. The customer ID or account number and customer vendor number are required.
5. Process the customer vendor response data returned from BASE24 and convey the results of the transaction to the customer.
6. If this is the last transaction for this customer, clear the customer context information.

Updating a Customer Vendor

The following is a recommended transaction sequence for updating a customer vendor.

Assumptions: The customer is allowed to perform customer vendor list inquiry (3H), customer vendor mask verify inquiry (BE), and customer vendor update (9Q) transactions.

1. Verify the customer, if required. See “Verifying the Customer” elsewhere in this appendix. If the customer has already been successfully verified in the current call to BASE24 or if customer verification is not required, proceed with the next step.
2. Send a customer vendor list inquiry (3H) transaction request to BASE24 to retrieve the information for the customer vendor to be updated. The default search direction is ascending.
 - A customer ID or account number and customer vendor name or number is required for this request.
 - Data element 48 is required for all BASE24-billpay list inquiry transactions.
 - Include the customer PIN if PIN verification is required and the PIN has not yet been verified in the current customer session.
3. Process the customer vendor list inquiry response data returned from BASE24.
 - a. If the customer vendor to be updated is included in the response, select it and proceed with step 4.
 - b. If the customer vendor to be updated is not included in the response and the response indicates that additional customer vendor items exist, capture the continuation key returned in the response, determine whether the search is to continue in ascending or descending order, and repeat step 2. Refer to the topic “Transaction Inquiry Continuation” in section 1 for more information on continuing inquiry transactions.
4. If updating the account with vendor, send a customer vendor mask verify inquiry (BE) transaction request to retrieve a list of all vendor masks for the vendor and vendor branch identified in steps 2 and 3. The customer ID or account number and the vendor branch by name or vendor branch by number keys are required. This inquiry request can be performed in ascending order only.

Data element 48 is required for all BASE24-billpay list inquiry transactions. A customer account with vendor is optional with this request. If included, the account number is validated against each of the vendor masks returned.

5. Process the customer vendor mask verify list response data returned from BASE24. If a customer account with vendor was provided with the request, each vendor mask item indicates whether the account number was valid for that mask.
 - a. If no more vendor masks exist, present the mask information to the customer and proceed with step 6.
 - b. If more vendor masks exist, present the mask information from the current response to the customer, capture the continuation key returned in the response, and repeat step 4. Refer to the topic “Transaction Inquiry Continuation” in section 1 for more information on continuing inquiry transactions.
6. Send a customer vendor update (9Q) transaction request to BASE24. Use the information retrieved from steps 2 through 5 to build the request.

The Vendor User Remit Info 1 and 2 fields are updated only if the Update Vendor Info flag in the request is set to Y. The Vendor Branch User Remit 1 and 2 fields are updated only if the Update Vendor Branch User Info flag in the request is set to Y.

The following fields cannot be left blank in this request:

- Vendor Name
 - Address 1
 - City
 - State
 - Postal Code
 - Account With Vendor
7. Process the customer vendor response data returned from BASE24 and convey the results of the transaction to the customer.
 8. If this is the last transaction for this customer, clear the customer context information.

Retrieving Previous Transactions

This subsection contains the transaction sequences for the following functions:

- Determining whether a particular check has cleared
- Stopping payment on a check or range of checks
- Retrieving debit transactions
- Retrieving credit transactions
- Retrieving debit and credit transactions for a specified time period
- Retrieving check transactions
- Retrieving transactions entered from a particular source (e.g., a remote banking device)
- Retrieving unscheduled transfer transactions
- Retrieving scheduled payments and transfers
- Retrieving BASE24-billpay transaction history

Determining Whether a Particular Check has Cleared

The following is a recommended transaction sequence for determining whether a particular check has cleared.

Assumptions: The customer is allowed to perform check clearance inquiry (3A) transactions.

1. Verify the customer, if required. See “Verifying the Customer” elsewhere in this appendix. If the customer has already been successfully verified in the current call to BASE24 or if customer verification is not required, proceed with the next step.

2. Send a check clearance inquiry (3A) transaction request to BASE24.
 - A customer ID or account number, and check number are required for this request.
 - A check amount and account number or account type are optional. The check amount must match exactly or no matching record will be found. If an account number or account type is not provided and more than one account exists, multiple account select information is returned.
 - Include the customer PIN if PIN verification is required and the PIN has not yet been verified in the current customer session.
3. Process the transaction history response data returned from BASE24 and convey the check clearance information returned in the response to the customer.
 - a. If the response includes multiple account select information and indicates that no more accounts exist, select the account to be used for the transaction and repeat steps 2 and 3.
 - b. If the response includes multiple account select information, indicates that more accounts exist, and does not include the account to be used for the transaction, capture the continuation key and repeat steps 2 and 3 until a particular account is selected. Refer to the topic “Transaction Inquiry Continuation” in section 1 for more information on continuing inquiry transactions.
 - c. If no more check clearance items need to be listed, proceed with step 4.
 - d. If more check clearance items need to be listed, capture the continuation key returned in the response and repeat steps 2 and 3. Refer to the topic “Transaction Inquiry Continuation” in section 1 for more information on continuing inquiry transactions.
4. If this is the last transaction for this customer, clear the customer context information.

Retrieving Debit Transactions

The following is a recommended transaction sequence for retrieving debit transactions.

Assumptions: The customer is allowed to perform last count of debit transactions inquiry (3E) transactions.

1. Verify the customer, if required. See “Verifying the Customer” elsewhere in this appendix. If the customer has already been successfully verified in the current call to BASE24 or if customer verification is not required, proceed with the next step.
2. Send a last count of debit transactions inquiry (3E) transaction request to BASE24.
 - A customer ID or account number is required for this request.
 - An account number or account qualifier are optional.
 - Include the customer PIN if PIN verification is required and the PIN has not yet been verified in the current customer session.
3. Process the transaction history response data returned from BASE24 and convey the debit transaction information returned in the response to the customer. Debit transaction items are returned in reverse chronological order.
 - a. If the response includes multiple account select information and indicates that no more accounts exist, select the account to be used for the transaction and repeat steps 2 and 3.
 - b. If the response includes multiple account select information, indicates that more accounts exist, and does not include the account to be used for the transaction, capture the continuation key and repeat steps 2 and 3 until a particular account is selected. Refer to the topic “Transaction Inquiry Continuation” in section 1 for more information on continuing inquiry transactions.
 - c. If no more debit transaction items need to be listed, proceed with step 4.
 - d. If more debit transactions need to be listed, capture the continuation key returned in the response and repeat steps 2 and 3. Refer to the topic “Transaction Inquiry Continuation” in section 1 for more information on continuing inquiry transactions.
4. If this is the last transaction for this customer, clear the customer context information.

Retrieving Credit Transactions

The following is a recommended transaction sequence for retrieving credit transactions.

Assumptions: The customer is allowed to perform last count of credit transactions inquiry (3F) transactions.

1. Verify the customer, if required. See “Verifying the Customer” elsewhere in this appendix. If the customer has already been successfully verified in the current call to BASE24 or if customer verification is not required, proceed with the next step.
2. Send a last count of credit transactions inquiry (3F) transaction request to BASE24.
 - A customer ID or account number is required for this request.
 - An account number or account qualifier are optional.
 - Include the customer PIN if PIN verification is required and the PIN has not yet been verified in the current customer session.
3. Process the transaction history response data returned from BASE24 and convey the credit transaction information returned in the response to the customer. Credit transaction items are returned in reverse chronological order.
 - a. If the response includes multiple account select information and indicates that no more accounts exist, select the account to be used for the transaction and repeat steps 2 and 3.
 - b. If the response includes multiple account select information, indicates that more accounts exist, and does not contain the account to be used for the transaction, capture the continuation key and repeat steps 2 and 3 until a particular account is selected. Refer to the topic “Transaction Inquiry Continuation” in section 1 for more information on continuing inquiry transactions.
 - c. If no more credit transaction items need to be listed, proceed with step 4.
 - d. If more credit transaction items need to be listed, capture the continuation key returned in the response and repeat steps 2 and 3. Refer to the topic “Transaction Inquiry Continuation” in section 1 for more information on continuing inquiry transactions.
4. If this is the last transaction for this customer, clear the customer context information.

Retrieving Debit and Credit Transactions

The following is a recommended transaction sequence for retrieving debit and credit transactions.

Assumptions: The customer is allowed to perform last count of debit/credit transactions inquiry (3B) transactions.

1. Verify the customer, if required. See “Verifying the Customer” elsewhere in this appendix. If the customer has already been successfully verified in the current call to BASE24 or if customer verification is not required, proceed with the next step.
2. Send a last count of debit/credit transactions inquiry (3B) transaction request to BASE24.
 - A customer ID or account number is required for this request.
 - An account number or account qualifier, as well as a start date and end date, are optional.
 - Include the customer PIN if PIN verification is required and the PIN has not yet been verified in the current customer session.
3. Process the transaction history response data returned from BASE24 and convey the transaction information returned in the response to the customer. Debit and credit transaction items are returned in reverse chronological order.
 - a. If the response includes multiple account select information and indicates that no more accounts exist, select the account to be used for the transaction and repeat steps 2 and 3.
 - b. If the response includes multiple account select information, indicates that more accounts exist, and does not contain the account to be used for the transaction, capture the continuation key and repeat steps 2 and 3 until a particular account is selected. Refer to the topic “Transaction Inquiry Continuation” in section 1 for more information on continuing inquiry transactions.
 - c. If no more debit or credit transaction items need to be listed, proceed with step 4.
 - d. If more debit or credit transaction items need to be listed, capture the continuation key returned in the response and repeat steps 2 and 3. Refer to the topic “Transaction Inquiry Continuation” in section 1 for more information on continuing inquiry transactions.
4. If this is the last transaction for this customer, clear the customer context information.

Retrieving Check Transactions

The following is a recommended transaction sequence for retrieving check transactions.

Assumptions: The customer is allowed to perform last count of check transactions inquiry (3D) transactions.

1. Verify the customer, if required. See “Verifying the Customer” elsewhere in this appendix. If the customer has already been successfully verified in the current call to BASE24 or if customer verification is not required, proceed with the next step.
2. Send a last count of check transactions inquiry (3D) transaction request to BASE24.
 - A customer ID or account number is required for this request.
 - An account number or account qualifier, as well as a check amount, are optional.
 - Include the customer PIN if PIN verification is required and the PIN has not yet been verified in the current customer session.
3. Process the transaction history response data returned from BASE24 and convey the check transaction information returned in the response to the customer. Check transaction items are returned in reverse chronological order.
 - a. If the response includes multiple account select information and indicates that no more accounts exist, select the account to be used for the transaction and repeat steps 2 and 3.
 - b. If the response includes multiple account select information, indicates that more accounts exist, and does not contain the account to be used for the transaction, capture the continuation key and repeat steps 2 and 3 until a particular account is selected. Refer to the topic “Transaction Inquiry Continuation” in section 1 for more information on continuing inquiry transactions.
 - c. If no more check transaction items need to be listed, proceed with step 4.
 - d. If more check transaction items need to be listed, capture the continuation key returned in the response and repeat steps 2 and 3. Refer to the topic “Transaction Inquiry Continuation” in section 1 for more information on continuing inquiry transactions.
4. If this is the last transaction for this customer, clear the customer context information.

Retrieving Transactions By Source

The following is a recommended transaction sequence for retrieving transactions by source.

Assumptions: The customer is allowed to perform last count of account by source transactions inquiry (3C) transactions.

1. Verify the customer, if required. See “Verifying the Customer” elsewhere in this appendix. If the customer has already been successfully verified in the current call to BASE24 or if customer verification is not required, proceed with the next step.
2. Send a last count of account by source transactions inquiry (3C) transaction request to BASE24.
 - A customer ID or account number, and source code are required for this request.
 - An account number or account qualifier are optional.
 - Include the customer PIN if PIN verification is required and the PIN has not yet been verified in the current customer session.
3. Process the transaction history response data returned from BASE24 and convey the transaction information returned in the response to the customer. Transaction items are returned in reverse chronological order.
 - a. If the response includes multiple account select information and indicates that no more accounts exist, select the account to be used for the transaction and repeat steps 2 and 3.
 - b. If the response includes multiple account select information, indicates that more accounts exist, and does not contain the account to be used for the transaction, capture the continuation key and repeat steps 2 and 3 until a particular account is selected. Refer to the topic “Transaction Inquiry Continuation” in section 1 for more information on continuing inquiry transactions.
 - c. If no more transaction items need to be listed for the specified source, proceed with step 4.
 - d. If more transaction items need to be listed for the specified source, capture the continuation key returned in the response and repeat steps 2 and 3. Refer to the topic “Transaction Inquiry Continuation” in section 1 for more information on continuing inquiry transactions.
4. If this is the last transaction for this customer, clear the customer context information.

Retrieving Unscheduled Transfers

The following is a recommended transaction sequence for retrieving unscheduled transfers. For this sequence, unscheduled refers to any transfers that have already been performed and are no longer scheduled.

Assumptions: The customer is allowed to perform last count of account transfer transactions inquiry (3G) transactions.

1. Verify the customer, if required. See “Verifying the Customer” elsewhere in this appendix. If the customer has already been successfully verified in the current call to BASE24 or if customer verification is not required, proceed with the next step.
2. Send a last count of account transfer transactions inquiry (3G) transaction request to BASE24.
 - A customer ID or account number is required for this request.
 - An account number or account qualifier are optional.
 - Include the customer PIN if PIN verification is required and the PIN has not yet been verified in the current customer session.
3. Process the transaction history response data returned from BASE24 and convey the transfer transaction information returned in the response to the customer. Transaction items are returned in reverse chronological order.
 - a. If the response includes multiple account select information and indicates that no more accounts exist, select the account to be used for the transaction and repeat steps 2 and 3.
 - b. If the response includes multiple account select information, indicates that more accounts exist, and does not contain the account to be used for the transaction, capture the continuation key and repeat steps 2 and 3 until a particular account is selected. Refer to the topic “Transaction Inquiry Continuation” in section 1 for more information on continuing inquiry transactions.
 - c. If no more transfer transaction items need to be listed for the specified source, proceed with step 4.
 - d. If more transaction items need to be listed for the specified source, capture the continuation key returned in the response and repeat steps 2 and 3. Refer to the topic “Transaction Inquiry Continuation” in section 1 for more information on continuing inquiry transactions.
4. If this is the last transaction for this customer, clear the customer context information.

Downloading Statement Information

The following is a recommended transaction sequence for downloading statement information on a particular account.

Assumptions: The customer is allowed to perform statement download (3R) or statement closing download (3Q) transactions.

1. Verify the customer, if required. See “Verifying the Customer” elsewhere in this appendix. If the customer has already been successfully verified in the current call to BASE24 or if customer verification is not required, proceed with the next step.
2. Send a statement download (3R) or statement closing download (3Q) transaction request to BASE24.
 - A customer ID or account number is required for this request.
 - An account number or account qualifier, as well as a start date and end date, are optional.
 - Include the customer PIN if PIN verification is required and the PIN has not yet been verified in the current customer session.
3. Process the statement download (3R) or statement closing download (3Q) response data returned from BASE24 and convey the statement information returned in the response to the customer. Transaction items are returned in the order determined by the host.
 - a. If the response includes multiple account select information and indicates that no more accounts exist, select the account to be used for the transaction and repeat steps 2 and 3.
 - b. If the response includes multiple account select information, indicates that more accounts exist, and does not contain the account to be used for the transaction, capture the continuation key and repeat steps 2 and 3 until a particular account is selected. Refer to the topic “Transaction Inquiry Continuation” in section 1 for more information on continuing inquiry transactions.
 - c. If no more transaction items need to be retrieved for the account, proceed with step 4.
 - d. If more transaction items need to be retrieved for the account, capture the continuation key returned in the response and repeat steps 2 and 3. Refer to the topic “Transaction Inquiry Continuation” in section 1 for more information on continuing inquiry transactions.

4. If this is the last transaction for this customer, clear the customer context information.

Retrieving Completed Payments and Transfers

The following is a recommended transaction sequence for retrieving completed payments and transfers.

Assumptions: The customer is allowed to perform history inquiry – payments and transfers (3L) transactions.

1. Verify the customer, if required. See “Verifying the Customer” elsewhere in this appendix. If the customer has already been successfully verified in the current call to BASE24 or if customer verification is not required, proceed with the next step.
2. Send a history inquiry – payments and transfers (3L) transaction request to BASE24. The inquiry defaults to search by business date. Transaction items can be listed in ascending or descending order. The default is to search by business date in descending order.
 - A customer ID or account number is required for this request.
 - A business date or customer reference number are optional.
 - Data element 48 is required for all BASE24-billpay list inquiry transactions. The history inquiry – payments and transfers request can also include a flag indicating whether only approved, denied, or both approved or denied transactions are returned.
 - Include the customer PIN if PIN verification is required and the PIN has not yet been verified in the current customer session.
3. Process the history inquiry response data returned from BASE24 and convey the transaction history information returned in the response to the customer.
 - a. If the response indicates that no more payment or transfer transaction items exist for the specified criteria, proceed with step 4.
 - b. If the response indicates that additional payment or transfer transaction items exist for the specified source, capture the continuation key returned in the response and repeat step 2. Refer to the topic “Transaction Inquiry Continuation” in section 1 for more information on continuing inquiry transactions.
4. If this is the last transaction for this customer, clear the customer context information.

Retrieving BASE24-billpay History Transactions

The following is a recommended transaction sequence for retrieving all BASE24-billpay transactions.

Assumptions: The customer is allowed to perform a history inquiry – all transactions (9X) transaction.

1. Verify the customer, if required. See “Verifying the Customer” elsewhere in this appendix. If the customer has already been successfully verified in the current call to BASE24 or if customer verification is not required, proceed with the next step.
2. Send a history inquiry – all transactions (9X) transaction request to BASE24. The inquiry defaults to search by business date. Transaction items can be listed in ascending or descending order. The default is to search by business date in descending order.
 - A customer ID or account number is required for this request.
 - A business date or customer reference number are optional.
 - Data element 48 is required for all BASE24-billpay list inquiry transactions. The history inquiry – all transactions request can also include a flag indicating whether only approved, denied, or both approved or denied transactions are returned.
 - Include the customer PIN if PIN verification is required and the PIN has not yet been verified in the current customer session.
3. Process the history inquiry response data returned from BASE24 and convey the transaction history information returned in the response to the customer.
 - a. If the response indicates that no more financial, nonfinancial, or maintenance transaction items exist for the specified criteria, proceed with step 4.
 - b. If the response indicates that additional financial, nonfinancial, or maintenance transaction items exist for the specified source, capture the continuation key returned in the response and repeat step 2. Refer to the topic “Transaction Inquiry Continuation” in section 1 for more information on continuing inquiry transactions.
4. If this is the last transaction for this customer, clear the customer context information.

Stopping Payment on a Check or Range of Checks

The following is a recommended transaction sequence for stopping payment on a check or range of checks.

Assumptions: The customer is allowed to perform check stop payment add (A0) transactions.

1. Verify the customer, if required. See “Verifying the Customer” elsewhere in this appendix. If the customer has already been successfully verified in the current call to BASE24 or if customer verification is not required, proceed with the next step.
2. Send a check stop payment add (A0) transaction request to BASE24.
 - A customer ID or account number, check number or range of numbers, and expiration date are required for this request.
 - A range of check numbers, check amount, check date, description, and the date and time the stop payment request was made are optional. If an account number or account type is not provided and more than one account exists, multiple account select information is returned. If the stop payment date and time is not provided, it defaults to the current date and time.
 - Include the customer PIN if PIN verification is required and the PIN has not yet been verified in the current customer session.
3. Process the check stop payment add response data returned from BASE24 and convey the results of the transaction to the customer.
 - a. If the response includes multiple account select information and indicates that no more accounts exist, select the account to be used for the transaction and repeat steps 2 and 3.
 - b. If the response includes multiple account select information, indicates that more accounts exist, and does not include the account to be used for the transaction, capture the continuation key and repeat steps 2 and 3 until a particular account is selected. Refer to the topic “Transaction Inquiry Continuation” in section 1 for more information on continuing inquiry transactions.
 - c. If the response does not include multiple account select information, proceed to step 4.
4. If this is the last transaction for this customer, clear the customer context information.

ACI Worldwide Inc.

Glossary

This glossary defines common terms you will encounter when using the BASE24 Remote Banking Standard Interface.

account mask — *See* vendor mask.

account qualifier — A user-defined code that can be used in place of the application account number and account type to identify and access an account. The account qualifier uniquely identifies an account to the customer. For example, if a customer has two savings accounts, account qualifiers of SAV and SAV1 could be used to differentiate the accounts. Account qualifiers serve as a convenient means of identifying accounts when performing transactions without having to provide the full account number.

account status — A code indicating the current status of the customer's application account. For some transactions, only application accounts with a verify status that allows the requested transaction can be used in transaction processing. For example, one account may have an open status, which allows for any transaction, while another account may have a status of restricted to deposits, which means the account accepts all transactions except transfers, scheduled transfers, payments, and scheduled payments from this account.

account type — A code identifying the type of customer application account. BASE24 provides a default account type code for each of the following types of accounts (the default account type is provided in parentheses): savings (SAV), checking (DDA), credit (CR), line of credit (LOC), certificate (CD), retirement account (IRA), interest-bearing checking account (NOW), commercial loan (CMRCL), installment loan (INSTL), and mortgage loan (MRTGL).

account with vendor — The customer's account with a vendor that is credited in a payment transaction.

account with vendor verify status — A code indicating the current status of the customer's account with a vendor. Only customer vendors with an account with vendor verify status of verified can be used in transaction processing.

action code — A three-character code that indicates the reason why a transaction was approved or declined. For descriptions of all possible action codes, refer to the *BASE24 Remote Banking Transaction Processing Manual*.

ASCII — A code that follows the American Standard Code for Information Interchange.

American National Standards Institute (ANSI) — An organization that defines standards for the United States, including standards for the exchange of electronic messages relating to financial transactions. ANSI is the primary source and official sales agent for ISO standards in the United States.

approval code — A random number generated by the authorizer for an approved financial transaction. An approval code is not generated for a financial transaction that is denied. An approval code is provided only if one is generated by the financial transaction authorizer. Some authorizers, such as a host, may not generate an approval code for an approved financial transaction.

approved transaction — Any transaction for which authorization is granted.

automated clearing house (ACH) — The Federal Reserve Bank or other organization operating as a clearing house for automated entries (electronic payments). The ACH processes entries between an originating depository financial institution and a receiving depository financial institution. Electronic payment remittance to vendors is always sent through an ACH.

billing group — Defines the customer billing statement cycle for a customer.

billing statement cycle — The length of period (for example, monthly or quarterly) and dates (for example, the first or tenth of each month) for which transaction activity is billed to the customer.

billing type — Defines the service level and fee structure for the billpay service.

binary — A number system with 2 as its base.

bit — In the binary number system, either of the digits 0 or 1. A bit is the smallest unit of memory or data stored in memory.

bit map — A series of 64 bits that identifies the presence or absence of a data element in an ISO message. BASE24 uses 16 hexadecimal characters to represent these 64 bits. Each bit identifies the presence (denoted by 1 or bit on) or absence (denoted by 0 or bit off) of a data element in the message. The primary bit map identifies the presence or absence of data elements 1 through 64. An optional secondary bit map identifies the presence or absence of data elements 65 through 128.

BSTM — *See* Telebanking Standard Internal Message (BSTM).

budget category — A one to four character alphanumeric code used to group payments to vendors for reporting purposes. For example, a category of UTIL could be used to group all payments a customer makes to vendors for utility services while a category of MRTG could be used to group all payments made to vendors for a home mortgage.

canceled transaction — Any transaction that is approved and then subsequently canceled. When a transaction is canceled, a reversal is generated to back out the originally approved transaction. An approved nonfinancial transaction can only be canceled if allowed by the customer's financial institution. A time limit may also apply as to how long after a transaction is approved it may still be canceled.

closed — A verify status value for customers and customer accounts that indicates that a customer ID or customer account cannot be used in processing unless it is reactivated.

completed transaction — Any transaction for which all actions (including any retry processing) have been completed. For scheduled future and recurring payments and transfers, the payment or transfer is not considered to be complete until the actual withdrawal or transfer of funds has been made or after all retries have been exhausted.

cryptographic service message (CSM) — A free-format ANSI X9.17 standard data element that is used to move key management data between two entities when the keys are distributed automatically.

customer — An individual or business who uses BASE24-billpay or BASE24-telebanking services to manage their bill payment or remote banking needs. Customers can perform remote banking transactions from remote banking devices such as a touch-tone telephone, or they can call a customer service representative to perform remote banking transactions on their behalf.

customer ID — A value that uniquely identifies a remote banking customer to the system. A customer can have an unlimited number of application accounts accessed from his or her customer ID.

customer profile — A code used to group the services (i.e., allowed transactions) available for the customer's ID. A customer profile allows customers to be associated with a particular level of service (e.g., basic, introductory, premium, etc.) defined by the customer's financial institution.

customer vendor — Each vendor to receive payments from a customer is considered a customer vendor. If a customer makes payments to more than one account with the same vendor, each account relationship with that vendor is considered to be a separate customer vendor.

customer vendor number — A one- to five-digit code used by the customer to identify the customer vendor in payment transactions. The customer vendor number is automatically assigned by the system to each of the customer's vendors. Each customer vendor number is related to an internal vendor number.

customer verify status — A code indicating the current status of the customer's ID. Only customer IDs with a customer verify status of verified can be used in financial transaction processing. Customer information can be added, activated, deactivated, updated, and inquired upon, regardless of the current customer ID verify status.

An active customer has a customer ID with a verify status of verified and an effective date range that includes the current date.

An inactive customer is defined by either of the following:

- A customer ID with a verify status of issued but inactive or closed
- An ending date for the effective date range that is less than the current date or a beginning date that is after the current date

declined transaction — Any transaction for which authorization is denied.

default account number — The number of the account for which balance information is returned to the customer with PIN verify transactions.

effective date range — A range of dates for which a customer ID, customer application account, customer account with a vendor, financial institution identifier, or vendor can be used in transaction processing.

financial transaction — Any transaction that credits or debits a customer account. Payments and transfers are financial transactions; however, transactions that schedule payments or transfers are nonfinancial transactions.

from account — The account debited in a transfer or payment transaction. For the Remote Banking Standard Interface, the *from* account is always defined as account 1.

future transaction — A payment or transfer transaction scheduled to be performed once on a future date.

group-item check — A written order from multiple customers belonging to the same financial institution to pay a vendor an amount from funds on deposit. A list is attached to the check that indicates the amount paid by each individual customer. The amount of the check is the total amount paid by all customers listed.

hexadecimal — A number system with 16 as its base.

initial — A verify status value for customers, customer accounts, customer vendors, and vendors that indicates that customer ID, customer account, customer vendor, or vendor information has been entered into the database but has not yet been verified. For this verify status value, information can be added, updated, inquired upon, or activated, but cannot be used in financial transaction processing. When customer IDs, customer accounts, and customer vendors are activated, their verify status is changed from initial to verified.

inquiry — A historical query on previous transactions performed (for example, to determine whether a specific check has cleared) or a query on current configured information (for example, listing all the customer vendors for a customer).

Internal Transaction Data (ITD) — The internal message format used to pass BASE24 Remote Banking transaction information between objects in integrated server processes. The ITD is placed in the BSTM for movement to procedural processes.

International Organization for Standardization (ISO) — An international organization that defines international standards, including standards for the exchange of electronic messages relating to financial transactions. Refer to section 1 for information on ordering ISO standards documents.

ISO — See International Organization for Standardization (ISO).

multiple account selection — If a transaction is initiated and two or more customer accounts with the same specified account number or account type exist, BASE24 responds with a list of all accounts that match the specified data. The specific accounts on which the transaction is to be performed can then be selected by the customer. This is known as multiple account selection and is available only if it is enabled on the BASE24 Remote Banking system. For a discussion of how multiple account selection is enabled, refer to the “Multiple Account Selection” discussion provided in appendix C.

nonfinancial transaction — Any transaction that does not credit or debit a customer account. All inquiries, as well as transactions that schedule payments or transfers, are nonfinancial transactions.

paid transaction — Any payment or transfer that was originally scheduled and has now been successfully paid or completed. For automated clearinghouse (ACH) payments, a configurable number of days must also elapse after the ACH payment has been approved and logged to the History Table (HIST) before the payment is considered to be paid. If the final disposition for a scheduled payment or transfer indicates that it failed after all retries are exhausted, it is not considered to be a paid transaction.

payment — The payment of funds from a customer account to a vendor (also known as a biller or payee). External transfers between two accounts owned by different financial institutions must be performed as payment transactions, where the financial institution owning the *to* account is configured as a customer vendor and the *to* account is configured as the customer’s account with the vendor.

personal identification number — A confidential code consisting of 1 to 16 numeric characters known as a PIN that provides secure access to a customer's ID. A customer can establish a different PIN for BASE24-telebanking and BASE24-billpay transactions than the one he or she uses for transactions performed with card-based BASE24 products such as BASE24-atm and BASE24-pos.

preloaded customer data — Customer information that is automatically loaded into the customer database from an existing database or tape. Financial institutions determine whether customer information is preloaded or not. If customer information is not preloaded, it must be manually entered after default customer data is applied.

recurring transaction — A payment or transfer transaction scheduled to be performed on a recurring period (for example, weekly or monthly) for a set number of periods.

remittance — The payment provided to a vendor. Remittance can be in the form of single-item checks, group-item checks, or electronic automated clearinghouse (ACH) payments. BASE24-billpay generates the raw data for remittance in files that can then be input to another application to generate the actual remittance.

reversal — A transaction generated to back out an originally approved transaction. *See also* canceled transaction.

scheduled payment or transfer — A payment or transfer that is scheduled to be performed on a one-time future date or on a recurring basis. *See also* future transaction and recurring transaction.

single-item check — A written order from a single customer's financial institution to pay a vendor an amount from the customer's funds on deposit.

Telebanking Standard Internal Message (BSTM) — The internal message format used between the object-oriented Integrated Authorization Server process and procedural (i.e., nonobject-oriented) processes. The ITD, which carries all transaction data, is placed within the BSTM.

to account — The account credited in a transfer transaction. For the Remote Banking Standard Interface, the *to* account is always defined as account 2.

transfer — A BASE24-telebanking transaction that moves funds from one customer account to another customer account owned by the same financial institution. Transfers can be performed immediately or can be scheduled to be performed on a one-time future date or on a recurring basis. This type of transfer is defined as an internal transfer.

An external transfer is defined as the movement of funds from one account to another when the two accounts are owned by different financial institutions. Note that external transfers can only be performed as BASE24-billpay payments, where the *to* account is defined as the account with vendor and the financial institution owning the *to* account is defined as the vendor. The BASE24-telebanking product and Enhanced Telebanking product modules only allow internal transfers.

unscheduled transfer — Any transfer that has already been performed and is no longer scheduled.

user-defined — Any transaction outside of the BASE24-telebanking transaction set that is defined by the user (that is, financial institution or service provider). BASE24-billpay does not support user-defined transactions.

vendor — The recipient of a payment transaction (also referred to as the payee). When adding customer vendors, a vendor that is already defined to the Vendor Table (VNDR) is considered to be a registered vendor. A vendor that is not already defined to the VNDR is considered to be an unregistered vendor. The BP-MASTER-VEND-ADD-ALLOWED param in the LCONF determines whether customer service representatives or branch employees are allowed to add customer vendors for unregistered vendors.

vendor mask — An edit check for verifying the account numbers that customers have with vendors. These masks are used when customer accounts with vendors are first entered into the system and help customers and customer service representatives to enter the customer account with the vendor correctly. Each vendor can have multiple masks. An edit check can be any of the following types: account number minimum and maximum length check, account mask matching (i.e., an account number template with wildcard characters), or user-defined algorithm check. Account mask matching checks the entered account number against a template of text and wildcard characters. For detailed information on account masks, refer to the *BASE24-billpay Tables Maintenance Manual*.

verify status — A value that indicates the current status for customers, customer accounts, customer vendors, and vendors. This value indicates how a customer ID, customer account, customer vendor, or vendor can be used in processing. Only customer IDs, customer accounts, customer vendors, or vendors with a verify status of verified can be used in financial transaction processing. *See also* closed, initial, and verified.

verified — A verify status value for customers, customer accounts, customer vendors, and vendors that indicates that a customer ID, customer account, customer vendor, or vendor information has been entered and verified. For this verify status value, information can be fully used in both financial and nonfinancial transaction processing.

ACI Worldwide Inc.

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